ALL IN ONE

VETERINARY CAPSULE 2021

A SUBJECTIVE TYPE MANUAL

For the Preparation of Veterinary Written Test Exams and Interviews



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DEDICATION:

“IT IS OUR GENUINE GRATEFULNESS AND WARMEST REGARD THAT WE DEDICATE THIS WORK TO THOSE PARALYZED MINDS WHO CAN’T EVEN IMAGINE THE OTHER’S WELL-BEING.”

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HISTORY OF VETERINARY MEDICINE:

- ✓ The word "Veterinary" comes from the Latin word ***veterinae*** meaning **"working animals"**. **"Veterinarian"** was first used in print by **Thomas Browne** in 1646.
- ✓ Archeological evidence, in the form of a cow skull upon which **trepanation** had been performed, shows that people were performing veterinary procedures in the Neolithic (3400–3000 BCE) (stone age). Persons serving as doctors to animals have existed since the earliest recorded times, and veterinary practice was already established as a specialty as early as 2000 BCE in Babylonia and **Egypt.**
- ✓ The **ancient Greeks** had a class of physicians who were called "horsedoctors," and the Latin term for the specialty, *veterinarius* ("pertaining to beast of burden"), came to denote the field in modern times.
- ✓ In ancient times, trepanation was thought to be a treatment for various ailments, such as head injuries. It may also have been used to treat pain. Some scientists also think that the practice was used to pull spirits from the body in rituals. Many times, the person would survive and heal after the Surgery
- ✓ The first veterinary school was founded in **Lyon**, France, in 1761 by **Claude Bourgelat**. 2nd school in 1776 in Paris.
- ✓ First mention of Rabies was reported in Eshuna code of 2300 B.C.
- ✓ Ashoka (269-232 B.C.) made compulsory provision of charitable animal hospitals.
- ✓ Royal charter (1844) recognized the Veterinary Art as a profession.
- ✓ After observing the devastation being caused by **cattle plague** to the French herds, Bourgelat devoted his time to seeking out a remedy. This resulted in his founding a veterinary school in Lyon in 1761, from which establishment he dispatched students to combat the disease; in a short time, the plague was stayed and the health of stock restored, through the assistance rendered to agriculture by veterinary science and art.
- ✓ First written record of veterinary medicine from ancient Egypt was provided by Kahum Papyrus (1900 BC).

LIVESTOCK SHARE 2020-21

Livestock having a share of 60.1 percent in Agriculture 11.53 percent in GDP achieved a growth of 3.06 percent. More than 8 million rural families are engaged in Livestock production and deriving more than 35-40 percent of their income from this source. Gross value addition of livestock increased to Rs. 1505 billion (2020-21). The fishing sector, with a share of 2.01 percent in agriculture value addition and 0.39 % in GDP, grew by 0.73 percent, while forestry sector having share of 2.10 percent in Agriculture and 0.40 percent in GDP grew by 1.42 percent.

ESTIMATED MILK AND MEAT PRODUCTION (2020-21):

Parameter	Production in Tons
Milk	63,684
Meat	4,955

ESTIMATED LIVESTOCK POPULATION (2020-21):

Species	Population No. (In Million)
Cattle	51.5
Buffalo	42.4
Sheep	31.6
Goat	80.3
Camels	1.1
Horses	0.4
Asses	5.6
Mules	0.2

FATHERS OF FIELDS:

1. Father of Blood Groups: Landsteiner
2. Father of Taxonomy: Carolus Linnaeus
3. Father of Surgery: Sushruta
4. Father of Medicine: Hippocrates
5. Father of Veterinary Medicine: Renatius
6. Father of Zoology: Aristotle
7. Cardinal Signs: Celsius
8. Antemortem: Galen
9. Modern Pathology: Morgagni
10. Father of Histology: Bichat
11. Father of Cellular Pathology: Virchow
12. Experimental Pathologist: Carl
13. Necropsy Technique: Hunter
14. Father of Microbiology: Leeuwenhoek
15. Father of Modern Microbiology: Pasteur
16. Father of Bacteriology: Koch
17. Modern Experimental Pathology: Julius
18. Inclusion Body: Negri
19. Inventor of Electron Microscope: Knoll and Ruska
20. Demonstration of Arthropoda: Smith
21. Father of Pathology: Benevieni
22. Phagocytosis Discovery: Metchnikoff
23. Father of Immunology: Edward Jenner
24. Nucleus: Bang
25. ELISA: Engwal and Parlman
26. Discovery of “Restriction Endonuclease": Kelly & Smith
27. Father of Nutrition: Antoine Lavoisier
28. Father of extension education: J.P. Leagans

SCIENTIST AND THEIR CONTRIBUTION:

- ✓ Gregor Johann Mendel: Father of Genetics
- ✓ Bateson: Term Genetics
- ✓ Johannsen: Term Gene, Genotype, Phenotype
- ✓ Lamarck: Theory of inheritance of acquired characters, Use and disuse of organ
- ✓ August Weismann: Theory of continuity of germplasm
- ✓ Charles Darwin: Theory of natural selection, Theory of Pangenesis
- ✓ Hugo de Vries: Mutation theory of evolution
- ✓ Jacob & Monod: Operon concept (gene regulation in prokaryotes)
- ✓ Beadle & Tatum: One Gene One Enzyme hypothesis
- ✓ Schleiden & Schwann: Cell Theory
- ✓ G.H. Shull: Term Heterosis/ Hybrid vigour
- ✓ Karl Landsteiner: Blood group and Rh factor
- ✓ Morgan: Sex-linked inheritance, chromosome theory of sex determination
- ✓ C.B.Bridge: Genic balance theory of sex determination
- ✓ Har Gobind Khorana, Nirenberg & Holley: Genetic code
- ✓ Barbara McClintock: Transposons /Transposable elements of "jumping genes"
- ✓ Knoll & Ruska: Electron microscope
- ✓ Sutton & Boveri: Chromosomal theory of heredity
- ✓ Waldeyer: Term chromosome
- ✓ Robert Hooke: Discovery of cell
- ✓ Robert Brown: Discovery of cell nucleus
- ✓ Watson and Crick: Double helical model of DNA
- ✓ CIB method- The method devised by Muller for detecting X-linked mutations in Drosophila
- ✓ Ascorbic acid was first isolated by Szent Gyorg.
- ✓ Scandinavian feeding system based on barley as the standard is introduced by Hanssen.
- ✓ Physiological fuel values were devised by Atwater.
- ✓ First accurate respiration calorimeter was constructed by Rubner.
- ✓ Starch equivalent system was designed by Kellner.
- ✓ The term “protein” is coined by Mulder.
- ✓ Balance and thermometer into nutrition studied for the first time by Antoine Lavoisier.
- ✓ Chemical method of DNA sequencing was developed by Maxam and Gilbert.
- ✓ Genetic slippage term was given by G. E. Dickerson.

HISTORIC MILESTONES/ DISCOVERIES:

- ✓ Louis Pasteur - coined term vaccine, Fowl cholera vaccine, Rabbies vaccine, Anthrax vaccine and concept of attenuation.
- ✓ Paul Ehrlich - Antibody production, Humoral theory, Acid fast staining, Father of chemotherapy.
- ✓ Edward Jenner – Small pox vaccination
- ✓ Waksman – Streptomycin discovery
- ✓ Bruce – Malta fever
- ✓ Loeffer and Shutz – Glander
- ✓ Moore – Fowl typhoid
- ✓ Nicolaier – Tetanus
- ✓ Ricketts – Rocky spotted mountain fever
- ✓ Bordet – Complement discovery
- ✓ Loffer – Swine erysipelas
- ✓ A. Flemming – First antibiotics as Penicillin
- ✓ J. Lister – Aseptic surgery, Carbolic acid
- ✓ R. Bucchim – Father of Pharmacology
- ✓ M.J.B. Orfla – Father of Toxicology
- ✓ Otto-lewi – Neurotransmitter discovery
- ✓ O.W. Holmes – Coined term Anaesthesia
- ✓ Rudolphi – Father of Parasitology
- ✓ Virchow – Father of Cellular Pathology
- ✓ Kohler & Milstein – Hybridoma (Monoclonal antibodies) technique
- ✓ E. Porter – Structure of antibodies
- ✓ Landsteiner – Blood Group
- ✓ Celsus – Four cardinal signs of inflammation

SCIENTIFIC NAMES OF ANIMALS:

- ✓ Alpaca- Llama pacos
- ✓ Ass- Equus asinus
- ✓ Buffalo- Bubalus bubalis
- ✓ Camel- Camelus dromedarianus (Single hump); Camelus bacterianus (Double hump)
- ✓ Cat- Felis domesticus
- ✓ Cattle- Bos indicus / Bos taurus
- ✓ Dog- Canis familaris
- ✓ Goat- Capra hircus
- ✓ Guanaco- Lama guanicoe
- ✓ Horse - Equus caballus
- ✓ Llama- Llama glama
- ✓ Mithun- Bos gaurus
- ✓ Pig- Sus scrofa
- ✓ Rabbit- Oryctolagus cuniculus
- ✓ Reindeer- Rangifer tarandus
- ✓ Sheep- Ovis aries
- ✓ Vicuna- Vicugna vicugna
- ✓ Yalk- Bos grunniens

SCIENTIFIC NAMES OF BIRDS:

- ✓ Chicken- Gallus domesticus
- ✓ Duck- Anas platyrhynchos
- ✓ Geese- Anser anser
- ✓ Golden eagle- Aquila chrysactes
- ✓ Guinea fowl- Numida meleagris
- ✓ Muscovy- Cainna maschata
- ✓ Ostrich- Struthio camelus
- ✓ Pea fowl (peacock)- Pavo cristatus
- ✓ Pigeon- Columba livia
- ✓ Pheasants- Phasians colchrices
- ✓ Quail- Coturnix spp.
- ✓ Red horned owl- Bubo bubo
- ✓ Screech owl- Tyto albo
- ✓ Swan- Cygnus spp.
- ✓ Turkey- Meleagris gallopovo

LIFE SPAN OF ANIMALS:

Sr. No.	Animal	Life Span (Years)
01.	Bull	28
02.	Cow	22
03.	Chicken, Goat and Sheep	15
04.	Quail	6
05.	Horse	40
06.	Dog	22
07.	Cat	25
08.	Elephant	70
09.	Parrot	80
10.	Mouse	4
11.	Camel	50
12.	Rabbit	9
13.	Ox	20

GESTATION PERIOD:

Animal	Gestation Period
Cattle	283 days (9 month+ 9days)
Buffalo	310 days (10months+10days)
Sheep	145 days (5 months−5days)
Goat	155 days (5months+5days)
Bitch	62 days (2months+2days)
Sow	114 days (3months+3weeks+3days)
Mare	341 days (11months+11days)
Camel	390 days (1year+1month+1day)
Rabbit	32-35 days
Cat	56-64 days

PHYSIOLOGICAL PH VALUES:

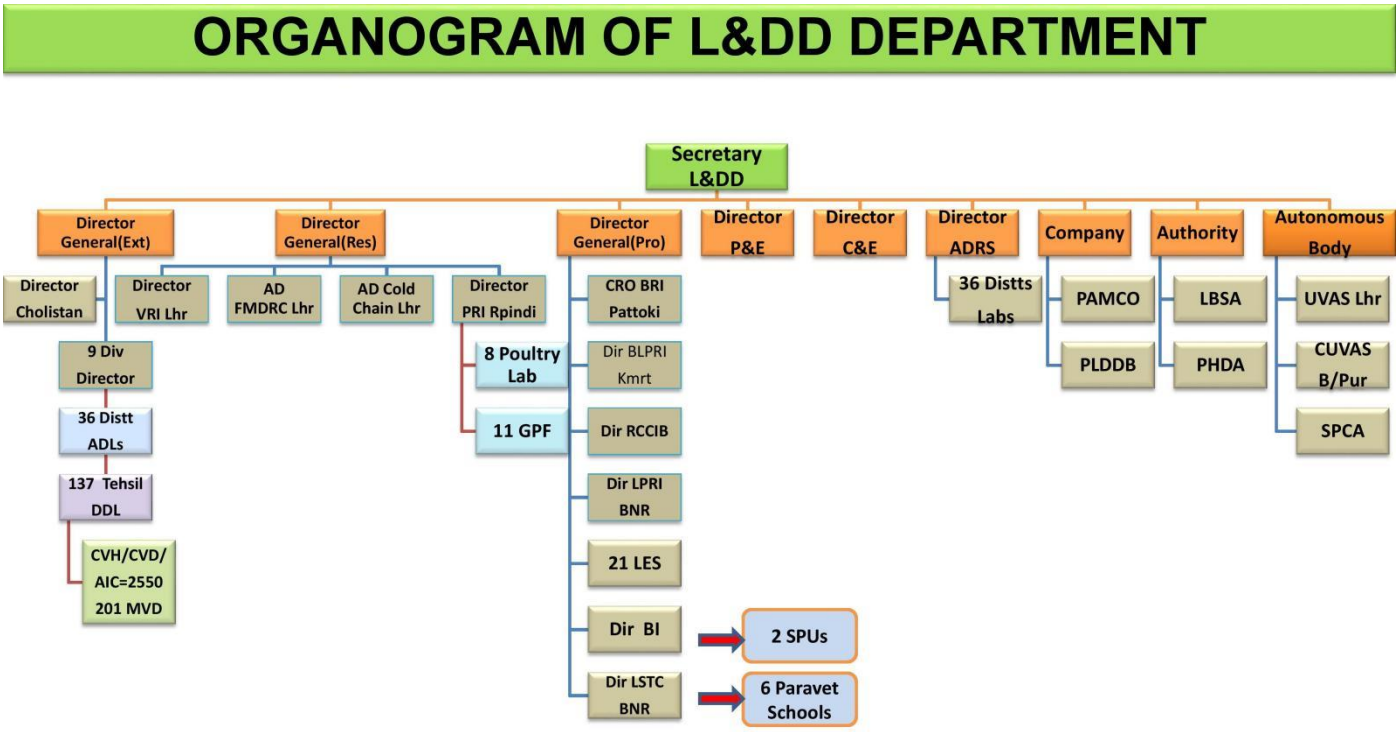
Sr. No.	Parameter	pH Value
1)	Water	6.5-8.5
2)	Carbonic Acid/ Bicarbonate Ions	6.1
3)	Silage	3.7-4.2
4)	Bull Semen	6.6-6.7
5)	Poultry Semen	7.0-7.6
6)	Milk	6.5-6.8
7)	Vagina	4.5
8)	Bovine Blood and Pathogenic Bacteria	7.2-7.4
9)	Abomasal Content	2.0
10)	Muscle	6.8-7.3
11)	Meat	5.5
12)	Microbes	6.5-7.5

IMPORTANT DAYS:

- ✓ January 12: National Youth Day
- ✓ March 24: World TB Day
- ✓ April 7: World Health Day
- ✓ April 22: World Earth Day
- ✓ April 25: World Malaria Day
- ✓ Last Saturday of April: World Veterinary Day
- ✓ June 5: World Environment Day
- ✓ July 1: Doctor's Day
- ✓ July 6: World Zoonoses Day
- ✓ July 11: World Population Day
- ✓ July 16: ICAR Foundation Day
- ✓ September 16: World Ozone Day
- ✓ September 28: World Rabies Day (Death anniversary of Louis Pasteur who developed the first efficacious rabies vaccine.
- ✓ October 4: World Animal Welfare Day
- ✓ Oct 6: World Wildlife Day
- ✓ 2nd Friday of October: World Egg Day
- ✓ Nov 12: National Bird Day (Birthday of Dr. Salim Ali)
- ✓ December 1: World AIDS Day

COLOUR REVOLUTIONS:

- ✓ Blue revolution: Fish Production
- ✓ Green revolution: Food grains (wheat, Rice)
- ✓ Red revolution: Meat/Tomato production
- ✓ Silver revolution: Egg/Poultry Production
- ✓ White revolution: Milk/Dairy production



VERNACULAR NAMES OF COMMON DISEASES:

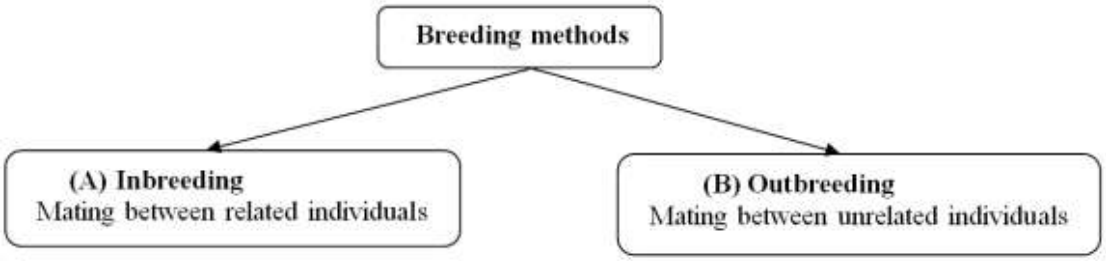
Disease/ Item	Vernacular Name
Toxemia	میٹھا باد زہر
Hydropsy	زہر عام باد
Panting	ہلا مارنا
Ingredients Mixing	گتاوا کرنا
Uterine torsion	ول
Prolapse	پیچھا مارنا
Debility	سک ٹک
Canine Distemper in Dog	پھیٹا
Surra in Camel	پھیٹا
Strangle	کنار
Glander	بد کنار
Diarrhea	موک
Bovine Ephemeral Fever	ول
Drench	کاڑھا
Pregnant	گبھن
Non-pregnant	پھنڈر
Cow heifer	وہڑی
Yearling Cow Bull	وہڑا
Female Cow Calf	وچھی
Male Cow Calf	وچھا
Female Buffalo Calf	کٹی
Male Buffalo Calf	کٹا
Exotic	ولائتی
Early Lactation	لیاری
Edema	پل پڑنا
Near to End of Lactation	توکڑ
Pyometra	پا مارنا
Docile	سیل
Halter	مورک / چبو
Muzzle	چھکو
Maize Fodder	گاچا
Three Teater	تلٹر
Two Teater	ڈھلٹر
Foal	وچھیرا
Black Buffalo with Blue Eyes	مانکی بھینس
Black Buffalo with Black Eyes	مشکن
Tumor on Third Eyelid	چندری
Worms/ Parasite	ملپ
Lesions at the Teat Opening	منہ سڑی
Panj Kalyan + Blue Eyes	کمری بھینس

Leather Halter	**تلارہ**
Constipation	**بن پڑنا**
Red Water	**سرکن**
FMD	**منہ کھر**
Abortion	**سٹنا بچہ /ترونا**
Abomasal Displacement	**اوجھری ہلنا**
Actinobacillosis	**ہجیراں**
Actinomycosis	**جبڑی**
Alopecia	**وال ڈگنا**
Anemia	**رت مکی**
Anoestrus	**ٹھنڈی**
Anthrax	**پھڑکی**
Arthritis	**سوج جوڑ جانا**
Aural Hematoma	**کنی**
Bed Sore	**زمینی پھٹ**
Skipping of one Milking	**ایکڑا**

BASICS OF ANIMAL BREEDING AND GENETICS:

CHROMOSOMES NUMBER OF VARIOUS ANIMALS/ORGANISMS:

Animal/ Organism	Chromosome No. (Diploid or 2N)
Drosophila	08
Cat, Pig	38
Rabbit	44
Swamp buffalo	48
River buffalo/water buffalo	50
Sheep	54
Goat, Cattle, Yak	60
Horse	64
Camel	74
Dog, Chicken	78



DIFFERENCE B/W CLOSE BREEDING AND LINE BREEDING:

Close breeding	Line breeding
More intensive form of inbreeding	Less intensive form of inbreeding
1. Parent-offspring mating: a. Sire to daughter mating b. Son to dam mating 2) Full sib mating:	1. Half-sib mating 2. Cousin mating 3. Grand parent-grand son/granddaughter mating

Note:

1. Full sib/full brother& sister: whose both parent are common
2. Half-sib/half brother& sister: whose one parent is common
3. Double first cousin: whose four grandparents are common
4. Single first cousin: whose two grand-parent are common
5. Half-first cousin: whose one grand-parent is common.

Mating	Coefficient of relationship (Rxy)	Inbreeding coefficient (Fx) (value of Fx = 1/2 Rxy)
Close breeding: a) Sire to daughter mating b) Son to dam mating c) Full sib mating	**0.50 or 50%**	**0.25 or 25%**
a) Half-sib mating b) Grand parent-grand son/granddaughter mating c) Double first cousin mating	**0.25 or 25%**	**0.125 or 12.5%**
Single first cousin mating	0.125 or 12.5%	0.0625 or 6.25%
Half first cousin mating	0.0625 or 6.25%	0.03125 or 3.125%

Note:

- ✓ **Inbreeding** is also known as **genetic assortative mating** while **out-breeding** is also known as **genetic dis-assortative mating**.
- ✓ **Inbreeding increases homozygosity while out-breeding increases heterozygosity.**
- ✓ **Geep** is a **chimera** produced by **combining the embryos of a goat & a sheep.**
- ✓ A sheep-goat chimera should not be confused with a sheep-goat hybrid. Chimera has four parents but a hybrid has only two parents.
- ✓ Selective breeding- **It is the out-crossing within a herd** with the help of selected sires. The **best method for improvement of buffalo breeds is selective breeding**.
- ✓ **Inbred line** - developed from two generations of full-sib mating. A line is called inbred line, if it has **minimum inbreeding coefficient of 0.375 or 37.5%**
- ✓ **In crossing** - Crossing of two different inbred lines derived from the **same breed**.
- ✓ **In cross breeding** - Crossing of the two different inbred lines derived from **different breeds.**

Hardy-Weinberg law:

In a **large, random-mating population**, in **absence of mutation, migration & selection**, the **gene & genotypic frequencies remain constant** from generation to generation.

Central tendency, measures of dispersion & relative measures of dispersion:

1. Central tendencies: It tells about the central position of any series

Examples: **Mean** (arithmetic mean, geometric mean, harmonic mean& weighted mean), **Mode & Median**

2. Measures of dispersion: Degree to which data tends to spread around an average

Examples: **Range, Variance, Standard deviation& Mean deviation**

3. Relative measures of dispersion:

Examples: **Standard error (SE); Coefficient of variance (CV)**

Note:

- ✓ **Range:** The difference between the smallest and largest values in set of data
- ✓ **Mean deviation:** Mean absolute deviation from an average. Its value is least affected by extreme values of a series.
- ✓ **Standard deviation (S.D.): -** It is the ideal measure of dispersion. Its value is always greater than mean deviation(M.D.)
- ✓ **Variance:** It is equal to square of standard deviation

Mendel's Laws:

1) **First law/law of segregation of gametes/law of purity of gametes**: -It is universally accepted Mendel‟s law.

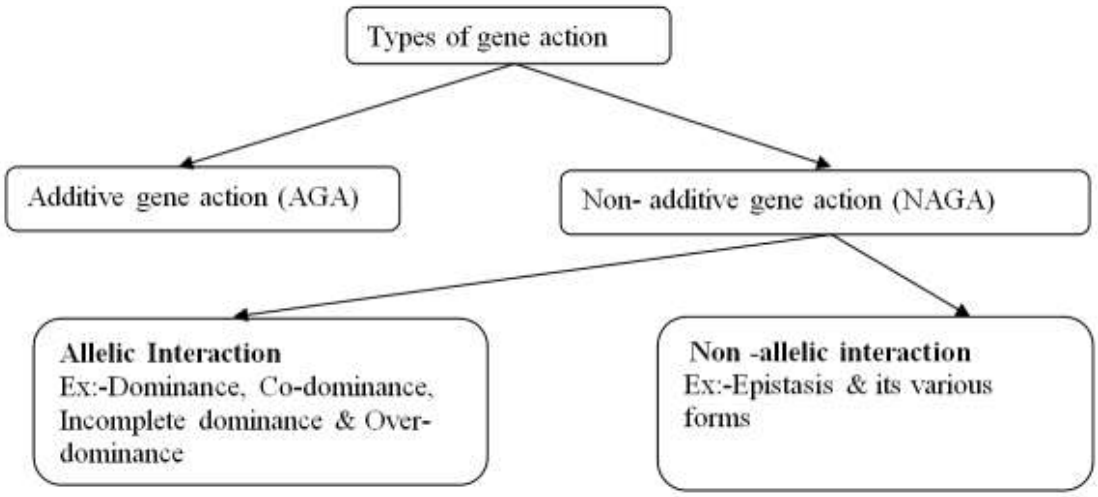
2) **Second law/law of independent assortment**: Its exception is linkage.

Note: Dihybrid cross = 2 × monohybrid cross

- ✓ F2 phenotypic ratio in dihybrid cross = (3:1) × (3:1) = 9:3:3:1
- ✓ F2 genotype ratio of dihybrid cross = (1:2:1) × (1:2:1) 1:2:1:2:4:2:1:2:1

Note:-

- ✓ Back cross is two types: Test cross & Out cross
- ✓ Back cross = F1 hybrid x Any of the homozygous parent (Tt × TT or tt)
- ✓ Test cross = F1 hybrid x Homozygous recessive parent (Tt × tt)
- ✓ Out cross = F1 hybrid x Homozygous dominant parent (Tt × TT)



Note:

- ✓ Allelic interaction is also known as intragenic interaction or intra-allelic interaction.
- ✓ Non-allelic interaction is also known as intergenic interaction or inter-allelic interaction.

Types of allelic interactions:

1) **Dominance**: When dominance is complete, then phenotype of heterozygote & dominant homozygote is same.

2) **Incomplete dominance/Partial-dominance**: The phenotypic value of heterozygote is lies between dominant & recessive homozygous.

3) **Over-dominance:** When the phenotypic value of heterozygous is superior to either of homozygous.

Note:

1) Pleiotropic gene: A gene with multiple phenotypic effects or a single gene that controls more than one trait

Examples:-

- ✓ Sickle cell anemia gene in human
- ✓ Double muscling gene in cattle

2) Polygenes: when one character is control by more than one gene.

3) Multiple allelism- More than two alternative form of a gene located on the same locus of the homologous chromosome are known as multiple alleles.

- ✓ Multiple alleles are produced by mutation
- ✓ Multiple alleles are found in the population, not in the single individual
- ✓ **Examples:** ABO blood group in human (3 alleles); Coat colour in Rabbit (4 alleles); Eye colour in Drosophila (15 alleles).

Heterosis/Hybrid-vigour:

Increase performance of offspring over the parents

- ✓ It is depended upon extent of heterozygosity & also depend upon Non-additive gene action (Dominance, Over-dominance & Epistasis)
- ✓ Traits with low h2 show high degree of heterosis or vice-versa.
- ✓ At each generation, heterozygosity reduced by 50% and heterosis also, so we can say that maximum hybrid vigour is obtained in F1 generation.
- ✓ Heterosis in F2 is diminished due to inter-se mating.

Note:

Repeatability is used in making culling decisions.

<u>The basis of selection/Aids to selection:</u>

1) Individual selection/Mass selection: Based on performance of individual

2) Pedigree selection: Based on performance of ancestors

3) Progeny testing: Based on performance of their progeny

4) Family selection: Based on performance of their collateral relatives.

- ✓ **Sire: Half of the herd**
- ✓ **Progeny testing** is used for the **selection of sire.**

Types of correlation:
1) Phenotypic correlation 2) Genetic correlation 3) Environmental correlation

Causes of genetic correlation:
a) Pleiotropy: Permanent cause of genetic correlation
b) Linkage: Temporary or transient cause of genetic correlation
c) Heterozygosity

Correlation coefficient(r):
1) If r = +1 (perfect positive correlation)
2) If r = −1 (perfect negative correlation)
3) If r = 0 (no correlation)

Examples of correlation:
1. Positively correlated traits:-
- ✓ Size of eggs and weight of eggs
- ✓ Milk yield and fat yield of dairy animal
- ✓ Heavier sheep produce more wool
- ✓ Heavier hen produce large sized eggs
- ✓ Birth weight and weight at weaning in cattle

2. Negatively correlated traits:
- ✓ Milk yield and fat%
- ✓ Number of eggs produced and size of eggs

3. No correlation:
- ✓ Milk yield& horn length
- ✓ Normal body temperature& body size

1. **Crossing over**: Exchange of segment between non-sister chromatids of homologous chromosomes in pachytene stage of Meiosis-1
2. **Translocation**: Exchange of chromosomal segments between non-homologous chromosomes
3. **Penetrance:** Percentage of individuals with a given genotype that expresses the particular phenotype
4. **Expressivity:** Degree or extent to which a given genotype is express phenotypically
5. **Variation**: (a) Genetic variation (b) Environmental variation

Causes of Genetic variation:
1. Segregation (Biggest cause of genetic variation)
2. Crossing over (2nd biggest cause of genetic variation)
3. Mutation, Migration, Selection, Random genetic drift

Mutation:

- ✓ Sudden heritable change in genetic material
- ✓ Term mutation: Hugo de Vries
- ✓ Hugo de Vries proposed mutation theory of evolution

Types of mutation:

A) **Chromosomal mutation**:

1) Heteroploidy 2) Chromosomal aberration

B) **Gene mutation**:

1) Substitution 2) Frame shift mutation

Heteroploidy is two types:

1) **Euploidy**: Change in number of sets of chromosomes

2) **Aneuploidy**: Change in number of chromosomes in a set

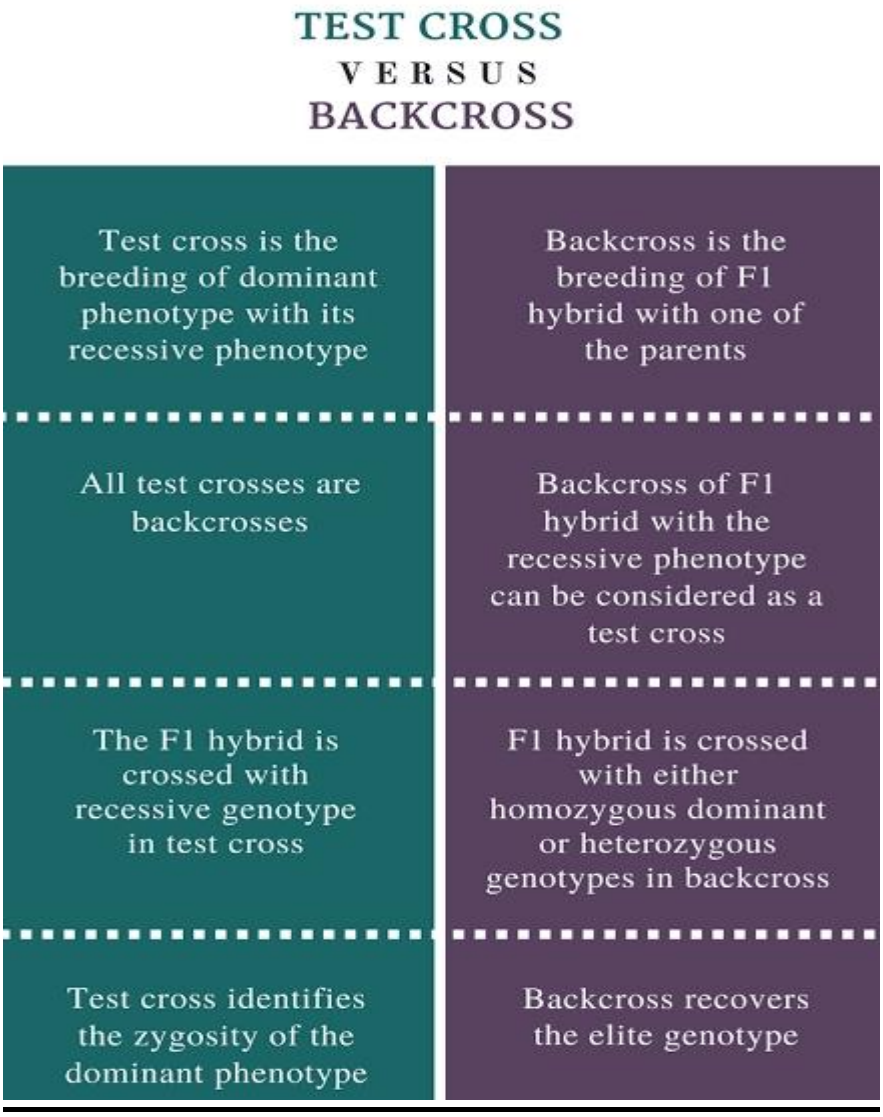
Two types of Substitution:

1) Transition: Replacement of purine by purine or replacement of pyrimidine-by-pyrimidine

2) Transversion: Replacement of purine by pyrimidine or replacement of pyrimidine by purine.

Genetic Terms	Definition	An example
Allele	Different forms of a gene, which produce variations in a genetically inherited trait.	Different alleles produce different hair colors—brown, blond, red, black, etc.
Genes	Genes are parts of DNA and carry hereditary information passed from parents to children.	Genes contain blue-print for each individual for her or his specific traits.
Dominant	Dominant version (allele) of a gene shows its specific trait even if only one parent passed the gene to the child.	When a child inherits dominant brown-hair gene form (allele) from dad, the child will have brown hair.
Recessive	Recessive gene shows its specific trait when both parents pass the gene to the child.	When a child inherits recessive blue-eye gene form (allele) from both mom and dad, the child will have blue eyes.
Homozygous	Two of the same form of a gene—one from mom and the other from dad.	Inheriting the same blue eye gene form from both parents result in a homozygous g
Heterozygous	Two different forms of a	Inheriting different eye color gene

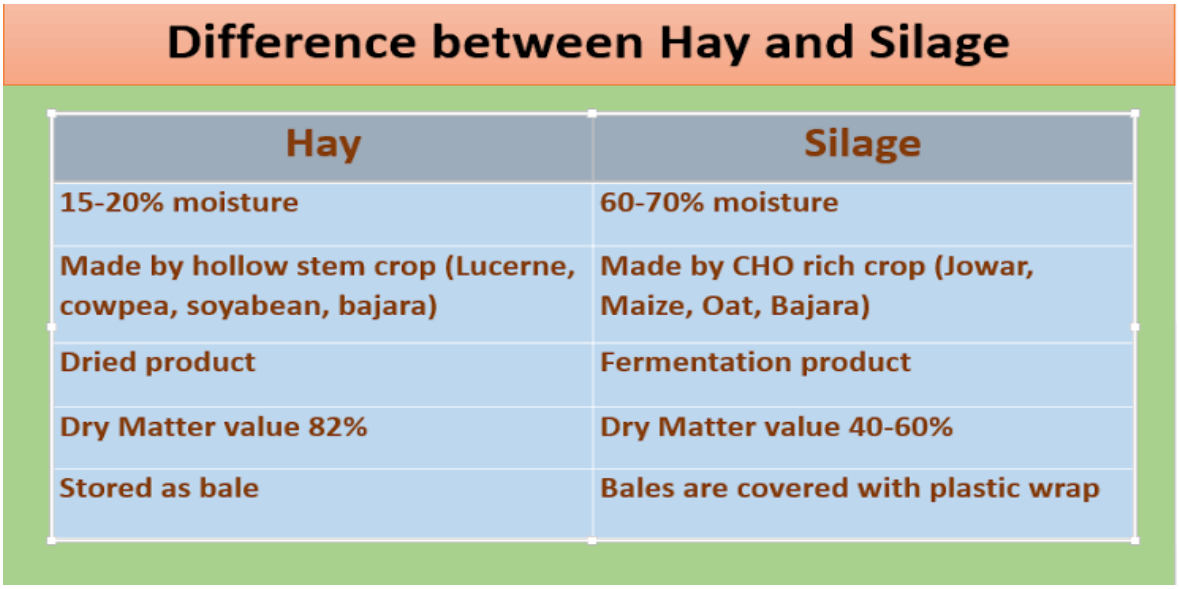
	gene—one from mom and the other from dad are different.	forms from mom and dad result in a heterozygous gene.
Genotype	Internal heredity information that contain genetic code.	Blue eye and brown eye have different genotypes—one is coded for blue and the other for brown.
Phenotype	Outwardly expressed traits or characteristics.	Both having or not having a widow's peak are phenotypes
Mendelian Inheritance	A simple genetic rule where a gene only comes in dominant or recessive forms.	Some genetic traits follow Mendelian Inheritance, while other genetic traits follow different inheritance patterns or rules.



BASICS OF ANIMAL NUTRITION:

DIFFERENCE B/W HAY AND SILAGE:

Item	Hay	Silage
Define	Sun dried forage	Method of preserving green fodder for a long period of time
Moisture	**12-14%**	**60-70%**
Best Crop	**Oat** (***Avena sativa***)	**Maize (Zea mays)**
Preferred plant	**Leafy Plant with thin stem**	Plant with **thick stem**
Harvesting time	Flowering stages(1/3 to ½ blossom)	Between flowering and milk stage



Note:

- ✓ **Best hay** is prepared with **Oat (*Avena sativa*)**
- ✓ **Best legume hay** is prepared with **Lucerne (alfa-alfa)**

Advantages of haymaking:

- ✓ **Sorghum (Jowar) lost its cyanide toxicity** during drying due to denaturation of enzymes.
- ✓ **Vitamin D content is very high** in hay due to sun drying.

Disadvantages of haymaking:

- ✓ **Vitamin A content of forage is lost up to 80%** due to sun drying.
- ✓ If hay made with high moisture legume, it results in fermentation & heat production leads to the formation of **Brown hay or mow-burnt hay**.
- ✓ **The brown colour of hay** is due to **Maillard-type of reaction**.

- ✓ **Lysine amino acid** is mainly susceptible to Maillard type reaction.
- ✓ **Nutritive value of hay is less** than that of silage

Important points for haymaking:

The crop should be harvested at the flowering stage (when flowering is initiated) because when the crop matures, its lignin content increases and nutritive value decreases. As far as time is concerned, the crop should be harvested early in the morning because at this time the dew has dried of.

Silage:

(1) Ensiling/Silage making: Process of **preserving green fodder** by **anaerobic fermentation.**

Note:

(1) Colour of good silage: **Yellowish or Brownish-green**

(2) Haylage: When grass & legume which are meant for hay are ensiled

- ✓ Moisture content of haylage is **45-55%**
- ✓ Having characteristics of both hay & silage

(2) Wastelage: Anaerobically fermented animal waste (poultry dropping, poultry litter, swine excreta, and bovine dung) with the help of **lactic acid producing bacteria**

Special points about silage making:

- **Crops rich in soluble sugar** such as maize, sorghum, bajra are most **suitable for ensiling**.
- ✓ Crops with less soluble carbohydrate can also be used for ensilage by the addition of **3-3.5% molasses.**
- ✓ Mixture of **grass/cereal fodder& legume** used for ensilage should be in the **ratio of 3:1**
- ✓ **Urea @1% & salt @0.5%** are added to cereals & grasses in order to improve N2 content& palatability
- ✓ **Pit silos are more common.**
- ✓ **Flieg-index** is commonly used method for **evaluation of silage quality.**

Note:

- ✓ **Avidin**: **Egg white injury factor**
- ✓ **Biotin: Anti- Egg white injury factor**
- ✓ Oral contraceptives & drug therapy of tuberculosis act as antagonistic to Vit. B6
- ✓ Horse is resistant to gossypol poisoning
- ✓ A fodder crop having retarded growth due to drought & regain its growth after rains will cause **Cyanide poisoning.**

Note:

- ✓ **Vitamin A, D, E&K are fat-soluble** while Vita.B-complex& Vita.C are water-soluble.
- ✓ **Vitamin E & selenium** show a **synergistic effect.**
- ✓ **Vitamin D acts as a hormone**.It synthesized in skin & cause calcium absorption through GIT.

- ✓ **Choline** acts as **"lipotropic factor"** which is responsible for prevention of fatty liver.

Note:

- ✓ Perosis/Slipped tendon in chickens: due to deficiencies of **Mn, Biotin, Choline, Folic-acid&Vita.B12**
- ✓ Ca & P act as antagonistic to Mg.
- ✓ Sheep is most susceptible to Cu toxicity & Co deficiency.
- ✓ Sheep is less prone to phosphorus deficiency due to selective grazing
- ✓ **Chromium** is known as "**glucose tolerant factor**"

Piglet anemia: **2-4 weeks piglets are most susceptible**
Prevention:
(1) **Drenching of saturated solution of FeSO4**
(2) **Teat painting with FeSO4**
(3) I/M injection of **Iron dextran** on **4 th&14th day after birth**

Sulphur:
1. Sulphur containing vitamins are **thiamine & biotin**
2. Sulphur-containing amino acids are methionine, cysteine, homocysteine, and taurine
3. Sulphur containing hormones are **insulin, ADH/vasopressin, oxytocin**
4. Wool contains 4% sulphur while human hair keratin contains 14% sulphur
5. Recommended Nitrogen: Sulphur ratio in ruminant diet should be 10:1
6. Nitrogen: Sulphur ratio in wool is 5:1

Sodium (Na) & Potassium (K):
1. Major Cation of Extracellular fluid (ECF) is Na^+ while major anion is Cl^-
2. Major Cation of Intracellular fluid (ICF) is K^+ while major anion is PO_4^-

Calcium (Ca) &Phosphorus (P):
1. Normal **blood calcium** level in ruminants is **8-12 mg/dl**
2. Normal **blood phosphorus** level in ruminants is **4-8 mg/dl**
3. Bone Ca: P ratio should be 2:1
4. The main path of excretion of **Ca & P is via faeces**
5. The main path of excretion of **Na+ &K+ is via urine** (but in carnivores, the main path of P excretion is via urine).
6. Phosphorus play important role in nucleotides formation such as ATP, ADP.

Note:

- ✓ **Feed-stuff**: consists of roughage, concentrate & feed additives
- ✓ **Energy feeds/Basal feeds**: contains **less than 20% CP & less than 18% CF.**Generally, Energy feeds consist of cereal grains & it"s by product
- ✓ **Protein supplements**: contains **more than 20% Crude protein and less than 18% crude fiber**
- ✓ **Soilage**: Pasture, when cut & fed green to an animal in its own stall

Feeding requirements:

Dry matter requirement (per animal per day):

- ✓ Cattle: 2- 2.5 Kg/100 Kg body weight
- ✓ Buffalo and crossbred cow: 2.5 - 3Kg/100 Kg body weight
- ✓ Sheep: 2.5 - 3 Kg per animal per day
- ✓ Lactating sheep: 4% of body weight
- ✓ Meat goat: 3-4% of their live weight
- ✓ Dairy goat: 5-7% of their live weight
- ✓ Camel: 2% of body weight
- ✓ Working & lactating camel: 3% of body weight

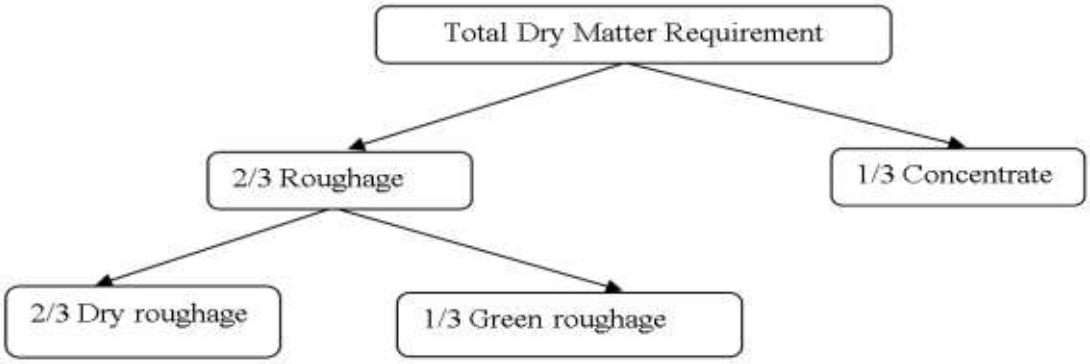
Production ration:

- ✓ Cattle: **1Kg concentrate/2.5 Kg milk**
- ✓ For Buffalo & Cross-bred cow: **1Kg concentrate/ 2 Kg milk**

Gestation ration during last trimester of pregnancy:

- ✓ For Zebu Cattle: **1.25 Kg concentrate**
- ✓ For Buffalo & Cross-bred cow: **1.75 Kg concentrate**

Note: Salt and mineral mixture should be given @ 1% of the concentrate mixture.



Note: If green fodder is legume, then roughage supplied **1/4 by green roughage& 3/4 by dry roughage.**

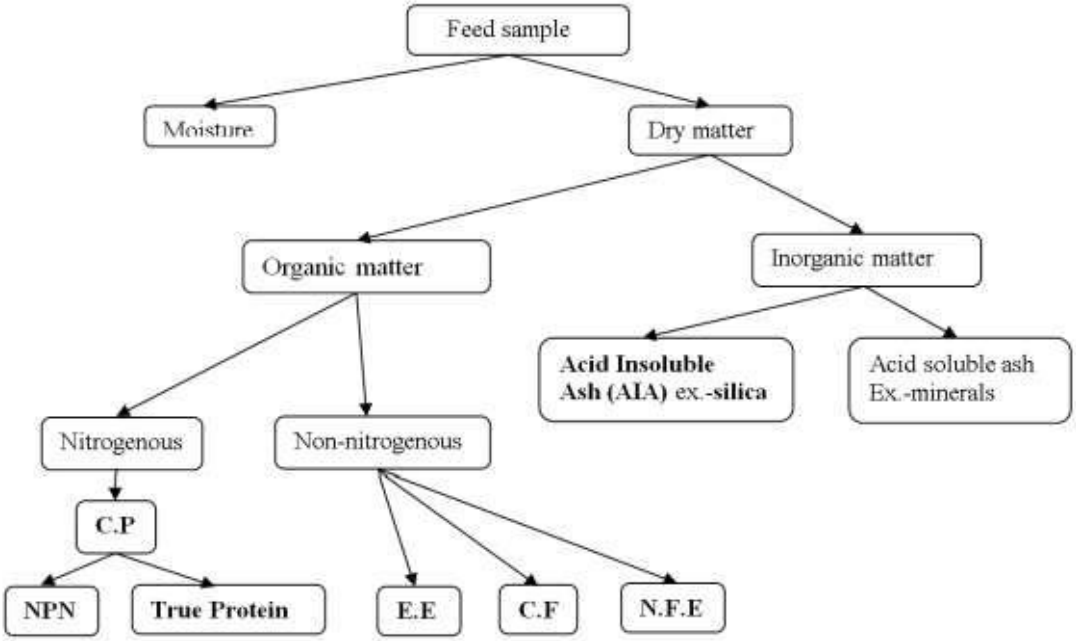
The composition of animal body:

- ✓ Water, protein& ash content of a **fat-free animal body are in the ratio of 19:5:1 (75% water, 20-22% protein & 3%ash).**
- ✓ **Total body water of an animal body is 60-70%**
- ✓ The total mineral content of the animal body is 3% of total body weight.
- ✓ **Calcium & phosphorus** are the major inorganic components of **animals**
- ✓ **Potassium &silicon** are the major inorganic components of **plants**

Weende system of proximate analysis:-

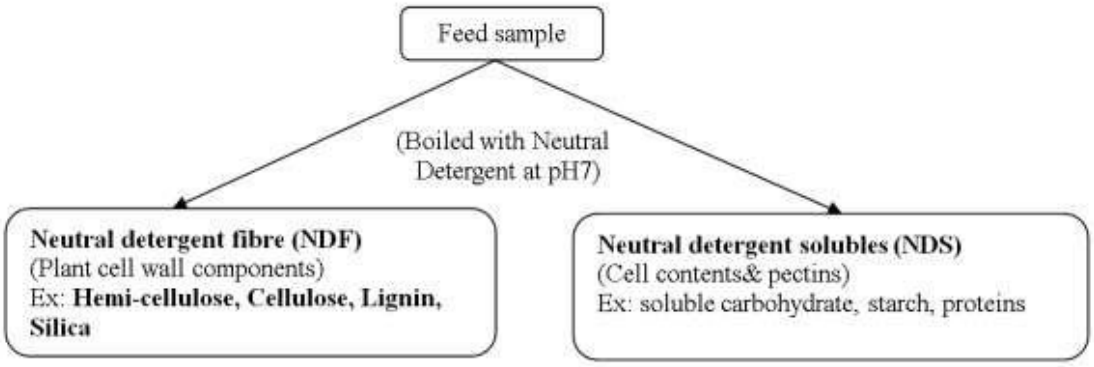
1. Proximate principles-are various **nutrients of feed**, which are required for the nourishment of the animal body**.**

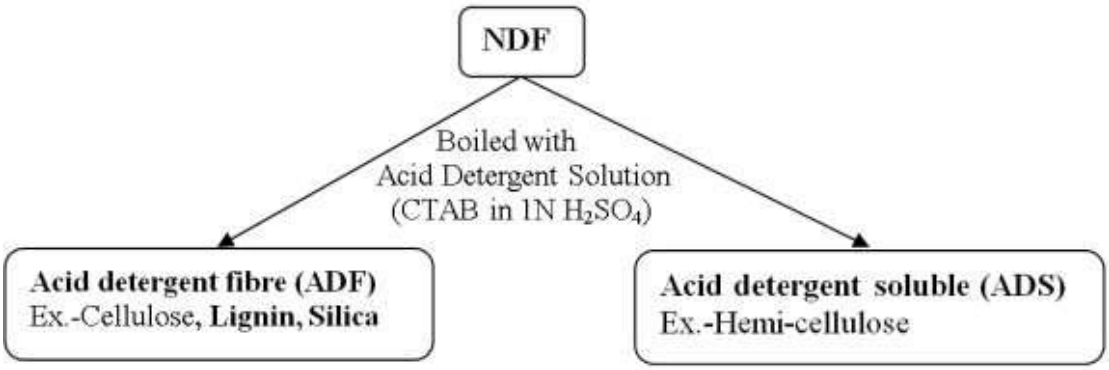
2. "System of analysis" proposed by Henneberg & Stohmann at **Weende experimental station**, **Germany in 1865**.



<u>The detergent method of forage analysis/Van-Soest method:-</u>

Partition of carbohydrates by Weende system into CF & NFE is not realistic so to overcome this Detergent method of forage analysis was propose by Van-Soest.





Note:

- ✓ **NFE is only proximate principle that cannot be determined by analysis**
- ✓ NFE is simply calculated by subtracting CP, EE, and CF &Ash from 100
- ✓ **NFE% (DM basis) = 100 – (CP%+EE%+CF%+Ash %)**
- ✓ **Feed containing moisture > 14 percentage cannot be stored** for a long time.
- ✓ Crude protein = True Protein + Non-Protein Nitrogen
- ✓ For estimating protein content of the feed, **Stutzer"s reagent (alkaline CuSO4)** is used.
- ✓ For calculating CP, we have to assume 2 things:
  a) **All food protein contains 16% N2** (factor 6.25 is coming from 100/16 = 6.25)
  b) All N2 of food comes from true protein.
- ✓ TDN% = DCP% + DCF% + DNFE% + (DEE × 2.25)
- ✓ 1Kg TDN = 4.4 Mcal/g Digestible Energy
- ✓ **1Kg TDN = 3.6 Mcal/g Metabolic Energy**
- ✓ 1 Kg TDN = 0.86 Starch Equivalent
- ✓ **0.32Kg TDN** is required by a cow for **1 litre of milk production**
- ✓ **400-500 ml blood** is circulating through out udder for the production of **1ml of milk.**
- ✓ **Calf starter contains 18.5-19.5%DCP, 23-26% CP & 75% TDN**
- ✓ Calf starter should be introduced from 2nd week of age

Metabolic water:

Amount of water produced in the body during various metabolic activities. It comprises of **5-10% of total intake of the nutrient.**

Straw& Stovers:-

- ✓ Deficient in all minerals like Ca, P etc. but rich in silica
- ✓ Poor in protein content (DCP of straw is 0%) & poor in energy content (TDN is about 40%)
- ✓ Have low digestibility & high Crude fiber content
- ✓ **Wheat straw have 0%DCP, 3-4%CP & 40% TDN**
- ✓ **We can improve N2 content, DCP, CP, TDN& digestibility of straw by urea treatment.**

Note: During urea treatment of straw, **water: urea ratio should be 10:1** for optimum utilization.

Urea as an NPN source to the ruminants:

1. Most widely used NPN substance in ruminant ration
2. Nitrogen content of urea is 46.6% (in pure form)
3. Urea is only used in ruminant ration
4. Urea should never feed to young calves (up to 6 months age) and monogastric animals (ex. horse, pig, dog etc.)

Characteristics of Urea:

1. It is deficient in all minerals.
2. It has no energy value
3. 100% degradable in rumen (quickly converted into NH3 in rumen)

Factors affecting urea utilization:-

1. For optimum use of urea, **Nitrogen: Sulphur (N:S) ratio should be 10:1**
2. **Starch: Urea ratio** in the diet should be **10:1**(1Kg starch require 100g urea)
3. Rumen bacteria are unable to use NH3 effectively if:-

(a) Diet contain CP>13% or

(b) **Rumen NH3 concentration >5-8mg/100ml**

4. **Adaptation period of rumen microbes** to dietary urea is **2-4 weeks.**

Urea toxicity in cattle:

Urea toxicity results when **rumen NH3 concentration exceeds 80mg/100ml** (or) when **blood NH3**
concentration exceeds 1mg/100ml.

Treatment of urea toxicity in cattle:

1. **Drenching 20-40 liters cold water** (cold water inhibits ureolytic activity)
2. **Drenching 4-5 liters of 10% acetic acid** (acetic acid neutralize the releasingNH3).

Metabolism of carbohydrate, protein & fat:

Net gain of ATPs by complete oxidation of one mole of substance is as follows:

1. Glucose (aerobic oxidation) 38 ATPs
2. Glucose (anaerobic oxidation) 02 ATPs
3. Glycogen (aerobic oxidation) 39 ATPs
4. Glycogen (anaerobic oxidation) 03 ATPs
5. Glycerol=22 ATPs
6. Acetate (VFA)=**10 ATPs**
7. Propionate (VFA)=**17 ATPs**
8. Butyrate (VFA)=**25 ATPs**
9. Palmitic acid=129 ATPs

Note:

1. **Defaunation**: Selective removal of protozoa from the ruminal ecosystem
2. **Chemical defaunation**: **Copper sulfate & Sodium lauryl sulfate**

Uses of volatile fatty acids (VFAS) produced during ruminal fermentation:

1. VFAS provides 70% of ruminant energy needs
2. **Acetate:** It is responsible for milk fat synthesis
3. **Propionate:** It is **gluconeogenic**. It is responsible for milk sugar synthesis
4. **Butyrate:** Ruminal epithelial cells convert it into β-hydroxy butyrate (a ketone body)

Note:

1. High roughage diet: high milk fat% (because acetate production is high)
2. High concentrate or grain diet: high milk yield (because propionate production is high).

Indicator method of digestibility determination:

- ✓ **Internal indicators** (ex:- **lignin, silica, acid-insoluble-ash/AIA)**
- ✓ **External indicators** (ex: **-chromic oxide**, ferric oxide, Mg-ferrite, radioisotopes like Cr-EDTA & Cerium

The experimental period for digestibility determination:

1) **Preliminary period/Adjustment period**: 10-12 days for ruminant & 7-10 days for horse & swine
2) **Collection period:** 7-10 days for ruminants as well as non-ruminants

Note:-

- ✓ In digestibility trial, male animals are preferred over female because in male both faeces & urine are collected separately.
- ✓ Castrated males are preferred over normal males
- ✓ In the digestibility trial, only faeces are used but in the metabolic trial, both faeces & urine are used.
- ✓ In poultry, digestibility determination is difficult because faeces & urine are excreted by same opening cloaca
- ✓ In poultry, digestibility is determined by two ways: -surgically & chemically

AMINO ACIDS:

1. Essential Ammino acid:

Limiting Ammino acid

- ✓ Lysine
- ✓ Methionine
- ✓ Tryptophan
- ✓ Threonine

Branch Chain Ammino acid

- ✓ Valine
- ✓ Leucine
- ✓ Isoleucine

Conditional Ammino acid

- ✓ Phenylalanine
- ✓ Arginine
- ✓ Histadine

2. <u>Non-Essential Ammino acid:</u>

They are called non-essential because they can be synthesized by other substances in the body:

- ✓ Glycine
- ✓ Serine
- ✓ Alanine
- ✓ Glutamic acid
- ✓ Proline
- ✓ Asparctic acid
- ✓ Asparagine
- ✓ Tyrosin
- ✓ Cysteine

Points:

1. The only true ketogenic amino acid is leucine.
2. A dietary excess of Tyrosine causes eye lesions.
3. Dietary excess of Methionine produces inhibition of ATP synthesis.
4. Main EAA in **Pig: Lysine**
5. Main EAA in **Poultry: Glycine**
6. Main EAA in **Sheep: Methionine**
7. Main EAA in **Cat: Taurine**

<u>Essential Fatty Acids (EFAs)</u>:

Linoleic acid, Linolenic acid & Arachidonic acid

1. EFA in Cat: Arachidonic acid
2. EFA in Pig & Poultry: Linolenic acid

<u>Caecotrophy/ Coprophagy/ Pseudo-rumination:</u>

Ingestion of specially produced soft faecal pellets, which are usually produced at night in the domestic rabbit.

- ✓ **In the rabbit, caecotrophy begins** at the **age of three weeks**.
- ✓ Domestic rabbits produce **hard faeces during the day**& soft faeces during night.
- ✓ In rabbits, the prevention of coprophagy leads to:

a) Decrease digestibility of food
b) Decrease protein utilization
c) Decrease nitrogen retention

<u>Methods of improving the nutritive value of poor quality roughage like straw:</u>

A. Supplementation with deficient nutrients:

1. Enrichment with urea and molasses
2. Silage making
3. Supplementation with green fodder (Legume or non-legume)
4. Supplementation with legume straw
5. Supplementation with urea-molasses liquid supplements

B. Treatments:

1. Physical method:

Examples: soaking, chopping, grinding, pelleting, irradiation, steam under pressure

2. Chemical method:

Examples: Alkali (NaOH, Ca(OH)2, KOH, NH4OH); Ammonia (gaseous, aqueous, urea-ammonia); Acids (H2SO4,HNO3); Salts (NaCl,Na2CO3); Gases (chlorine, SO2); Oxidizing agents(H2O2, ozone)

3. Physio-chemical methods: combination of physical& chemical treatment. Example: NaOH/Pelleting

4. Biological methods:

Examples: Enzymes (cellulase), white-rot fungi, mushrooms

Function and Temperature time protocol of some instruments:

1. Autoclave: It kills microorganisms and spores. The Temperature is 121 Degree Celsius/15 lbs/15 min
2. Hot air oven: It uses dry heat to sterilize. The Temperature is 160 Degree Celsius/1hour
3. Muffle furnace: It is used to determine Ash content in sample. The Temperature is 600 Degree Celsius/1-2 hours.
4. Kjeldahl Apparatus: It is used to determine protein content. The Temperature is 360-410 Degree Celsius.

DIFFERENCE B/W LACTIC ACID AND BUTYRIC ACID FERMENTATION:

SILAGE

Parameters	Lactic Acid Fermentation	Butyric Acid Fermentation
Condition	When fodder is rich in **soluble sugar or carbohydrate**	**When fodder is rich in protein content**
Quality	**Good & High quality**	**Bad odour - good silage**

CLASSIFICATION:

Parameters	Very Good Silage	Good Silage	Fair Silage
pH	**3.5 - 4.2**	4.2-4.5	4.8 or above
Ammonical N2 in total N2	**Less than 10%**	10-15%	20% or above
Butyric acid	Nil	Trace	Some amount

INCLUSION LEVELS OF COMMON FEED INGREDIENTS:

Name of the Ingredients	The maximum safe Quantity	Reasons
Cotton seed cake	20-25	Price and quality; taste; eaten in low quality; proteins and price
Rapeseed cake	10-13	Protein level; milk does not taste good
Sunflower cake	10-13	Quality and taste
Cotton seed meal	15-20	Price, protein
Rapeseed meal	15-20	Price and the quality of protein
Canola meal	15-20	Not much poisonous; less production
Peanut meal	20-25	Bad taste
Maize carbohydrate 20 and 30%	20-25	Increase urination
Maize carbohydrate 60%	5-10	Diarrhea
Soyabean	10-15	Price
Urea	1-2	Less quantity of calcium
Rice Polishing	20	High price; not very digestible
Wheat bran	25	According to requirement
Molasses	10-15	
Wheat grains	15 -20	
Maize grains	40-50	
Sorghum	15-20	
Oil	2-3	
Salt	1-2	
DCP	1-2	
Salt powder	2	
Maize gluten 20%	20-30	
Maize gluten 30%	20-25	
Maize gluten 60%	5-10%	
Common salt	1-2	
Mineral Mixture	2	

ANTI NUTRITIONAL FACTORS:

Method of detoxification of antinutritional factor in feed		
Feed	**Antinutritional factor**	**Processing for detoxification**
Oil cake	Mould	Add propionate or NH_3
Groundnut cake	Aflatoxin	Add NH_4OH, Add fungicidal
All feed	Salmonella	Pelleting of feed
Soya bean	Trypsin inhibitor	Autoclaving at 15 psi pressure for 15 min at 120 ^0C
Castor bean	Ricin	By moist heat
Lucerne	Saponin	Feed with cotton seed and cholesterol
Sorgum	Tannin	Soaking and cooking anaerobic storage
Soya bean, seasame	Phytate	Supplementation of Mn, Zn, Fe
Mollases	Oxalate	Supplement $CaCO_3$
Brassica	Glucosinolate	Soaking and cooking Supplement with I_2
Cotton seed cake	Gossypol	Supplement with Fe or $FeSO_4$
Soya bean	Antivitamin A	Autoclaving
Soya bean	Antivitamin D	Autoclaving
Subabool	Mimosin	Drying of leaves treatment with $FeSO_4$
Jowar	Dhurin	Nitrate and thiosulphate
Raw fish	Thiaminase	Heating

DAILY SALT AND DCP REQUIREMENTS OF FARM ANIMALS

SALT REQUIREMENT:

Salt	Maintenance Requirement(g/d)	Lactation Requirements (g/d)	Total requirements(g/d)	Requirements as% of total dry matter
Sodium	32.8	33.7	66.5	0.30
Potassium	93.5	83.8	177.3	0.80
Chloride	43.5	56.5	100.0	0.45

- ✓ Salt (Common salt) contains 0.42 % Sodium (Na), 1.08 % Potassium (K) and 0.63 % chloride (Cl).

DCP REQUIREMENT:

Catagory	B.wt(kg)	DCP(KG)
Calf	45	0.170
	60	0.195
	70	0.220

Catagory	B.wt(kg)	DM(KG)	DCP(KG)
Growing cattle and Buffalo (Gain 0.5 kg/d)	150	4.20	0.310
	200	5.20	0.350
	250	6.25	0.380
	300	6.90	0.410

Catagory	B.wt(kg)	DM(KG)	DCP(KG)
Mature cow and buffalo	300	4.50	0.200
	350	5.00	0.230
	400	5.50	0.250
	450	6.00	0.280
	500	6.50	0.300
	550	7.00	0.330
	600	7.50	0.350

Catagory	B.wt(kg)	DM(KG)	DCP(KG)
Maintenance and Pregnancy (last 2 months of gestation)	300	5.60	0.290
	350	6.40	0.320
	400	7.20	0.350
	450	7.90	0.400
	500	8.60	0.430
	550	9.30	0.470
	600	10.0	0.500

Catagory	B.wt(kg)	DM(KG)	DCP(KG)
For Milk production(per kg milk)	Fat%		
	4	-	0.045
	5	-	0.051
	6	-	0.057
	7	-	0.063

Catagory	B.wt(kg)	DM(KG)	DCP(KG)
Breeding bull	500	-	0.450
	600	-	0.530

Animal	Liters/day	Pounds/day
Beef cattle	22–66	48–145
Dairy cattle	38–110	84–242
Sheep and goats	4–15	9–33
Horses	30–45	66–99
Swine	11–19	24–42
Chickens	0.2–0.4	0.4–0.9
Turkeys	0.4–0.6	0.9–1.3

FEED PHASES:

There re “FIVE” feeding phases:

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
First 10 weeks after calving. Milk production increases rapidly. Cow is usually not able to eat enough and draws on body reserves to maintain production. A protein content of **18% crude protein** is recommended in rations for high-yielding cows.	**Second 10 weeks** after calving Maximum level of feed intake reached. **15–18% crude protein** is recommended in the whole ration	**Last half of lactation** Milk production decreases 12-15% CP is recommended Cow might be conceived at this stage	Dry period ranges from 7 to 8 weeks Time for restoring cow’s body reserves A minimum of **11 percent CP is** recommended	**(Transition period or closeup period**) Includes last two weeks of parturition. Nutritional requirement is higher for fetal growth **CP% should be exceeded up to 15-18%.**

CLASSES OF FEEDS:

Class	Traits	Example
Dry forages or roughage	> 18% fiber	Hay, straw, seed hulls, fodder, stover
Succulent forages or roughage	> 18% fiber	Pasture, green chop, cannery residues
Silages	> 18% fiber	Whole-plant grain crops, wilted or low-moisture grasses or legumes
Energy feeds	< 20% protein and < 18% fiber	Cereal grains, milling by-products, roots and tubers, brewery by-products
Protein supplements	< 20% protein and < 18% fiber	Animal by-products (meat scraps) Marine by-products (fish meal) Avian by-products (hydrolized feathers) Plant by-products (soybean meal, cottonseed meal, linseed meal, corn gluten meal)

Mineral supplements	Guaranteed analysis	Steamed-bone meal Dicalcium phosphate Iodized salt Trace mineralized salt
Vitamin supplements	Guaranteed potency	Vitamin A acetate Vitamins A, D, E

VERNACULAR NAMES OF COMMON GRASSES:

Scientific name	Vernacular Name
Cenchrus ciliaris	Buffel grass/ Dhaman
Chrysopogon Aucheri	Khar
Cynodon dactylon	Bermuda grass/ khabbal
Eleusine flagellifera	chhimber
Digitaria bicornis	Crabgrass/pharion
Cymbopogon jwarancusa	khawai
Diconthium annulatum	Murgha

CLASSIFICATION OF ANIMALS ON THE BASIS OF FEEDING HABITS:

There are three main classes in which animals can be classified on the basis of their eating habits-**Herbivores, Carnivores and Omnivores**

Classes	Definition	Example
Herbivores	Animals that eat plants. They have sharp teeth called molars and premolars for chewing food, herbivorous animals have mouth parts adapted to rasping or grinding.	Cow, deer, goat and giraffe
Carnivores	Animals which fed on flesh of other animals, carnivores meaning “meat eaters”	Lion, tiger and fox
Omnivores	Animals that eat both animals and plants	Bear, cow, squirrel

Grazing systems:

Three types of grazing systems

Rational grazing system	Zero grazing system	Continuous grazing system
Rotational grazing is a **system where a large pasture is divided into smaller paddocks allowing livestock to be moved from one paddock to the other easily**. Using this method cattle are concentrated on a smaller area of the pasture for a few days then moved to another section of pasture	Zero grazing is a **system where the grass is harvested and fed fresh to housed cattle on a daily basis**. It allows farms to increase fresh grass utilisation and milk yield from grass/forage where that farm may not have the infrastructure to graze cattle.	In the continuous grazing system the **pasture is not divided** into sub-pastures or paddocks. Livestock are allowed access to all the pasture area at any given time. Continuous grazing often creates overgrazed areas, areas where unfavorable plants go to seed, and weed issues.

BASICS OF LIVESTOCK PRODUCTION MANAGEMENT:

LIVESTOCK PRODUCTION SYSTEMS:

1. Rural Subsistence Small Holdings (55-60%)
2. Rural Market Oriented Small Holdings (20-25%)
3. Rural Commercial Medium Size Farming (10-15%)
4. Peri-urban Commercial Large Size Farming (6-8%)

Metabolic water:

Amount of water produced in the body during various metabolic activities. It comprises of **5-10% of total intake of the nutrient**.

Note:

Water requirement is directly proportional to dry matter intake

Exotic breeds of cattle:

- ✓ Dairy breeds: -Jersey, Holstein-Friesian (HF), Brown Swiss, Ayrshire, Guernsey
- ✓ Beef breeds: -Hereford, Angus, Shorthorn.

Milking Methods:

A. Hand milking: A common practice among individual farmers
B. Machine milking: Commonly practised in an organized farm.

Note:

- ✓ Cow are milked from the left side
- ✓ The hindquarters of the udder are slightly larger than the front ones and contain more milk.
- ✓ The approximate ratio of milk is 60:40 (Hindquarter: Forequarter) for cattle.

Methods of Drying:

1. **Incomplete milking**: **Best method of drying off a cow**
2. Intermittent milking
3. Complete cessation: Wrong method of drying off a cow

Methods of age determination of animals:

1. By record keeping
2. By physical appearance of animal
3. By counting no. of horn rings (**Age=N+2**; where N is no. of horn ring)
   - ✓ This method is applicable to cattle/buffalo whose calving interval is 1year; in **cattle, first horn ring appears at the age of 3 years.**

3. By dentition: M**ost commonly used method for age determination**

Note:-

- ✓ Full mouth condition: -Age at which all permanent teeth appear

[(a) Sheep: 4 years (b) Horse: 4.5 years (c) Goat, Cattle & Buffalo: 5 years (d) Camel: 7 years]

- ✓ Broken mouth condition: -Age at which one or more teeth have disappeared.
- ✓ **Gummer: - an animal that has lost it's all teeth.**
- ✓ Cheek teeth: Premolars& molars together known as cheek teeth
- ✓ Age of the animal is determined by incisors of the lower jaw
- ✓ **Dental Star** (mark on central pulp cavity of incisors) & **Galvayne's groove** (a groove in upper corner incisor) **are related to incisors of horses**.

Note:

- ✓ Branding is most suitable for marking of cattle, buffaloes, camels &horses. It causes partial burning of tissue & produces a permanent scar. The lower part of the thigh is the best site for branding.
- ✓ Tattooing is the best method for identification of sheep, goat &newborn calves. Best site for tattooing is inside the ear between large veins. Tattooing is not suitable for black coloured animals.
- ✓ Ear notching is commonly used in pigs. It is done in a V-shape manner on the borders of the ear
- ✓ Wing tagging is commonly used in poultry birds &done on the 1 st day of hatching
- ✓ In the case of Sheep, tattooing & ear tagging are used for identification

DIFFERENCE B/W TATTOO AND TAG:

Tattoo	Tag
Tattoos are permanent identification of the animal that can not be read from the Distance	Ear tags are easy to read but not permanent.



Cattle housing system:

1. Loose housing system

2. Conventional dairy barn/ Stanchion dairy barn

Cow-shed arrangement:

1. Single row- If herd is small (less than 10 cows)
2. Double row- If herd is large (more than 10 cows)

Methods of castration:

A. Open method/surgical method- Removal of both testicles through incision

B. Closed method-

1. Burdizzo castrator: It separately crushes each spermatic cord.

- ✓ It is a **bloodless castration**
- ✓ **It is the best method of castration**

2. By Rubber ring method

Special points:

- ✓ Castration in female is known as **spaying or ovariohysterectomy** (removal of both ovaries & uterus)
- ✓ Castrated female is known as **Spayed**
- ✓ Castrated fowl is known as Capon & process is caponisation
- ✓ Castrated Tom (male cat) is known as Neuter & process is neutering

Weaning: The process of separating the baby from its mother at a very early age, up to this age whole milk feeding is desired.

Note: Early weaning is difficult in buffaloes due to high motherly instinct.

CASTRATION AND WEANING AGE:

Animal	Age of Castration	Age of weaning
Calf	8-10 weeks	90 days (8-12 weeks)
Lamb	2 weeks	14 weeks
Kid	2-4 weeks	16 weeks

PREHENSILE ORGANS IN ANIMALS:

Specie	Organ
Cow and buffalo	Tongue
Dogs and cats	Forelimbs
Camel	Sensitive Mobile Lips
Monkey	Appendages and Tail
Fowl	Beak
Swine	Pointed lower lip
Horse	Upper lip, Tongue and the incisor teeth
Sheep	Clefted upper lip

Dehorning/Disbudding:

The process of removing horns/horn buds of an animal after birth.

- ✓ **Best age of dehorning in calf: before 10th day old** (up to this time horn button does not attach to the skull)

Methods of dehorning:

1. Chemical method - **KOH (caustic potash)** is used
2. Electrical method - **Electrical dehorner (1000 Degree F/10 seconds)**
3. Mechanical-by clipper saws or by rubber bands.

Important Points:

- ✓ It has been estimated that about 6-7 million buffalo/cattle male calves are available for fattening in Punjab. But the majority of these calves are sent to slaughter at 1-3 weeks of age.
- ✓ Some calves are raised to 60-80 kg on extremely poor and unbalanced diets.
- ✓ If these calves are saved and raised on balanced, fattening diets based on crop residues and agro-industrial by-products to gain a live weight of 250-300 kg, it is expected that total beef production could be doubled.
- ✓ Though livestock production is very fragmented and most farm units are small and only 10 percent of the farms in Punjab hold from 10-20 buffalos/cows and 5 percent over 20 heads each.

- ✓ A very effective way to prevent scours in calves is to vaccinate the dry cow with a scour vaccine before calving.
- ✓ Mature cows should be vaccinated four to six weeks before freshening.
- ✓ Heifers should be vaccinated two months before calving and boostered again one month before freshening.
- ✓ Scour vaccines can include the rota and corona virus, E. coli, and/or Clostridium perfingens.
- ✓ Calves Pens should be square and should provide 150 to 200 square feet of space.
- ✓ After birth, the navel cord should be dipped (not sprayed) with a 7% tincture of iodine solution. True colostrum is obtained only from the first milking.
- ✓ After the first milking and for the next two and a half days, the cow's milk is called transition milk.
- ✓ True colostrum contains twice as much dry matter and total solids, two to three times as many minerals, and five times as much protein as whole milk. Colostrum is lower in lactose, thus decreasing the incidence of diarrhea.
- ✓ Calves are born with little defense or immunity against disease.
- ✓ Holstein calves should be hand-fed 5 to 6 pints or 3 quarts of good quality colostrum within an hour of birth and again within 12 hours or the next regular feeding.
- ✓ Within six hours after birth, the ability of the gut to absorb antibodies decreases by one-third. By 24 hours, the gut can absorb only 11% of what it originally could have absorbed at birth.
- ✓ Also, at 24 hours of age, digestive enzymes break down and digest all of the antibodies.
- ✓ Colostrum should be very thick and creamy. The quality of the colostrum can be determined using a colostrometer.
- ✓ Superior quality colostrum contains greater than 50 mg/ml of immunoglobulins.
- ✓ Newborn calves should not be fed colostrum that is obtained from cows testing positive for Johne's disease.
- ✓ Three to 5 half-gallon containers of quality colostrum should be stored in a non-frost-free freezer for up to one year for feeding to calves born to dams with mastitis or no milk.
- ✓ After the initial feeding of colostrum, for the next three days, calves should receive transition milk from their dam or other cows in the herd all of which test negative for Johne's disease.
- ✓ Calves should be fed approximately 10% of their birth body weight.
- ✓ Calves with diarrhea should receive an electrolyte solution and their regular allocation of milk.
- ✓ Scouring calves can lose 10 to 12% of their body weight in water losses. Electrolytes should be given 30 minutes to 1 hour after feeding milk.
- ✓ Calf starter and water should be fed to calves starting at four days of age.
- ✓ Calves can be weaned from milk when they are eating daily 1.5 to 2 pounds of starter.
- ✓ Hay should not be fed until calves are weaned and/or they are eight weeks of age.

WATER REQUIREMENT:

Cows require at least **60liters** of water/head/day and may need **100 liters** or more depending upon yield. Water can be obtained from different sources like feed, ad-libitum consumption and from the body processes.Lactating cows in the tropic require **60-70L** water/day for maintenance, plus an extra **4 to 5L** for each litre of milk produced. Water requirement rise with air temperature. An increase of 4 degree centigrade will increase water requirements by **6 -7l/d**. High yielding milking cows can drink **150-200** L water/d during the hot season. Expected water consumption of various classes and species of adult livestock in a temperate climate

Stages of Dairy Cow:

- ✓ Stage 1 – Pre-calving. 3 weeks before calving (Transition)
- ✓ Stage 2 – Post-calving. Days 1 – 30 (Fresh cows)
- ✓ Stage 3 – Early lactation. Days 31 – 130 (Peak milk production)
- ✓ Stage 4 – Mid lactation. Days 131 – 230 (Settled period after mating for churning out milk solids)
- ✓ Stage 5 – Late lactation
- ✓ Stage 6 – Dry cow.

Differences b/w Wool and Hair

Parameter	Wool	Hair
Medulla	Almost absent	Present
Cuticle	Irregular	Regular and smooth
Sides	Scaly projections	Smooth
Diameter	Less	More
Growth	Continuous if not sheared	Reaches a maximum and then shed
Softness	More	Less
Elasticity	More	Less
Heat retention	More	Less
Moisture retention	More (12-17%)	Less (7-12%)
Dye retention	Permanent	Temporary
Luster	More	Less

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وزیرِاعظم پاکستان کے زرعی خوشحالی منصوبہ کے تحت

کٹا بچاؤ اور کٹا فربہ پروگرام

پہلے اور دوسرے مراحل کی کامیاب تکمیل کے بعد

حکومتِ پنجاب کی جانب سے گوشت کی پیداوار میں اضافے کے لئے

پنجاب بھر میں تیسرے مرحلے کا آغاز

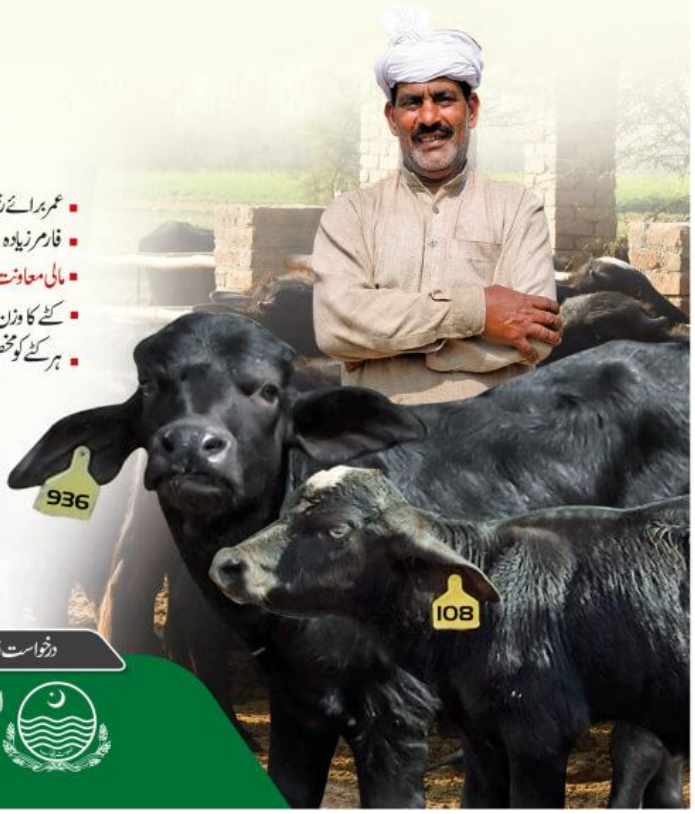
اس منصوبے کے تحت

کٹا بچاؤ پروگرام

- عمر برائے رجسٹریشن: 1 ماہ ▪ دورانیہ: 4 ماہ
- فارمر زیادہ سے زیادہ 10 کٹے رجسٹر کروا سکتا ہے
- مالی معاونت: 6,500 روپے فی کٹا
- کٹے کا وزن اوسطاً کم از کم 300 گرام روزانہ بڑھنا لازمی ہے
- ہر کٹے کو مخصوص ٹیٹو نمبر لگایا جائے گا

کٹا فربہ پروگرام

- عمر برائے رجسٹریشن: 1 سے 1½ سال ▪ دورانیہ: 3 ماہ
- فارمر زیادہ سے زیادہ 25 کٹے رجسٹر کروا سکتا ہے
- مالی معاونت: 4,000 روپے فی کٹا
- کٹے کا وزن اوسطاً کم از کم 700 گرام روزانہ بڑھنا لازمی ہے
- ہر کٹے کو مخصوص ٹیگ نمبر لگایا جائے گا



قریبی سرکاری شفاخانہ حیوانات سے دستیاب مفت سہولیات

- ☑ رجسٹریشن (درخواست فارم، شناختی کارڈ کی فوٹو کاپی کے ساتھ)
- ☑ علاج معالجہ کی سہولت
- ☑ فارمرز کو کٹے پالنے کی تربیت

درخواست فارم کی فوٹو کاپی بھی قابلِ قبول ہے

مزید معلومات و رہنمائی کے لیے 08000-9211 پر مفت کال کریں

لائیوسٹاک اینڈ ڈیری ڈویلپمنٹ ڈیپارٹمنٹ، حکومت پنجاب

@PunjabLivestock @Livestockpunjab @Livestockpunjab

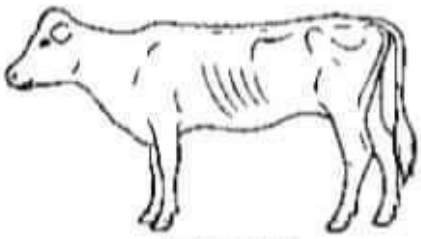
www.livestock.punjab.gov.pk

BREEDS OF INDIGENOUS DAIRY CATTLE

BULLS		COWS
	**Gir** **Native tract:** Junagadh, Rajkot, Bhavnagar and Amreli districts in Gujarat	
	**Hariana** **Native tract:** Rohtak, Hissar, Sonepat, Gurgaon, Jind and Jhajjar districts in Haryana	
	**Sahiwal** **Native tract:** Ferozpur and Amritsar district of Punjab & Shri Ganganagar district of Rajasthan	
	**Rathi** **Native tract:** Bikaner & Shri Ganganagar districts of Rajasthan	
	**Red Sindhi** **Native tract:** In Pakistan, also found in Punjab, Haryana & Rajasthan & Uttarakhand	
	**Tharparkar** **Native tract:** Jaisalmer, Barmer and Jodhpur districts in Rajasthan	
	**Kankrej** **Native tract:** Kutch, Mehsana & Banaskantha districts of Gujarat	

C. BODY SCORING

- Body scoring is very important in assessing the health status of an animal.
- A low score may indicate diseases or improper feeding while a high score may indicate a high probability of breeding and metabolic problems.



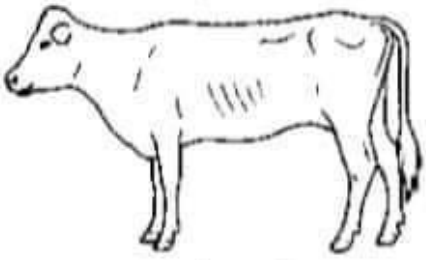
Body score 1

BODY SCORE-1

- Extremely thin. No fat in brisket or tail docks.
- All skeletal structures are visible.
- Dull hair
- May be diseased and survival during stress is doubtful



Cattle with body score 1



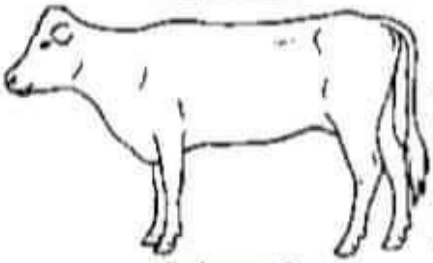
Body score 2

BODY SCORE-2

- Thin. Vertebrae, hips and pin bone prominent.
- Some tissue cover around tail dock, hip bones and the flank.
- Muscle tissue evident but not abundant, health may be OK.



Cattle with body score 2



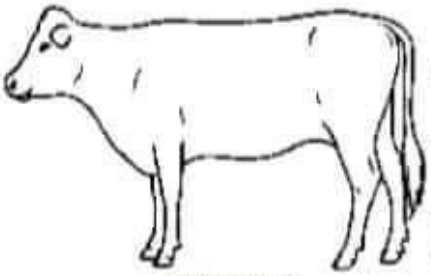
Body score 3

BODY SCORE-3

- Ribcage only slightly visible.
- Fat deposit behind shoulder obvious, ideal condition for calving.
- Fat deposit in brisket area.
- Hook and pin bone visible, but not prominent.



Cattle with body score 3



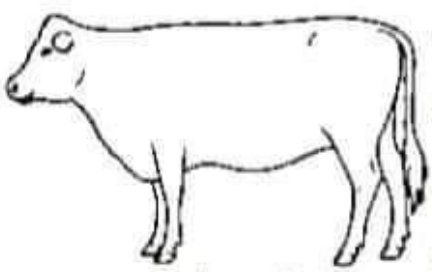
Body score 4

BODY SCORE-4

- Skeletal structure difficult to identify.
- Obvious fat deposits behind shoulder and tail head.
- Flat appearance to the top line.
- Folds of fat starting to develop over ribs and thighs.



Cattle with body score 4 may have metabolic problems at calving.



Body score 5

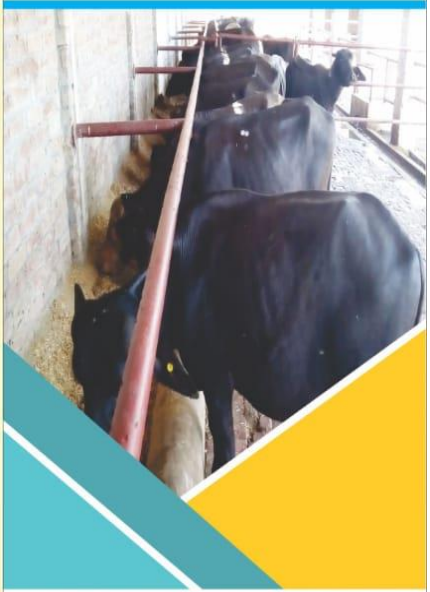
BODY SCORE-5

- Animal is obese, flat appearance dominates.
- Brisket is heavy and bone structure not noticeable.
- Tail head and hip bones completely buried in folds of fat.
- Back is flat and completely covered by fat.
- Mobility impaired by large fat deposits.



Cattle with body score 5 and above have high probability of metabolic and breeding problems.

کٹوں کی خرید و فروخت کے لیے ہدایات



محکمہ لائیوسٹاک اینڈ ڈیری ڈیولپمنٹ حکومت پنجاب

تعارف

اللہ تعالیٰ نے جہاں انسان کو بے شمار نعمتوں سے نوازا ہے، وہاں کھانے پینے کی اشیاء میں دودھ کے بعد گوشت ایسی نعمت ہے جس میں صحت کے لیے تمام ضروری اجزاء موجود ہیں۔ خوراک میں گوشت کا استعمال جسمانی بڑھوتری میں فائدہ مند ہونے کے ساتھ ساتھ انسانی جسم میں بیماریوں کے خلاف قوت مدافعت بڑھانے میں بھی مددگار ہے۔ ہمارے ملک میں ایک اندازے کے مطابق سالانہ لاکھوں کٹے پیدا ہوتے ہیں جنہیں مناسب دیکھ بھال اور متوازن خوراک دے کر فربہ کریں تو وافر مقدار میں گوشت (Beef) پیدا کیا جاسکتا ہے۔

جانوروں کا چناؤ (خرید اور ابتدائی عوامل)

کٹے فربہ کرنے کے لیے مقامی منڈیوں سے خریدے جاسکتے ہیں۔ اس کے علاوہ دور دراز علاقوں سے جانور نسبتاً سستے داموں مل جاتے ہیں۔ جانور خریدتے وقت مندرجہ ذیل باتوں کا خیال رکھیں:۔

(i) تندرست ہوں اور بہت لاغر نہ ہوں۔

(ii) اپنے ہم عمر جانوروں سے وزن میں زیادہ ہونے والے جانوروں کی نہ صرف شرح بڑھوتری نسبتاً بہتر ہوتی ہے، بلکہ ایسے جانور بیماری کے خلاف قوت مدافعت زیادہ رکھتے ہیں۔

(iii) نئے جانور خرید کر کم از کم 14 دن تک دوسرے جانوروں سے علیحدہ رکھیں تاکہ ان میں کسی بھی قسم کی بیماری کے اثرات ہوں تو وہ دوسرے جانوروں تک نہ پہنچ سکیں۔

(iv) شناخت کے لیے کانوں میں (Ear Tag) لگائیں۔

(v) مختلف بیماریوں مثلاً گل گھوٹو، منہ کھر کے لیے حفاظتی ٹیکے بروقت لگوائیں اور اندرونی کرموں کے لیے کرم کش دوائی کا استعمال کریں۔

(vi) جانوروں کو 24 گھنٹے صاف اور تازہ پانی کی وافر مقدار میں فراہم کریں۔

(vii) نئے خریدے گئے جانور چونکہ چارہ کھانے کے عادی ہوتے ہیں لہٰذا انہیں چارے سے بتدریج متوازن خوراک کی طرف لانا چاہئے 10 سے 15 دن کے بعد جانوروں کو مکمل طور پر راشن پر لایا جاسکتا ہے۔

متوازن راشن کی اہمیت

ترقی یافتہ ممالک میں گوشت کے لیے جانوروں کو فربہ کرنے کے لیے خوراک میں متوازن ونڈے کا استعمال عام ہے۔ ونڈا جانور کو فربہ کرنے کے لیے توانائی مہیا کرتا ہے۔ ہمارے ہاں جانوروں کو زیادہ تر چارے پر ہی پالا جاتا ہے اور ان کو ونڈا نہیں دیا جاتا۔ اگر انہی جانوروں کی مناسب دیکھ بھال کی جائے اور ان کو متوازن ونڈا دیا جائے تو اس سے نہ صرف جانور کا وزن جلدی بڑھتا ہے بلکہ اس پر آنے والی پیداواری لاگت بھی کم ہو جاتی ہے جس کی وجہ سے مالک کو منافع بھی زیادہ ہوتا ہے جانوروں کو فربہ کرنے کے لیے متوازن خوراک کی ضرورت ہے۔ اگر جانور کو صرف سبز چارہ ہی دیا جائے تو اس کی بڑھوتری بہت ہی کم ہوگی ان تمام باتوں کو مدنظر رکھتے ہوئے ضروری ہے کہ جانوروں کو عمر کے مطابق ایک متوازن خوراک دی جائے۔ جس سے لاگت کم اور تھوڑے عرصہ میں جانور فربہ ہو جائے۔

ضرورت اس امر کی ہے کہ مسلسل اجتماعی اور قابلِ عمل کوششوں سے جانوروں سے فی کس اضافی گوشت کے حصول کو ممکن بنایا جائے۔

کٹوں کی فروخت

کٹوں کو پالنے کے بعد اگلا مرحلہ ان کی فروخت ہے۔ شہروں میں مخصوص دنوں میں مویشی منڈیاں لگتی ہیں۔ جہاں جانوروں کی فروخت کی جاتی ہے۔ نیز جانوروں کی خرید کے مراکز مختلف شہروں میں موجود ہیں۔ جہاں گوشت کے جانوروں کو فروخت کیا جاسکتا ہے۔ اس کے علاوہ پاکستان سے کچھ برآمد کنندگان دوسرے ممالک کو گوشت بھی برآمد کرتے ہیں۔ جو کہ اچھے جانوروں کو بہتر قیمت پر خریدتے ہیں۔ پنجاب میں منعقد ہونے والی مختلف منڈیوں کا شیڈول اور برآمد کنندگان کی تفصیل ذیل میں دی گئی ہے۔

نوٹ: بیف بائرز/ایکسپورٹرز کے رابطہ نمبرز اور ایڈریس فارمرز کی سہولت کے مدنظر مہیا کیے گئے ہیں۔ ان کے ساتھ فارمر حضرات کسی بھی قسم کا لین دین اپنی ذمہ داری اور سہولت پر کریں۔

صوبہ پنجاب میں ہفتہ وار شیڈول برائے مویشی منڈی

نمبر شمار	شہر کا نام	دن	نمبر شمار	شہر کا نام	دن
1	اسلام آباد/روات	منگل، بدھ	2	پنڈیاں	منگل
3	لاہور شاہ پور کانجرہ	بدھ	4	خوشاب	منگل
5	میانوالی	اتوار	6	جھنگ	ہفتہ
7	سرگودھا	بدھ	8	میاں چنوں	جمعہ
9	ساہیوال	بدھ	10	یزمان بہاولپور	ہفتہ
11	فتح جھنگ	جمعہ، ہفتہ	12	احمد پور شرقیہ	جمعہ
13	چنیوٹ	بدھ	14	بہاولپور	منگل، بدھ
15	بہاولنگر	بدھ	16	ملتان	بدھ، اتوار
17	گوجرہ	منگل	18	گوجرانوالہ	منگل
19	فیصل آباد	بدھ	20	گجرات	منگل
21	لودھراں	بدھ	22	جام پور	منگل
23	ڈی جی خان	بدھ	24	چوک اعظم	منگل
25	بھکر	ہفتہ	26	کوٹ چھوٹہ	جمعرات
27	سیالکوٹ	ہفتہ	28	سرائے مہاجر	جمعہ
29	چکوال	بدھ	30	تاندلیانوالہ فیصل آباد	بدھ
31	شکانہ صاحب	منگل	32	تلہ گنگ	سوموار، جمعہ
33	اٹک	اتوار	34	ملہموڑ جھنگ	ہفتہ
35	منڈی بہاؤدین	بدھ	36	مُدراسہ بہاولنگر	ہفتہ
37	کھاریاں	سوموار، بدھ	38	پل کمبر ڈی جی خان	منگل
39	رحیم یار خان	اتوار	40	عارف والہ	ہر ماہ ایک دن
41	شیخوپورہ	ہفتہ	42	سیدار منڈی فیصل آباد	بدھ
43	راجن پور	منگل، اتوار	44	کونجر گجرات	ہفتہ

بیف بائرز/ایکسپورٹرز لسٹ

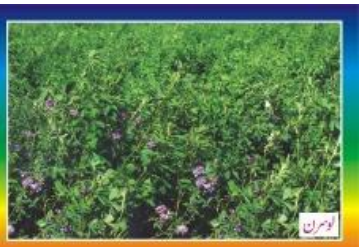
نمبر شمار	کمپنی کا نام	ایڈریس اور رابطہ نمبرز
1	بیف سینسر (UVAS)	عبدالقادر جیلانی روڈ، نزد یونیورسٹی آف ویٹرنری اینڈ اینمل سائنسز لاہور۔ 0311-6810997
2	زینتھ ایسوسی ایٹس	3.5 کلومیٹر مانگا روڈ رائیونڈ، لاہور 03008464607-/04235391490 zeenithasso@hotmail.com/
3	لحم ٹریڈرز	B-134 بلاک ماڈل ٹاؤن، لاہور۔ 03224855555
4	سید ٹریڈرز	177- بی بلاک جوہر ٹاؤن، لاہور۔ 03008401680,04235169450, 04235111575, 04235169449 info@syedtraders.com
5	الشائزہ انٹرنیشنل	مکان نمبر 1، گلی نمبر 4 شاداب کالونی روڈ 18 کلومیٹر فیروز پور روڈ لاہور۔ 03028490001- asit68@gmail.com
6	عابدین انٹرنیشنل	670- اے ٹو سیکٹر 4، ٹاؤن، لاہور۔ 03334214108-04235111575 Info@zain-abedin-intl.com www.zain-abedin-intl.com
7	تاجز میٹ اینڈ فوڈز	3.5 کلومیٹر مانگا روڈ رائیونڈ، لاہور 04235393465/04235393467/03219545570 a.hannan@tajizgroup.com
8	کیلکو انٹرنیشنل	632- بی، سبلائٹ ٹاؤن نزد رانا اقبال ایڈووکیٹ، گوجرانوالہ 03009642067
9	ال امان ٹریڈنگ کمپنی	23- سمال انڈسٹریل اسٹیٹ، کوٹ لکھپت، لاہور 03014005003,04235150724-25 Livestock.meat@aliman.com.pk-
10	کارنیگر کوئیکشن	88- اے، مین بلیورڈ گلبرگ لاہور 03004743330
11	اکمل ٹریڈرز	633- جی گلشن راوی لاہور 04237465741 / 03009447051 Akmaltraders@hotmail.com
12	ال معراج انٹرنیشنل	4- مانگا روڈ نزد نشاط چوبیاں، لاہور۔ 03216433800,04235391911 04235391912.ant@yahoo.com

بیف بائرز/ایکسپورٹرز لسٹ

نمبر شمار	کمپنی کا نام	ایڈریس اور رابطہ نمبرز
13	کاشف ٹریڈرز	مکان نمبر 180-5 اے، ٹو، ریظل رنچ روڈ لاہور، اقبال پارک مین بلیورڈ ڈی ایچ اے 04236672266/03004289567- mrehan@brain.net.pk
14	گرین میڈوز	Q- بلاک مکان نمبر 201 ڈی ایچ اے لاہور۔ 03008415142
15	انیس ایسوسی ایٹس	4- کلومیٹر مانگا روڈ، رائیونڈ۔ 04235393510/03334379923 - Saifigroup@hotmail.com
16	ابدالی ٹریڈرز	405- نشتر بلاک علامہ اقبال ٹاؤن لاہور۔ 03334257813 - abdalitraders@hotmail.com
17	ایس ایم انٹر پرائزز	607- لینڈ مارک پلازہ، جیل روڈ لاہور۔ 04235877395/04235079693, 03008416739 04237980036/sm enterprisespk@yahoo.com
18	سیف انٹر پرائزز	94- ایچ، فیز، اڈی ایچ اے 04235745 62/04235742562/03214741837 Suk-ghuman@yahoo.com
19	کول فوڈز	5th فلور کمرہ نمبر 219 میگا ٹاور مین بلیورڈ گلبرگ لاہور۔ 04235777684/03004743330
20	طلکار پوریشن	15- کلومیٹر ماڈل ٹاؤن لاہور۔ 04235916712 /04235916711/ 03218440011 tahatrading@live.com
21	شفیع فوڈز	51- جی ٹی روڈ، چھڑہ منڈی پوسٹ بکس 1676 لاہور۔ 042357257089 /03008478472- info@everfreshmeat.com info@shafifoods.com
22	الصبا انٹرنیشنل	337- عسکری 10 لاہور کینٹ۔ 04236503030 /03009411723 Aftab_waseem@hotmail.com
23	اشاء لائیوسٹاک اینڈ میٹ کمپنی	6- اے بلاک جوہر ٹاؤن لاہور۔ 04235150725 /0304005003 Livestock.meat@aliman.com.pk
24	ایمک ٹریڈنگ کمپنی	586- کیو بلاک ایم اے جوہر ٹاؤن لاہور۔ 03009408539 / 04235310940
25	اسٹاک فیڈ	2A- احمد بلاک گارڈن ٹاؤن، لاہور۔ UAN #111-111-220



موسم سرما کے چارے



نمبر شمار	نام فصل	اقسام	قسم زمین، تیاری	شرح بیج فی ایکڑ	وقت کاشت	طریقہ کاشت	کھاد فی ایکڑ		مقدار عناصر		وقت برداشت	پیداوار فی ایکڑ	کیمیائی تجزیہ	
							وقت استعمال	مقدار بوری	ناٹروجن	فاسفورس			لحمیات	خشک مادہ فیصد
1	برسیم	1- اگیتی برسیم 2- پنجابی برسیم 3- پشاوری 4- انمول 5- سپریٹ فیصل آباد	بھاری میرا زمین ہلکی کلراٹھی زمین پر بھی کاشت کی جاسکتی ہے۔ تین چار دفعہ دیسی ہل اور سہاگہ چلائیں	8 تا 10 کلوگرام	اکتوبر تا نومبر	بذریعہ چھٹہ کھڑے پانی میں	بجائی چار پتوں پر	5 بوری ایس ایس پی یا 1.5 بوری ڈی اے پی ½ 1 بوری یوریا	35 23 23	50	نومبر تا مئی	30 تا 35 ٹن	19	12
2	لوسرن	سرگودھا لوسرن	میرا زمین ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ ہل اور سہاگہ چلائیں	6 تا 8 کلوگرام	اکتوبر تا نومبر	بذریعہ چھٹہ کھڑے پانی میں یا بذریعہ ڈرل قطاروں کا فاصلہ 30 سینٹی میٹر	بجائی چار پتوں پر	5 بوری ایس ایس پی یا 1.5 بوری ڈی اے پی ½ 1 بوری یوریا	35 23 23	50	دسمبر سے کٹائی دائمی فصل تین سال تک اچھی فصل دیتا ہے	50 تا 60 ٹن	23	20
3	جئی	1- اے ون کینٹ 2- ایس 2000 3- پی ڈی 2 ایل وی 65 4- سی کے 1 5- راوی	بھاری میرا زمین ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ ہل اور سہاگہ چلائیں	30 کلوگرام بیج کیلئے 20 کلوگرام	اکتوبر تا دسمبر	بذریعہ ڈرل 25 تا 30 سینٹی میٹر	بجائی	½ 2 بوری ایس ایس پی 1 بوری ڈی اے پی 1 بوری یوریا بیج کیلئے 2 بوری ڈی اے پی 1 بوری ایس او پی 1 بوری یوریا	12 23	25	جنوری تا مئی	25 تا 35 ٹن	14	23.81
4	رائی گراس سالانہ اقسام دائمی اقسام	اٹلین، پلین، ٹیٹران ملٹیو اور ویسٹرولڈ ہائبرڈ یوں، موریٹی فیوگو اور مسٹرال	بھاری میرا زمین ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ ہل اور سہاگہ چلائیں	8 کلوگرام	اکتوبر تا دسمبر	بذریعہ چھٹہ کھڑے پانی میں	بجائی پر پہلے پانی پر اور ہر کٹائی پر	½ 2 بوری ایس ایس پی 1 بوری ڈی اے پی 1 بوری یوریا بیج کیلئے 2 بوری ڈی اے پی 1 بوری ایس او پی 1 بوری یوریا	23 23 23	25	دسمبر تا اپریل	30 تا 40 ٹن	15	19
5	برسیم + جئی		بھاری میرا زمین ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ ہل اور سہاگہ چلائیں	برسیم 6 کلوگرام جئی 10 تا 12 کلوگرام	اکتوبر تا دسمبر	جئی سہاگہ کے نیچے اور برسیم کھڑے پانی چھٹہ	بجائی پر پہلی کٹائی دوسری کٹائی	5 بوری ایس ایس پی یا 1.5 بوری ڈی اے پی 1.5 بوری یوریا 1 بوری یوریا 1 بوری یوریا	35 23 23	50	دسمبر تا مئی	30 تا 35 ٹن	17.90	12.80
6	برسیم + رائی گراس		بھاری میرا زمین ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ ہل اور سہاگہ چلائیں	6 کلوگرام برسیم 4 کلوگرام رائی گراس	اکتوبر تا دسمبر	بذریعہ چھٹہ کھڑے پانی میں	بجائی پر پہلی کٹائی دوسری کٹائی	5 بوری ایس ایس پی یا 1.5 بوری ڈی اے پی 1.5 بوری یوریا 1 بوری یوریا 1 بوری یوریا	35 23 23	50	دسمبر تا مئی	35 تا 40 ٹن	18.66	14.26



نوٹ

1- زمین کی زرخیزی کو مدنظر رکھتے ہوئے کھاد کے استعمال میں کمی بیشی کی جاسکتی ہے۔

2- زہریلی ادویات مجبوری کی صورت میں استعمال کریں اور ایسی ادویات استعمال کریں جن کا اثر شام سے کم وقت میں ختم ہو جائے گا۔

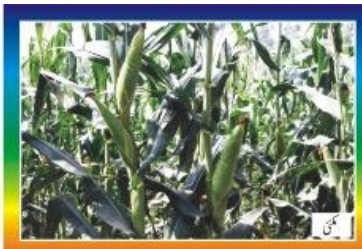
3- سپرے شدہ چارہ کی فصل کو کم از کم پندرہ دن کے بعد بطور چارہ استعمال کریں۔



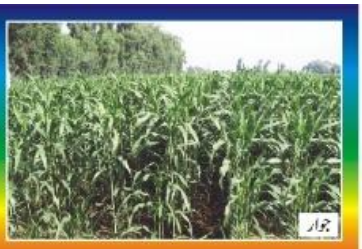
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مزید معلومات کیلئے شعبہ چارہ جات سے رجوع فرمائیں

شائع کردہ: ادارہ تحقیقات افزائش حیوانات بہادر نگر اوکاڑا



موسم گرما کے چارے



نمبر شمار	نام فصل	ورائٹی	قسم زمین، تیاری	شرح بیج فی ایکڑ	وقت کاشت	طریقہ کاشت	کھاد فی ایکڑ		مقدار عناصر (کلوگرام)			وقت برداشت	پیداوار فی ایکڑ	کیمیائی تجزیہ	
							وقت استعمال	مقدار بوری	فاسفورس	نائٹروجن				لحمیات	خشک مادہ فیصد
1	مکئی	2-او کے ایس (OK.M.A.S) 3- ساہیوال 2002	میرا زمین، ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ دیسی ہل اور سہاگہ چلائیں۔	40 کلوگرام بیج کیلئے 20تا15 کلوگرام بذریعہ ڈرل	مارچ تا ستمبر	بذریعہ ڈرل 25 تا 30 سینٹی میٹر	بجائی پر پہلے پانی پر	3 بوری ایس ایس پی 1/2 بوری یوریا یا 1 بوری ڈی اے پی 1 بوری یوریا بیج کیلئے 2 بوری ڈی اے پی 2 بوری ایس او پی، 3 بوری یوریا	30	12 23		مئی تا دسمبر	20تا24 ٹن	8.3	24.0
2	جوار	1- کے-94 2- ہے ایس 2002 3- ہنگاری 4- نور 5- پی ای سی 1 6- ساتھ کٹ بار	گہری زرخیز زمین ، ہلکی کلراٹھی زمین پر بھی کاشت ہوسکتی ہے۔ ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ دیسی ہل اور سہاگہ چلائیں	30 کلوگرام بیج کیلئے 10تا8 کلوگرام بذریعہ ڈرل	مارچ تا اگست	بذریعہ ڈرل 25 تا 30 سینٹی میٹر	بجائی پر پہلے پانی پر	3 بوری ایس ایس پی 1/2 بوری یوریا یا 1 بوری ڈی اے پی 1 بوری یوریا بیج کیلئے 2 بوری ڈی اے پی 1 بوری ایس او پی، 2 بوری یوریا	30	12 23		جون تا نومبر	24تا28 ٹن	7.35	29.09
3	سدابہار	میٹھی سدابہار لاجواب	بھاری میرا زمین ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ دیسی ہل اور سہاگہ چلائیں	15 کلوگرام	مارچ تا مئی	بذریعہ ڈرل 45 سینٹی میٹر	بجائی پر پہلے پانی پر	3 بوری ایس ایس پی 1/2 بوری یوریا یا 1 بوری ڈی اے پی 1 بوری یوریا	30	12 23		مئی تا اکتوبر	50تا60 ٹن	7.35	24.0
4	ماٹ گراس	1- ملائم 2- بہادر ماٹ	بھاری میرا زمین ایک دفعہ مٹی پلٹنے والا ہل اور پانچ دفعہ ہل اور سہاگہ چلائیں	5000 قلمیں	فروری تا مارچ	90x90 سینٹی میٹر	بجائی پر پہلے پانی پر ہر کٹائی کے بعد	5 بوری ایس ایس پی یا 2 بوری ڈی اے پی 1 بوری یوریا 1½ بوری یوریا	30	12 23 35		مئی تا اکتوبر	80تا100 ٹن	7.30	14.31
5	پارا گراس	پارا گراس	سیلاب اور سیم زدہ زمین پر اچھی فصل دیتا ہے ایک دفعہ مٹی پلٹنے والا ہل اور چار دفعہ دیسی ہل اور سہاگہ چلائیں	22000 قلمیں	فروری تا مارچ	25x25 سینٹی میٹر	بجائی پر پہلے پانی پر ہر کٹائی کے بعد	3 بوری ایس ایس پی 1/2 بوری یوریا 1 بوری یوریا 1½ بوری یوریا	30	12 23 35		مئی تا نومبر	40تا50 ٹن	6.96	21.06
6	باجرہ	1- بی ایس 2002 2- ایم بی 87 3- ڈی بی 2001 4- ایف 786	زرخیز زمین، ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ دیسی ہل اور سہاگہ چلائیں	5 کلوگرام بیج کیلئے 3 کلوگرام	اپریل تا ستمبر	بذریعہ ڈرل 25 تا 30 سینٹی میٹر	بجائی پر پہلے پانی پر	3 بوری ایس ایس پی 1/2 بوری یوریا 1 بوری یوریا بیج کیلئے 2 بوری ڈی اے پی 1 بوری ایس او پی، 2 بوری یوریا	30	12 23		جون تا نومبر	20تا22 ٹن	8.26	16.80
7	روانحہ	1- رواں 2003 2- رواں قسم نمبر 1	بھاری میرا زمین، ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ دیسی ہل اور سہاگہ چلائیں	20 کلوگرام بیج کیلئے 10تا8 کلوگرام بذریعہ ڈرل	مارچ تا اگست	بذریعہ ڈرل 45 سینٹی میٹر	بجائی پر	3 بوری ایس ایس پی 1/2 بوری یوریا بیج کیلئے 2 بوری ڈی اے پی 1 بوری ایس او پی	30	12		مئی تا اکتوبر	15تا20 ٹن	18.3	18.57
8	گوارہ	1- بی آر-99 2- بی آر-90	میرا زمین، ایک دفعہ مٹی پلٹنے والا ہل اور تین چار دفعہ دیسی ہل اور سہاگہ چلائیں۔	20 کلوگرام	اپریل تا جولائی	بذریعہ ڈرل 45 سینٹی میٹر	بجائی پر	3 بوری ایس ایس پی 1/2 بوری یوریا	30	12		جون تا نومبر	15تا20 ٹن	20.0	17.10











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مزید معلومات کیلئے شُعبہ چارہ جات سے رجوع فرمائیں

شائع کردہ: ادارہ تحقیقات افزائش حیوانات بہادر نگر اوکاڑا

BASICS OF DAIRY SCIENCE:

Note:
- ✓ **HTST is the modern method of pasteurization** and used at commercial level
- ✓ In Vacuum pasteurization, milk is pasteurized **under reduced pressure by direct steam**
- ✓ Vacreator is used to remove feed & other volatile flavors from cream & pasteurize it for **butter making**

Advantages & disadvantages of pasteurization:
- ✓ Diminishes the nutritive value of milk
- ✓ Reduces cream line or cream volume
- ✓ It causes complete destruction of **Phosphatase enzyme**
- ✓ It causes **100% destruction of pathogenic bacteria**
- ✓ It fails to destroy bacterial toxins in the milk
- ✓ Rennet cannot clot the pasteurized milk.

Note:
- ✓ **Index organism for pasteurization is *Coxiella burnetti*** (causative agent of Q-fever)
- ✓ Phosphatase enzyme in normally present in raw milk, but it is destroyed by pasteurization so phosphatase test for pasteurized milk is negative hence **phosphatase test is used to detect inadequacy of pasteurization.**
- ✓ **Plate count for pasteurized milk should not exceed 30,000 per ml of raw milk.**

Methylene blue reduction test:
Objective: To determine the extent of bacterial contamination in milk
Principle: Time taken to reduce M.B. is a measure of its bacterial contamination.

Cow milk composition:
- ✓ **It contains 87% water & 13% total solids (TS/dry matter)**
- ✓ Total Solids is categorized under (a) Fat/Lipid (b) Solid Not Fat (SNF)
- ✓ Fat includes: Phospholipids, Cholesterol, Carotene & Fat soluble vitamins (A,D,E,K)
- ✓ **SNF includes: Lactose(milk sugar), Proteins, Ash(minerals)** &others
- ✓ Milk proteins includes- (a) Casein protein (b) Whey proteins/milk serum proteins
- ✓ **Whey proteins includes: α-lactalbumin & β-lactoglobulin**
- ✓ Riboflavin is a greenish- yellow pigment that provides characteristic colour to whey.

Note:
- ✓ Cow milk contains **3% casein** while buffalo milk contains **4.3% casein**
- ✓ Casein contributes about **80%** of total milk protein
- ✓ Casein exists in milk in the form of **Calcium-caseinate-phosphate complex**
- ✓ Casein is responsible for **viscosity & white colour** of milk
- ✓ There are three types of casein protein: α, β, γ

- ✓ α -casein consists of two components namely αs-casein (calcium-sensitive casein) and K-casein (calcium-insensitive casein)
- ✓ αs-casein is precipitated by calcium ion while K-casein is not precipitated by calcium ion.

Milking methods:

A. **Hand milking:** Most common practice among individual farmers

B. Machine milking: Practiced in organized government, Commercial/ Corporate & military dairy farms.

Note:

- ✓ The cow should be **milked from the left side**
- ✓ First few strips of milk should not mix with rest of the milk because **fore- milk contains large no. of bacteria.**

Machine milking:

- ✓ It performs two basic functions:

(a) It opens the streak canal with partial vacuum, allowing the continuous flow of milk

(b) It massages the teats that **prevent congestion of blood in the teat.**

Note:

- ✓ Minimum herd size for machine milking is **20**
- ✓ Recommended vacuum in machine milking is **10-15 inches**
- ✓ The optimum pulsation rate in machine milking: **60 cycles/minute**
- ✓ Pulsation rate should not less than 40 cycles/minute & not more than 120 cycles per minute
- ✓ Pulsator is also known as "heart of milking machine"
- ✓ Pulsation ratio of milking machine- (a) For cattle: 60/40 (b) For buffalo: 50/50
- ✓ Ideal vacuum of milking machine- (a) For cattle: 352 mmHg (b) For buffalo: 400 mmHg

Colostrum/1st milk/Beestings:

- ✓ Colostrum is fed to newborn calf **@ 10 % of its body wt. (or) 2-2.5 liters per day up to 3-4 days.**
- ✓ Placental transfer of antibodies does not occur in ruminants so their neonates depend upon colostrum as a source of antibodies.
- ✓ **Colostrums provide passive immunity** to the calf.
- ✓ It act as **laxative & clears the meconium (1st faeces voided by new born calf)**
- ✓ Colostrum of ruminant contains a **trypsin inhibitor** that protects the immunoglobulins from digestion in G.I.T. of the calf.
- ✓ Milk lost its colostral property within 3-4 days after parturition.

Note:

- ✓ Predominant immunoglobulin in bovine colostrum is IgG while in human colostrum is IgA
- ✓ Immune factors present in bovine colostrum are lactoferrin & immunoglobulins like IgG > IgM > IgA > IgG 2 (according to decreasing order of their concentration).
- ✓ Lactoferrin is an iron-binding protein, very important during mammary involution.
- ✓ High yielding breed produce lower fat %
- ✓ **Final stripping contains higher fat percentage than foremilk**.
- ✓ Evening milk contains higher fat than morning milk.

Recknagel phenomenon:

- ✓ The specific gravity of freshly drawn milk is low. It increases by 0.001 as time advances.

Note:

(1) Specific gravity:

 Specific gravity of milk = 1+ CLR/1000 (where CLR is Corrected Lactometer Reading)

- ✓ **The specific gravity of water is one**
- ✓ Specific gravity of milk is lowered by addition of water or by addition of cream/fat
- ✓ Specific gravity of milk is increased by addition of skim milk or by removal of cream/fat

(2) Freezing point:

- ✓ The freezing point of pure distilled water is 00C while freezing point of milk is slightly lower than that of water due to the presence of soluble constituents like lactose, minerals etc., which depress the freezing point.
- ✓ Lactose and chloride are primarily responsible for depressing the freezing point.
- ✓ Freezing point of cow or buffalo milk ranges from -0.535 Degree Celsius to -0.550 Degree Celsius **with an Avg. of -0.545 Degree Celsius.**
- ✓ The addition of 1% water to milk will raise its freezing point by 0.0060C.
- ✓ Mastitis milk shows the normal freezing point.

Udder (Mammary gland):

- ✓ Mammary glands are modified sudoriferous (sweat) glands.
- ✓ Udder is ectodermal in origin
- ✓ Mammary glands are innervating with only sympathetic fibers
- ✓ Parasympathetic fibers are absent in udder**.**
- ✓ **Inter milking interval:** time interval between subsequent milking.
- ✓ The ideal inter- milking interval is 12 hours.

The proportion of milk stored in cistern/alveolus:

- ✓ **Goat-80:20**, **Sheep:-50:50 , Cattle:-30:70 , Buffalo & Camel :-5:95 , Sow:- 0:100**
- ✓ Dairy cows store only 30% of the total milk yield volume in the cisterns after a normal milking interval. However, the **cisternal milk accounts for up to 80% in dairy goats**

that is why we can draw milk at any time from goat because there is no oxytocin reflex is required in case of goat for a letdown.

Circulatory system:

- ✓ **Arterial blood supply: -by external pudic artery(mammary artery)**
- ✓ **Venous return(3 primary routes)**

(a) **Subcutaneous abdominal vein (milk vein)-1 st route**
(b) External pudic vein-2 nd route
(c) Perineal vein-3 rd route

- ✓ **Subcutaneous abdominal vein** act as an **"index for milking ability"**

Suspensory structures of udder:

a) Median suspensory ligament
b) Two lateral suspensory ligament
c) Skin

- ✓ Median suspensory ligament is elastic while laterals are non-elastic
- ✓ The median suspensory ligament is primarily responsible for udder "break-down".
- ✓ There is no communication among all 4 quarters of the udder. It will minimize the spread of infection within the udder.

Note:

400-500 ml of blood must pass through udder for producing 1 ml of milk.

Note:

Milk normally contains only 2−4 ppm of iron. But the high iron content of khoa is because of karahi is made up of iron.

Special points about Khoa:

- ✓ Buffalo milk is preferred
- ✓ A minimum fat level of 4% in cow milk & 5% in buffalo milk is necessary to obtain good quality khoa.
- ✓ Normal yield of khoa: 17-19% from cow milk & 21-23% from
- ✓ buffalo milk.

Dahi/Curd:

It is a fermented milk product

Types of Dahi:

(a) Sweet/Sour dahi: 0.5-1% starter
(b) Sweetened dahi: Prepared by adding 6.25% cane sugar to milk.

Some special points:

- ✓ **Buffalo milk** is **white** in colour due to **casein protein** while **cow milk is yellow** in colour due to **β- carotene**.
- ✓ **Least variable** component of milk is **lactose** while a **most variable** component of milk is **fat.**

- ✓ **Most commonly used adulterant in milk is water.**
- ✓ **The real acidity of milk** is due to **lactic acid** formed by bacterial action.
- ✓ Milk sourness is due to Volatile acids.
- ✓ Milk sweetness is due to lactose while bitterness or saltiness is due to chloride.
- ✓ Milk – Milk Serum= Coagulum
- ✓ Milk –Milk Fat= Skim milk
- ✓ Fishy odour in milk is due to kephalin while fishy odour in the egg is due to E.coli.
- ✓ Fishy odour in butter is due to rancidity
- ✓ Butter contain all fat soluble vitamins (Vita.A, D, E & K)
- ✓ Flavoring agent used in butter is **Diacetyl (not more than 4ppm)**
- ✓ Yellow colour of butter is due to **Annatto**
- ✓ Colouring agents used in milk are: -Annatto, Coal tar dye & Turmeric
- ✓ Cooked flavor of heated milk is due to formation of **Sulfhydryl compounds**
- ✓ **Thickening agents** used in milk are: **Cane sugar (sucrose), Starch, Gelatin**
- ✓ **Preservatives used in milk** are: Boric acid, Borax, Benzoic acid, Formalin, H2O2, Salicylic acid, Na2CO3, NaHCO3
- ✓ Natural preservatives present in milk are -Lactoperoxidase, Lactoferrin, and Conglutinin.
- ✓ **Homogenization of milk**: The process of making a stable emulsion of milk serum & milk fat by use of a homogenizer.

(a) We cannot separate cream from homogenized milk

(b) Size of fat globule in homogenized milk is **2 microns**.

- ✓ **Bactofugation**: the process of removing 99.99% bacteria by use of centrifugal forces.It **triples the shelf life** of market milk
- ✓ Milk letdown hormone is **oxytocin** while hormone responsible for **holding up of milk is epinephrine.**
- ✓ Lactation period of dairy cattle divided into three phases: Early phase (1 to 100 days), Mid-phase (101 to 200 days) and Late phase (201 to 305 days)
- ✓ Peak milk yield in cattle is in **6-8 weeks after parturition**. Milk yield increases over lactation.
- ✓ Maximum milk yield occurs between 4th -6 th lactation in cow & fourth lactation in buffalo.
- ✓ In dairy cattle, dry matter intake is at peak during mid-phase of lactation.
- ✓ **Residual milk**: It is the amount of milk left in the udder after normal milking.It can obtain **only after oxytocin injection or by remilking.**

(a) Older cows have a higher percentage of residual milk than first-calf heifers.

(b) % of residual milk is more in low yielders than high yielders

(c) A cow with a high percentage of residual milk shows a lower persistency of lactation.

- ✓ **Acetate** (VFA) is responsible for **milk fat synthesis.**
- ✓ **Vitamin B1 &Vitamin C** are lost during process of pasteurization & sterilization of milk.
- ✓ **Irradiated milk:** Vitamin D content of the milk is increased by U.V.rays treatment.
- ✓ **Stainless steel and aluminium alloy** mainly used for the manufacturing of the dairy utensils.

✓ Milking utensils should be sanitizing with **hypochlorite** before use.

TYPES OF EMULSION:

Item	Type of Emulsion
Milk, Cream	Fat in Water
Butter	Water in Fat
Meat	Oil in Water

MILK CONSITUENTS:

Milk Constituent	Form in which exists
Fat	Emulsion
Casein protein	Colloidal suspension
Whey protein	Colloidal Solution
Lactose, Ash	True Solution

MILK DIRECT AND INDIRECT TESTS:

Sr. No.	Test Name	Use	Reagent	Result
1	**Organoleptic test**	To detect taste, sight & smell of milk	----------	----------
2	**Clot on boiling**	To detect acid milk or abnormal milk	----------	----------
3	**The alcohol test**	To detect instability of proteins	Ethanol	If coagulation test will be positive
4	**Phosphate test for pasteurized milk**	To detect milk is adequately pasteurized or not.	Alkaline phosphatase	----------
5	**The Gerber butter fat test**	To detect fat content of milk	H_2SO_4 & amyl alcohol	----------
6	**SNF of milk**	To detect milk adulterations	Lactometer is used	----------
7	**Salt detection test**	To detect salt adulter	5% Potassium chromate,	Presence of red color indicate

		ant in milk	0.1 N Silver nitrates	absence of chloride
8	**Sugar detection test**	Detect Lactose	Conc. HCl, Resorcinol	If brick red color test will be positive
9	**Urea detection test**	To detect urea in milk	Dimethyl aminobanzal dehyde	Yellow color test will be positive
10	**Determination of milk pH**	To determine milk pH	Indicator paper strips used for this	Cow milk pH 6.7 Buffalo 6.8 If pH < 6.7 indication of disease If pH >6.7 indicate neutralization of milk by NaOH, Na2 CO3 & NaHCO3 .
11	**Freezing Point determination**	----------	Cryoscope is used for this purpose	Normal value -0.54°C
12	**Detection of Detergents**	----------	Methylene blue & Chloroform	If blue color in lower layer test will be positive
13	**Rapid test for detection of aflatoxin M1 in milk**	To detect Asper gillus spp toxin	----------	----------
14	**Formaline test or Hehner's test**	----------	90% H2So4 & formaldehyde	If formation of purple ring test will be positive

15	**Holde's test**	Detection of mineral oil in ghee	4% Potassium hydroxide	Turbidity indicate test will be positive
16	**Milk ring test**	Detection of brucellosis	----------	Formation of dark blue ring test will be positive
17	**California mastitis test**	----------	----------	LCC is < 200,000 test – ve if >5000,000 test will be strong +ve
18	**Resazurin test**	To detect milk hygiene & also detect microbial activity	----------	If blue color, then quality of milk will be excellent; While white color shows bad quality of milk
19	**Viable count**	Plate count method	Ringer solution is used	Incubate and no. of colonies are calculated
20	**Ammonium Sulfate detection test**	----------	Barium chloride 5%, trichloroacetic acid	Formation of milky white precipitates indicate test will be positive

BASICS OF MEAT SCIENCE:

The composition of animal meat:

- ✓ 72% water & remaining 28% DM (DM consists of **21% Protein, 6% Fat & 1% Ash)**

Note:

- ✓ **Bright red** color of meat is due to **oxymyoglobin**
- ✓ **Brown color** of meat is due to **metmyoglobin**
- ✓ **Pink color** of cured meat is due to **nitro-haemochromogen**
- ✓ **Variety meat/organ meat: -**organs such as tongue, brain, thymus, heart, liver & kidney are known as variety meat
- ✓ **Pork contains 5-10 times more vita.B1 (thiamin) than beef& mutton.**
- ✓ Folic acid & vita.B12 is higher in beef & mutton
- ✓ **Bacon:** meat from the sides and the back of a pig
- ✓ Bacon is preserved by curing with brine
- ✓ **Ham:** meat of the thigh or buttocks of a pig
- ✓ **Prime pork**: carcass of castrated male & virgin female of a pig
- ✓ Poultry meat is known as white meat while the meat of other animal is known as red meat
- ✓ **Vita. B12**: Vitamin found in highest amount in animal meat.
- ✓ **Niacin:** Vitamin found in highest amount in poultry meat.
- ✓ Vitamin A is present in beef and mutton but absent in buffalo meat, chevon or pork

Pale, soft & exudative meat/PSE meat/Watery pork:

- ✓ It commonly occurs in **stress-susceptible animal‟s where pH fall very quickly to 5.8-5.6**
- ✓ **Most commonly found in pork**
- ✓ PSE meat has lower water binding properties and loses weight rapidly during cooking resulting in a decrease in processing yields.

DFD meat/Dry, Firm &Dark meat/Dark cutting in beef animals:-

- ✓ It commonly occurs in **animals which have not been fed for a period before slaughter, or which have been excessively fatigued during transportation and lairage.**
- ✓ Most of the muscle glycogen has used already. Thus, **muscle pH does not fall below pH 6.0**.
- ✓ The high pH causes the muscle proteins to retain most of their bound water and they absorb most of the light striking the meat surface, giving a dark appearance.

Note:

Casing: Sub-mucosal lining of intestine cleaned & used for stuffing sausages.
Sausage: Meat filled casing is known as sausage

Terminology related to casing industry:

- ✓ Rounds: Casing from small intestines of sheep, goat

Cattle casings:

- ✓ Runners: Casing from small intestine of cattle
- ✓ Middles: Casing from large intestine of cattle

Bung: Casing from caecum of cattle
Weasand: Casing from esophagus of cattle

The optimum growth temperature of bacteria:
- ✓ **Psychrophilic bacteria: Below 20 Degree Celsius**
- ✓ **Mesophilic bacteria: 20-45 Degree Celsius**
- ✓ **Thermophilic bacteria: Above 45 Degree Celsius**

Note:
- ✓ Physical changes in stored meat: (1) Shrinkage (2) Sweating (3) Loss of bloom
- ✓ Freezing losses in meat: (1) Bone darkening (2) Freezer burn (3) Drip loss

Methods of disposal of carcass:
1) Burial method: Most commonly used method of disposal of carcass
2) Burning or Incineration method: Preferred method of disposal for anthrax carcass
- ✓ Incinerator temperature for carcass disposal is 600-8000C

2) **Chemical treatment**: Lime water (1:20) or suspension of bleaching powder (1:20) is mainly used.

Methods of preservation of meat:
(1) Chilling/Refrigeration: At **40C**,
- ✓ Method of choice for short-term preservation because it slows down the microbial growth and enzymatic activities

(2) Freezing: At **-18 Degree Celsius**,
- ✓ Method of choice for long-term preservation because it stops the microbial growth and enzymatic activities.

The shelf life of frozen meat at -18 Degree Celsius:
- ✓ For mutton, chevon, beef, and carabeef: 6 months
- ✓ For pork and chicken meat: 4 months
- ✓ For cured and salted meat: 2 months

(3) Curing: **Oldest method of preservation of meat**
- ✓ Commonly used curing agents: Salt, sugar, nitrite, nitrate etc
- ✓ The permissible level of **nitrate is 500 ppm & nitrite is 200 ppm**

(4) Smoking: to develop flavor, aroma, and color in meat
(5) Thermal processing: Pasteurization &Sterilization
(6) Canning: Sterilization of product in thermatically sealed container at **121 Degree Celsius.**
(7) Dehydration: Removal of water from meat
(8) Irradiation:
- ✓ **Gamma rays are cheapest & most frequently used radiation** in food preservation.

Abattoir/Slaughter house:

Area requirement for abattoir:

- 1 Buffalo/Cattle = 1 Livestock unit
- 1 Sheep/Goat = 0.2 Livestock unit
- 1 Calf = 0.3 Livestock unit
- 1 Livestock unit = 1 buffalo/cattle= 2 pigs=3 calves=5 sheep/goat

- ✓ **As per FAO (1978) water requirement for both clean & unclean operations in abattoir per animal per day:**
- ✓ Cattle/Buffalo: 1000 liters
- ✓ Sheep/goat: 100 liters

The light requirement in abattoir:

- ✓ For inspection & packing point: 550 Lux or 50 ft candle
- ✓ For work room: 220 Lux or 20 ft candle
- ✓ For chilling room: 110 Lux or 10 ft candle

Note:

These intensities are measured at a height of 0.9 meters from the floor but in case of inspection point, it is measured at a height of from 1.5 meters.

Chill rooms:

- ✓ Chilling temperature should not exceed 7 Degree Celsius for carcasses and 4 Degree Celsius for offals.
- ✓ Rail spacing in chill room should be 0.9 m for buffalo,0.7 m for pig and 0.5 m for sheep/goat carcasses

Note:

For proper drainage, abattoir building should have a floor gradient of 2 cm per meter.

Lairage:

- ✓ Before slaughter, animals need a rest of nearly 24 hours to restore their normal physiological condition.
- ✓ Arrangement of feeding, watering & antemortem inspection
- ✓ Animals are protected from cold, rain and heat
- ✓ Space required in lairage per animal: 2.8 m^2 for buffalo/cattle and 0.6 m^2 for small animals.

Race: It is a narrow space between lairage & slaughter hall. It should be at least 10 meters.

Methods of slaughter:

There are two methods (A) Scientific/humane (B) Ritual

(A) Scientific /Humane slaughter: Scientific slaughter avoids unnecessary pain and suffering to the animals.

Stunning: Process use to create a state of immobility or unconsciousness at the time of slaughter

Stunning time: 30 seconds to 1 minute:

- ✓ Stunning is performing only in case of humane slaughter, not in ritual slaughter.

Stunning techniques are three types:

1) Mechanical stunning:

✓ Instruments used: Captive bolt pistol, percussion stunner etc

 Commonly used in cattle & buffaloes

2) Electrical stunning:

✓ Commonly used in stunning of Sheep, Goat, Pigs & Poultry

✓ Instrument used: Elther apparatus (alternating current of high frequency (250mA) and low voltage (75 volts) is passed through the brain for 10 seconds.

3) Chemical stunning/Gas stunning:

✓ 70% CO2 & 30% O2 used for 45 seconds

✓ CO2 gas blocks the nerve endings.

Note:

1) Sticking/bleeding in animals: By severing carotid arteries & jugular veins

2) Back bleeding/over sticking: If knife reaches too far in the chest during sticking, blood may be aspirated in the thoracic cavity and lungs become contaminated.

The average yield of blood per animal & their bleeding time:

(1) Buffalo/cattle: **10-12 Kg** in **6 min**

(2) Sheep/Goat: **1-1.5 Kg** in **5-6 min**

(3) Fowl: **30-50g** in **1.5-2 min**

(B) Ritual slaughter:

(1) Jewish method/Schechita method

(2) Muslim method/Halal method

(3) Sikh method/ Jhatka method

(4) Evernazine method/Neck Stab method

Note:

✓ Feeding & watering is done before slaughter in both Jhatka & Halal method of ritual slaughter

✓ Jugular veins & carotid arteries are severed in the case of Jewish & Muslim methods of slaughter.

BASICS OF PHYSIOLOGY:

Note:

- ✓ Common grazing animals - Cattle, Sheep
- ✓ Common browsing animals- Goat, Camel
- ✓ The **gall bladder** is **absent** in **horse as well as in all pseudo-ruminants** like-camel, llama, alpaca, vicuna.

Classification of the nervous system:

(A) According to location:

- ✓ Central Nervous System (CNS): Brain & Spinal cord
- ✓ Peripheral Nervous System (PNS): Cranial nerves & Spinal nerves

(B) According to function:-

- ✓ Autonomic Nervous System (ANS): Sympathetic& Parasympathetic
- ✓ Somatic Nervous System (SNS): Somatic sensory nerves & Somatic motor nerves

Note:

- ✓ The largest endocrine gland of the body is thyroid while smallest is pineal
- ✓ Largest endocrine organ of the body is GIT
- ✓ The largest gland of the body is liver
- ✓ The largest organ of the body is the skin
- ✓ Total body water of animal body: **60-70%** of body weight
- ✓ Total blood volume in animal body: **8%** of body weight
- ✓ Plasma volume in animal body: 5% of body weight
- ✓ Mineral content of animal body: 3% of body weight
- ✓ The weight of animal heart: **0.6%** of body weight

Total Body Water (TBW) is classifying under two categories:

- ✓ Intracellular fluid (ICF): **40%** of animal body weight
- ✓ Extracellular fluid (ECF): **20%** of animal body weight

ECF is further classified into three categories:

- ✓ Interstitial fluid (ISF): 15% of animal body weight
- ✓ Plasma volume/Intravascular fluid(PV/IVF): 5% of animal body weight
- ✓ Transcellular fluid (TCF): Found in body cavities.It includes cerebrospinal fluid, synovial fluid, intraocular fluid, urine, bile & fluid of digestive tract.

Note:

- ✓ Volume of different fluid compartments: ICF > ECF > ISF > PV > TCF
- ✓ Heavy water (deuterium oxide) & radioactive water (tritium oxide) are use to measure total body water (TBW)
- ✓ Inulin is use to measure **ECF** & G**lomerular Filtration Rate (GFR)**
- ✓ **Evans blue dye** (*T*-1824) is use to measure **plasma volume**
- ✓ ICF cannot be estimated directly but it is measured by subtracting ECF from TBW.

Water loss from the body:
(1) Sensible water loss: urine, stool
(2) Insensible water loss: sweating, respiratory water loss

Blood:
- ✓ Fluid connective tissue consists of formed elements suspended in plasma
- ✓ Normal pH=**7.4**
- ✓ Venous blood is more acidic than arterial blood because of CO2
- ✓ Blood cells are more acidic than blood plasma
- ✓ 55% of total blood volume is plasma & remaining 45% is formed elements.
- ✓ Red colour of blood is due to hemoglobin present in RBC
- ✓ Colour of plasma is due to presence of bilirubin

Formed elements & their main functions:
- ✓ RBC/Erythrocytes: Oxygen transport
- ✓ WBC/Leukocytes: Body defense
- ✓ Platelets/Thrombocytes: Blood clotting

Blood profile on centrifugation:
- ✓ Upper layer- Plasma (55%)
- ✓ Middle layer/interface/**buffy coat: Leukocytes & Platelets (< 1%)**
- ✓ Lower layer: Erythrocytes (45%)

Leukocytes are two types:
- ✓ **Granulocytes**: Neutrophils, Eosinophils & Basophils
- ✓ **Agranulocytes**: Monocytes, Lymphocytes (B-cells, T-cells & Natural Killer cells).

Note:
- ✓ Serum = Plasma – (Fibrinogen + Other clotting factors)
- ✓ Immature/young neutrophils: **Band cells**
- ✓ Mature neutrophils: **Segmented cells**
- ✓ Birds have nucleated erythrocyte & nucleated thrombocytes while mammals not have.
- ✓ Order of size or diameter of blood cells: WBC > RBC > Platelet
- ✓ No. of blood cells per µl of blood: RBC > Platelets > WBC
- ✓ Monocyte: Largest blood cell or largest leukocyte
- ✓ Monocytes transformed into macrophage on entering the tissues
- ✓ **Most numerous leukocytes in ruminants**, pig & poultry: **Lymphocyte** (60-65%)
- ✓ Most numerous leukocytes in **horse, dog & cat**: **Neutrophils** (60-65%)
- ✓ **Plasma cells derived from B-Lymphocytes** & produce antibodies against infections.
- ✓ Plasma cells have **Cartwheel shape nucleus.**
- ✓ **Agranulocytes** are also known as **Peripheral Blood Mononuclear Cells** (**PBMCs**)
- ✓ **Granulocytes** are also known as **Poly Morpho Nuclear Leukocytes** (**PMN**/**PML**/**PMNL**)

Macrophages with their anatomical locations:

- ✓ Monocyte: Bone marrow/Blood
- ✓ Kupffer cell: Liver
- ✓ Dust cell/Alveolar macrophage: Lungs
- ✓ Microglia: Brain
- ✓ Glomerular mesangial cell: Kidney
- ✓ Hofbauer cell: Placenta
- ✓ Osteoclast: Bone

Note:

- ✓ The largest size of RBC found in the dog (7 μm) & smallest in goat (4 μm)
- ✓ Heterophils: Avian neutrophils
- ✓ Red colour of blood is due to hemoglobin present in the RBC
- ✓ Basophils resemble with mast cell histologically. Both cells produce heparin, histamine, serotonin, bradykinin & lysosomal enzymes in areas of inflammation.
- ✓ Basophils enhance allergic reactions, whereas eosinophils tend to dampen them.
- ✓ Spleen is the largest lymphoid organ of the animal body

Lymphoid organs:

1. **Primary lymphoid organs (**sites of lymphopoiesis):
Places where the B & Tlymphocytes differentiate from stem cells**.**
Examples: Thymus, Bone marrow, Bursa of Fabricius

2. **Secondary lymphoid organs**:
Examples: Spleen, Lymph nodes, Tonsils &adenoids, Peyer‟s patches & appendix

Antigen Presenting Cells (APCs):

It includes **dendritic cells, macrophages & B-lymphocytes** & they present antigens to T-helper cells.

Anemia: Abnormal reduction in the number of RBCs or hemoglobin content or both
A) According to size of RBC & concentration of hemoglobin

- ✓ **Normocytic & Normochromic**: occurs in chronic disease
- ✓ **Microcytic & Hypochromic**: due to deficiency of iron
- ✓ **Macrocytic & Hyperchromic**: due to deficiency of cobalt, vitamin B12 & folic acid

B) **Aplastic anemia:** due to lack of functional bone marrow

Immunity:

(1) Innate immunity/Non-specific immunity: By-birth immunity**.** It consists of a first & second line of defenses.

- ✓ **First line of defense**: Skin, mucous membranes, secretions like gastric HCl, tears, saliva etc
- ✓ **Second line of defense**: Inflammation, fever, interferons, complement system, natural killer cells (NK cells) phagocytes such as neutrophils, macrophages & eosinophils

(2) Acquired/Adaptive/Specific immunity:

- ✓ Develops during the course of life when any foreign antigen is entering in the body.
- ✓ It consists of the **3rd line of defense** that includes B and T Lymphocytes & antigen presenting cells.

Types of acquired immunity:

(A) Active Immunity: It develops when the body cells produce antibodies in response to infection or vaccine.

- ✓ Natural active immunity: By any infection
- ✓ Artificial active immunity: Vaccination, Tetanus toxoid/ *Tetanus* vaccine

(B) Passive immunity: By direct antibody inoculation

- ✓ Natural passive immunity: Antibodies transfer from mother to foetus via placenta or via colostrum
- ✓ Artificial passive immunity: Injection of antiserum that contains antibodies e.g. Anti-Tetanus Serum (ATS)

Note:

- ✓ Tetanus Toxoid (TT) has prophylactic action while anti-tetanus serum (ATS) has curative action.
- ✓ TT is used to prevent tetanus while ATS is used to cure tetanus.

Hemoglobin (Hb) & its forms:

(1) Oxy-haemoglobin/HbO2/Bright red colour: formed by oxygenation of Hb

(2) Carboxy-haemoglobin/HbCO/Bright cherry red colour: Hb combines with carbon monoxide.

(3) Met-haemoglobin/Chocolate brown colour: formed by oxidation of Hb (eg: Nitrate poisoning, chlorate poisoning). It is true oxidation product of Hb.

(4) Carbaminohaemoglobin: Hb combines with Carbon dioxide (HbCO2)

Terminology:

- ✓ **Anemia: D**eficiency of RBCs or hemoglobin in the blood
- ✓ **Polycythemia:** An increase in the total RBC mass of the blood
- ✓ **Anisocytosis: Unequal size of red blood cells**
- ✓ **Poikilocytosis: Abnormally shaped red blood cells.**

Note:

- ✓ A Suffix eg.-**cytosis** or **-philia** indicates an **increase in a number** of that particular cell type.
- ✓ A suffix eg.-**penia** indicates a **decrease in the number** of that particular cell type.

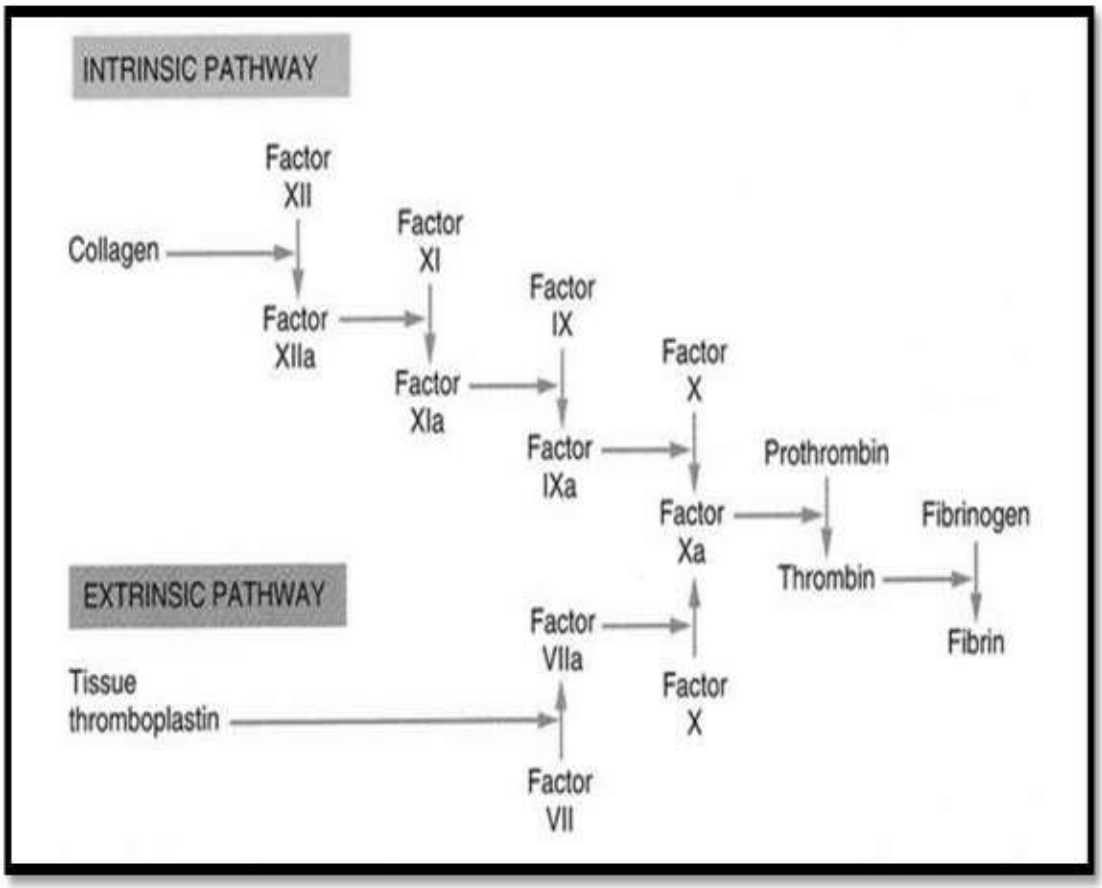
Note:

- ✓ The value of packed cell volume (PCV) is generally 3 times that of hemoglobin value.
- ✓ Erythrocyte sedimentation rate (ESR) is highest in the case of horse & lowest in the case of ruminants

- ✓ Stroke volume: Amount of blood pumped out by the left ventricle in one contraction
- ✓ Cardiac output: Amount of blood pumped by heart per minute
- ✓ Cardiac output = Stroke volume × Heart rate
- ✓ Pulse pressure = Systolic pressure – Diastolic pressure
- ✓ Mean arterial pressure (MAP) = Diastolic pressure + 1/3 Pulse pressure
- ✓ MAP = Cardiac output × Total peripheral resistance

Blood clotting factors:

- ✓ Factor I: Fibrinogen
- ✓ Factor II: Prothrombin
- ✓ Factor III: **Tissue factor/ Tissue thromboplastin**
- ✓ Factor IV: **Calcium**
- ✓ Factor V: Proaccelerin
- ✓ Factor VII: Proconvertin
- ✓ Factor VIII: Antihemophilic factor A/**Anti Hemophilic Globulin (AHG)**
- ✓ Factor IX: Antihemophilic factor B/ **Christmas Factor**/Plasma Thromboplastin Component **(PTC)**
- ✓ Factor X: Stuart factor
- ✓ Factor XI: Plasma Thromboplastin Antecedent **(PTA)**
- ✓ Factor XII: Hageman factor
- ✓ Factor XIII: Fibrin-stabilizing factor.



Note:

- ✓ Blood clotting pathways: Intrinsic pathway & extrinsic pathway
- ✓ Extrinsic pathway/tissue factor pathway: initiated due to release of tissue factor (factor III) at the site of injury
- ✓ Extrinsic pathway is faster than intrinsic pathway
- ✓ Both pathways are common after activation of factor X to factor Xa
- ✓ Enzyme responsible for clot lysis: Plasmin
- ✓ Key enzyme in blood coagulation or hemostasis: Thrombin

Blood anticoagulants:

- ✓ Heparin: **Natural** or ***In-vivo*** **anticoagulant**, produced by mast cells& basophils
- ✓ Sodium citrate: used in **blood transfusion** in animals & for study of blood coagulation
- ✓ EDTA: used in study of **blood morphology**, RBC index determination & PCV determination
- ✓ Sodium fluoride: used for **glucose estimation**

Vasodilators:

✓ Nitric oxide (most potent); Atrial natriuretic peptide (ANP); histamine, heparin, bradykinin,
vasoactive intestinal peptide (VIP), Platelet activating factor (PAF), Substance P, Prostacyclin (PGI2), PGE2

Vasoconstrictors:

✓ Endothelin (most potent), vasopressin(ADH), Angiotensin II, norepinephrine, dopamine, thromboxane, neuropeptide Y(NPY)

✓ Epinephrine acts as both vasoconstrictors as well as a vasodilator.

✓ When epinephrine acts on alpha receptors it causing vasoconstriction & on beta receptors causing vasodilation. The affinity of epinephrine for beta-receptors is greater than its affinity for alphareceptors.

Note:

- ✓ **Defaunation**: Selective removal of protozoa from the ruminal ecosystem
- ✓ **Chemical defaunation**: **Copper sulfate & Sodium lauryl sulfate**

Uses of Volatile Fatty Acids (VFAS) produced during ruminal fermentation:

- ✓ VFAS provides 70% of ruminant energy needs
- ✓ **Acetate:** It is responsible for **milk fat synthesis**
- ✓ **Propionate:** It is **gluconeogenic**. It is responsible for milk sugar synthesis
- ✓ **Butyrate:** Ruminal epithelial cells convert it into β-hydroxy butyrate (a ketone body)

Note:

- ✓ High roughage diet: high milk fat% (Acetate production is high)
- ✓ High concentrate or grain diet: high milk yield (Propionate production is high)

Note:

- ✓ **Renin**(secreted by kidney) while **Rennin**(secreted by stomach)
- ✓ **Ileum**(part of small intestine) while **Ilium**(part of hip bone)
- ✓ **Kidney hormones**: Erythropoietin, calcitriol (1,25- dihydroxycholecalciferol) and renin.

Types of respiration with examples:

- ✓ Costal/Thoracic respiration: dog & cat
- ✓ Abdominal respiration: ruminants
- ✓ Thoraco-abdominal respiration: horse, mule, donkey

Transport of O2& CO2 in the body:

(1) Transport of oxygen:

- ✓ 98.5% of total O2 in the form of oxyhemoglobin
- ✓ 1.5% in dissolved plasma

(2) Transport of CO2:

- ✓ 70% in the form of bicarbonate ($HCO3^-$)
- ✓ 23% in the form of carbaminohemoglobin(inside RBC)
- ✓ 7% in dissolved plasma

Note:

- ✓ One molecule of hemoglobin contains four atoms of iron (in Fe++state) & can transport four molecules of O2
- ✓ 1 gram of Hb carries 1.34 ml of oxygen
- ✓ Normal Hb concentration in the blood is 15g/dl hence 100 ml of blood carries 20 ml of oxygen.

Note:

- ✓ **Bohr effect**: effect of CO2 & H+ on the ability of hemoglobin to yield or receive oxygen
- ✓ **Haldane effect**: The effect of O2 on H+ & CO2 loading & unloading from hemoglobin

Terminology related to respiration:

- ✓ Eupnea: Normal breathing
- ✓ Dyspnea: Difficult breathing
- ✓ Apnea: Complete cessation of breathing
- ✓ Polypnea: Rapid shallow breathing (frequency increase but depth decrease)
- ✓ Hyperpnea: Breathing characterize by increase depth, frequency or both
- ✓ Tachypnea: Rapidity of breathing
- ✓ Bradypnea: Slowness of breathing

Mechanism of breathing:

- ✓ Respiration cycle of animals consists of an inspiratory phase & an expiratory phase.

✓ During inspiration, the diaphragm & external intercostal muscle contract
✓ During expiration, these muscles relax.
✓ Expiration is a passive process

The difference between mammalian &avian respiration

✓ **Diaphragm is absent in birds**
✓ In birds, inspirations, as well as expiration both, are active process while in the case of mammals,
expiration is a passive process & inspiration is active
✓ Birds have a larynx, but it is not used to make sounds.In birds **"syrinx"** serves as the "voice box."
✓ Birds have lungs and **nine air sacs** (two cervical, two anterior thoracic, two posterior thoracic, two abdominal, **unpaired clavicular**)
✓ Site of gaseous exchange in birds: **Parabronchi or tertiary bronchi**
✓ Two types of parabronchi: **paleopulmonic(ancient lung) & neopulmonic (new lung)**
✓ Paleopulmonic parabronchi is found in all birds
✓ Air flow in paleopulmonic parabronchi is unidirectional
✓ Air flow in neopulmonic parabronchi is bidirectional
✓ **In penguin & emu, neopulmonic parabronchi is absent**
✓ **In fowl & songbirds, neopulmonic is more developed than paleopulmonic**
✓ There is no gaseous exchange in air sacs because **air sacs are avascular**
✓ Respiratory system of birds is more efficient than mammals
✓ In mammals, alveoli of lungs are lined by surfactant-secreting cells known as **type-II pneumocytes** while birds have **granular cells**
✓ Surfactant: surface tension lowering agent
✓ **Dipalmitoylphosphatidylcholine (DPPC)** or Dipalmitoyl lecithin, act as a pulmonary surfactant.

(B) Chemoreceptors:

(1) Central chemoreceptors (H+ & *P*CO2 sensitive receptors)

✓ Located in the medulla oblongata
✓ They are stimulated by increased *P*CO2 via an associated change in H+ concentration of an interstitial fluid of brain

(2) Peripheral chemoreceptors (*P*O2, H+ & *P*CO2 sensitive receptors)

✓ These are **carotid body& aortic arch**, located at the bifurcation of carotid arteries & aortic arch respectively
✓ They are stimulated by decreased *P*O2, increased *P*CO2 & increased H+ concentration in the arterial blood
✓ **Carotid body** send their signal to respiratory centers via **Glossopharyngeal nerves**
✓ **Aortic arch** sends their signal to respiratory centers via **Vagus nerves**

<u>Renal physiology:</u>

Functional unit of kidney: Nephron

Two types of mammalian nephron:

1) Cortical nephron: Short loop of Henle located in cortex & its main function is reabsorption & secretion of substances
2) Juxtamedullary nephron: Long loop of Henle extended deep in the renal medulla & its main function is concentrating the urine
✓ Transport of tubular fluid from Bowman‟s capsule to renal pelvis occurs due to hydrostatic pressure gradient
✓ Transport of urine from renal pelvis to urinary bladder occurs due to peristalsis in the ureters.

Two countercurrent mechanism of the kidney:
✓ Countercurrent multiplier (loops of Henle)
✓ Countercurrent exchanger (vasa recta)

Juxta Glomerular (JG) apparatus:
✓ Specialized structure formed at the junction of DCT & Glomerular afferent arteriole
It consists of three cells:
1) Macula densa: Specialized epithelial cells in DCT that detect sodium concentration in tubular fluid
2) Juxtaglomerular cells (JG cells): smooth muscle cells of afferent arterioles that secrete **renin** hormone
3) Extraglomerular mesangial cells (**Lacis cells**)

Note:
✓ Generally, protein molecules are not filtered through glomerulus because of their large molecular size & polyanionic nature
✓ Micturition/Urination: process of emptying urinary bladder
✓ Micturition is parasympathetic activity, sympathetic nerve has no effect on micturition.
✓ Yellow colour of urine is due to urobilin
✓ Principal nitrogenous component of mammalian urine is Urea (NH3 converted to urea in liver)
✓ Principal nitrogenous component of avian urine is Uric acid (NH3 converted to uric acid in liver & kidney)

Glomerular Filtration Rate (GFR): Volume of the plasma filtered by all nephrons of both kidneys per unit time.
✓ Normal GFR of human = 125ml/min or 180 L/day
✓ GFR can be measured by **creatinine clearance or inulin clearance**
✓ High plasma concentration of creatinine is an indication of kidney disease

Diuretics: agents/drugs that increase urine output, these are useful for treating edema & hypertension.
✓ **Loop diuretics (furosemide): Inhibiting Na+-K+-2Cl - cotransport** in luminal membrane of **thick ascending limb of loop of Henle**

✓ **Osmotic diuretics (mannitol):** inhibit water & solute reabsorption **by increasing effective osmotic pressure of tubular fluid of PCT**

Types of diabetes insipidus:

✓ Neurogenic diabetes insipidus/**central diabetes insipidus**: due to **deficiency of ADH**

✓ **Nephrogenic diabetes insipidus**: when **kidneys do not respond** normally to ADH

Note:

✓ Plasma concentration (ECF) of Na^+ is regulated by **Osmoreceptor-ADH-Thirst system**

✓ Plasma concentration (ECF) of K^+ is regulated by **Aldosterone**

✓ Plasma concentration (ECF) of Cl^- is regulated secondarily to regulation of Na^+ & HCO^{3-}

Important instruments and their uses

✓ **Sphygmomanometer**: to measure blood pressure

✓ **Stethoscope**: to measure heart rate

Instruments of weather station are used to measure:

✓ **Stevenson screen**: It is used to protect meteorological instruments against rainfall and heat radiation from outside sources

✓ **Anemometer**: Wind speed

✓ **Wind vane:** to detect the direction of the wind

✓ **Hygrometer**: Humidity

✓ **Sling Psychrometer: Relative humidity**

✓ **Pyranometer**: Solar radiation

✓ **Barometer**: Atmospheric pressure

✓ Thermometer: Air and sea surface temperature

✓ **Rain gauge**: Rainfall or precipitation

✓ **Infra-red thermometer**: **Skin surface temperature** without touching the surface.

<u>Pheromones</u>:

Chemical messengers that carry information between individuals of the same species

✓ These are **olfactory stimulus** detected by **Vomeronasal organ** (VNO) or organ of Jacobson

✓ During estrus, vaginal secretion of bitch contain **methyl-p-hydroxy benzoate** (a pheromone)

✓ Boar saliva contain **5α-androstenone** (a pheromone)

<u>Bruce effect</u>:

Termination of pregnancy due to **blockage of prolactin release** (when a male mouse introduced into a group of recently bred mice).

<u>Note</u>:

✓ **Ram effect** is similar to Whitten effect (but occur in ewes)

- ✓ **Blood Brain Barrier:** Brain endothelial cells, which are connected by tight junctions
- ✓ **Blood CSF barrier**: Epithelial cells of choroid plexus
- ✓ **Blood testes barrier**: Sertoli cells along with myoid cells.

CLASSIFICATION OF RECEPTORS:

Receptors	Types	Sub-types
Cholinergic	Muscarinic/nicotinic	M1, M2/ Skeletal muscle
Adrenergic	Alpha + Beta	Alpha 1,alpha 2 & beta 1,beta 2
steroid	P4, E2, Glucocorticoids,T4	

DIFFERENCE B/W SOMATIC & AUTONOMIC NERVOUS SYSTEM:

Features	Somatic N.S	Autonomic N.S
Organ	Skeletal muscle	All other organs
Distal synapse	Within CNS	Outside CNS (in ganglia)
Transmitter	Acetylcholine	Nor adrenaline
Effect	Paralysis & atrophy	Activity maintained ,no atrophy

AUTACOIDS AND LOCAL HORMONES:

Autacoids	Local hormones
Amine autacoids	Histamine + 5-h-t (Serotonin)
Peptides	Plasma kinins + Bradykinins + Angiotensin
Lipid derivatives	Eicosanoids & platelets activating factor (PAF) + prostaglandins+ leukotriene.

ANTI HORMONE EXAMPLES:

Anti	Example
Anti-androgen	Cyproterone acetate + flumtamide + finasteroide
Anti-progestin	Mifomiphene
Anti estrogenisis	Centchroman + Clomiphene citrate
Anti-spermatogenesis	Gossypol

HORMONE AND THEIR SECRETION:

Hormone	Secreted from	Action
Calcitonin	Parafollicular cells of **thyroid gland**	Decreases blood Ca level

Parathormone (PTH)	Parathyroid gland	Increases blood Ca level
Vita.D3 (act as hormone)	Synthesized in skin on exposure to sunlight	Absorption of Ca through intestine

EXOCRINE GLANDS (DUCT GLANDS):

Sweat	Salivary
Mammary	Ceremonious
Lacrimal	Sebaceous
Mucous	Prostate

ENDOCRINE GLANDS (DUCTLESS GLANDS):

Hypothalamus	Pituitary
Parathyroid	Thyroid
Adrenal	Pineal
Ovaries	Testes

TERMS RELATED TO HEART:

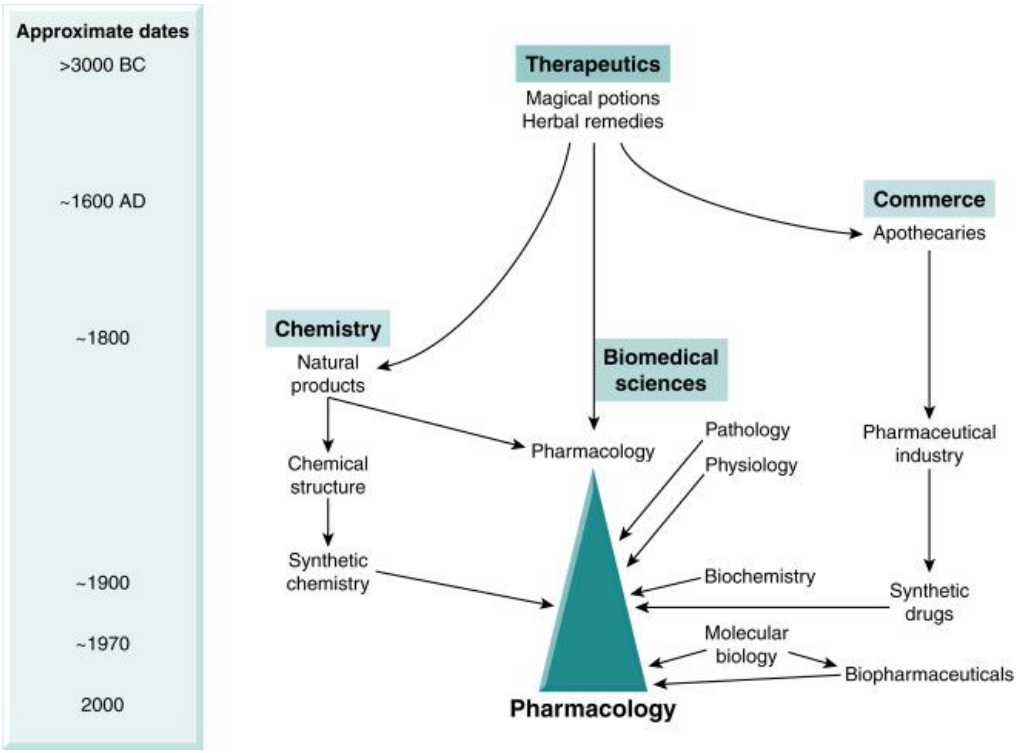
Right Atrium	Deoxygenated blood from body part.
Left Atrium	Oxygenated blood from lungs.
Tricuspid Valve	B/w right atrium & right ventricle.
Bicuspid Valve (Mitral Valve)	B/w left atrium and left ventricle
Pulmonary Valve	B/w Right ventricle & pulmonary artery.
Aortic Valve	B/w left ventricle & aorta.

BASICS OF PHARMACOLOGY:

Pharmacology:

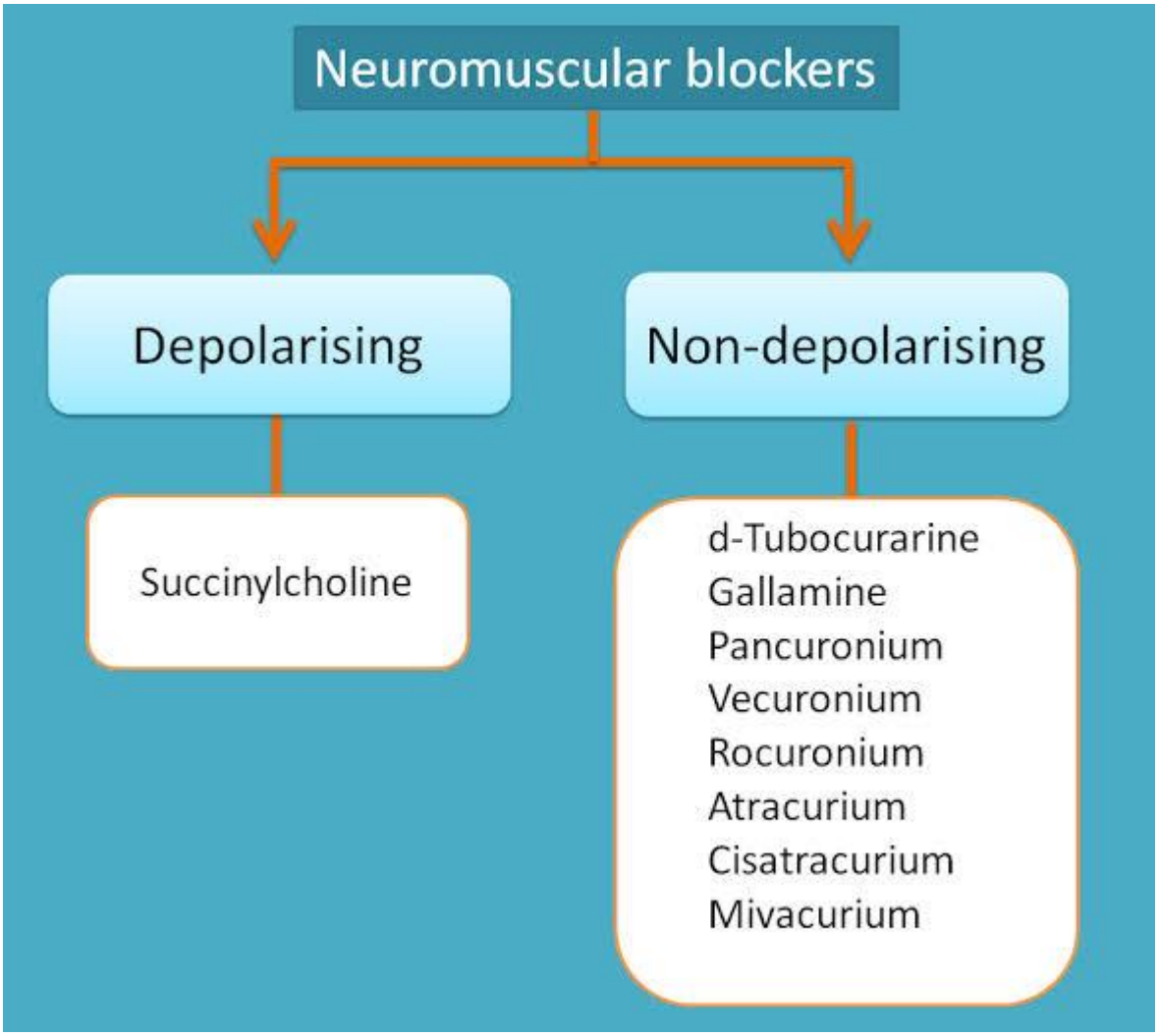
- ✓ It is the study of drugs action on organisms.
- ✓ It deals with the 4 objects;

A- Absorption (Taken in to body)
B- Distribution (Moved in to tissues)
C- Metabolism (Changed so can be excreted)
D- Excreted (Removed from the body)



Chemotherapy:

- ✓ Use of certain chemical in the treatment of infections.
- ✓ Father of chemotherapy is Paul Ehrilich.
- ✓ Anti-cancer drug is antimetabolites, it is antagonists of folic acid, purine & pyrimidine.



Toxicology:

It is the science of poison and incorporate detection, evaluation and their treatment.

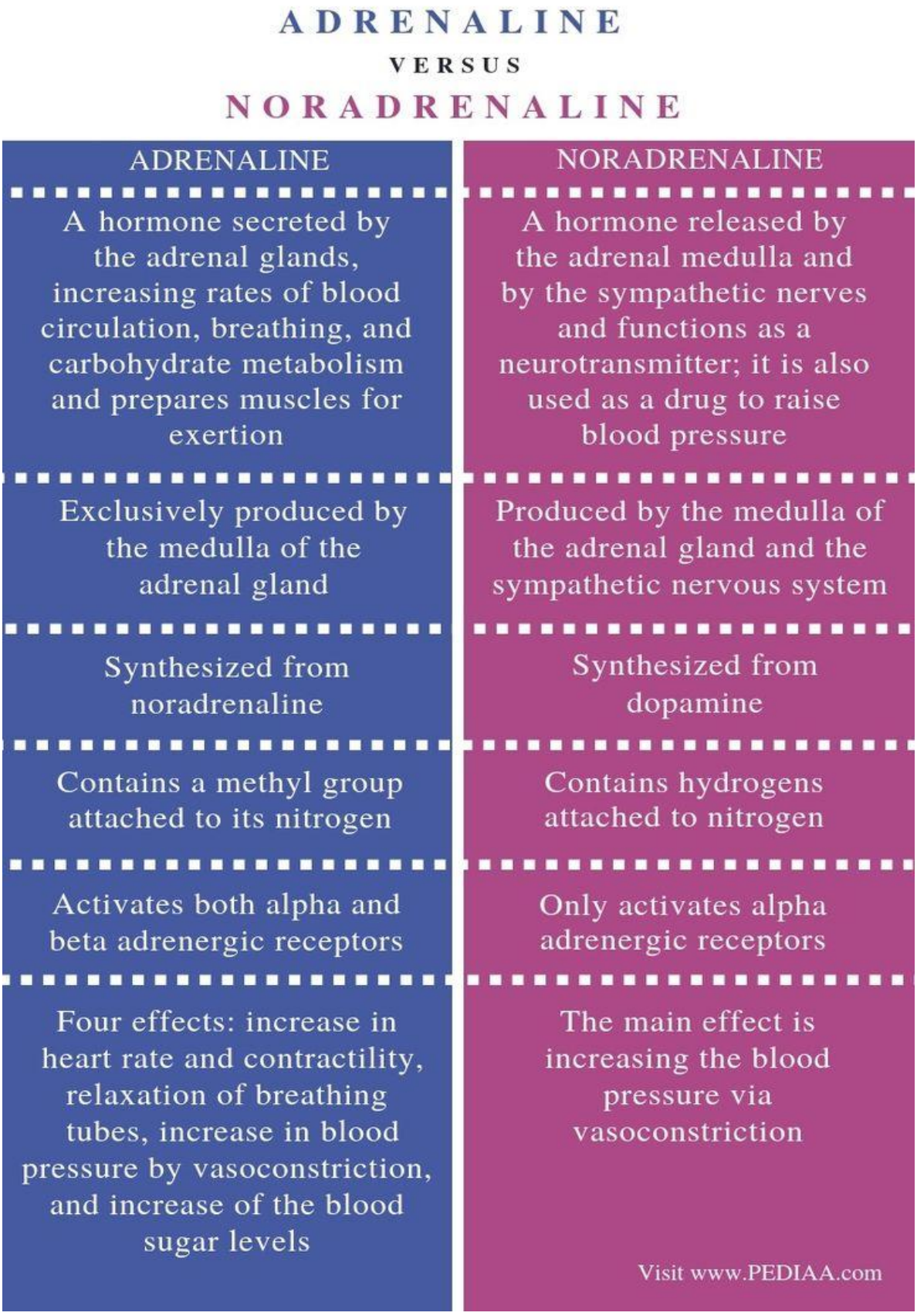
PLANT TOXICOLOGY:

Toxicity	Antidote
No3/ No2 toxicity	Methylene blue
Sweet clover	Vit K
Organochlorine compound	Activated charcoal
Organophosphate compound	Atropine sulphate
Urea	5 % acetic acid

Drugs and their Mode of Actions:

Mnemonic	Antibiotic Class	Examples	Coverage	MOA	Uses
Antibiotics	Aminoglycosides	Streptomycin Gentamicin	Gram (-)	Inhibit Protein Synthesis (30s)	Bacteremia, Abdominal Infections
Can	Cephalosporins	Ceftriaxone Cefepime	Gram (+)/(-)	Inhibit Cell Wall Synthesis	Skin, Urinary, Resp. Infections
Terminate	Tetracyclines	Tetracycline Doxycycline	Gram (+)/(-)	Inhibit Protein Synthesis (30s)	Lyme Disease, PID, STIs
Protein	Penicillins	Ampicillin Amoxicillin	Gram (+)/(-)	Inhibit Cell Wall Synthesis	ENT, Skin, Urinary Infections
Synthesis	Sulfonamides	Sulfasalazine Sulfamethoxazole	Gram (+)/(-)	Inhibit Folate Synthesis	UTIs, Burns, Eye Infections
For	Fluoroquinolones	Ciprofloxacin Levofloxacin	Gram (+)/(-)	Inhibit DNA Replication	Respiratory & Urinary Infections
Microbial	Macrolides	Azithromycin Erythromycin	Gram (+)	Inhibit Protein Synthesis (50s)	Pneumonia, Sinus, ENT, STIs
Cells	Carbapenems	Meropenem Ertapenem	Gram (+)/(-)	Inhibit Cell Wall Synthesis	Urinary, Abdom. Infections
Like	Lincosamides	Clindamycin	Gram (+)	Inhibit Protein Synthesis (50s)	Skin, Bone, Lung Infections
Germs	Glycopeptides	Vancomycin	Gram (+)	Inhibit Cell Wall Synthesis	MRSA, Skin, Endocarditis

EPINEPHRINE AND NOREPINEPHRIE:



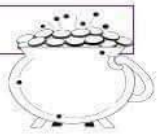
PLANT SOURCE



Source	Plant	Drug	Use
Leaf	Digitalis	Digoxin	CHF
Bark	Cinchona	Quinine	Malaria
Fruit	Opium	Morphine	Analgesic
Seed	Eserin	Anticholinestrase	M.G

Contd.,

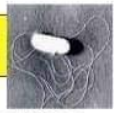
Mineral Sources



- Minerals

Example	Trade Name	Classification
Magnesium	Milk of Magnesia	Antacid, Laxative
Zinc	Zinc Oxide Oint.	Sunscreen, Skin Protectant
Gold	Solganal, Auranofin	Anti-inflammatory; Used in tx of Rheumatoid Arthritis

MICROORGANISM SOURCE



- Bacterial, Fungi, Moulds imp source of many life saving drugs.
- These obtained from MO and used to kill Microorganisms.

Drug	Microorganism
• Penicillin	*Penicilium notatum*
• Chloramphenicol	*Streptomyces venezuelace*
• Griseofluvin	*Pencillin grisofullivum*
• Streptomycin	*Streptomyces griseus*
• Neomycin	*Streptomyces fradiae*

ANIMAL SOURCE



- Obtained from animal

Drug	Animal
Heparin	Leech
Insulin	Pork pancreas
Thyroxin	Thyroid
Vit. B_{12}	Liver extract
Cod liver oil	
Anti toxic sera	

Contd.,

DRUGS CONSIDERED SAFE IN PREGNANCY

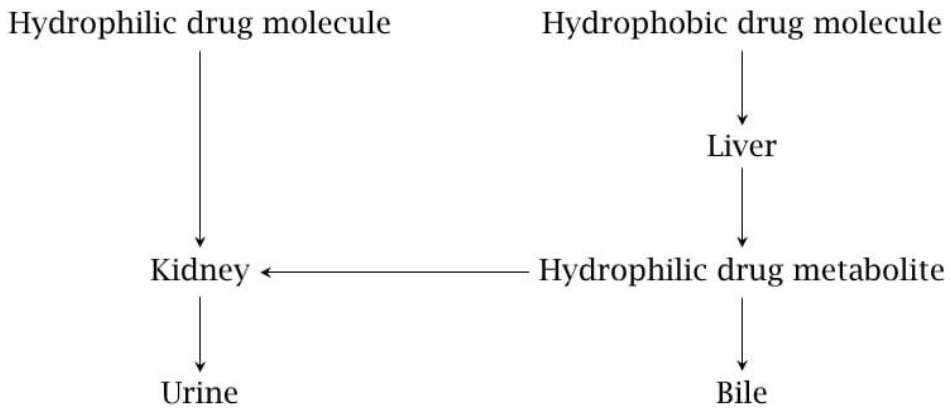
- cephalosporins
- fenbendazole
- penicillins
- ivermectin
- amoxicillin with clavulanic acid
- praziquantel
- clindamycin
- pyrantel pamoate
- erythromycin
- diethylcarbamazine
- milbemycin oxime

SELECTED DRUGS TO AVOID DURING PREGNANCY

- aminoglycosides
- enalapril
- tetracyclines
- theophylline
- sulfonamides
- misoprostol
- chloramphenicol
- diethylstilbestrol
- metronidazole
- organophosphates
- griseofulvin
- omeprazole
- ketoconazole
- aspirin
- ketamine
- acepromazine
- diazepam
- barbiturates
- glucocorticoids
- mitotane

Sympathetic and Parasympathetic Effects of Different Drug Types ([redacted])				
Drug type	**Example(s)**	**Sympathetic effect**	**Parasympathetic effect**	**Overall result**
Nicotinic agonists	Nicotine	Mimic ACh at preganglionic synapses, causing activation of postganglionic fibers and the release of norepinephrine onto the target organ	Mimic ACh at preganglionic synapses, causing activation of postganglionic fibers and the release of ACh onto the target organ	Most conflicting signals cancel each other out, but cardiovascular system is susceptible to hypertension and arrhythmias
Sympathomimetic drugs	Phenylephrine	Bind to adrenergic receptors or mimics sympathetic action in some other way	No effect	Increase sympathetic tone
Sympatholytic drugs	β-blockers such as propanolol or metoprolol; α-agonists such as clonidine	Block binding to adrenergic drug or decrease adrenergic signals	No effect	Increase parasympathetic tone
Parasymphatho-mimetics/muscarinic agonists	Pilocarpine	No effect, except on sweat glands	Bind to muscarinic receptor, similar to ACh	Increase parasympathetic tone
Anticholinergics/muscarinic antagonists	Atropine, scopolamine, dimenhydrinate	No effect	Block muscarinic receptors and parasympathetic function	Increase sympathetic tone

Hydrophilic antibiotics		Lipophilic antibiotics	
Class	Examples	Class	Examples
β-Lactams • Penicillins • Carbapenems • Monobactams • Cephalosporins	 Piperacillin Meropenem Aztreonam Ceftriaxone	Fluoroquinolones	Ciprofloxacin Moxifloxacin
Glycopeptides	Teicoplanin Vancomycin	Macrolides	Clarithromycin Azithromycin
Aminoglycosides	Gentamicin Amikacin	Glycylcyclines	Tigecycline
Oxazolidinones	Linezolid	Lincosamides	Clindamycin
Characteristics			
Mostly renal excretion		Mostly hepatic excretion	
Risk of underdosing with third-spacing, hypoalbuminaemia and inotropes		Minimally effected by third-spacing, hypoalbuminaemia and inotropes	



BASICS OF ANIMAL REPRODUCTION:

Gynaecology- Study of female reproductive system
Andrology- Study of male reproductive system

Types of animals based on estrous cycle:
(1) Monoestrous animals: Bitch, wild animals
(2) Polyestrous animals: Cow, Sow
(3) Seasonal polyestrous animals are two types:
- ✓ Long day breeder: Mare, cat
- ✓ Short day breeder: Sheep, goat, buffalo, camel

Note:
1) Syndesmochorial in sheep & goat
2) Epitheliochorial in cattle & buffalo

Terminology related to ruminant placentation:
- ✓ **Cotyledon**: fetal side of the placenta
- ✓ **Caruncle**: maternal side of the placenta
- ✓ **Placentome**: a cotyledon and caruncle together

Follicular Phase:
- ✓ Proestrus and estrus periods are collectively called "Follicular phase" or "Estrogenic phase".
- ✓ During this phase, estrogen level is more than progesterone.

Luteal Phase:
Metestrus and Diestrus periods are collectively called "Luteal phase" or "Progestrone phase".
Natural C.L. diameter is about 2-3 cm

COMPARISON OF FSH AND LH:

FSH (eCG/ PMSG)	LH (hCG)
Its action is like FSH in all animals except in mare where its action is LH like.	Acton is LH like. Embryo starts production of HCG by day 10th of pregnancy.
Less half Life	Long Half Life
Costly	Cheap
Low CH2O Content	High CH2O Content

Stage of gestation in bovines
1) **Period of ovum**: fertilization to 10-12 days of pregnancy
2) **Period of embryo**: 13 to 45 days of pregnancy
- ✓ Heart is formed & starts beating on 22nd day of pregnancy

3) **Period of foetus**: 45 days onwards until birth
✓ Differentiation of organs and systems

Site of fertilization:
1) In birds: Infundibulum
2) In domestic animals: Ampullary-isthmic junction of the oviduct

Stages of parturition:
1) First stage: dilation of the cervix
2) Second stage: delivery of the newborn (birth)
3) Third stage: expulsion of fetal membranes & involution of uterus

HEAT SIGNS:

Buffalo	Goat	Camel
Twitching tail to one side	Wagging of tail up and down	Extrusion of soft palate

COMMON CAUSES OF DYSTOKIA:

Buffalo	Mare	Camel
Uterine Torsion	Feto postural abnormality	Feto postural abnormality

Methods of artificial insemination in various animals:
1) Cattle & Buffalo - Rectovaginal method
2) Sheep& Goat- Vaginal speculum method
3) Swine - Corkscrew pipette method
4) Mare - Vaginal method

SEMEN QUALITY:

Vey Good	Good	Fair	Poor	Vey Poor
80-100	60-80	40-60	20-40	0-20

Fertile Life:

Sr. No.	Specie	Sperm	Oocyte
01.	Cattle	24-48 hours	12-24 hours
02.	Sheep	24-48 hours	16-24 hours
03.	Mare	72-120 hours	6-8 hours

Methods of pregnancy diagnosis in various animals:

1) Cattle & buffalo -Rectal palpation
2) Sheep & goat - Abdominal palpation
3) She-camel - Cocking of tail, rectal palpation
4) Mare- Cuboni test & Aschheim-Zondek (A-Z) test

STAGES OF PARTURITION:

First Stage	Second Stage	Third Stage
Dilation of Cervix	Expulsion of fetus	Expulsion of Placenta
3-8 hours (Cattle/Buffalo), 2-6 hours (Sheep/Goat) 1/2-4 hour (Mare)	1/2 -1 hour (Cattle/Buffalo), 4 hours (Heifer) 1/2-2 hour (Sheep/Goat), 10-30 min (Mare)	2-8 hours (Cattle/ Buffalo), 1/2-3 hours (Mares)

Note:

1) Most commonly used **cryogen** in bull semen is **liquid nitrogen/LN2 at -196 Degree Celsius.**
2) Most commonly used **cryoprotective agent** in bull semen is **glycerol**
3) **Thawing temperature** of frozen semen is **37- 40 Degree Celsius for 30 seconds**

Types of the uterus:

1) Bipartite uterus- mare, ewe, cow etc
2) Bicornuate uterus- sow, she-camel etc
3) Duplex uterus - rat, rabbit etc

The shape of the uterus:

1) Y-Shape uterus - ruminants
2) T-Shape uterus - mare
3) Between Y & T Shape - She-camel

The shape of the ovary:

1) Cattle, sheep, goat- Almond shape
2) Mare-Bean shape or kidney shape
3) Sow & poultry -Bunch of grapes shape

Male reproductive system:

1) Primary sex organs (gonads) - testicles
2) Secondary sex organs- epididymis,vas-deferens,penis
3) Accessory sex glands- seminal vesicles, prostate, Cowper's gland
4) Copulatory organ- penis

Note:

✓ The male reproductive system develops from the Wolffian duct or Mesonephric duct.

✓ The female reproductive system develops from the Mullerian duct or Paramesonephric duct.

Testes:

✓ Each testis is composed of mass of seminiferous tubules

✓ Seminiferous tubule is made up of two types of cells

a) Germ cells - forms spermatozoa

b) Sertoli cells/sustentacular cells-act as "sperm mother cells" or "nurse cells"

✓ In between seminiferous tubules, interstitial cells or Leydig cells are present which produce testosterone hormone

✓ Temperature of testes is 4-50C less than body temperature in case of domestic animals

Epididymis:

Functions of epididymis:

✓ Maturation of sperms

✓ Concentration of diluted sperm suspension

✓ The tail (cauda) acts as storage house of sperms

Note:

✓ Semen = Sperms + Seminal plasma

✓ Seminal plasma is the composite secretions of testicles, epididymis and accessory sex glands.

Accessory sex glands:

(1) Ampullae: Stallion ampullae secretion contains- ergothioneine

(2) Seminal vesicles: - contributes 60% of total semen volume.

✓ Secretion contains- fructose, prostaglandins, citric acid, riboflavin, vitamin C etc.

✓ The fructose is the main source of energy for the spermatozoa motility.

✓ Boar seminal vesicles secretion contains- inositol & ergothioneine

✓ Buck seminal vesicles secretion contains- lysolecithin

(3) Prostate gland- only unpaired accessory gland, found in all domestic animals

✓ Prostate fluid is alkaline; it helps to neutralize the acidity of the vaginal tract & provide the motility & fertility.

(4) Bulbourethral glands/Cowper"s glands

Note:

✓ Seminal vesicles & bulbourethral glands are absent in dog

✓ Preputial diverticulum is present in boar penis

✓ Double folded prepuce is found in stallion

✓ Glans penis is absent in boar& tom

✓ Os-penis is a visceral bone found in dog penis

✓ Corona glandis is found in stallion

- ✓ Bulbus glandis is the enlarged part of glans in dog

Sigmoid flexure: "S"-shaped curve in the body of **non-erect penises**

- ✓ Pre-scrotal sigmoid flexure-found in boar& camel
- ✓ Post- scrotal sigmoid flexure-found in bull, ram &buck
- ✓ Sigmoid flexure is absent in stallion & dog penis

FEMALE REPRODUCTIVE SYSTEM:

1) Primary sex organs (Gonads): Ovaries
2) Reproductive tracts: Fallopian tube, uterus, cervix, vagina
3) Copulatory organ: Vagina

Note:

- ✓ Cervix is also known as „neck of the womb"
- ✓ Vagina acts as **birth canal & copulatory organ**

Terminology:

- ✓ **Abortion**: expulsion of a dead foetus before completion of the gestation period
- ✓ **Stillbirth**: expulsion of dead foetus at the time of parturition (means it completes gestation period)
- ✓ **Premature birth**: The expulsion of a live foetus before completion of the gestation period

Trichomoniasis in cattle:

- ✓ It is a **venereal disease (transmitted by coitus)**
- ✓ Causative agent: ***Tritrichomonas foetus*** (protozoan)
- ✓ It cause **post-coital pyometra** (potato soup like pus in uterus)
- ✓ **Tritrichomonas foetus** causes **abortion in the first trimester of pregnancy.**

Note:

Brucella, leptospira, mycobacterium and listeria cause abortion mainly in last trimester of pregnancy.

Maternal Recognition of Pregnancy (MRP):

1) Cattle: **bovine interferon tau**/bovine trophoblast protein-1(bTP-1), **16-17 days**
2) Sheep: ovine interferon tau/ovine trophoblast protein-1(oTP-1), 12-14 days
3) Mare: **transuterine migration of conceptus**, 12-14 days
4) MRP is not necessary in case of bitch & cat.

Important Points:

- ✓ Cholestrol: 27 Carbons (Precursor of E2, P4 and T4).
- ✓ Hypothalamus: GnRH (Mother Releasing Hormone)- Anterior Area.
- ✓ Three type of Reproductive Cells:

1. Sertoli/ Supporting Cells
2. Leiding/ Interstitial Cells-T4 production

3. Spermatogonia-Gamete formation.

- ✓ FSH receptors on Granulosal cells of ovary and Sertoli cells of testes.
- ✓ Gonadotropins: FSH and LH- Prolactin suppressive effect.
- ✓ Oxytocin is produced by GnRH and stored and released by posterior lobe of pituitary.
- ✓ GnRH receptors on Reproductive smooth muscles.
- ✓ Teats have tactile receptors.
- ✓ Estrone (E1) and Estradiol (E2)- Principle Estrogen.
- ✓ Equilin and Equilinen- Equines.
- ✓ Immediate precursor of Estrogen-Androgen (T4) by Aromatose enzyme.
- ✓ CIDR: Controlled Internal Drug Release
- ✓ PRID: Progesterone Releasing Intravaginal Device.
- ✓ 5 Alpha- Dihydrotestosterone (Metabolite of T4)
- ✓ T4 causes baldness and RBC's synthesis (10-15 %).
- ✓ Decidua- Modified Mucosal lining of Uterus.
- ✓ Involusion time is 45-60 days

Ovaries:
In Cattle, Buffalo, Sheep and Goat, ovaries are oval-shaped. In Mare, kidney-shaped and in Camel, these are like a bunch of grapes.

Cervix:
In mare and camel, no cervical rings. In mare, there are conspicuous folds in mucosa.

Ovulation Time:
In Cattle, Buffalo, 10-15 hrs after the end of estrus and in Sheep, Goat, 24-36 hrs before the end of estrus.

Heat Signs:
(i) Dirty rumps
(ii) Streaks of saliva by licking on flanks
(iii) Rises her tail and move side wise
(iv) Animal licks vulva of other animal and would like to be licked by other animals and put chin on the back of other
(v) Frequent urination
(vi) Refuse to eat
(vii) Bellowing
(viii) More walking 2-4 times more than normal
(ix) Discharge of mucous from vulva
(x) Decreased milk production
(xi) Swelling of vulva
(xii) Butting

Most reliable sign of heat is "Stands to be mounted". Heat without any estrus sign (external) is called "Silent Heat".

In Cattle and Camel, length of estrus period depends upon mating, it stops within 1-2 days after mating.

Seasonally Polyestrus:
These animals which have many estrous cycles during particular season of the year e.g. Mare, Sheep and Goat.

Non-Seasonal Polyestrus
In these animals, estrus cycle continue through out the year e.g. Cattle and Buffalo

Mono-estrus
Estrus cycle occurs once in a season. E.g. Bitch, Cat and Wild animals

CL dependant animals for Progesterone
Cattle, Buffalo, Goat -- > If we remove CL, abortion will occur

Placenta dependant animals for Progesterone
Mare

Different Stages of Estrus Cycle
- ✓ Proestrus 2-3 days
- ✓ Estrus
- ✓ Metestrus 3-5 days
- ✓ Diestrus: It is the longest period of cycle; 10-14 days. This is the period of CL.

Nymphomenia
- ✓ There is hypersensitivity due to Cystic Ovarian Disease (COD). Female appear like bull, pitch of voice is also bullish, and there is relaxation of sacroschiatic ligament.

OVULATION:
- ✓ In Bovine, ovulation can occur at any point on the ovary except hilus and part of the ovary which is attached to the broad ligament.
- ✓ In Mare, ovulation can only occur at "ovulation fossa".
- ✓ In Camel, ovulation can occur alternatively right and left ovary.
- ✓ In Cattle, 60% ovulations occur from right ovary.

Basic role in the ovulation process is of LH (Leutinizing Hormone).

Sponatenous Ovulators:
Ovulation take place automatically (special stimulations are not required)
Examples:
Cattle, Buffalo, Mare, Bitch

Induce Ovulators:
These animals require certain special stimulus for ovulation

Examples:
Camel, Cat, Rabbit.

In these animals, ovulation needs first copulation as stimulus (Physical vaginal stimulation). It is required for surge release of hormone.

Nulli-parous Animals:
The animals which has not given birth to any young one
Primiparous Animals:
The animals that calved for first time in their life

Differential Point:

- ✓ **In Rabies, animal start bellowing continuously but in estrus, intermittent bellowing.**
- ✓ Restlessness in both, off feed, decreased milk production in both.
- ✓ Estrogen along with progesterone cause appearance of pronounced estrus.
- ✓ First estrus at puberty and after parturition is silent because there is no or very low progesterone level at that time.

Average Age of Puberty (Range)

Species	Male	Female
Bovine	11 mo (7-18)	11 mo (9-24)
Ovine	7 mo (6-9)	7 mo (4-14)
Porcine	7 mo (5-8)	6 mo (5-7)
Equine	14 mo (10-24)	18 mo (12-19)
Human	13 yr	12 yr
Canine	9 mo (5 - 12)	12 mo (6 - 24)
Feline	9 mo (8 - 10)	8 mo (4 -12)

Signs of Approaching Parturition:
(i) Relaxation of sacroschiatic ligament
(ii) Vulva size increased 2-4 times
(iii) Enlargement of udder
(iv) Opaque, yellow turbid discharge from teats

(v) Stringy mucous discharge from cervix (vulva)
(vi) Anorexia
(vii) Reslessness
(viii) Frequent sitting and standing
(ix) Arching back
(x) Kicking at belly
(xi) Sweating in flank region and below elbow before 4-6 hrs

Role of Hormones in Parturition:

Estrogen:

i) Stimulate release of PGF2a from the uterus
ii) Primes the uterus for effect of oxytocin
iii) Primes the cervix for effect of relaxin

PGF2 alpha:

i) CL lysis
ii) Strong uterine contraction

Relaxin:

i) Relaxation of cervix

Oxytocin:

i) Strong uterine contraction for fetal expulsion

Puerperal Period (Involution period):

Time to complete uterine involution.

Foal heat:

Mare comes into heat 10-13 days after parturition, Mating at this time may have very less chances of conception.

REPRODUCTIVE BIOTECHNOLOGY

Artificial Insamination (AI)

It is started in 1954 in UVAS. In 1962, technology was accepted by government. A major semen production units in Pakistan.

Advantages:

i) Increase bull efficiency
ii) Increase potential for genetic selection
iii) Decrease inbreeding chances
iv) Discourage disease transmission
v) It provides facility at the door step
vi) Semen can be stored for years
vii)Helps in progeny testing

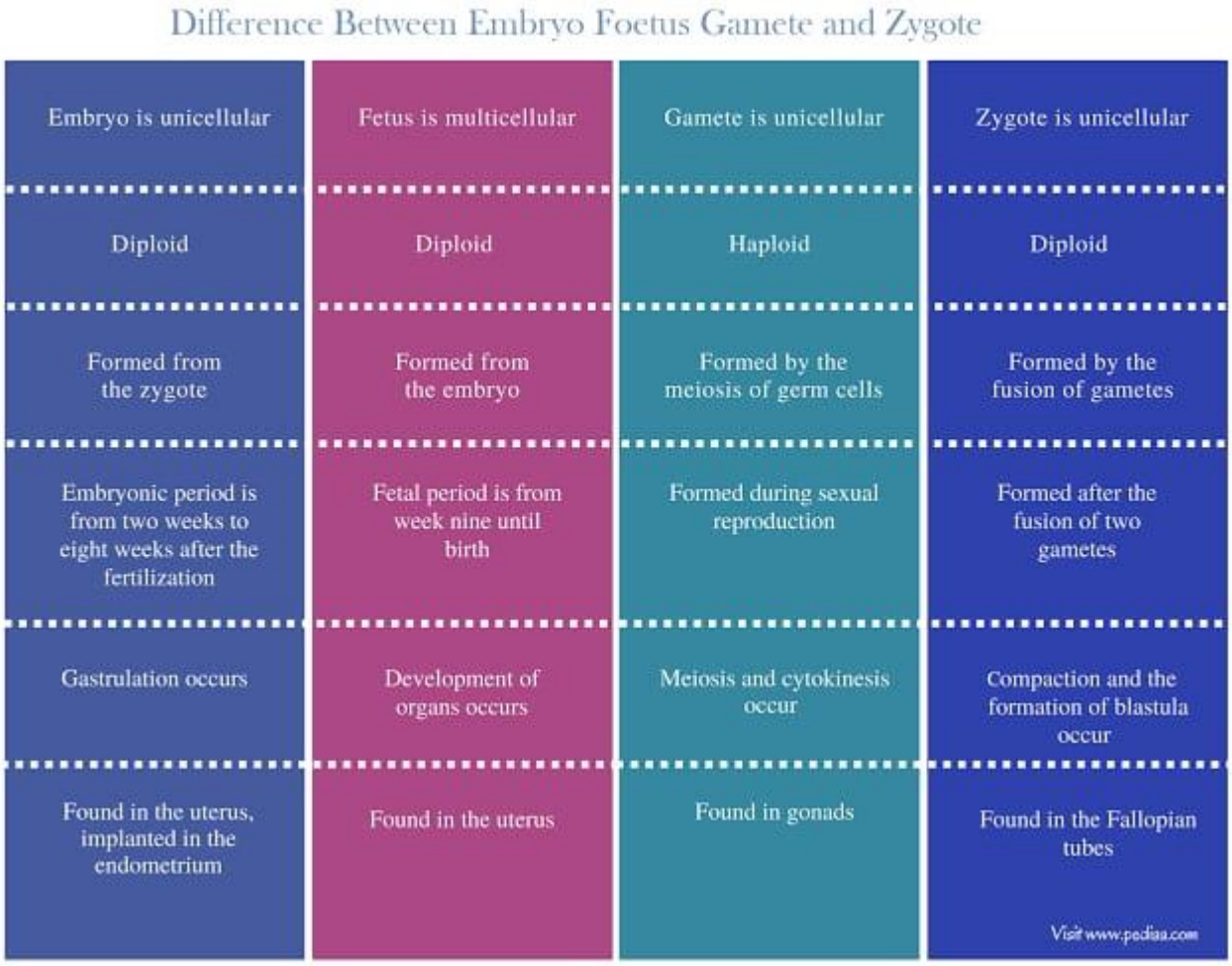
Success of A.I.:
a) Time of insemination
b) Proper heat detection
Mare: Insamination done 36 hrs before ovulation time and ovulation occurs 24-48 hrs before end of estrus. Follicle size: 35-40 mm

No. of Sperms per Semen Dose:
Fresh: 10 million
Frozen: 20-25 million

Methods of Semen Collection:
1. Recovery Method:
    Advantage: Maximum ejaculate can be obtained
Disadvantage:
    Sperm loss and deterioration
2. Massage Method:
Disadvantage:
    Semen is not balance in composition.
3. Electro-ejaculation:
    Mostly in ram/buck
Disadvantage:
    Urine contamination
4. Using Artificial Vagina (AV) – Temperature (42-44 C)

Gross Evaluation of Semen
i) Volume:
Depends on the sexual excitement and activity
    Decreased: Young male, excessive use of bull, incomplete ejaculation, failure of ejaculation, bilateral seminal vesiculitis
    Increased: Volume relates with: Maturity of bull, body size of animal, reproductive health and vigor of male animal
ii) General appearance:
    Creamy (Grade 2), Light Cream (1)
iii) Presence of any foreign particle
iv) pH of semen (6.5-7.2 is ideal one)

Macroscopic Examination
i) Mass Activity:
    It reflects sperm concentration and viability
ii) Motility percentage:
    2.9% Sodium citrate solution is used.
iii) Concentration of sperms; by using hemocytometer
Extenders for Semen:
    MYG, LYG, LFGY, FYG and Sodium citrate – egg yolk extender



Cystic Ovarian Disease (COD)

There are three types of cysts on ovary:

(i) Follicular cyst:

Follicles develop on ovary and after attaining a size of maturity (2 cm) persist on ovary for a period of 1 week or more in the absence of functional CL. There is complete deficiency of CL in this case.

Rx:

LH injection, (hCG has function like LH) – Inj. Physex given IV @ 300 IU

(ii) Luteal cyst:

It is unilateral single cyst on the ovary; leathery appearance due to lutinization. It is due to problem in release of LH.

(iii) Cystic Corpora Lutea:

It is a CL with centrally fluid filled cavity.

Causes:

(a) Hormonal imbalance (insufficiency in release of preovulatory LH surge).
(b) Stress factors
(c) Toxins and infections
(d) Secondary to uterine infection

(e) Exotic breeds are more prone to COD
(f) Nutritional stress
(g) Increased milk production

Treatment:

PMSG or hCG (5,000 to 10,000 IU) or use GnRH preparations.

Metritis

Inflammation of metrium and pus form, no persistent CL, cervix is partially open.

Pyometra

It is due to infection, damage to endometrium, PGs not released and CL persist, cervix is closed.

Infection of Uterus at Open Period

Predisposing factors are unhygienic mating, ascending infection, wrong time insemination and other malpractices.

Types:

Three categories based on exudation, load of infection.

i) Endometritis

It is of three types:

1st degree endometritis:

Inflammation changes occur, mucus + pus in estrus period, most infection occur in this period, uterus less hard.

Dx:

At the time of insemination

Rx:

Lugol's iodine is not given in pus condition but if 1st degree becomes chronic and wall becomes thick then lugol's iodine can be given.

Post A.I:

Single antibiotic non irritating inj can be given.

2nd degree endometritis:

Mucopurulent discharge even after estrus period, swelling of uterus is more resembling a pregnant (1- 1 ½ month) uterus, uterine wall thickened and swollen.

Rx:

Antibiotic given for 7 days, Rest for a cycle, then Lugol's iodine can be given after 10-15 days to generate new endometrium, endometrium has secretory function. Antibiotic @ 40-50 ml given I/U in separate horns (half + half) – Streptomycin, Oxytetracycline may be used for this purpose.

3rd degree endometritis:

Uterus size increased like 2 to 3 months pregnancy, cervix is closed, pyometra, persistent CL.

Rx:

PGF2 alpha given for luteolysis of persistent CL. First E2 (stilboestrol) given for flushing @ 1ml in buffalo -- > make 50 ml solution and given intrauterine.

Important Points:

 Calving interval in buffalo is more than cow which is 470 days while it is 365 days in cow.
 1 cm size graffian follicle going to ovulate
 1 cc = 1 ml = 10 IU

Infertility

Temporary failure of reproductive efficiency due to any abnormality e.g. endometritis, COD.

Sterility

Absolute loss of reproductive ability in male or female e.g. Bilateral hypoplasia (congenital), Fallopian tube blockade, ovario-bursal adhesion.

Fertility

Ability to conceive, to carry viable fetus. By normal calving interval, calving rate, we measure the fertility.

Freemartin

In case of Freemartin, ovum contains both ovarian and testicular tissue. Freemartin is a classical example of intersex, in cattle when co-twin produced -- > male develop normal while female does not.

Salpingitis

Inflammation of fallopian tube

Endometritis

Uterine mucosa is involved; most common cause in open period.

Metritis

When entire thickness of uterus is involved (uterine mucosa + myometrium)

Perimetritis:

When serosa is also involved

Parametritis

Adjacent tissue + uterus is involved in inflammatory process.

SYNCHRONIZATION:

Causing animals to be in heat at the same time. It is done by two methods:

(i) Termination of Functional CL:

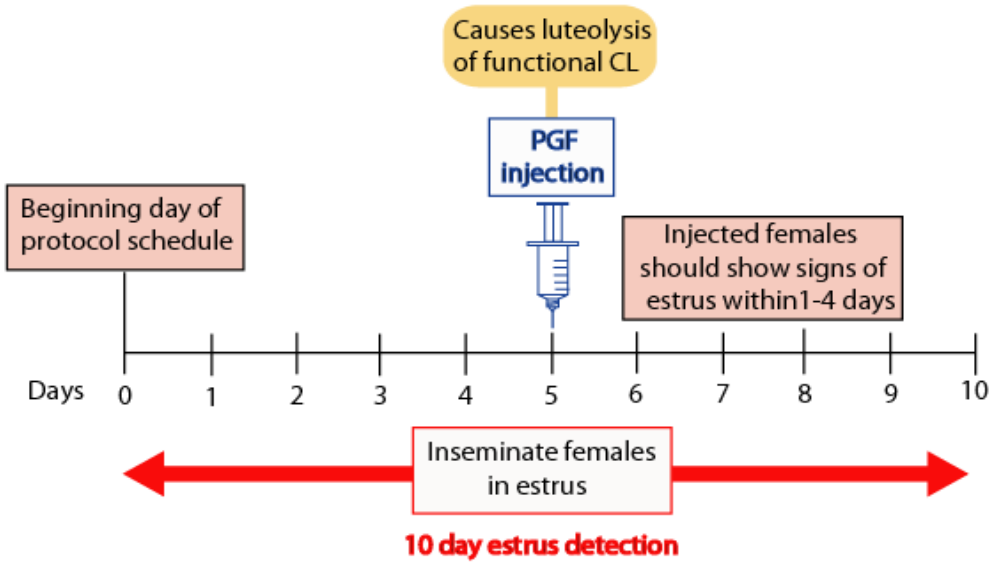
Use following drugs; Lutalyse @ 5 ml / animal, Estromate @ 2ml / animal, Dalmazine @ 2 ml IM, Dinoprost (PGF2 alpha0

(ii) P4- containing Devices:

These are used which block the release of gonadotropins (GnRH and FSH, LH) for 10-14 days. When progesterone is spontaneously removed the gonadotropins are released in more amount and cause ovarian rebounce.

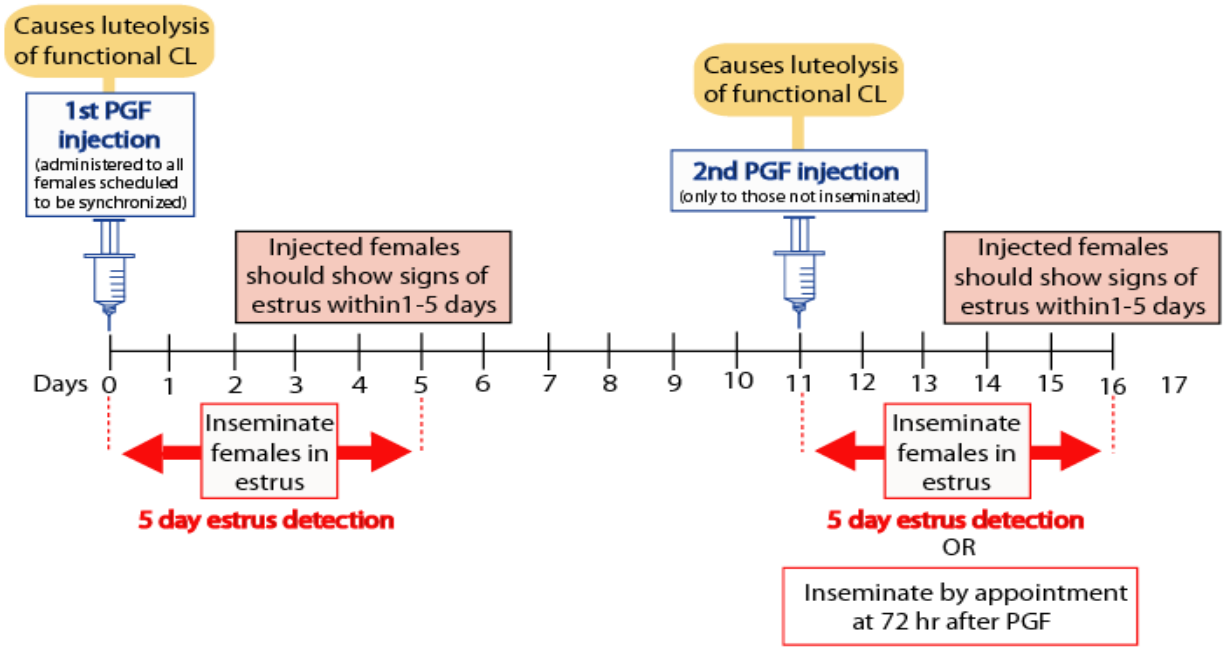
Devices: PRID , CIDR – Animals comes to heat after 2-3 days after removal.

Protocol for One Injection of Prostaglandin (PGF)

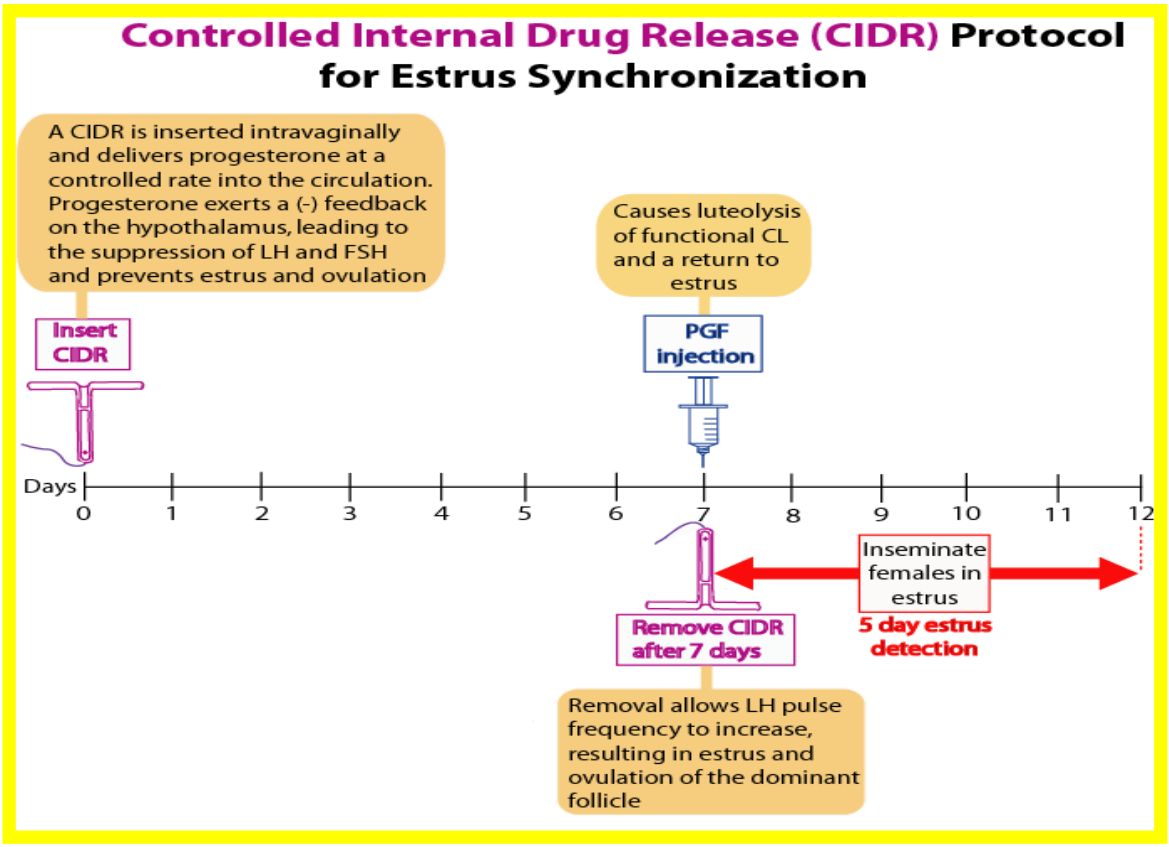


Females that have been inseminated prior to day 5 should not be administered PGF injection unless abortion is desired.

Protocol for Two Injections of Prostaglandin (PGF) with Split Insemination



Females that have been inseminated prior to day 11 should not be administered PGF injection unless abortion is desired.



Methods of Estrus Detection:

(i) Visual observation

(ii) K-Mar heat mount patch Capsules having dye in it are placed on rump of animal, when animal is jumped by other animal, it rupture and impart color to back of the animal.

(iii) Putting of yellow mark on back of animal

Color is rubbed off when animal is jumped by other animal

(iv) Use of teaser bull : penis deviated

(v) Chin ball: can be used on bulls, when it jumps it colours the hook of female.

(vi) Pedometer: When animal in heat, it travel more (2-4 times), activity is observed through this device.

(vii) Trained dogs: doing their job by smelling pheromones

(viii) Use of close circuit cameras

(ix) Detection of hormones:

Progesterone concentration decreased in milk during heat. It should be < 5 ng for heat. P4 conc. is also used for early pregnancy diagnosis.

Misconception and Abortion (Hormone and Pregnancy Timeline)

- ✓ 24-48 hrs --- > 40-30 mg DES (Diethylstilboestrol) or 4-8 mg Estradiol
- ✓ 2- 7 days --- > oxytocin @ 100- 200 IU, IM
- ✓ After development of CL , we can give PGF2 alpha, it causes regression of CL.
- ✓ 1- 5 month --- > PGF2 alpha
- ✓ 2- 6-8 month --- > PGF2 alpha + Corticosteroid (Dexamethasone)

After 5 month, PG for CL and Dexa for placental source of progesterone is destroyed because at that time, animal has two sources of P4.

Procreation:

Creation of new individual from existing one.

Early Embryonic Death:

Mostly occur between 8-19 days after mating

Factors:

(i) Nutrition
(ii) Age
(iii) Cytogenic abnormalities i.e. chromosomal abnormalities.
(iv) Immunological factors: embryo is considered as foreign body. Abortion occur due to not developing immunosuppression.
(v) Environment of uterus:
(vi) Hormonal imbalance
(vii) Time of A.I. and Sire Infertility

Important Note:

In case of hydroallantoic placenta is affected but in hydroamnios, fetus is affected.

Causes of Abortion:

1- Non-infectious i.e. Genetic (Chromosomal abnormalities)
2- Infectious --- > Different diseases
3- Nutritional --- > Starvation, Deficiency of vit. A, E, D, selenium, iodine. Toxication of plants and weeds.
4- Hormonal imbalance: ---- > P4 defficiency, E2 increased, Costricosteroids
5- Physical Causes:
a) Rectal palpation
b) Transportation
c) Making animal to run
d) Major surgery
e) Hyperthermia
f) Allergies
g) Shock

BACTERIAL DISEASES WHICH CAUSE ABORTION:

1. Contagious abortion

Time of abortion:

Abortion occur in second half of pregnancy.

Clinical findings:

There is placentitis, retention of fetal membranes after abortion, uterine atonicity causes secondary bacterial infection. Brucellosis causes infertility by introduction of infection from infected bull (bacteria resides in testicles, epididymus, seminal vesicles).

Treatment:

Vaccination, 4-8 mon calf should be vaccinated with Brucella 51 strain vaccine.

2. **Trichomoniasis**

It is venereal disease which is transmitted by coitus of infected bull; remains carrier throughout the life but cow recovers spontaneously after infection. Infection is characterized by:

(i) Low pregnancy rate
(ii) Mucopurulent vulvular discharge
(iii) Early abortion
(iv) Pyometra
(v) Edometritis
(vi) Vaginitis

Visible abortion occur at 2 to 4 months of gestation.

Treatment:

Metronidazole is best choice for treatment.

3. **Vibrioisis**

Bulls remain permanently infected. Infertility, increased no. of services per conception, early embryonic loss/mortality. Disease spread through coitus or improperly handled semen.

Clinical findings:

Abortion is less common (occasionally occur). Organism is found in female genital tract, fetus, placenta, prepuce and semen. After 13 weeks, local immunity is developed.

Diagnosis:

Agglutination test

Treatment:

Streptomycin at dose rate 25 mg/kg, Dihydrostreptomycin also gives good results.

4. **Leptospirosis**

Caused by L. Pomona, L. canicola, L. icterohemorrhagica. There is elevated body temperature, fetal death, abortion, still birth and birth of weak calf. Increased fever with icterus , hemoglobinurea. In some herds, abortion occurs after "leptospira mastitis". Transmission is through mucus membrane, abraded skin and urine of infected cattle.

Treatment:

Vaccination of all pregnant animals. Dihydrostreptomycin @ 25 mg/kg is effective.

Bacterial Infectious Agent	Time Of Abortion	Lesion
Brucellaabortus	6-9 months. Abortion or stillbirth 2 week to 5 month after infection	Placenta: retained, cotyledons necroticred-yellow, area between thickened.
Brucellosis-Bang's disease		Calf: normal or autolytic with bronchopneumonia.
Campylobacter foetus	5-8 months	Placenta: mild placentitis, hemorrhagic cotyledons and an edematous intercotyledonary area.
Venerealis Vibriosis		Foetus: fresh or autolysed; mild fibrinous pleuritis, peritonitis, bronchopneumonia.

VIRAL DISEASES WHICH CAUSE ABORTION:

1. **Bovine Viral Diarrhea (BVD)**

Caused by Pestivirus of Family Flaviviridae. It causes abortion and fertility in the animals. Infection can be characterized by pyrexia, leucopenia, viremia, watery diarrhea, buccal ulcers, nasal discharge.

Effect of reproduction:

Venereal transmission, and Transplacental infection. There may be abortion or mummification. Transmission is through oculo-nasal discharge, saliva, urine and feces.

Treatment and Control:

Extensive culling is recommended.

2. **Infectious Bovine Rhinotracheitis (IBR)**

Caused by bovine herpes virus (BHV). It causes acute respiratory disease with conjunctivitis. It causes infectious pustular vulvovaginitis, abortion and infertility. Infection transmitted venereally.

Sweling of vulva -- > congested -- > vesicles -- > pustules -- > ulcers formation and release of muco-purulent discharge. Similar lesions are found on penile integument and perpetual mucosa and prepitual discharge. Abortion occurs in late pregnancy.

Treatment:

Amolient creams on vulva and prepuce.

Viral Infectious	Time Of Abortion	Lesion
Bovine Viral Diarrhoea Virus (BVDMD)	Abortion usually up to 4 months	Placenta: retained, no specific lesions. Foetus: no specific lesions, autolysed, Mummified
Bovine Herpesvirus type I (BHV I) Infectious Bovine rhinotracheitis virus	Possibly any stage but most commonly from 4 months to term	Placenta: necrotizing vasculitis Foetus: autolysed, foci of necrosis in the liver
Blue tongue virus	Variable	Foetus: autolysed
Blue tongue Aspergillussp (60-80%Mucorsp,Absidia, orRhizopussp	4 months to term most common in winter	Cotyledons enlarged, necrotic, intercotyledonary area is thickened and leathery. Foetus: autolysed~30% have gray ring worm like skin lesions principally involving the head and shoulders.
Tritrichomonas (Trichomonas) foetus Trichomoniasis	First half of gestation	Placenta: retained, mild placentitis with hemorrhagic Cotyledons and thickened intercotyledonary areas covered with flocculent exudates. Foetus: no specific lesions.
Neosporacaninum Neosporosis	Any stage, but most often 5-6 months	Placenta, foetus: no specific gross lesions, autolysed. Microscopic: focal encephalitis with necrosis and Nonsuppurative inflammation, hepatitis in

IMPORTANT TERMINOLOGY:

Gynecology

It is a branch of science which deals with female reproduction during non-pregnant period.

Andrology

It covers physiological and pathological aspects of male reproduction (word “Andrology” is derived from androgens).

Theriogenology

It is the branch of veterinary medicine which covers all aspects of male and female reproduction. Dr. Bartlett introduced this term for the first time.

Dystokia

It means "Difficult birth"

Eutokia

When delivery of fetus is normal and spontaneous

Primiparous

Animal giving birth to young one for first time

Multiparous

Animal going to parturite 2nd or subsequent delivery

Nulliparous:

Female which yet has not given birth to any young one.

Causes of Dystokia:

1. Maternal

(i) Uterus: There may be:

a) Primary uterine inertia

b) Secondary uterine inertia

c) Rupture of uterus

d) Torsion of uterus

(ii) Cervix: There may be extensive fibrosis

(iii) Vagina: Young age, Any tumor, injury or fibrosis

2. Fetal

(i) Oversize

(ii) Disposition

To treat uterine inertia: we inject Ca- boronate + OT, 20 IU if position of fetus is right

Schaffer's method of rotation is used to correct the torsion.

Schaffer's plank dimension: 15 feet long, 1 feet wide and 2 inch thick – wooden plank

REPRODUCTIVE PROLAPSES OF CATTLE:

- ✓ A prolapse is the abnormal repositioning of an organ from its normal anatomical position. Cattle occasionally develop problems with prolapses near the time of calving.
- ✓ Two distinct types of prolapses occur in the reproductive tract of cattle:

Vaginal or **Uterine** or Vagino-cervical prolapse

1. Vaginal Prolapse:

- ✓ A vaginal prolapse occurs due to increased pressure in the abdominal cavity during the latter stages of Pregnancy.
- ✓ This type of prolapse is more common than a uterine prolapse and typically looks like a pink bulge of tissue ranging in size from a large grapefruit to a soccer ball.
- ✓ The bulge often retracts when the cow gets up and pressure is reduced.
- ✓ Older cows are more prone to Vaginal prolapse.

- ✓ Vaginal Prolapse is not itself life threatening
- ✓ Vaginal prolapse can be an inherited trait.

2. Uterine Prolapse:

- ✓ It is seen immediately following or within few hours of calving. Compared to vaginal prolapse, Uterine prolapse is larger, longer, deep red in color and covered with 'Buttons'.
- ✓ Uterine prolapse is considered Medical Emergency. This condition is life threatening.
- ✓ Uterine prolapses are not heritable like vaginal prolapses.

For correction, we can use:

1% solution of KMnO4 for antiseptic wash, Apply cream like Kenadex, it contains Neomycin + Dexamethasone, Replacement of mass, OT injection given after replacement about 50 ml, Then Use Truss for prevention. Give pain killer (Diclofenic sodium) and antibiotics

TORSION:

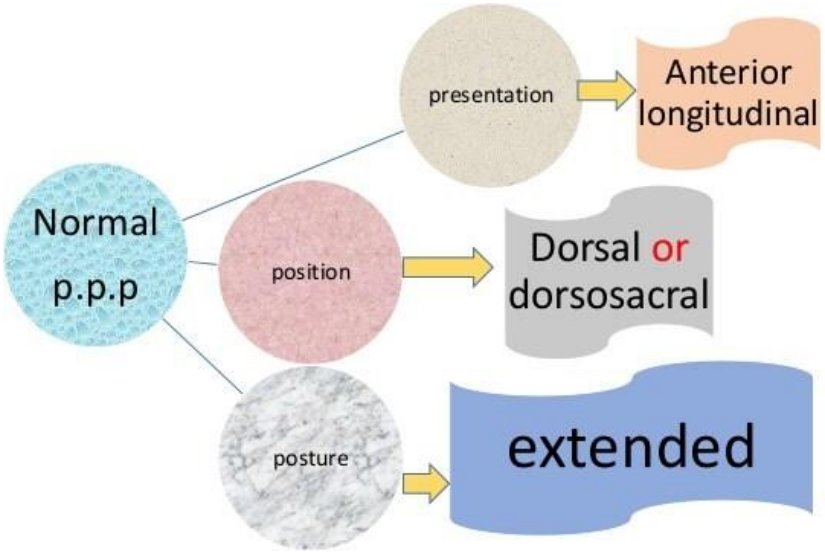
Rotation of the uterus on its long axis with twisting of anterior vagina. Uterine torsion is the complication of 1st stage of labor or early 2nd stage of labor.

Causes:

Instability of bovine uterus which results from greater curvature of the organ being dorsal and the uterus being disposed anteriorly to its suspension of broad ligament.

(i) Incordinate fetal movement in which fetus try to adjust itself to its normal posture in response to myometrial contraction.
(ii) Uterine instability (180 degree).
(iii) Frequent sitting and standing behavior of water buffalo
(iv) Jumping of buffalo/cow

NORMAL PPP IN ANIMALS:



Types:

1. Right-sided torsion:

Right ligament moves downward

2. Left-sided torsion:

Left ligament moves downward

Anatomical Structures:

Oviduct:

It has three parts;

a) Infundibulum b) Ampula c) Isthmus

Uterus:

It has three layers;

a) Endometrium b) Myometrium c) Serosa

Placentome:

Cruncle (from uterus) + Cotyledon (from placenta) + Villi

Cervix:

Cervical rings varies from 2-5

Broad Ligament

It consists of following parts:

a) Mesosalpinx: Part of B.L. which is attached or support the fallopian tube

b) Mesometrium: Part of B.L. which supports the uterus (especially horns and body)

c) Mesovarian: Part of B.L. which supports the ovaries.

d) Inter-cornual: It is present between the horns

Pregnancy Diagnosis by Rectal Palpation

1) Asymmetry of horns

2) Fluctuation of pregnant horn; At day 30, the quantity of fluid is 30-60 ml

3) Conceptus (embryo): On day 30, conceptus can be felt, it ranges from 0.8 to 1 cm

4) Fetal Membranes: 3 distinct membranes (from outside)

i) Chorion

ii) Allantois c) Amnion

Placentomes

Placentomes are more prominent at the base of the horn. These are present throughout the uterus in 4 rows; 2 dorsal and 2 ventral.

Fetal Ballottement

Strike the horn, the fetus will strike back your hand.

Feeling of Freemitus

Feeling of buzzing bee sound on uterine artery at 4-5 months of pregnancy.

Teratology

Study of developmental defects of the growing fetus

CLINICAL USE OF IMPORTANT REPRODUCTIVE HORMONES:

GnRH

(For release of FSH and LH)

Buserelin @ 10-20 g IM [Dalmerilin]**Indications:**

a) Follicular cyst

b) Acyclicity

c) Anovulation

PMSG/eCG

(FSH-like action)

Inj. Folligon @ 1500-3000 IU

Indication:

a) Superovulation in embryo transfer

hCG/LH

Stimulates follicular maturation and leutinization

Physex Leo @ 1500-3000 IU I/M

Indications:

a) Anovulation

b) Ovarian cysts

c) Repeat breeder

Oxytocin

Oxytocin @ 10 IU I/M

Indications:

a) Milk let down

b) Myometrial contraction during parturition and post-parturition

PGF-2α

Luteolytic effect

Dinoprost @ 25-35 mg IM

Indications:

a) Induction of parturition

b) Induction of abortion and mummified fetus

c) Treatment of pyometra

d) Treatment of endometritis

e) Ovarian cyst

Hormone:

Hormone is a chemical substance that regulates certain activity of its target tissue (target is specific tissue which has receptors for the effect of the hormone).

Receptor:

Receptor is a special molecule present on the cell that binds with hormone to bring the required effect of the hormone. Hormones are organic physiological substances. All

reproductive processes are driven by hormones. When problem is there in the effect of hormone function then it also indicates that the problem may lie due to the number of receptors present at target site.

GnRH (Gonadotropin Releasing Hormone)

 GnRH is the mother reproductive hormone. It is a deca-peptide hormone.

 It is secreted from hypothalamus which is 1/300 part of the brain. It is the part of *diencephalon* present at the base of the brain.

 Hypothalamus is made up of neurons, secretion occur from cluster of cell bodies and gonadotrops in the pituitary are the target cells.

 It controls hunger, anger, emotions and sexual process and pleasure.

Factors which STIMULATE GnRH secretions

These factors can be divided into two categories, viz.: External or environmental and Internal or endogenous.

External Factors:

- ✓ Good balanced nutrition
- ✓ Viewing sexually attractive individual of opposite sex
- ✓ Weather (pleasant) especially in seasonal breeders. Long day is favourable for mare. Short day is favourable for sheep.
- ✓ Favourable, comfortable and stress free environment.

Internal Factors:

 Epinephrine & Nor-epinephrine.

 Melatonin favours the release of GnRH

 Coitus in Cat (physical stimulation to vagina)

 In cats and camels the sexual stimulation favors the release of GnRH as they are induced ovulators.

Factors which SUPPRESS GnRH secretions:

External Factors:

- ✓ Stress and Depression
- ✓ Starvation
- ✓ Prolonged sickness and Acute pain

Internal Factors:

- ✓ *Prolactin* (a hormone) suppress the secretion of GnRH.
- ✓ *Opoids* i.e. β-endorphins, secreted while suckling, thus suppress the release of GnRH
- ✓ *Progesterone* (P4) will lead to low *GnRH* production.
- ✓ Low Estrogen (*estradiol 17β*) will lead to low production of *GnRH*. *Estradiol 17β* is needed for animal to be in heat.

Gonadotropins

These are of two types:

i) Pituitary gonadotropins: FSH & LH

ii) Placental gonadotropins: hCG & eCG

FSH, LH, eCG and hCG are gonadotropic hormones. These are members of family glycoprotein.These hormones are made up of alpha and beta carbohydrates molecule chains.

Half life of Gonadotropins:

Pituitary gonadotropins:
FSH: 2 hours
LH: 2 hours
Placental gonadotropins: eCG: >3 days; hCG: >12 hours

Comparative Features of pituitary and placental gonadotropins:

1. **FSH:**

Follicotopin
Growth of follicle from secondary to tertiary follicle
It plays major role in synthesis of estradiole
Target is ovary
Source of natural FSH is pituitary of porcine or bovine. It is also produced now a days through genetic engineering by inserting FSH producing gene into the E.coli organism.

2. **LH:**

Final growth of follicle, ovulation, formation of CL, maintenance of CL. Target is ovary and CL. Source of natural LH is porcine or bovine pituitary.

Clinical Use of FSH:

- ✓ Infertility due to failure of follicular growth
- ✓ It is used for superovulation

Clinical Use of PMSG:

- ✓ Infertility due to failure of follicular growth
- ✓ It is used for superovulation

Clinical Use of LH:

- ✓ For treatment of follicular cyst
- ✓ For treatment of delayed ovulation

Clinical Use of hCG:

- ✓ For treatment of follicular cyst
- ✓ For treatment of delayed ovulation
- ✓ Used in the Rig test

Rig Test

Purpose: To confirm the suspected cases of cryptorchidsm

Prostaglandins

It is group of hormone like substances, derived from fatty acids mainly from arechidonic acid (essential FA). Arechidonic acid is 20 C polyunsaturated fatty acids with a cyclopentane ring in its structure.

Function:

- ✓ Cyclic regression of CL (At 17-18 day of cycle abundant PGF2α is released
- ✓ PGE2α is luteotropic, favours the growth of CL. It is vasodilator and increases blood supply
- ✓ Helps in transmission of semen in duct system
- ✓ In male prostaglandin causes smooth muscle contraction for ejaculation of semen.
- ✓ Involved in ovulation. (In endometrium release of PGF2 is activated by estrogen and oxytocin).
- ✓ At the time of parturition it softens the cervix and cervix is easily is expandable.
- ✓ Involved in first phase of parturition. PGF2α produced in placentome under the effect of increased level of estrogen.
- ✓ Fetal placental expulsion is also by the role of PGF2 α.

Sources of PGF2*α*:

- ✓ In female endometrium
- ✓ Fetoplacental unit during parturition
- ✓ Myometrium (parturition)
- ✓ Grafian follicle (during ovulation)

Uses:

- ✓ Treatment of acyclic animal due to persistent CL
- ✓ Treatment of luteal cyst
- ✓ Estrus synchronization (2 injections of PGF2 11 days apart)
- ✓ Induction of parturition. We use corticosteroids also along with it to finish all sources of progesterone.
- ✓ Termination of pathological pregnancies (as fetal mummification, fetal maceration, hydroamnios, aydroallontois)
- ✓ Termination of unwanted pregnancies. CL remains for 5-6 days insensitive to PGF2α, so give PGF2α after 7 days of ovulation to avoid pregnancy. Response of CL to PGF2α is better for 10-11 days aged CL than 7-8 days aged. Because receptors are not much effective on CL. Upto 150 days alone PGF2α can be used but after that corticosteroids are also used because placenta is also the source of progesterone and both CL and placenta should be regressed.
- ✓ Prevention of pregnancy.
- ✓ For the treatment of pyometra because CL persists in this problem; so the CL is regressed by PGF2α, cervix is relaxed and E2 is then given.
- ✓ In mild or first degree endometritis - uterus is soft, cycle is normal, history of repeat breeding; so check for small flakes, so give some PGF2α.
- ✓ In silent estrous. Estrous without behavioral signs of estrous or weak heat signs.

Estrogen

Chemically it is steroid (having 4 ringed 17 C structure). It is female sex hormone. Cholesterol (27 C) → Progestin (P4) (21 C) → androgens (19 C) (male sex hormone) → estrogen (18 C)

All steroid hormones are derived from the cholesterol which is a 27 C molecule. In stallion normally large amount of estrogen is secreted in urine which is converted from testosterone.

Functions:

- ✓ Androgens, if in high concentration, bring secondary sex characters prominent in the female.
- ✓ In fetus responsible for neural development which is different from male. LH surge release centre in brain is only in female and this is because of neural development difference.
- ✓ E2 is responsible for secondary sex characteristics
- ✓ E2 causes behavioral heat signs in all animals
- ✓ E2 causes release of pre ovulatory surge of LH
- ✓ E2 softens the cervix
- ✓ Pelvic ligament dilation
- ✓ Mammary gland development
- ✓ Decrease the chances of infection
- ✓ During pregnancy, E1 (estrone) level increases, so indicate pregnancy.
- ✓ At the time of estrus it causes increase blood flow to uterine lumen, promotes the growth of endometrium, favors production of mucous through mucous glands, causes relaxation of cervix, increases water content in reproductive tract, increases tonicity of myometrium (uterine contractability increases) and increases capillary bed of uterine wall.
- ✓ At the time of parturition it favours or stimulates the secretion of PGF2α, increases contractability of uterus and softening of cervix and relaxation of ligaments.

Clinical Uses:

- ✓ Prevention of pregnancy
- ✓ Termination of pregnancies
- ✓ It may help in the treatment of torsion to avoid twist in fresh cases of less degree for purpose of cervical dilation.

Progesterone

It is steroid hormone. Its main source is CL. Before puberty it comes from adrenal cortex. Its effects are seen after puberty. When animal becomes cyclic, its production starts. Just after ovulation its concentration gradually increases. In cow ovulation occurs at day two. Then becomes carpus hemorrhagic and ruptured follicle is converted into CL and complete CL after 4 days and at day 5 CL is embedded in ovarian tissue and is soft. Normally 0.2-0.5 ng/ml of plasma on heat day. At day 17-18 CL is regressing and progesterone concentration starts decreasing. Cyclic cow on day 17 has more than or equal

to 5 ng/ml of plasma in cyclic cow. PGF2α when comes then CL start decreasing abruptly. Animal having high concentration of progesterone at estrus have not good conception rate. It affects the transport of gametes.. At day 5 there is more than or equal to 2 ng/ml of plasma. In cattle buffalo maximum concentration of progesterone is at day 10-12.

Functions:

- ✓ Progesterone and estrogen are antagonist. They work together during gestation period. As the pregnancy advances for uterus growth estrogen is accelerating. Progesterone suppresses the heat. Estrogen is in higher concentration at the time of estrus, stimulates GnRH and causes the release of LH surge. In proestrus estrogen increasing and progesterone decreasing and it causes negative effect on GnRH.
- ✓ P4 causes the maintenance of pregnancy, stops cyclic activity by suppressing gonadotropin release, puts negative effect on the hypothalamus and anterior pituitary and does not let the ovulation to occur. Progesterone also suppresses the heat signs. **Progesterone released from the CL in non pregnant animals and in pregnant animals it also comes from placenta.**
- ✓ P4 suppresses the uterus contraction.
- ✓ P4 suppresses the local immune system response (suppresses infiltration).
- ✓ P4 stimulates secretary activity of endometrium that helps in nourishment of early embryo as the fetus get nutrition from uterine milk (in form of nutrients).
- ✓ P4 helps in closure of cervix and formation of the pregnancy seal (thick gummy mucus plug formed within the cervix).
- ✓ It helps in maternal behaviour in females.

Clinical Uses:

o Estrus synchronization PRID, CID

OBSTETRICAL INSTRUMENTS:

Obstetrical instruments are grouped into various categories namely:

A. Instruments used for traction or extraction of foetus.
B. Instruments used for incising or excising.
C. Instruments used for repulsion and rotation.
D. Instruments used in foetotomy.
E. Miscellaneous equipment.

A. **Instruments used for traction or extraction of foetus:**

- ✓ **Ropes**: Made of nylon or cotton.It has a loop on one end and the other end is loopless.
- ✓ **Chains**: Obstetrical chains are made up of metal used for the traction of fetus.
- ✓ **Wooden Handle**: It is about a foot in length.
- ✓ **Whelping forceps**: It is used to apply traction in small animals like bitches and does.
- ✓ **Fetal extractor:** Used for the traction of fetus.
- ✓ **Hooks**: Hooks are used to hold the foetal parts where hand cannot reach and surface are slippery

- ✓ **Krey schottler's hook**: (multi-joint traction hook)this hook has 4 hinge joints(moveable joints) with a moveable ring.This is used for force traction of dead foetus.It is placed in head or eye-ball. During traction, its hooks enter the tissues of foetus and avoid laceration of birth canal.
- ✓ **Harris eye hook**:The hook is used for traction of dead foetus.The hook is inserted behind the incision or in the inner canthus of eye.
- ✓ Obermayer's anal hook:It consists of a long handle and an eye on one side and hook on other side..It is used at time of breech presentation and inserted in anus or at croup region for extraction of foetus.
- ✓ Robert's hook: Used for traction of the dead foetus.
- ✓ Freyburger's hook: This is used for traction of live foetus.These hooks have three paerts viz.hook proper,handle and eye for passing ropes. After passing the rope, hooks are attached behind orbital fossa and the foetus is oulled outside.
- ✓ Sharp hooks:Frequently cause injury either to operator or the animals. Therfore it should be used in dead foetus.In live foetus mostly blunt hooks are used.
- ✓ Kellers semisharp spatula:It is especially useful for decortication of limb and subsequent division of muscle attached to the truk.
- ✓ Short hooks: Have an eye on one end through which cords run while long hooks have handle to operate. Hence long hooks are easy to handle from outside.
- ✓ Eye Hooks: Applied at the inner canthus of the eye for traction.
- ✓ Anal hooks: Are applied at the pubis.anal hooks are slightly longer than the eye hooks.

B. Instruments used for incision and excision:

- ✓ Guarded embyotome: These are used to make incisions on the fetus within the uterus to drain fluid or gas for reduction in size and easy delivery in ascitic, anasarca or emphysematous fetuses.
- ✓ Gunther's finger knife: Used for intra-uterine embryotomy operation.
- ✓ Willium's long Obstetrical hook: Used for embryotomy operation. For getting hold on the foetal part where hand cannot reach and the surface are slippery. It can be applied on Head, palatine arch, spine and pelvis.
- ✓ Chisel:It is used for separating skin during abdominal and thoracic evisceration and it is aslo used in emphysematous foetus.
- ✓ Spatula:For subcutaneous embryotomy,Keller's semi sharp spatula is most commonly used.

C. Instruments used for retropulsion & rotation:

Repellers:

a. Gunthers repeller
b. Binz's repeller.
c. Reindle repeller (three pronged)
d. Reindle repeller (two pronged)
e. Kuhn's crutch.

Rotator:

a. Cornell's detorsion cord

b. Cammer's tortion fork.

D. Foetomy Equipments:

- ✓ Embryotome:Consist of two metal tubes 70 cm long firmly fixed together. The tubes are connected by three transverse bridges to give the instrument rigidity. It is used to protect the birth canal from cutting action of wire saw and allow cuts in different planes.
- ✓ Saw introducer: To introduce fetotomy wirethrough tubes of embryotome.
- ✓ Fetotomy saw (wire):For cutting of fetus.
- ✓ Hand grips of fetotome wire: For graspingof fetotome wire firmly during fetotomy
- ✓ Sand director:For passing of fetotomy wirearround part of fetus to be sectioned.
- ✓ **Ute**rine Pump:Used for theintroduction of artificial fluid into uterus.

E. Miscellaneous Equipments:

Buckets, soaps, several types of antiseptics, common surgical instruments including syringes, needles, scalpel, scissors, forceps, suture needles, sterile drapes, towels, razor. Cotton etc.

DIFFERENT TYPES OF CL:

Cl Verum	CL Spirium	CL Albicans	CL Hemmorhagicum
During Pregnancy	During Cycle	Regress	After Ovulation

CL AND P4 PRODUCTION LEVEL:

Developing CL	Growing CL	Mature CL	Regressing CL	Regressed CL
2-5 Day	6-10 Day	10-17 Day	18-21 Day	Hat Day
-	2-5 ng P4	>5 ng P4	<5ng P4	<0.5 ng P4

SEMEN COLOR:

Brownish Color	Yellow Color	Pinkish/ Reddish Color	Greenish Color
Degenerative Blood Tissue	Pus and Cells	Fresh Blood	Purulent Degeneration

PRIMARY, SECONDARY AND TERTIARY ABNORMALITIES OF SPERMS:

Sr. No.	Primary Abnormality	Secondary Abnormality	Tertiary Abnormality
01.	Disorders of seminiferous or germinal epithelium	Sperms left germinal epithelium	During or after ejaculation and Mishandling of semen
02.	E.g: Head abnormalities I.e Microcephalic,	E.g: Detached normal head, Proximal and distal	E.g: Excessive agitation, overheating,

	Macrocephalic, Elongated narrow, Short broad, pyriform, Double head, Double middle piece and tail, swelling, Abaxially attached middle piece and Highly coiled middle piece and Tail.	protoplasmic droplets, Spermatozoa with bent tail and Detached and loosened galea capitis.	too rapid cooling and presence of water and urine etc.

AGE AND SIZE OF FETUS (RESEMBLANCE):

2 Months	3 Months	4 Months	5 Months	6 Months	7 Months
Mouse	Rat	Small cat	Large Cat	Beagle Dog	Fetus Palpation

FETAL SLIP MEMBRANE TEST (TIME TO PERFORM):

32 Days	45 Days	60 Days	70 Days
Small thread in 1 horn	Small strings in 1 horn	Strings in both horns	Large strings in both horns

SIZE OF PLACENTOMES (GESTATION):

75 Days	100 Days	115 Days	125 Days	150 Days	180 Days
1-1.5	1.5-2.5	2.5-3.0	3.0-4.0	4.0-5.0	5.0-6.0
Pea Size	Dime	Nickle	Quarter	Half Dollar	Silver Dollar

FOLLICULAR AND LUTEAL CYST:

FOLLICULAR CYST	LUTEAL CYST
Multiple in nature	Single in Nature
Persist for more than 10 days	Persist less than 10 days
E2 at peak	P4 at peak
Causing Nymphomenia	No heat/silent heat
Exp: Adrenal virilism	**Exp**: Swiss Cheese appearance and advanced virilism
Rx: GnRH	**Rx**: PGF2 Alpha

REPRODUCTIVE HORMONES, FUNCTIONS AND THEIR NATURE:

1. Anterior lobe of Pituitary:
a. FSH (Ovarian Follicle and E2 production)
b. LH (Ovulation+ CL formation+ T4 support in male)
c. Prolactin (Secretions inhibition by Ergot Alkaloid)

2. Posterior lobe of Pituitary:
a. Oxytocin
b. GnRH (Storage site)

3. Granulosal Cells (Female)
a. Inhibin
b. Activin

4. Sertoli Cells (Male)
a. Inhibin
b. Activin

5. Trophoblast of blastocyst (Chorion)
a. hCG

6. Chorionic Girdle Cells:
a. eCG (Human placenta)

7. Neuropeptide:
a. GnRH
b. Oxytocin

8. Hormonal Peptide:
a. Leptin

9. Protein:
a. Prolactin
b. Placental Lactogen

10. Glycoproteins:
a. LH
b. FSH
c. Inhibin
d. Activin
e. hCG
f. eCG
g. Alpha proteo protein

h. Relaxin
i. Interferone

11. Steroids:
a. E2 (18 carbons)
b. P4 (21 Carbons)
c. T4 (19 Carbons)
d. Arachdonic Acid (20 Carbons Poly Unsaturated Fatty Acid)

12. Peptide:
a. ACTH

13. Polypeptide:
a. Placental lactogen (Somatomammotropin)

14. Placenta Origin:
a. Relaxin
b. Lactogen
c. PGF2 Alpha

15. Fetal origin:
PMSG (Endometrium)

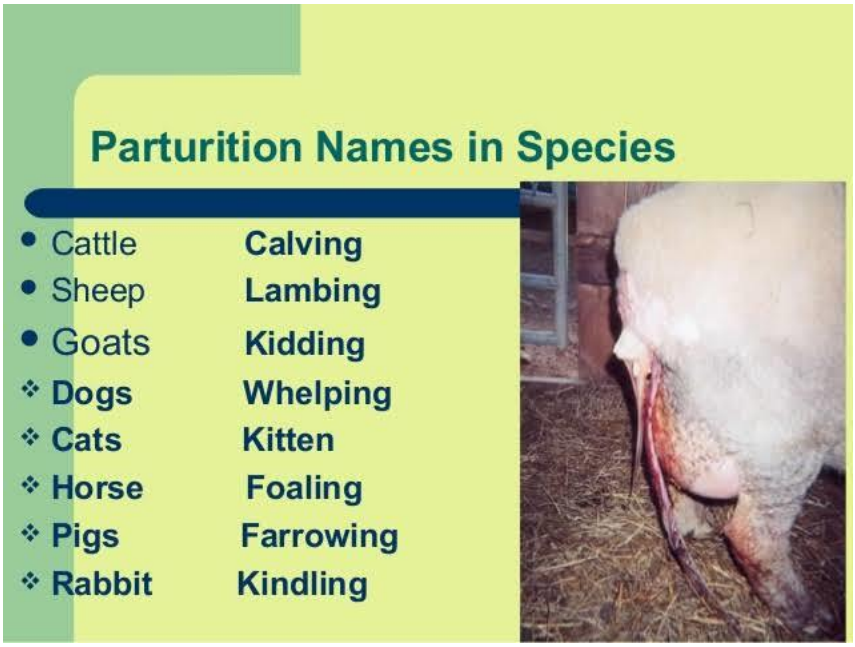
VIVIPAROUS, OVOVIVIPAROUS AND OVIPAROUS:

Viviparous	Ovoviviparous	Oviparous
Development of an embryo inside the body of the parent.	Producing young by means of Egg which are hatched within the body of the parent.	Producing young by means of Eggs which are hatched after they have been laid by the parent.
E.g: Cattle, Horse etc.	E.g: Strongyloides, Snakes etc.	E.g: Birds and Trematodes etc.

ULTRASOUND, TRANSDUCER AND ITS TYPES:

✓ Ultrasound: High frequency sound waves (20-20,000 Hz)
✓ Transducer contains Piezoelectric crystals (Lead Zirconate or Titanate)
✓ Transducer has 2 types:
1. Linear Transrectal (5-10 MHz)
2. Sector Transabdominal (1-4 MHz)
✓ There are 3 modes of Ultrasound machine:
1. A Mode: Pinpoint View
2. B Mode: Two dimensional View (Cathode Ray Tube- Real Time)
3. M Mode: Doppler Effect/ Moving Objects

- ✓ Acoustic Impedance (Z): Physical property of tissue how much resistance an Ultrasound beam encounter.
- ✓ Z=d*c (d=density, c=speed of sound wave)
- ✓ Bone has maximum Acoustic Impedance=$7.8*10^6$
- ✓ Air has minimum Acoustic Impedance=$0.0004*10^6$
- ✓ Hyper-echoic:White image-Stones/Bones/Air (High dense)
- ✓ Iso-echoic: Grey-Normal tissue
- ✓ Hypo-echoic: Black-Fluid Filled Cavity/Follicles/CL (Low dense).



Name of species	Age of Puberty	Breeding Season	Estrous cycle			Time of Ovulation	Breeding Time	Gestation Period	Name of Young One	Parturation Term	wt. at Birth	Uterine Involution time
			Estrous Cycle Nature	Estrous Cycle Length	Estrus Duration							
Cattle •Sahiwal •Desi •Exotic	24-30 Months 8-11 Months	March April	Polyestrous	21 Days	12 - 18 Hours	In Metestrus 8-16hr after end of estrus	8-12 Hrs after ownset of estrus	9 Months ± 9 Days	Calf	Calving	30 kg	45-50 Days
Buffalo	28-30 Months	November to December	Polyestrous	21 Days	18 - 24 Hours	In Metestrus 8-16hr after end of estrus	= 10 hrs after end of estrus	10 Months ± 10 Days	Calf	Calving	28 kg	45-50 Days
Camel	36-48 Months (3-4 Years)	November to March	Seasonal Polyestrous	23 - 29 Days	2 - 5 Days	36 - 48 Hours after mating	Breed 2 times with in interval of 12 Hours	1 Years + 20Days	Calf	Calving	36 kg	30-50 Days
Mare	16-24 Months	15 Feb, to 1st July	Seasonal Polyestrous	22 Days	4 - 6 Days	on 5th -6th Day of Estrus	1 - 2 mating onward 4th days of estrus	11 Months ± 11 Days	Foal	Foaling	10 % of Mother Weight	21-28 Days
Sheep	9-14 Months	Sep-Nov	Seasonal Polyestrous	17 Days	22- 36 Hours	24-36 hrs after ounset of estrus	12-18 hrs after ounset of estrus	5 Months ± 5 Days	Lamb	Lambing	≃05 kg	30 Days
Goat	9-14 Months	Sep-Nov	Seasonal Polyestrous	21 Days	30 - 36 Hours	24-36 hrs after ounset of estrus	12-18 hrs after ounset of estrus	5 Months ± 5 Days	Kid	Kidding	18 -36 Kg	30 Days
Dog	6-12 Months	March April	Seasonal Polyestrous (In Large Breed) Seasonal Monoestrous (In Small Breed)	6 Months	7 - 10 Days	3 - 4 Days of estrus	2 - 3 time mating onward 3rd day of estrus	58-65 Days	Pup	Whelping	400 - 800 Grams	90 Days
Cat	6-12 Months	March Sep	Seasonal Polyestrous	16 Days	6 - 7 Days	after 2-3 Mating with in 24- hrs	3 - 4 Services Should done	65 Days	Kitten	Kittening	99 Grams	30 Days

Term related to different species of Avians

Animal	Male	Female	Young one	Group name	Act of Parturition
Fowl	Cock	Hen	Chick	Flock	Hatching
Goose	Gander	Goose	Gosling	Gaggle	Hatching
Ostrich	Cock	Hen	Chick	Flock	Hatching
Owl	Owl	Jenny/ Howlet	Owlet	Parliament	Hatching
Duck	Drake	Duck	Duckling	Flock	Hatching
Turkey	Tom	Hen	Poult	Flock	Hatching
Swan	Cob	Pen	Cygnet	Flock	Hatching

Term related to different wild herbivores

Animal	Male	Female	Young one	Group name	Act of Parturition
Elephant	Bull	Cow	Calf	Herd	Calving
Rhinoceros	Bull	Cow	Calf	Herd	Foaling
Hippopotamus	Bull	Cow	Calf	Herd	Foaling
Blue-bull	Bull	Cow	Calf	Herd	Foaling
Zebra	Stallion	Mare	Foal/ Colt	Pace	Foaling
Giraffe	Bull	Cow	Calf	Herd	Calving
Deer	Buck	Doe	Fawn	Herd	Calving

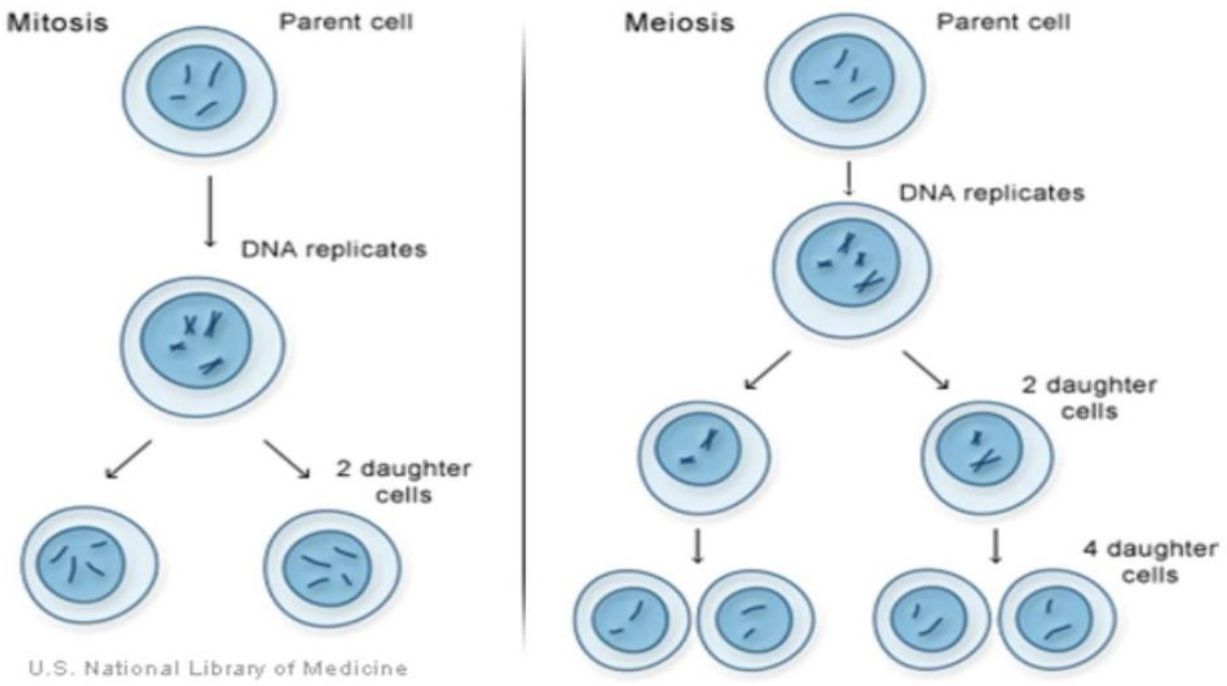
The most important hormones influencing ovarian activity

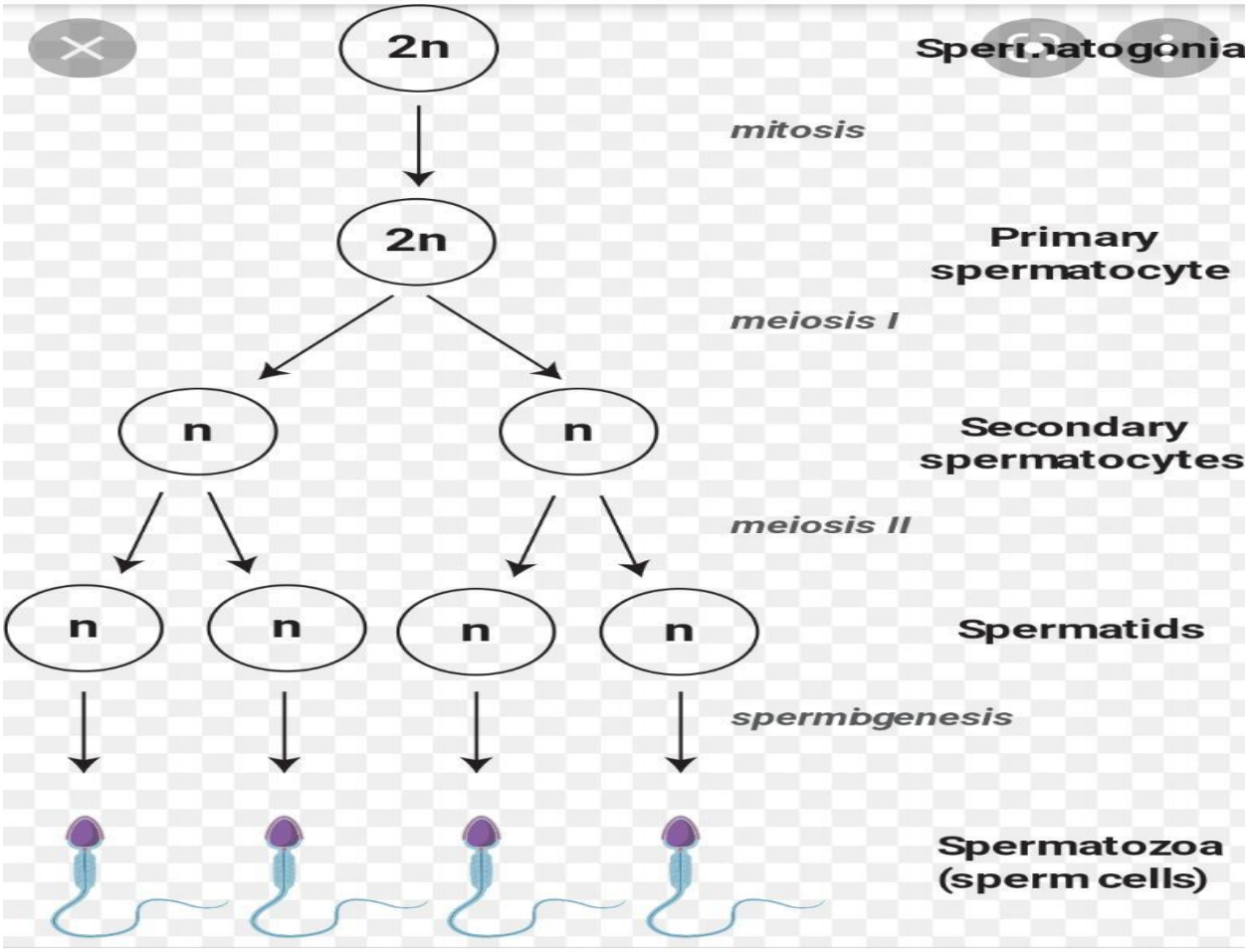
Hormone	Biochemical classification	Source	Action in female	Effect on ovary
GnRH	Decapeptide	Hypothalamus	Release FSH and LH	Follicular development; ovulation
LH	Glycoprotein	Pituitary	Stimulates ovulation and P4 secretion	Formation of CL
FSH	Glycoprotein	Pituitary	Follicular development, E2 synthesis	Development of follicles
Progesterone	Steroid	CL, placenta	Maintenance of pregnancy	Inhibits GnRH release
Estradiol	Steroid	Follicle, placenta	Sexual behavior	-
hCG	Glycoprotein	Chorion	↑ovarian P4 synthesis	-
eCG	Glycoprotein	Chorion	Formation of accessory CL's	-
$PGF_{2\alpha}$	Prostaglandin	Endometrium	Destruction of CL	Luteolysis
Inhibin	Glycoprotein	Granulosal cells	Inhibits FSH secretion	Inhibits follicle development

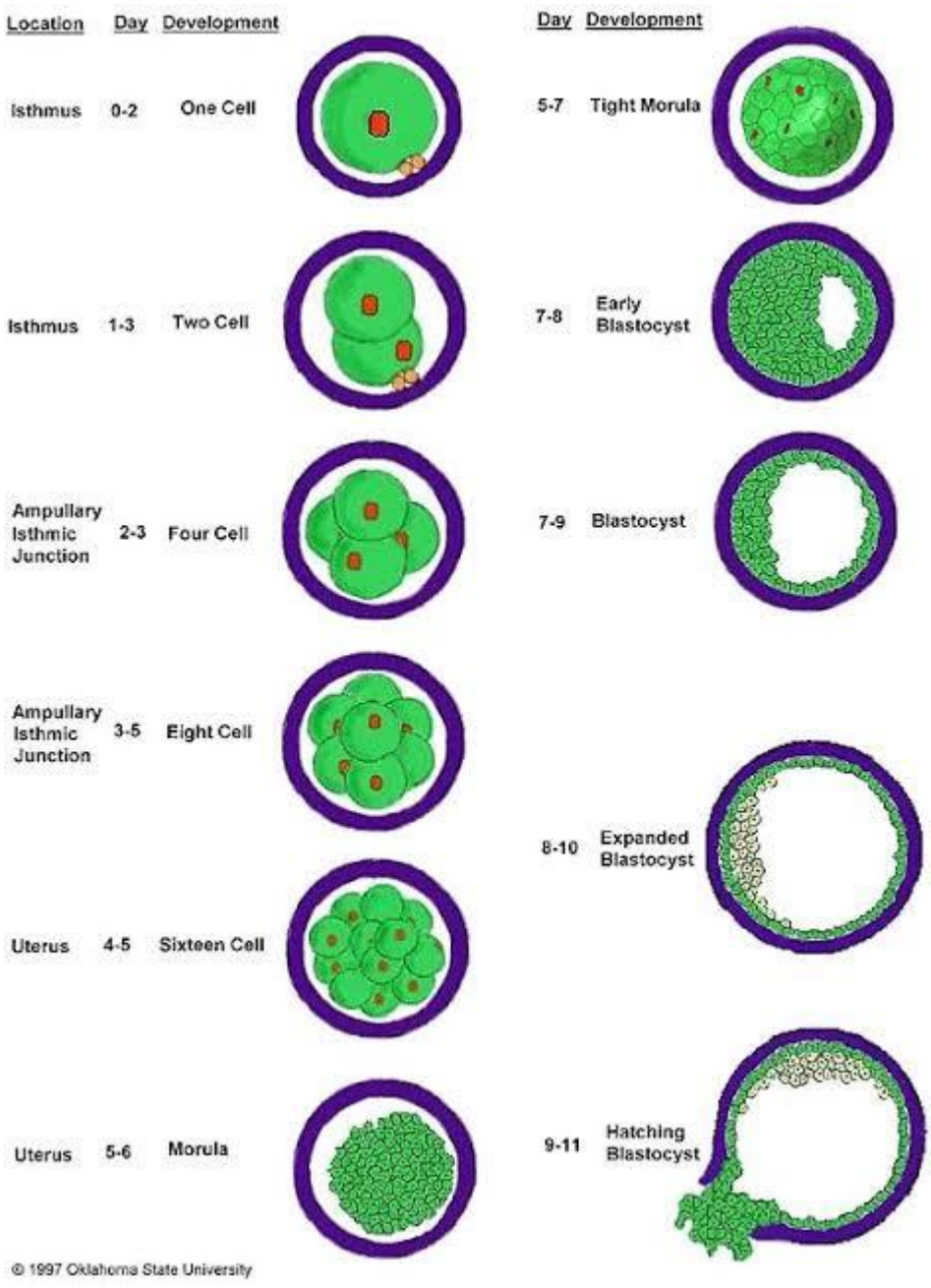
Male, Female, Young one of domestic animals with their group names

Animal	Male	Female	Young one	Group name
Cow	Bull	Cow	Calf	Herd
Buffalo	Buffalo bull	She buffalo	Buffalo calf	Herd
Goat	Buck	Doe	Kid	Trip
Sheep	Ram	Ewe	Lamb	Flock
Camel	Bull	Cow	Calf	Flock
Horse	Stallion	Mare	Foal	Stable/Herd
Donkey	Jack	Jenny	Foal	Drove
Pig	Boar	Sow	Piglet	Drove
Dog	Dog	Bitch	Pup	Pack
Cat	Tom	Queen	Kitten	Clowder

mitosis and meiosis, the two types of cell division.






Location
Day
Development
Isthmus
0-2
One Cell
Isthmus
1-3
Two Cell
Ampullary Isthmic Junction
2-3
Four Cell
Ampullary Isthmic Junction
3-5
Eight Cell
Uterus
4-5
Sixteen Cell
Uterus
5-6
Morula
Day
Development
5-7
Tight Morula
7-8
Early Blastocyst
7-9
Blastocyst
8-10
Expanded Blastocyst
9-11
Hatching Blastocyst
© 1997 Oklahoma State University

BASICS OF POULTRY PRODUCTION:

Note:

- ✓ **Oviposition**: Act or process of **egg laying in birds**
- ✓ Total time taken: **24-26 hours**
- ✓ Hormone responsible: **Arginine vasotocin (AVT)** from posterior pituitary
- ✓ **Incubation period:** time-period from oviposition to hatching of an egg
- ✓ Chicken egg of 58 g (standard size) has **74% water & 26% dry matter**(dry matter includes-**13% protein**,11% fat,1% carbohydrate&1% ash)
- ✓ A whole egg of chicken consists of **11% shell & shell membrane, 58% albumin & 31% yolk**
- ✓ **Eggshell consists of 96% inorganic matter** (94% calcium carbonate, 1% magnesium carbonate& 1% calcium phosphate**) 4% organic matter**
- ✓ **Vitamin C is absent in eggs, meat & milk**

Terminology related to poultry:

- ✓ **Layer:** -poultry raised for the purpose of egg production
- ✓ **Broiler/Fryer**: -young chicken of either sex up to 8 weeks of age used for meat purpose.
- ✓ **Roaster**: - young chicken of either sex up to 12-16 weeks of age used for meat purpose& heavier than the broiler
- ✓ **Capon**: -castrated male fowl
- ✓ **Cockerel**: -male fowl under 1 year age
- ✓ **Pullet**: -female chicken lay 1st time
- ✓ **Loafer**: -unproductive layer
- ✓ **Cock /Rooster**: -adult male fowl
- ✓ **Hover:** -heat source for brooding
- ✓ **Clutch**: -a batch of an egg laid by the hen on consecutive days.

Candling of eggs: Done twice by electric Candler in a dark room
(1) **First candling**: on the **7 th day** of incubation **to discard infertile eggs**
(2) **Second candling**: 0n **18th day** of incubation **to discard dead-in-shell**

Types of incubator: (2 types)
(1) Natural draft incubator/flate type: -it is usually of **small capacity**. It is further two type (a) hot water type (b) hot air type
(2) Forced draft incubator/cabinet type: Used commercially.
Factor essential for proper incubation:
(A) Temperature:
(a) During first 18 days of incubation: 37.6 Degree Celsius
(b) During last 3 days of incubation: 37.2 Degree Celsius
(B) Humidity:
(a) During first 18 days of incubation: 60% R.H.
(b) During last 3 days of incubation: 70% R.H.

Note:

(1) Position of eggs in the incubator: - position of the egg with respect to setter& hatcher tray

(a) **In Setter(first 18 days of incubation): -Upright position** (**broad end upward**)

(b) In Hatcher (last 3 days of incubation): - Horizontal position

(2) Turning of eggs: - occur at 450 vice-versa

(a) Objective: -to prevent embryo sticking to the shell membrane

(b) Turning: for optimum hatchability at least 8 times in a day during first 18 days of incubation

Egg and its properties:

(1) Haugh unit:

- ✓ **Albumin quality** of egg is measured in term of haugh unit
- ✓ It measures the height of albumin.
- ✓ **Haugh unit of standard chicken egg is 72** (range=65-85)

(2) Yolk index:

- ✓ Yolk quality is measured in term of yolk index
- ✓ It reflects the spherical nature of yolk
- ✓ **Yolk index of std.egg is 0.50** (range=0.40-0.50)
- ✓ Yolk index=Yolk height/Yolk width

(3) Egg Shape Index (ESI):

- ✓ It is measured by **Vernier Caliper**
- ✓ ESI= Maximum width/Maximum length×100
- ✓ ESI of chicken egg is 74, ESI of Duck egg is 72, ESI of quail egg is 78
- ✓ Interpretation: - (1) high-value ESI: -more rounded egg (2) lower value of ESI: - more elongated egg.

(4) Shell strength of egg:

- ✓ It is directly proportional to shell thickness.
- ✓ It is measured by **Screw gauge**
- ✓ Shell thickness or Strength of std. egg = 0.30-0.50mm

(5) The specific gravity of egg:

- ✓ Determined by brine flotation technology
- ✓ Specific gravity of egg = 1.060-1.090
- ✓ Higher specific gravity indicates thick-shelled egg.

Note:

- ✓ Cholesterol content of 58 g standard egg of a chicken is 230 mg.
- ✓ Albumin of egg consists of four layers: Outer thin, outer thick, inner thin, inner thick.
- ✓ An inner thick layer of albumin: Chalaziferous layer (which holds the yolk in position)
- ✓ Physiological zero temperature: The temperature at which embryonic growth is arrested. its value is 21.1 Degree Celsius or 70 Degree F
- ✓ Best age of debeaking: day-old chick

✓ Time of Marek‟s disease vaccination (HVT-MD Vaccine): day-old chick
✓ Broilers are vaccinating against only three diseases: Marek‟s disease, Gumboro & Ranikhet
✓ Brooding: Process of providing extra heat to the chickens until they become well feathered (0-8 weeks)
✓ One Brooding unit: 250 chicks, four 60-watt bulbs hanging at 6-inch height, a brooder guard of 5 feet radius, 3 feeders (egg tray) & 4 waterers

Classification of poultry breeds:
(1) American breeds: New Hampshire, Plymouth Rock
(2) Mediterranean breeds: Leghorn
(3) English breeds:
(4) Asiatic breeds
(5) Indian breeds

Note:
✓ All poultry breeds laid brown tinted eggs while **Mediterranean breeds laid white shelled eggs**
✓ White Leghorn is best layer breed in the world while Cornish is best broiler breed in the world.
✓ English breeds are known as table bird
✓ All American breeds are dual purpose breeds

Note:
✓ **Broiler ration should contain 22-24% crude protein and layer ration should contain 16-18% crude protein.**

Modern feed additives for poultry:
(1) Additives that promote feed intake or selection:
(a) Antioxidants
(b) Flavoring agents
(c) Pellet binder

(2) Additives that enhance the colour or quality of the marketed product:
Examples: - Xanthophylls, Maize, Canthaxanthin

(3) Additives that facilitate digestion &absorption:
(a) Grit- oyster shell & lime-stone
(b) **Chelates -** EDTA
(c) Enzymes – beta-glucanase enzymes preparations
(d) Probiotics – beneficial strains of lactobacillus & streptococcus
(e) Antibiotics – penicillins, streptomycins

(4) Additives that alters metabolism

(a) Hormones –**Diethylstilbesterol(DES); Thyroxine**
(5) Additives that alters health status

Methods of sexing of chicks:
(1) Japanese method/Vent method/Cloaca method: It is the **best method of sexing** in which genital eminence of **day old chick is seen**
(2) Proctoscope method
(3) Autosexing: **By sex-linked characters** such as
(a) Colour of plumage (barred plumage in poultry)
(b) feather growth (slow feathering or rapid feathering)

The reproductive system of poultry:
✓ At birth, two ovaries & two oviducts are there but in adult poultry **only left ovary& left oviduct remains functional**
The digestive system of fowl:
✓ **Fowl has two-chambered stomach** (proventriculus & Gizzard)
✓ **Crop**: It is the enlargement of esophagus having the function of **temporary storage of food**. It is **absent in Duck.**
✓ **Proventriculus**: **Glandular stomach**
✓ **Ventriculus/Gizzard/Muscular stomach: It is the largest single organ of the body.** Its secretions contain **"koilin"** which makes it hard.
Function: Crushing & grinding of food
✓ **Cloaca**: -It connects large intestine to the Vent. It is a common opening for the digestive, reproductive, and urinary tracts**.**

Cannibalism (eating of own species) & feather picking in poultry:-
✓ Reason: It occurs in flocks fed fibre-free diets
✓ The chicken diet should contain 3-4% crude fiber.
✓ This problem is more with pelleted& crumbled diet.

Egg producing carrier of a laying hen:
✓ Egg production starts at 22 weeks of age (5.5 months) then rise sharply
✓ Peak egg production: 32-35 weeks of age
✓ Then egg production gradually declines @ 0.5 per week
✓ Replace the layer at the age of 72 weeks (18 months)

Abnormal eggs:
(1) Double-yolked eggs:
✓ **Reason**: two ova ripening at the same time or one ovum pushed back into oviduct at the same time when another ovulation takes place.
✓ **More common in Pullet than older hens**
(2) Meat spot & blood spot: - result of hemorrhage in the ovary or in the oviduct
(a) If the spot is in yolk: **hemorrhage of ovary or infundibulum**
(b) If spot is in albumin: **hemorrhage of magnum**

(3) An egg within an egg condition: due to **reverse peristalsis of the oviduct**

Special points:

(1) Moulting: Process of shedding & renewing feathers

✓ During the moult, the bird has a complete rest from laying. It normally occurs once in a year.

✓ Normal order of moulting is: **Head(1st), Neck, Body, Wing & Tail (Last)**

✓ Three types of moulters: Early, medium & late

✓ **Early moulters are poor layer**

✓ **Late moulter are good layer**

✓ Early moulter takes a longer break from laying.

(2) Broodiness: external evidence of the maternal instinct

✓ It is a dominant sex-linked character

✓ **Good layers are less broody** such as Leghorn

✓ **Prolactin hormone** is responsible for broodines

(3) Depigmentation:

✓ Order of disappearance of pigment from the body is: Vent(1st), eye ring & ear lobes, beak, shanks(last)

✓ Depigmentation firstly occurs in those body parts having good blood circulation.

(4) Temperament: -A good layer is more active, alert & easy to handle than a poor layer.

(5) Persistency: - Bird continues to lay well towards the end of her first laying year.

Methods of feeding:

(a) Whole grain feeding: Selective feeding is maximum in this method of feeding

(b) Grain& mash feeding: Slightly better than whole grain feeding

(c) All mash feeding: Best method for all types of poultry housing. This method prevents selective feeding.

(d) Pellet feeding: Minimize the wastage during feeding but pellets are 10% more expensive than other feed.

(e) Restricted or controlled feeding

Different types of housing system of poultry:

(A) Free range system/Extensive system: oldest method of poultry housing

(B) Semi-intensive system

(C) Folding unit system:

✓ Floor space requirement per bird: 1square foot in house & 3 sq.ft. in run (total: 4 sq.ft.per bird)

(D) Intensive system:

1) Deep litter system: for both broiler & layer housing

✓ Depth of the litters: **8-12 inches**

✓ Floor space: 3.5 - 4 sq.ft. per bird

✓ To maintain dryness of the litter: Superphosphate or hydrated lime is used/

✓ **Turning of litter**: once weekly (at least)

2) Cage system (Battery system): **for layer housing only**

Preservation of eggs:

(A) Methods of preserving fewer numbers of eggs:

(1) **By using earthen pot**
(2) **Immersion in liquids**
(3) **Oil coating method**
(4) **Refrigeration**

(B) Methods of preserving more numbers of eggs (Commercial preservation):

(1) **Cold storage method**
(2) **Thermo-stabilization method**
(4) Pasteurization

Some important poultry diseases:

- ✓ **Viral:** Ranikhet, Gumboro, Marek‟s disease, Fowl pox, Infectious Bronchitis (I.B.)
- ✓ **Bacterial:** Pullorum, Chronic respiratory disease, Infectious coryza
- ✓ **Fungal:** Aspergillosis(Brooder pneumonia) & Aflatoxicosis
- ✓ **Vitamins& minerals deficiency diseases**: Nutritional roup (vita. A); Polyneuritis(vita. B1); Curled toe paralysis (vita. B2); Crazy chick disease/Encephalomalacia(vita. E); Perosis/Slipped tendon(Mn, Biotin, Choline, Folic acid**)**; Swollen hock syndrome(Zn); Fatty liver kidney syndrome/FLKS (Biotin); Femoral head necrosis (Mo).

Important Facts:

- ✓ Chicken breeds are 150.
- ✓ Chickens are omnivores
- ✓ Life span is 5-7 years
- ✓ 3types of feathers. A) Primary B) Secondary C) Axial.
- ✓ Plumage is feathers of bird.
- ✓ Pigment bleaching start from vent.
- ✓ Yellow color indicates less productive.
- ✓ White color indicates more productive.
- ✓ Abdominal capacity is 3-4 fingers width.
- ✓ Fowl pox occur in feather less area.
- ✓ Emeria has 9 spp in chicken and 7 spp in turkey.
- ✓ IBD as gelatinous bursa, hemorrhage in junction between proventriculus and gizzard.
- ✓ 4% sugar solution use in birds for flushing.
- ✓ Rural Poultry (RP) accounts for about 80% of the world's poultry stocks in many
- ✓ developing countries.
- ✓ RP production is a valuable asset to the local human population despite its relatively low productive performance of 40 to 160 eggs per year and 1.5 to 1.7 kg body weight at maturity.
- ✓ The poultry units are being distributed among the rural and peri-urban population at 40-60 % subsidized rates and steps are being taken to avoid constraints such as mismanagement, malnutrition, and diseases.

- ✓ Punjab Province contributes almost 73% of poultry farming in the country.
- ✓ About 2/3rd of its population lives in rural areas, which is comparatively more deficient in the nutrition of the animal protein foods.

BASICS OF PARASITOLOGY:

- ✓ A parasite is an organism that lives on or in a host organism and gets its food from or at the expense of its host.
- ✓ **PARASITISM:** A form of symbiosis in which one organism (called parasite) benefits at the expense of another organism usually of different species(called host).
- ✓ Main types of Parasites: There are three main types of parasites.

1- Helminths
2- Protozoa
3- Ecto-parasites

HELMINTHOLOGY
Study of HELMINTHS

Helminthology; consists of Study of
(i) Platyhelminthes: **[**Flukes, Tepworms, Flateworms]
(ii) Nemathelminthes [Roundworms]

Platyhelminthes are hermaphrodite. These include:
(a) Trematodes
(b) Cestodes
Nemathelminthes includes:
(a) Nematodes

Nematode	Trematode	Cestode
Round worm	Flat worms	Tapeworms
Spindle shaped	Leaf shaped	Ribbon shaped
No segment	No segment	Segmented
Unisexual	Hermaphrodite	Hermaphrodite
Alimentary canal present	Present	Absent

Trematodes: [Flatworms]

(i) ***Fasciola hepatica***: [Liver fluke] snail is intermediate host.
Condition is known as Bottle jaw (edema in intermandibular space)
(ii) ***Fasciola gigentica***:
(iii) Schistosoma japonicum: lives in mesenteric and portal vessels of man and animals.
Causes Hematuria.

Cestodes: [Tapeworms]

(i) Raillietina tetragona
(ii) Dipylidium caninum (common tapeworm of dogs)
(iii) Hymenolepis nana (dwarf tapeworm of dogs)
(iv) Taenia solium
(v) Taenia saginata
(vi) Echinococcus granulosis; form the hydatid cyst

(vi) Echinococcus granulosis; form the hydatid cyst

Nematodes: [Roundworms]

(i) Ascarus lumbricoids
(ii) Parascaris
(iii) Oxyuris equi
(iv) Toxocara vitulorum
(v) Strongyloides : palisade worms
Haemonchus contortus; stomach worm or wireworm of ruminants
(vi) Dictylocaulus pilaria [Lungworms]: in bronchi of sheep
(vii) Habronema: cutaneous habronema is also called "summer sores" or Bursati or "Granular dermatitis"; habronema larvae
(viii) Dicrofilaria immitus [Heartworms]; occurs in dog and cat, and right ventricle of fox, and there is pulmonary stasis.

Nematodes in Blood Exam

a) Dirofilaria immitus; Heart worm of dog, cat and fox
b) Dipetalonema reconditum; S/cut worm of dog
c) Stearia cervi; Peritoneal worm of cattle.

Pallisade worm:

- ✓ A kind of strongle which infests horses, *Strongylus armatus*; also, any roundworm or nematoid of large size, as *Eustrongylus gigas*, which grows to be over three feet long.
- ✓ Horse stomach worms Habronema spp. infect the stomachs of horses.

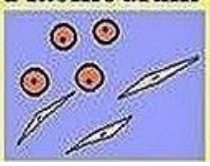
ARACHNIDS

(1) Mites
(2) Ticks
(a) Argus persicus (Fowl tick); causes Tick paralysis and transmits Borrelia ancirina (Spirochaetosis).
(b) Spinose ear tick
(c) Eyeless tampans; live in sands under trees.
(d) Hard Ticks [Ixodes]
Ixodes recinus transmits red water; caused by Babesia divergens and B.bovis or B.bigemina

Protozoology:

- ✓ Protozoology is the study of protozoa.
- ✓ The word protozoa is derived from the Greek words – "Proto" means first and "Zoa" means animals.
- ✓ The first animal life which appeared on this earth belonged to this category.
- ✓ Protozoa was discovered by Antoni Van Leeuwenhoek (Dutch scientist) and also known as Father of Protozoology.
- ✓ Protozoa are an unicellular, microscopic and eukaryotic organism having a distinct nucleus enclosed in a membrane.

Classification of Protozoa:

Phylum	Common Name	Locomotion	Examples
Sarcodina	sarcodines	pseudopodia	Amoeba 
Ciliophora	ciliates	cilia	Paramecium 
Sarco-mastigophora (Zoomastigina)	zooflagellates	flagella	Trypanosma  Giardia
Apicomplexa (Sporozoa)	sporozoans	none in adult form	Plasmodium 

INTRACELLULAR AND EXTRA CELLULAR PROTOZOA:

Extracellular	Intracellular
Trypanosoma	Leishmania
Entamoeba	Plasmodium
	Toxoplasma
	Babesia
	Theleria
	Anaplasma
	Ricketsia
	Ehrlichia

Protozoa	Bacteria
Eukaryotic organisms (nucleas)	Prokaryotic organisms(no nucleus)
Presence of distinct nucleus	Distinct nucleus is absent but DNA is present in the form of closed circle.
Capsule is absent outside the cell-wall	Capsule is present outside the cell-wall
Usually larger than bacteria	Smaller than protozoa
Examples: *Theileria annulata*	*Pasturella multocida*

Protozoa	Rickettsia
Eukaryotic (nucleas)	Prokaryotic
Intra or extra-cellular organisms	Intracellular organisms
Cytoplasm present	Cytoplasm absent
Examples: *Hepatozoon canis*	Examples: *Anaplasma* spp.

E) PROTOZOAN & RICKETTSIAL DISEASE

Name of Disease	Synonym	Causative Agent
Anaplasmosis	Gall Sickness	Anaplasma marginale
Ehrlichiosis	Canine rickettsiosis	Ehrlichia canis
Babesiosis	Tick fever; Texas fever; Splenic fever; Red fever	B. bigemina, B. bovis, B.equi, B. cabalii
Theileriosis	East coast fever; Rhodesian tick fever	Theileria parva, T. annulata, T.mutans
Trypanosomiasis	Surra	Trypanosoma evansi
Leishmaniasis	--	Leishmania donovani
Toxoplasmosis	--	Toxoplasma gondi
Cryptosporidiosis	--	Cryptosporidium parvum

ANTIPROTOZOALS

Babesiosis: Imidocarb @ 3mg / kg bwt
Diminazine @ 3-5 mg/kg bwt S/C
Quinuronium sulphate [Acaprine, Babesan] @ 0.4 mg/kg bwt
Tryan Blue [Fresh] @ 1-4 gm / kg bwt

Trypanocidals:
Suramine [Antrypol] @ 10 mg/kg bwt
Diminazine @ 8 mg / kg bwt S/C

Anti- Giardia, Amoeba, Blantidtium, Trichomonas:
Metronidazole; in canine dysentery @ 50 mg/kg wt orally for 5 days
In bovine trichomoniasis: @ 75 mg/kg bwt IV 3 times a day

Anitcoccidals:
Suplhaquinoxalone
Amprolium

IMPORTANT PROTOZOAL DISEASES:

Protozoal Diseases

(1) Babesiosis: Red water fever; caused by B.bovis, B.bigemina
(2) Anaplasmosis: rickettsial disease caused by A.marginale& A.ovis(Sheep/Goat
(3) Coccidiosis: Caused by Eimeria bovis, E. arloingi (sheep & goat)
(4) Toxoplasmosis: Caused by Toxoplasma gondii
(5) Dourine: Caused by Trypanosoma equiperdum
(6) Surra: Caused by Trypanosoma evansi

1. Babesia:

- ✓ Babesia species are haemoprotozoan parasites found inside the RBCs (intraerythrocytic) of the infected hosts.
- ✓ Babesia parasite has been observed first time by Babes in 1888 from blood of cattle.
- ✓ It is tick (hard tick Ixodida) transmitted parasites.
- ✓ In 1893, Smith and Kilborne first discovered that transmission of Babesia bigemina by the Boophilus annulatus tick.

Babesia species caused diseases are known as:

- ✓ **Babesiosis**
- ✓ **Red water disease**
- ✓ **Texas fever**
- ✓ **Biliary fever**
- ✓ **Piroplasmosis**
- ✓ **Spleenic fever**
- ✓ **Tick fever**

PIPE-STEM DIARRHOEA IN BABESIA:

- ✓ Due to constriction of the anal sphincter, the feces are evacuated with great force in a long thin stream.
- ✓ Hepatomegaly and splenomegaly
- ✓ Death in untreated case.
- ✓ **Diminazene aceturate and imidocarb dipropionate** is used in Babesia infection.

2. **Theleriosis**:

- ✓ Theileria is a genus of parasites that belongs to the phylum Apicomplexa, and is closely related to Plasmodium.
- ✓ Two Theileria species, T. annulata and T. parva, are important cattle parasites.
- ✓ T. annulata causes tropical theileriosis and T. parva causes East Coast fever.
- ✓ Theileria species are transmitted by ticks. *(Rhipicephalus appendiculatus*.)
- ✓ *Theileria* named in honor of Arnold Theiler (scientist).
- ✓ Theiler found that East coast fever was not the same as redwater but caused by a different protozoan.
- ✓ It is tick transmitted haemoprotozoan parasite of ruminants and captive ungulates. (Ungulates: Any of a number of mammals with hooves that are superficially similar but not necessarily closely related taxonomically)
- ✓ Its piroplasm stage found inside RBCs and schizonts stage inside lymphocytes.
- ✓ Theileria species are transmitted by Ixodid ticks.

PROTOZOAL DISEASES IN CATTLE

DISEASE	BEBIOSIS	ANAPLASMOSIS	TRYPNASOMIASIS	THELERIOSIS
Synonyms	Red water	Yellow Fever	Mental Fever	Lymph node Fever, Gadod ka bukhar
Causetive Agent	Babesia bigemina	Anaplasma marginale, Anaplasma bovis,	Trypnasoma congolense, trypnasoma vivax, T. brucei.	Theileria annulate, Theileria parva,
Symptoms	Anemia, Recurrent Fever 106F Heamoglobinuria, Increased respiration rate.	Jaundice, Yellow mucus membrane, No heamoglobinuria,	Head shaking , Moving body in circle, sudden falling,	Enlargement in lymph node, High Fever.
Treatment	Diamenazine	Oxytetracycline, Imidocarb	Isometamedium (Trypamedium), Diamenazine	Buparvaqone

Differential diagnosis of Haemoprotozoan diseases

Symptoms	Babesiosis	Theileriosis	Trypnosomiosis	Anaplasmosis
Fever	Present	Present	Present	Present
Lymph node	Normal	Swollen	Normal	Normal
Nervous sign	Absent	Absent	Present	Absent
Jaundice	Occasionally present	Present	Absent	Occasionally present
Abortion	Occasionally occurs	Absent	Absent	Occasionally occurs
Nasal discharge	Absent	Present	Absent	Absent
Lacrimation	Absent	Present	Absent	Absent
Pin point hemorrhages in conjunctiva	Absent	Present	Absent	Absent
Hemoglobinuria	Present	Absent	Absent	Absent

Differential Symptoms of Common Haemoprotozoan Diseases in Dairy Cattle

Mode of Transmission



Ticks Infestation



Biting Fly

Babesiosis



Deep Red to Coffee Color Urine



Pale conjunctiva

Anaplasmosis



Pale to yellow Coloration of Sclera

Theileriosis



Swelling of Lymph node



Lacrimation & Corneal Opacity



Sub-mandibular Edema

Trypanosomiasis

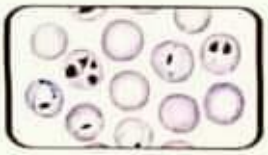


Circling Movement of Animal

&

Hitting head on hard object

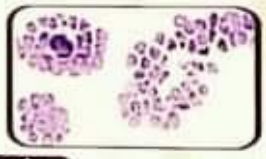
Microscopic view of Causative Micro-organisms



Babesia bovis
Babesia bigemina



Anaplasma marginale
Anaplasma centrale



Theileria annulata
Theileria parva



Trypanosoma brucei
Trypanosoma evansi

Common Species

Common Symptoms



Difficulty in Breathing (Dyspnea)



High Temp. (104°-107°F)

Major Outcome



Abortion



Low Milk Yield



Death of Animal

Recommended Treatment

Cattle/Dogs/Horses/Camel/Sheep/Goat:
Imidocarb dipropionate
(with Oxytetracycline/Doxycycline)

Cattle/Dogs/Horses/Camel/Sheep/Goat:
Imidocarb dipropionate

Cattle/Horses/Camel/Sheep/Goat:
Buparvaquone

Cattle: Isometamedium
Horse/Camel/Pig/Dogs:
Quinapyramine Sulphate

HELMINTHOLOGY

Study of HELMINTHS

Helminthology:

Consists of Study of

(i) Platyhelminthes: **[**Flukes, Tepworms, Flateworms]

(ii) Nemathelminthes [Roundworms]

Platyhelminthes are hermaphrodite. These include:

(a) Trematodes

(b) Cestodes

Nemathelminthes includes:

(a) Nematodes

DIFFERENCE AMONG NEMATODE, TREMATODE AND CESTODE:

Nematode	Trematode	Cestode
Round worm	Flat worms	Tapeworms
Spindle shaped	Leaf shaped	Ribbon shaped
No segment	No segment	Segmented
Unisexual	Hermaphrodite	Hermaphrodite
Alimentary canal present	Present	Absent

Trematodes: (Flatworms)

(i) ***Fasciola hepatica***: (Liver fluke) Snail is intermediate host.

Condition is known as Bottle jaw (edema in intermandibular space)

(ii) ***Fasciola gigentica***:

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Causes Hematuria.

Cestodes: (Tapeworms)

(i) Raillietina tetragona

(ii) Dipylidium caninum (common tapeworm of dogs)

(iii) Hymenolepis nana (dwarf tapeworm of dogs)

(iv) Taenia solium

(v) Taenia saginata

(vi) Echinococcus granulosis; form the hydatid cyst

Nematodes: (Roundworms)

(i) Ascarus lumbricoids

(ii) Parascaris

(iii) Oxyuris equi

(iv) Toxocara vitulorum

(v) Strongyloides : palisade worms

Haemonchus contortus; stomach worm or wireworm of ruminants

(vi) Dictylocaulus pilaria (Lungworms): in bronchi of sheep

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Nematodes in Blood Exam

a) Dirofilaria immitus; Heart worm of dog, cat and fox
b) Dipetalonema reconditum; S/cut worm of dog
c) Stearia cervi; Peritoneal worm of cattle.

CESTODES

Sexually mature form found in intestine of vertebrate
Segmented body, segment called proglottid
Larvae; 6 hooked larvae called hexacanth
Intermediate host; vertebrate and intervertebrate
Head is termed scolex

Larval Forms

Also called metacestodes
Non-budding forms

a) Procercoid (1st larval stage in arthropod)
b) Plerocercoid
c) Cysticercoid
d) Cysticercus
e) Strobilocercus
f) Coenurus
g) Hydatid

HAIR AND TRACHEAL WORM:

Hair worms in small intestine	1-Capilaria annulata (Chicken & turkey) 2-Capilaria columbae (Pigeon & turkey)
Tracheal worm	Syngamus trachea (Chicken & turkey)

- ✓ Red mite of poultry is dermanyssus gallinae.
- ✓ Tongue worm is linguatula serrate.

TAPEWORM OF DOG:

Diphylidium caninum	Taenia	Mesocestodes lineatum	Diphylobothridium latum

INTERMEDIATE HOST OF DISEASES IN ANIMALS:

Parasite	Intermediate Host
Plasmodium	Humans
Dirofilaria immitis	Mosquito *(Aedes* spp., *Culex* spp., *Anopheles* spp., *Mansonia* spp.)
Wuchereria bancrofti, Brugia timori & Brugia malayi	Mosquito *(Culex* spp., *Aedes* spp.& *Anopheles* spp.)
Liver Flukes (Clonorchis, Opisthorchis)	Fish of Cyprinidae family (carps and minnows)
Tapeworm (Taenia solium)	Humans, Cow, Pigs
Schistosoma mansoni	Freshwater snails *(Biomphalaria* spp.)
Toxoplasma gondii	Many mammals and birds
Trypanosoma cruzi, Trypanosoma gambiense	2 intermediate hosts: a) Invertebrate host (Triatomine insects like tsetse fly- Glossina palpalis) b) Vertebrate host (Humans)
SARS-CoV Virus	Masked palm civet
MERS-CoV Virus	Dromedary camel
Fasciolosis	Fresh water snail

ENTOMOLOGY:

i. Insects; Lices etc
ii. Arachnids: Ticks & Mites

INSECTS

(1) Lices:

(a) Hematopinus suis; Pig sucking lice
(b) Haematopinus eurysternus: Cattle lice
(c) Lignocanthus vituli: Long nose cattle louse

(2) Bugs:

(a) Cimex lecturalis; Bed bug

(3) Beetles

(4) Fleas

(a) Pulex irritans

(5) Biting midges or Punkies and Flies

(a) Mosquitoes
(b) Black flies or buffalo gnats
(c) Sand flies
(d) Horse flies; involve in the transmission of causative agents of anthrax, equine infectious anemia, Surra, Nagana.
(e) Common house flies: involve in the transmission of causative agents of typhoid fever, cholera, tuberculosis, dysentery, anthrax.
(f) Stable flies; Blood suckers e.g. Stomoxys; involve in transmission of habronema, equine infectious anemia, anthrax.
(g) Buffalo flies
(h) Horn flies
(i) Tsetse flies; causes Sleeping sickness of man and Nagana
(j) Blow flies
(k) Blue bottle flies
(l) "Strike" by Lucilia, Calliphora, Phormia.
- Callitroga causes myasis of man and is called screwworms
- Tumbu flies
- Sarcophaginae: Flesh flies
- Gastrophilus: Horse bots
- Oestrus ovis: Sheep nasal fly; Called as "False Gid".
- Hypoderma; Ox warbles

ARACHNIDS

(1) Mites
(2) Ticks
(a) Argus persicus (Fowl tick); causes Tick paralysis and transmits Borrelia ancirina (Spirochaetosis).
(b) Spinose ear tick
(c) Eyeless tampans; live in sands under trees.

(d) Hard Ticks [Ixodes]
Ixodes recinus transmits red water; caused by Babesia divergens and B.bovis or B.bigemina
Hard ticks are involved in:
- Tick borne fever of sheep
- Viruses of loping ill
- Coxiella burnetii
- Tick paralysis
- Staphylococcus aureus
- Anaplasma marginale
- Spirochaetosis
- Canine pirpoplasmosis [Rephicephalus sanguinus]
- Redwater billiary fever [Babesia equi]

Dermacenter reticularis (in Asia)
Transmits Babesia spp.

External Parasites:

1. Permanent	Depluming(body mange)	Scaly leg (Leg mange)
2. Intermittent		

Ticks:

- ✓ Cattle Fever Ticks, known scientifically as Rhipicephalus (formerly Boophilus) annulatus and R. microplus, are a significant threat to cattle industry.
- ✓ These ticks are capable of carrying the protozoa, or microscopic parasites, Babesia bovis or B. bigemina, commonly known as cattle fever.

DIFFERENCE B/W HARD TICK AND SOFT TICK:

Hard ticks	Soft Tick
Presence of rigid chitin scutum.	Scutum absent
Exp: Ixodidae family	Exp: Argasidae family (Bird ticks, tampans)

BURROWING AND NON BURROWING MITES:

Burrowing	Non burrowing
Sarcoptes	Psoroptes
Notoedres	Chorioptes
Knemidocotes	Otodectes
Demodex	Dermanyssus

ARTHROPODES:

Insecta Diptera (Flies and Mosquito) Phthiraptera (Lice) Siphonaptera(Fleas) Hemiptera (Red bugs)	Arachnids (Ticks + Mites), Acarina	Crustaceca + Pentastomida

MYIASIS:

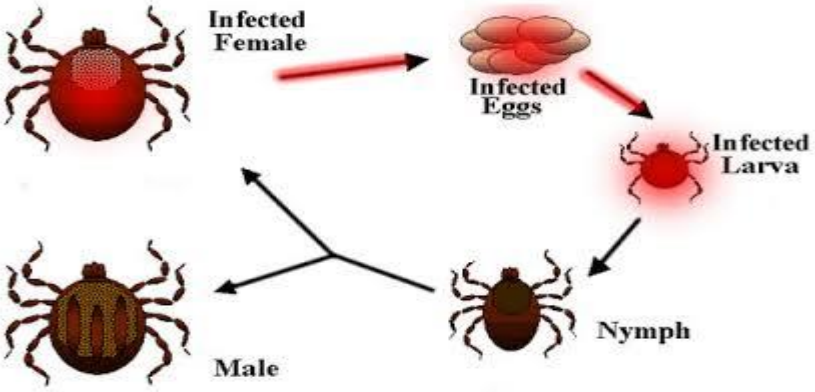
✓ Infestation of living animals with the larvae of diptera flies.

Facultative	Obligatory

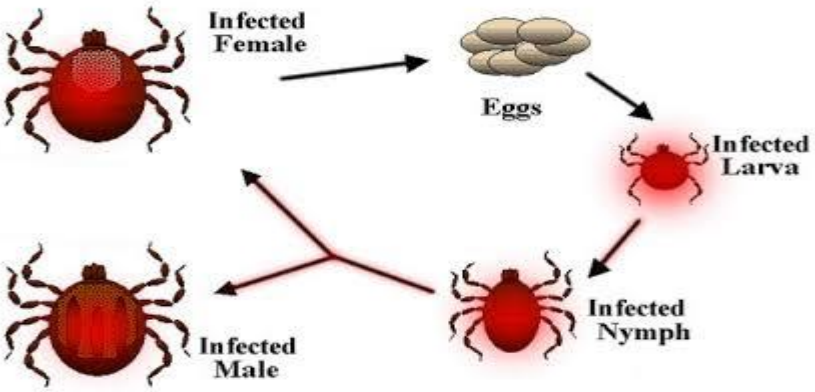
Cutaneous (Lucilia)	Nasal (Nasal bots)	Somatic (Hypoderma/ Warble fly)

PHTHIRAPTERA:

Anopleura (Sucking lice)	Mallophaga (Biting lice)



Transovarial transmission



Transstadial transmission

PROTOZOAL DISEASES:

(1) Babesiosis: Red water fever; caused by B.bovis, B.bigemina
(2) Anaplasmosis: rickettsial disease caused by A.marginale& A.ovis(Sheep/Goat
(3) Coccidiosis: Caused by Eimeria bovis, E. arloingi (sheep & goat)
(4) Toxoplasmosis: Caused by Toxoplasma gondii
(5) Dourine: Caused by Trypanosoma equiperdum
(6) Surra: Caused by Trypanosoma evansi

PROTOZOA

Entamoeba histolytica; causes amoebic dysentery in man, dog, cat etc.
Entamoeba coli:
Affects man, dog, pig and other primates.

Sarcocysts spp:

Found in striated and heart muscles

Cooccidiosis:

There are 3 pathogenic genera:
i) Isospora (2 sporocyst and each have 4 sporozoites)
ii) Eimeria (4 sporocyst and each have 2 sporozoites)
iii) Tuzzeria (8 free sporozoites)

Eimeria:

Affect sheep, goat, cattle, buffalo, cat, horse and poultry.
Chicken: E. tenella (Cecal cocci), Intestinal cocci (E.brunetti)
Cattle: E. bovis, E. zuerni
Dog: Isospora

Toxoplasma gondii

Affects rodents, cat and dog. Affects brain, heart, liver, lung and spleen.

Plasmodium

Mammals, reptiles, and birds
In erythrocytes and endothelial cells of inner organs
Vector:
Culex (birds), Anopheles (Mammals)

Leishminia

Man, dog, rodents, guinea pig
Found in macrophages, reticuloendothelial cells.
Vector:
Sand fly, phelbotomus and lutzomyia
Species: L.braziliansis, L.chagasi, L.maxicana, L.donovani

Trypanosomiasis

Parasites of circulatory system and tissue fluid

Vector:

Tsetse fly of Glossina species

Stercorarian:

Spread via faeces

Salivarian:

Spread via saliva

Species:

T. evansi, T. gambience, T. bruci, T. cruzi, T. equiperdum

Diseases:

i) Surra
ii) Chagas disease
iii) Sleeping sickness
iv) African trypanosomiasis
v) Dourine

(c) Parasitic Diseases

Parasite	Specie	Adult site	Larva site
Long strangyles	Strongylus vulgaris	Large intestine	Arteries/ liver
	Triodontophorus spp.	Large intestine	Intestinal wall
Small strangyles	Trichonma spp.	LI	Intestinal wall
Round worms	Parascarus equinum	SI	Liver, lung
Thread worms	Strongyloides westerni	SI	Lung
Lung worms	Dictylocaulus spp.	Respiratory passage	Lymphatics
Pinworms	Oxyuris equi	LI	Intestinal wall
Tapeworms	Anoplocephala spp	SI / LI	---
Bots	Gastrophilus spp.	Flies in environment	Stomach

<u>Important Points:</u>

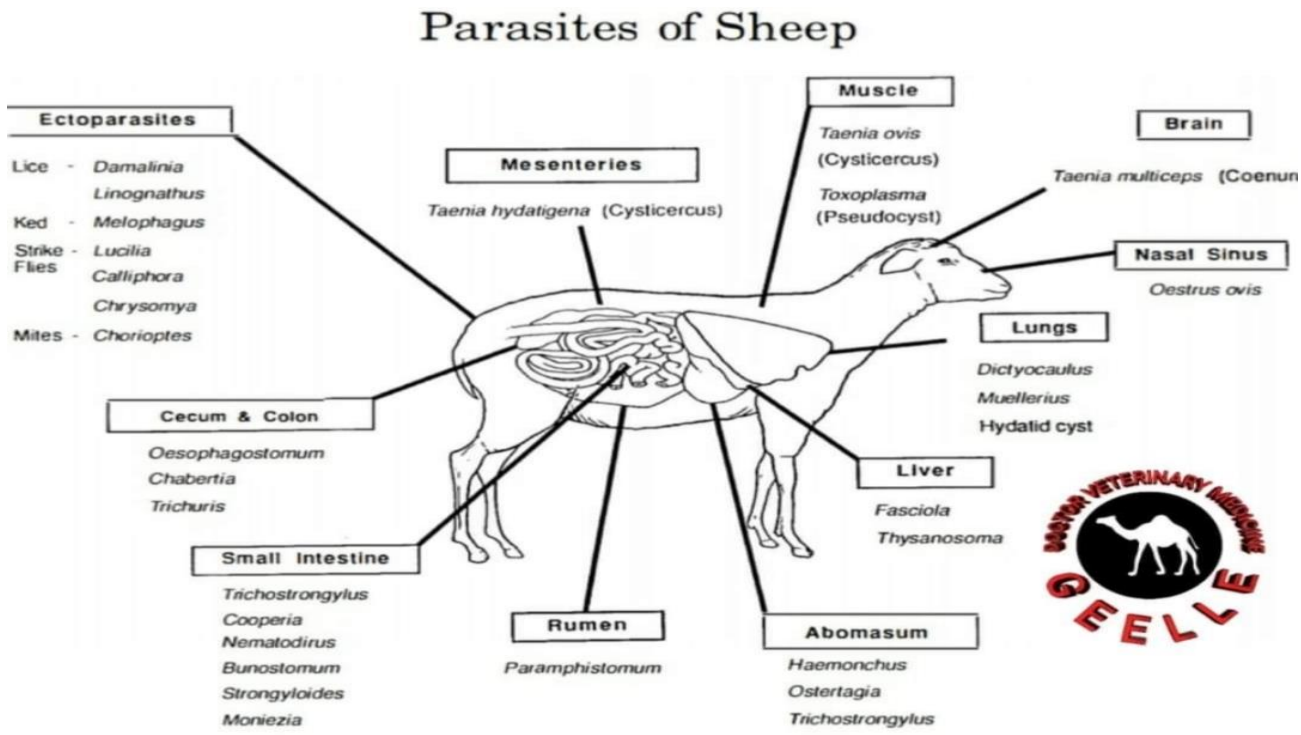
- ✓ Ticks & mites are larvae-nymph-adult, active in winter.
- ✓ Fleas are larvae-maggot-pupa-adult, active in warm.
- ✓ Blow is laying egg.
- ✓ Strike is development of larvae & damage.
- ✓ Calliphora is blue bottles.
- ✓ Lucila is green bottle.
- ✓ Hyoderma is sometic myiasis.
- ✓ Oestrus is nasal myiasis.
- ✓ Cochliomyia macellaris is screw worm myiasis.
- ✓ Bovis lineatum is warble fly hyoderma.
- ✓ Gastrophilus nasalis is nose bot fly of equine.
- ✓ Cephalopnia is nasal bot fly of camel.

- ✓ Red water increase in sulphur.
- ✓ Hippobosaca is forest flies.
- ✓ Anoplura is piercing type mouth parts, it cause anemia.
- ✓ Haematopinus is largest louse of domestic mammals.
- ✓ Lingonathus is longed nose louse of mammals.
- ✓ Haemotopinus is short nose louse of mammals.
- ✓ Felicola is lice of cat.
- ✓ Wing louse of domestic fowl is lipeurus.
- ✓ Head louse of domestic fowl is cuclologaster.
- ✓ Body of louse of domestic fowl is menacanthus.
- ✓ Fluff of louse of domestic fowl is goniocotes.
- ✓ Shape of louse of domestic fowl is menopones.

EIMERIA SPECIES:

There are 12 species in Cattle, 11 in Sheep, 9 in Goats and Chicken and 7 species in Turkey:

Sr. No.	Species	Location/ Organ
01.	**E.Tenella**	Caecum
02	**E.Necatrix**	Medium portion of SI
03	**E.Acervulina**	Duodenum of Small Intestine
04	**E.Brunetti**	Ileum of SI
05	**E.Maxima**	Medium portion of SI
06	**E.Mitis**	Lower SI
07	**E.Praecox**	Upper SI
08	**E.Adenoeides**	Caecum
09	**E.Meleagrimitis**	Upper and Lower SI



BASICS OF VETERINARY SURGERY AND RADIOLOGY:

- ✓ Branch of medicine which uses manipulation as well as other methods in the treatment of injury & removal of harmful & useless part.
- ✓ Death of tissue in absence of infection is known as Necrobiosis (Aseptic Infarct)
- ✓ X-rays were discovered by Roentgen.
- ✓ Repeated Exposure to X-rays delays healing process.
- ✓ Minimal Invasive surgery aims to minimal loss of blood during operation.
- ✓ COLIC means abdominal pain.
- ✓ The distended metatarsal vein in the horse is known as "Blood Spavin".
- ✓ Aseptic surgery was introduced by J. Lister
- ✓ Thick yellowish pus indicates involvement of staphylococci
- ✓ Blind fistula has only one opening
- ✓ Blunt bodies causes contused wound

Important Terminologies:

1. **Sit fast**: Dry gangrene due to arrests fof blood supply caused by pressure of collar on neck region
2. **Mud fever**: Papular form of dermatitis without pus
3. **Acne**: Inflammation of sebaceous gland and hair follicles affecting dog and horse
4. **Boil (Eczema)**: Skin appears like vesicle
5. **Collositis or horny induration**: Due to repeated injury on the hard ground
6. **Greese**: Chronic exudative hypertrophic dermatitis of lower part of limb of horses
7. **Elephantiasis**: Common in hind limb below the hock

Removal of Ribs:

- ✓ **4th Rib**: Apical lobectomy
- ✓ **5th Rib**: Left and right lobectomy
- ✓ **5th Rib**: Pericardial drainage
- ✓ **5th Rib**: Thoracic drainage
- ✓ **6th and 7th Rib**: Diaphragmatic hernia
- ✓ **7th Rib**: Diaphragmatic Abscess
- ✓ **8th Rib**: Oesophagotomy

Important Note:

- ✓ Ribs **9th, 10th. 11th and 12th** are not removed which make part of abdomen.

Important points:

- ✓ **Gall** is painful inflammatory swelling of skin & sub cutaneous tissues due to infiltration with serum and exudate caused by constant pressure movement & irritation of yolk & saddle.
- ✓ Contusion wound is 1st, second (hematoma) & third degree.
- ✓ Cancellous bone collapse and compress upon itself in comparative fracture.
- ✓ Anopthames is absence of eye ball.

- ✓ Micropthames is small size globe.
- ✓ Buphthalmes is bulging of eye ball due to excessive fluid in interior chamber.
- ✓ Exophthalmos is abnormal protrusion of eye ball.
- ✓ Endopthalmose is sinking of eye ball.
- ✓ Megalopthalmose is large size of eye without fluid.
- ✓ Nanopthalmia is restricted development of eye.
- ✓ Nystagmus is involuntary movement of eye ball.
- ✓ Strabismus is deviation of globe from normal axis.
- ✓ Chalazion is enlargement of tarsal gland.
- ✓ Chemosis is extensive oedematous condition of conjunctiva.
- ✓ Hyperemia is collection of blood is anterior chamber of eye.
- ✓ Hypopion is collection of pus in anterior eye chamber.
- ✓ Aniridia is absence if iris.
- ✓ Iridocyclitis is inflammation of iris and ciliary bodies.
- ✓ Papppilloma is benign tumor of surface epithelium.
- ✓ Melanoma is benign tumor of melanoblast.
- ✓ Lipoma is benign tumor of fatty tissue.
- ✓ Squamous cell carcinoma is horn cancer in bullock.
- ✓ Paraphimoss is failure of penis retraction in prepuce.
- ✓ Phimosis is failure of penis to come out from prepuce.
- ✓ Tendon healing theory is one wound one scar.
- ✓ Upward fixation of patella in bullock.
- ✓ Upward fixation of patella is known as string halt.
- ✓ Medial patella ligament is absent in camel.
- ✓ Mobilization of calcium from bone to serum is endo steal.
- ✓ Displacement of articular surface is luxation.
- ✓ Synovial hernia is out pouching of synovial membrane beneath the skin.
- ✓ Keratitis is inflammation of cornea.
- ✓ Simple fracture is when fracture site does not communicate with environment.
- ✓ Compound fracture is when skin broken with bone fracture.
- ✓ Incomplete fracture is when bone is not listing its continuity.
- ✓ Complete fracture is complete loss of bone continuity.
- ✓ Simple crack on bone is fissure fraction.
- ✓ Complete fracture is avulsion fracture.
- ✓ Impacted fracture is cortical end of fracture is forced into cancellous bone.
- ✓ Compound fracture is two or more fracture line interconnects each other at one point.
- ✓ Multiple fracture is two or more fracture line do not inter connect.

Paravertebral anesthesia indication:

- ✓ Laparotomy & ruminal fistula.
- ✓ Caesarian section.
- ✓ Operation for foreign body syndrome.
- ✓ Hernia.

Pudendal Nerve block:

- ✓ To examine or treatment & surgery of penis.

- ✓ Exsanguination:
  Complete loss of Blood.

Important Points:

- ✓ **Melena:** Presence of Blood in feces.
- ✓ Haematocele: Blood in Tunica Vaginalis
- ✓ Haematoma: Blood collected in newly formed cavity
- ✓ Aneurysm: Weakness in Blood Vessel
- ✓ Diathermy: Cauterization of vessels
- ✓ Vitamin C is required for collagen synthesis
- ✓ Sulphur Containing Amino Acids are Methionine and Cystine
- ✓ Zinc is essential for remodeling of extra cellular matrix.
- ✓ The middle layer of skin, the dermis, contains blood vessels, nerves, glands (Sweat and sebaceous) and hair follicles
- ✓ Astringent: Arrests haemorrhage by Styptic (External Blood) and Haemostatic (Within Vessel)
- ✓ Hemostasis, platelets aggregate and form a clot with the help of fibroblasts (fibrin).
- ✓ Proud flesh/excessive granulation tissue. Granulation continues largely healing.
- ✓ Connective tissue continues to proliferation (in horses genetic).
- ✓ Median suspensory ligament is elastic layers separate right & left halves.
- ✓ Lateral suspensory ligament is fibrous (non elastic).
- ✓ Furs ten bergs rosette is valve between teat cistern & teat canal.
- ✓ Treatment of fracture of scapula in small animal is valpeau bandage.
- ✓ P.H of tissue fluid in fraction is acidic.
- ✓ Corticosteroids delay healing.
- ✓ Inflammation of planter ligament over fibular tarsal bone is curbing.
- ✓ Involvement of sensitive & insensitive tissue at angle of wall & bar due to presence of shoe is corn.
- ✓ Ossification of collateral cartilage of digital joint is side bone.
- ✓ New bone growth of 1, 23 phalanx is ring bone.
- ✓ Appearance of three characteristics swelling is bog spavin.
- ✓ Contrast radiography of bronchus bronchial tissue is bronchography.
- ✓ Contrast radiography of renal architecture is intravenous pyelography.
- ✓ Contrast radiography of spinal cord is myelography.
- ✓ Contrast agent skoidan& pent opaque in myelography.
- ✓ Contrast agent canary 420 in intravenous pyelography.
- ✓ Contrast agent diagnosis is bronchograhy.
- ✓ Contrast agent Buso4 is oesophaography (immediate x-ray).
- ✓ Improper mineralization of osteoid tissue is osteomalacia.
- ✓ Uniform porocity of bone due to low Ca is osteoporosis.
- ✓ Increase osteoclast activity is osteolysis.

- ✓ Inadequate osteoid synthesis is osteopenia.
- ✓ Swelling of tissue of seasmoid bone is sesamoditis.
- ✓ Periostitis of 3rd metatarsal bone is bone spavin.
- ✓ Tenosynovitis of tarsal sheath via thorough pin.
- ✓ Acquired bursitis of point of elbow is hygroma of elbow.
- ✓ Hard & concrete floor is hygroma of carpal.
- ✓ Outward bearing of limb & inward rotation of digit is paralysis of pectoral nerve.
- ✓ Knuckling at carpus & fetlock is contraction of flexor tendon.
- ✓ Splint is low in vit A, D, Ca, P.
- ✓ Firing is treatment of splint.

HERNIA AND ITS TYPES:

A hernia happens when an internal organ pushes through a weak spot in your muscle or tissue

Types of hernias include:

- ✓ Inguinal hernia.
- ✓ Femoral hernia.
- ✓ Umbilical hernia.
- ✓ Incisional hernia.
- ✓ Epigastric hernia.
- ✓ Hiatal hernia.

1. Peritoneo Pericardial Diaphragmatic Hernias (PPDHs) occur in cats and dogs
2. Diaphragmatic hernia has been considered as a serious digestive disorder in buffaloes.
3. Higher prevalence of diaphragmatic hernia in buffaloes.
4. Inguinal or scrotal hernias are common in pigs, horses (particularly draft breeds and warm bloods), and many breeds of dogs and are suspected to be hereditary.
5. Femoral hernias in the groin commonly occur in female dogs that are pregnant with large litters
6. Umbilical hernias are a common finding in many puppies.
7. Most incisional hernias in dogs and cats
8. Congenital pleuroperitoneal hernias have been described in small animals, horses, and calves
9. Hiatal hernias occur through the esophageal hiatus

Important points:

- ✓ Polyglycolic acid is badly damaged by auto claving.
- ✓ Halothan vapours are used as 2-4%.
- ✓ Ranula is retention cyst.
- ✓ Dropped elbow flexion by radial nerve.
- ✓ Dentigerous cyst is in horse.

- ✓ Protrusion of inus is staplyons.
- ✓ Thick & inspissated pus granule in guttural pouch is chondroids.
- ✓ Schlotzs tonometer records intra ocular pressure.
- ✓ Dental fistula in dogs in 4th upper molar.
- ✓ Upward fixation of patella in bovine in ovaries & testes.
- ✓ Unit of radiation dose in rem is rads.
- ✓ Common mode of scan is B mode.
- ✓ Computerized topographic value of H2o is o.
- ✓ Electrocardiographic monitoring is electrical activity.
- ✓ After cardiac impulse leaves the SA node and deopolarizesthe aorta it goes to AV nodes.
- ✓ Lead 2nd measures left rear left & right foreleg.
- ✓ QRS complex is depolarization of ventricles.
- ✓ Negative QRS complex is pulmonic stenosis.
- ✓ SA node is normal physiological pacemaker.
- ✓ Most versatile crystalloid solution in hypovolemic in ringer.
- ✓ Minimum PCV is 20%.
- ✓ Dry heat sterilization is for fats.
- ✓ Organism collected from cat abscess is pasturella spp.
- ✓ Clean contaminated operation is for enterotomy.
- ✓ Gram –ve bacterial infection is in gastro intestinal tract.
- ✓ Heat destroys microorganism from protein.
- ✓ Autoclave use gravity displacement.
- ✓ Gamma radiation use for sterilization of surgical material.
- ✓ Epinephrine release by any injury.
- ✓ Mitosis is regeneration and hyper plasia.
- ✓ Roentgen is air unit.
- ✓ Radiolucent foreign body lodged in esophagus is barium swallow.
- ✓ Film store in dry & cool location.
- ✓ Temp of filament is for quantity of x-rays produced.
- ✓ Limiting factor in x-ray tube is anode.
- ✓ Increase atomic no. is probability of electronic interaction.
- ✓ Cystography is structural abnormalities of bladder.
- ✓ Grid is absorbing secondary radiation.
- ✓ Lead apron use outside near the neck.
- ✓ Fluid deficient body is 5 %.
- ✓ Mild clinical sign is 6%.
- ✓ Extra cellular fluid is higher sodium.
- ✓ Osmotic pressure of blood is albumin.
- ✓ Bound heal better temp is 30 degree c.
- ✓ Zeps operation is in ear (otitis media).
- ✓ Fetid odor due to spherohorous necrophrus +black material in lateral solei of foot is thrush.
- ✓ Crico thyroid ligament is laryngeal hemiplegic.

- ✓ One jugular furrow to another jugular or furrow is wired sucking.
- ✓ Carbonic anhydrase in erythrocyte speed up the break down of carbonic acid to carbon dioxide and water.
- ✓ Function of sodium in concentration of cardiac muscle to ensure rhythmicity.
- ✓ Long administration of ACTH or cortisone cause metabolic alkalosis.
- ✓ Blood store at 4 degree c.
- ✓ Horse safe collection of blood limit is 4 liters.
- ✓ Maximum safe period to store blood is 20 days.
- ✓ Dog safe collection of blood is 40ml/ kg b.wt.
- ✓ 15% body weight reduction dehydration is hyper osmolarity & circulating collapse.
- ✓ Main cause of over hydration is renal shut down.
- ✓ Augmented anti diuretic hormone come in severe pain & injury.
- ✓ Body sodium & chloride ratio is 3:2.
- ✓ Intra cellular magnesium concentration is ten times greater.
- ✓ Milk magnesium concentration is five times greater.
- ✓ Total body water %age is 55-65%.
- ✓ Extracellular body water is 15-20%.
- ✓ Sweat contains Na+K+Cl+Water loss.
- ✓ AV conduction time is PPR interval.
- ✓ Lymphoid cells are most sensitive for radiation.
- ✓ Inflammation of ligament is desmitis.
- ✓ Cold application is used in acute inflammation.
- ✓ Angioma is tumor of lymphatic /blood vessels.
- ✓ Teat surgery local anesthesia 2% lignocaine & General sedation with xylazine 4 mg /500kg b.wt (contraindicated in advance pregnancy)
- ✓ Degeneration of erythrocytes during graft material is pronounced in horse.
- ✓ Personal monitoring of radiation is pocket dosimeter.
- ✓ Strain is drainage to tendon.
- ✓ Simple fracture is always sterile.
- ✓ Hematoma should be opened after a period of 10-15 days.
- ✓ Fluctuating non inflammatory swelling with the wall of content which is slow in development & painless is cyst.
- ✓ Local anesthesia is xylocaine.
- ✓ Fracture of olecranon process & dislocation of elbow joint is called monteggia fracture.
- ✓ Bone has fractured but fractured fragment do not separate until or unless some extra force is applied is deferred fracture.
- ✓ Diagnostic sign of complete fracture is crepitus sound.
- ✓ Sequestrum is a dead bone piece.
- ✓ Pan arthritis is inflammation of complete wall of artery.
- ✓ Retrobulbular nerve block is for eye ball.
- ✓ Nitrous oxide concentration to produce anesthesia is 30%.
- ✓ Nerve to desensitize the left flank in cow is paravertebral.
- ✓ Myxoma is tumor of mucus tissue.

- ✓ Father of veterinarian radiology is R.EBERLEIN.
- ✓ High frequency wave are used is ultrasonic surgery.
- ✓ Cat gut is sub mucosa of S.I of sheep.
- ✓ Kangaroo tendon is tendon of tail.
- ✓ Maximum suture holding power of tissue is skin.
- ✓ Infection associated with over use of antimicrobial drug, faulty diagnostic procedure, advances of chronic infection is nosocomial infection.
- ✓ Rupture of larger blood vessels of lymphatic of s/c tissue is 2nd degree contusion.
- ✓ In gunshot wound exit opening is wider.
- ✓ Open circumscribed lesion located on free surface & showing no tendency to heel is ulcer.
- ✓ Tract leading from suppurative cavity to surface discharge chronic pus in sinus.
- ✓ Abnormal passage leading from internal organ to exterior.
- ✓ Amputation of penis to remove diseased part is Vinsots operation.
- ✓ Bread and butter appearance of heart in traumatic pericarditis.
- ✓ For repair rectovaginal fistula in mare is 6 bite patterns.
- ✓ Pyocele is pus in scrotum.
- ✓ Dry used to potentiate the effect of local anesthesia is adrenaline.
- ✓ Bicipital bursitis is shoulder.
- ✓ Involuntary flexion of hock is string halt.
- ✓ Arthritis of stifle joint is gonitis.
- ✓ Foul smell discharge ventral to the anus is anal gland abscess.
- ✓ Utrecht method is left flank omentopexy.
- ✓ Sebaceous cyst in false nostril of equine atheroma.
- ✓ Injury of tissue due to low temp (3-7 degree centigrade.) is frost bite.
- ✓ Chill burn is prolonged action of temperature near 0 degree c.
- ✓ Two fragments of fracture do not unite by body cells is non-worm pseudo arthritis.
- ✓ PH of hematoma is acidic 6.p in first 24 hrs.).
- ✓ Woven bone forms the primary callus.
- ✓ Dog flat bone neoplastic condition is chondrosarcoma.
- ✓ Spiral fracture is of humerus.
- ✓ Periosteum is absent in surface of patella.
- ✓ Fibroblast is pleuripotential cell.
- ✓ Best suture material for suturing tendon is stainless steel wire.
- ✓ Osteoblasts originate from intracanalicular level.
- ✓ Bone staple is used for epiphyseal fracture.
- ✓ Clinical union of fracture is the stage when fixation device can be removed.
- ✓ Blood vessels & nerve fiber enter the tendon via mesotenon.
- ✓ Repair of cranial crucial ligament of dog is carbon fiber.
- ✓ Shape of bone is controlled by Vit A.
- ✓ Caudal epidural anesthesia is needle downward & forward at the angle of 15 degree with vertical.
- ✓ Feline ovariohysterectomy approach is horizontal flanks.
- ✓ Protractors (perpetual muscles) arise from xiphoid region.

- ✓ Scrotal layers are 1-skin 2- dartos 3-secrotal fascia.
- ✓ Femoral head & neck excision orthoplasty in calve leggy perthes disease.
- ✓ Change in femoral neck is coxa valga.
- ✓ Sialography is salivary gland.
- ✓ Dacrocystorhinography is lacrimal apparatus without lacrimal.
- ✓ Contrast agent for cholangiogram is cholografin.
- ✓ Contrast material for cerebral angiography is Conroy – 280.
- ✓ Columns of spongy bone uniting the epiphyseal plate and diaphysis is known as Meta physics.
- ✓ The joining pressing a joint cavity with synovial membrane is called diaesthrosis.
- ✓ Epiphyseal closure time for proximal end of humerus in dog is 16-18 months.
- ✓ Radiopaque liver is abdominal side of diaphragm.
- ✓ Implant oxidizes in the body is stainless steel.
- ✓ Best material for arterial graft is Teflon.
- ✓ Best material for suturing vascular graft is Dacron.
- ✓ Ideal bone implantation material is porous & flexible.
- ✓ Silicon rubber implants can be sterilizes by boiling n distill water, it is ideal implant for joint space in synovial cavity.
- ✓ Calcium phosphate is absorbed from small intestine.
- ✓ Standard radiation is gamma rays.
- ✓ Embryo is radiation sensitive during organogenesis period.
- ✓ Cell sensivity to radiation increase during oxygenated condition is lymphoid cells.
- ✓ Mechanism of ultrasound is thermal, mechanical & gravitational.
- ✓ Temp during developing x-ray film in 68 degree f.
- ✓ Injury to suspensory ligament is desmitis.
- ✓ Chronic purulent inflammation of collateral cartilage of coffin joint is quitter.
- ✓ 1st & 2nd phalanx arthritis is high ring bone.
- ✓ 2nd & 3rd phalanx arthritis is low ring bone.
- ✓ Group of drugs which paralyze the voluntary muscle tone is muscle relaxant.
- ✓ Spray anesthesia (ethyl chloride).
- ✓ Barbiturates are derivative of barbituric acid.
- ✓ Luminal is phenobarbital.
- ✓ Nembutal is phenobarbital sodium.
- ✓ Ketamine use in cat.
- ✓ Ether is volatile liquid.
- ✓ Antagonist of xylazine is yohimbine.
- ✓ Atropine sulphate is 1-anticholinergic drug.2-obtained from atropa belladona3-dilation of pupil.
- ✓ Chlorpromazine is contraindicated in ruminants.
- ✓ Depolarizing agent is decamethonium.
- ✓ Lidocaine is amide & synthetic.
- ✓ Nitrous oxide is colorless & non-irritating gas.
- ✓ Positive juglar pulse is pericarditis, CHFstenosis of tricuspid valve.
- ✓ Negative juglar pulse is stenosis of tricuspid valve.

- ✓ Cardiac tapenade is sudden filling of pericardial sac.
- ✓ Cardiac syncope is transient attach of acute heart failure + psychic disease in men.
- ✓ Luccilia of Pigeon is E.columbae.
- ✓ Epluis is fibrous beningn tumor of gingiva.
- ✓ Range paralysis is Mareks disease.
- ✓ Ephemeral fever is transient fever.
- ✓ Roaring is laryngeal stenosis.
- ✓ Hofluned syndrome is vagus indigestion,bradychardia.
- ✓ Cor-pulmonal is colic and tachycardia.
- ✓ Kyphosis is dorsal bending of spinal column.
- ✓ Goose steeping gait in pig is low in pentothanic acid.
- ✓ Foot rot is ulcerative pododermatitis.
- ✓ Esterified form of vit a present in fish oil.
- ✓ Acceto Acetic acid ketone body is more toxic for ovine.
- ✓ Blue tongue in sheep has seasonal appearance.
- ✓ Phonocardiography is used for measurement of cardiac sounds.
- ✓ Common cause of venous thrombosis in calves is naval ill.
- ✓ Mediastinum is mesothelioma.
- ✓ Pericarditis is frictional sounds.
- ✓ Endocarditis is cardiac murmurs.
- ✓ Acute heart failure is SA node shifting.
- ✓ Arterial thrombosis in horse is iliac.
- ✓ Acute endocarditis is leukocytosis with shift to left.
- ✓ Congestive heart failure due to left side is lung.
- ✓ Congestive heart failure due to right side is liver.

DIFFERENT SUTURES IN DIFFERENT SURGERIES:

1. Bunnel sutures are used in tendon surgeries
2. Uterecht sutures are used in Left Flank Omentopexy
3. Overlapping sutures are used in Hernia
4. Quill sutures are used in Vaginal and uterine Prolapse
5. Lock stitch sutures are used in abomasopexy
6. Lambert and Cushing sutures are used un Rumenotomy
7. Connel sutures are used in esophageal obstruction
8. Schamidean suture scheme is used in Intestinal Obstruction
9. Lock stitch sutures are used in Horn surgery, Cystorapphy, and diaphragmatic hernia
10. In tail gangrene Horizontal Mattress sutures are used
11. In teat fistula vertical mattress sutures are applied.
12. Purse string sutures are used in Rectal Prolapse
13. Simple continuous sutures are used on simple skin lacerations
14. Lambert sutures are used on the uterus

INCISION AND BODY PART:

Incision	Body part
Criss cross	Atresia ani (Genetically related.)
Semilunar	Diaphragmatic hernia
Triangular	Phimosis
Circumscribe	Perpetual prolapse
Prepubic Para median approach	Cytorrhaphy
Ischial approach+ epidural anesthesia	Urethrotomy
Incision in front of occipital	Poll evil/fistulas withers

SURGICAL CONDITIONS:

Butter foot	1-Pyramidal disease 2-Low ring bone 3-Concussion 4-Swelling of coronary band with raised hair.
Canker	1-Chronic hypertrophy of frog of 3rd phalanx. 2-Lack of proper frog pressure. 3-$CuSO_4$ + ZnO cauterizing agent 4-Chronic hypertrophy of horn producing tissue of sole.
Thrush	1-Unhygenic building or muddy ground 2-Black necrotic fetid odor material in lateral solei of foot. 3-10% phenol + tincture iodine + 10%formaline
Wind puff	1-Synovial swelling of joint capsule and tendon sheath 2-No lameness 3-Poor nutritional status
Founder	1-Laminitis 2-Asetic spontaneous inflammation of sensitive lameness of foot. 3-Seedy toe condition 4-High amount of histamine secretion 5-Obesity & ingestion of grass 6-Hind limb udder body & forelimb forward

TYPES OF HAEMORRHAGES:

External	Internal	Time of Occurrence	Source

HAEMORRHAGE ON THE BASIS OF TIME:

Primary	Secondary	Reactionary
Immediate	After Weeks	After 24 hours

X-RAY:

Electron	High energy particle.
Collimator	Limit primary x-ray beam to size of film.
Grid	Absorbed scattered radiation
Filter	Absorb harmful & non penetrating x-rays.
Cassette film	Used to keep x-ray
Over exposure	Dark film
Under exposure	Light x-ray detail lost
Secondary radiation	Grey non contrast radiography
Fog film	Weak & old development
Dark spot	Spilling drop of developer
White sot	Metallic base medicine

TIMING FOR X-RAY FILM:

Developing time	**5 minutes**
Washing time	**30 sec**
Fixing time	**10 min**
Final washing time	**20 min**
X-ray time for colon	**3hrs**
X-ray time for rectum	**5hrs**
X-ray time for Nephrogram	**7-10 sec**
X-ray time for reticulum	**30 min**

TUMORS:

Leiomyoma	Benign neoplasm of smooth muscle.
Leiomyosarcoma	Malignant neoplasm of smooth muscle.
Rhabdomyoma	Benign tumor of striated muscle.
Chondroma	Benign tumor of cartilage.
Synovial sarcoma	Malignant tumor of synovial sheath & tendon.

TERMS RELATED TO TEATS:

Teat stenosis	Chronic inflammation treated with teat dilator(hug knife)introduced in to streak canal at 30 degree angle.
Teat leaker	Treat with phosphorous supplement & irritant,lugols iodine with tuberculin syringe to stimulate fibrosis.
Blind teat	Treat with teat dilator/introduce 18 guage needle into teat canal
Teat abscess	Treat with teat amputation in case of purulent masttis,putting udder permanently dry with 5%Cuso4 solution with two infusion at 3 days

	interval.
Teat spider	Membranous obstruction of teat at base of teat closed to udder castor, treat with teat bistoury.
Lactoliths/milk stone	Calculi of calcium salt in teat canal ,treat with foreces.

SURGICAL CONDITIONS OF TEAT SPHINCTER:

Teat stenosis/ hard milker	Teat leaker/ free milk	Blind teat/ congenital/ acquired

CONTENT OF TEAT TERMS:

Affections on epithelial surface	1-teat fissure (winter with tincture iodine & glycerin.) 2-accessory teat(super numerous teats joined with webbed or Siamese teats)
Teat papillomas/ warts	Benign viral growth like finger projection treats with cauterization excision & ligature.
Obstructive thelitis	In buffalo,frequent in ist order of lactation with in one month of calving,treated with anti histaminescorticosteroid&full hand milking.
Teat fistula	More in goat due to pedulus udder.Abnormal opening /passage treated with simple intrutted suture pattern in sub mucosa & mucosa ,simple ontrupted suture pattern skin (purstring suture n prolapse.)
Deep laceration of teat	Involve skin & muscle .simple continuous pattern in muscle.simple interrupted in skin.
Pre-pubic tendon rupture	Increase weight of gravid uterus +severe udder oedema +twinning. Poor prognosis
Teat sighon	To drain milk
Hug lancette	To dilate teat canal
Teat alligator forcep	To remove milk stone

TERMS RELATED TO LAMENESS:

Sweeny	Paralysis of pre-scapular nerve (shoulder slip.)
Bicipita bursitis	Inflammation of bicipital burse.
Omarthritis	Arthritis of shoulder joints
Carpitis/popped knee	Inflammation of carpal joints
Sore shins	Periostitis on anterior aspect of metacarpus or metatarsus of horses.
Splints	Swelling below carpus large swelling on medial aspect & few small swelling on lateral aspect.
Wind puff/wind galls	Distention of joint capsule tarsal sheath/bursa.
Bow legs	Lateral deviation of carpal joint.
Knocked knee	Media deviation of carpal joint.

Calf knee/sheep knee	Backward deviation of carpal joint.
Bucked knee/ Goat knee	Forward deviation of carpal joint.
String halt	Involuntary flexion of hock mostly nervous in origin.
Thrush	Chronic degeneration condition of frog.
Spavin	Arthritis of hock joint
Canker	Chronic hypertrophy of horn producing tissue of sole.
Coral paralysis	Femoral nerve paralysis.
Osselets/mouse joint	Arthritis of fetlock joint
Pedal ostitis	Inflammation of coffin joint.
Bowed tendon	Low tendo vaginitis Addison between superficial & deep digital flexor tendon. Asheim operation Cobalt 60 therapy

TERMS RELATED TO ANESTHESIA:

Analgesia	Relive from pain.
Sedation	Mild degree depression in which patient is awake and calm.
Narcosis	Moderate degree depression in which deep sleep & analgesia.
Hypnosis	Artificially induce sleep
Tranquilizer	Patient is relaxed & unconcerned with surrounding.

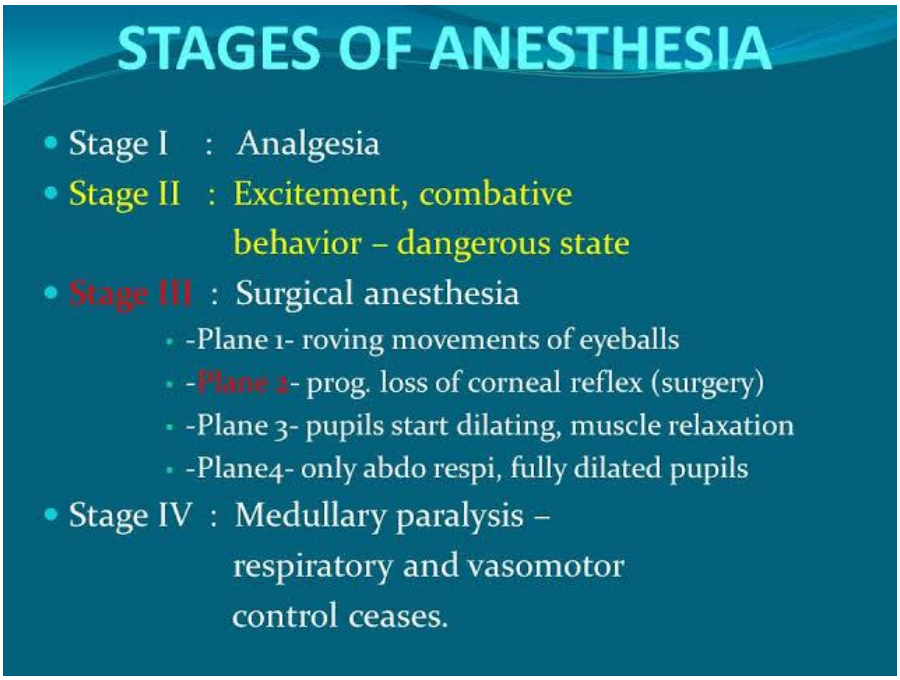
ANESTHESIA AND ITS TYPES:

Basal anesthesia	Light level of anesthesia produced by pre-anesthetic.
Balanced anesthesia	Two or more anesthetic are combined well.
Dissociative anesthesia	Loss of sensation due to functional disruption by stimulation.
Spinal anesthesia	Drug is injected in to subarachnoid space.

STAGES OF ANESTHESIA:

1st stage	1-Eye ball position is normal 2-stage of analgesia
2nd stage	1-Stage of delirium 2-Respiration is thoraco-abnormal
3rd stage	1-Swelling of reflex is lost 2-Stage of surgical anesthesia 3-Hypnotism
4th stage	1-Cough reflex is totally absent

	2-Shock 3-Mucous membrane pale/white 4-Gasping type respiration



Procaine	1-Ester of Para amine benzoic acid 2-Not used as surface anesthesia 3-Doping in horse 4-Synthetic

ANESTHETICS DOSE:

Small animal dose	High
Fatty animal dose	Less
Activated animal dose	More
Male animal dose	More, because increase BMR (70%)
Pregnant animal dose	More

TERMS FOR HERNIA:

Types 1-Reversible Type 2-Irreversible

Enterocoele	Intestine
Epiplocoele	Mesentery/omentum
Oscheocoele	Scrotum
Bubonocele	Inguinal ring
Hysterocele	Uterus
Vesiculocele	Bladder
Omphalocele	Umbilical
Enteroppiploceole	Intestine+omentum

ABNORMAL BODY DEVELOPMENTS:

Feature	Abscess	Hematoma	Cyst	Hernia	Tumor	Arthritis/ tendinitis
Development	Acute, often 2-5 days	Immediate after injury	Very slow development	Immediate after injury	Slow	Take 5-10 days
Nature	Painful & form at periphery and soft in nature	Painful doughy on palpation	Painless	Painless(compound,sac,ring,content)	Benign painless, malignant painful	Painful
Exploratory procedure	Pus	Blood/serum fluid	Straw colored	Never punctured	Nothing or blood	Synovial fluid

GANGRENE:

Dry gangrene	Moist gangrene	Gas gangrene
Dark, Black, Dry and Hair low	No color change	Port in hot &painful
Temp normal	Sub normal temp	High temp
No injury &swelling	Infection swelling present	Edematous swelling
Total gangrene in large animals+ERGOT poisoning	Moist gangrene of udder	Clostridia infection

SURGERY TYPES:

Palliative Surgery	Prophylactic Surgery	Extirpative Surgery	Physiological Surgery
Reducing/ Less severe clinical signs	Prevent Ontoward effect	Altering a disease process	
Arthrodesis	Dehorning	Amputation	Proto systemic Shunt

DIFFERENCE AMONG X-RAY, CT, MRI, MRA AND PET SCAN:

X-Ray	CT	MRI	MRA	Pet SCAN
It shows bone/skull only. Does not show the brain. Best	It is a quick test. Shows brain without great detail. Shows	Long test. Shows brain with great detail. Shows	It shows the blood flow in the vascular system of the	Pet SCAN shows how active different parts of the

used to detect if there are bone fractures.	bleeding, ischemia or tumors.	smaller bleeding, ischemia, or tumors.	brain. If there is vessel narrowing or blockage this test would show it.	brain are. Very sensitive to detect cancer cells.

CLINICAL CLASSIFICATION OF WOUND:

1- Clean wound/ 0 hour wound/ surgical wound/ incised wound	2- Contaminated wound (less than 6-8 hrs.)	3-Infected wound/ open wound (more then 6-8 hrs)

MICROBIOLOGY RELATED TERMS:

Asepsis	Disinfection	Sterilization	Antisepsis
Free of Microbes	Kill most microbes on non living things	Kill all microbes including spore	Kill microbes but not spores on living body

DIFFERENCE B/W ARTERY AND VEIN:

Artery	Vein
Bright Red	Dark Red
Pulsative	Continuous
Cardiac Side	Peripheral Side

TYPES OF SOLUTION:

Crystallized	Colloid
NaCl + Ringer Lactate	Albumin + Plasma Protein + Dextran

TYPES OF DEHYDRATION:

Type I	Type II	Type III
No Sign	Moderate	Severe
4-6 %	6-8 %	8-12 %
1-2 Sec Skin Fold Test	2-4 Sec Skin Fold Test	20-45 Sec Skin Fold Test

TYPES OF SHOCK:

Hypovolumic	Cardiogenic	Septic/ Endotoxic	Vasogenic
		Endotoxin due to Gram -ve Microbes (Heat Labile)	Pain (Vasoconstriction) and Deep Anaesthesia (Vasodilation)

DIFFERENCE B/W CASTING AND RESTRAINING:

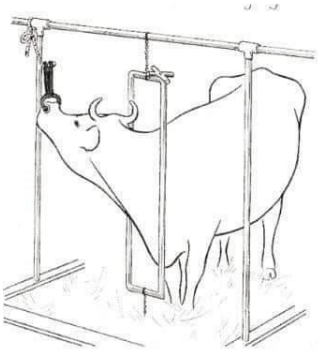
Casting	Restraining
Controlling Animal in Recumbent position	Controlling animal in standing position
1. Reuf/ Roop Squeeze Method (Cattle/ Buffalo) 2. Criss Cross/ Burlay Method (Cattle/ Buffalo) 3. Double Side Line (Equine)	1. Front Leg Hopple (Cattle/ Buffalo) 2. Nose Ring (Cattle/ Buffalo) 3. Single Side Line (Equine) 4. Ear Twitch and Lip Twitch (Equine)

§ **Nose lead:**

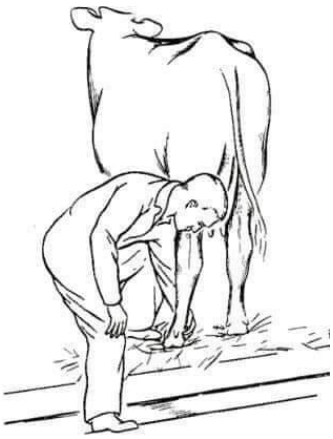


Using a nose lead it is possible to:

- give intravenous injection
- examine the hoof
- do udder surgery

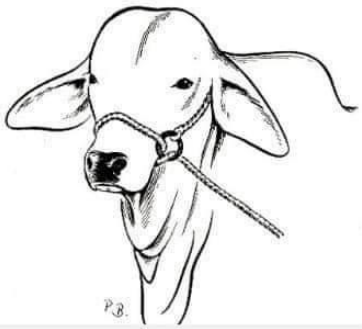


§ **Raising the Rear Leg Manually:** A nose lead is used and the cow's head pulled to the side opposite that of the foot to be lifted. The operator grasps the leg at the pastern with his left hand.



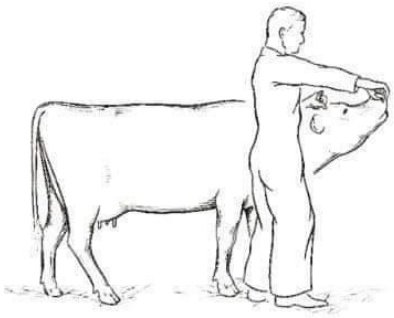
1. Halter

- The primary method of restraint used in cattle is the halter.
- It is mainly used to control the head and once the head is controlled, the animal can be handled with relative ease.



§ **Restraint of the Head:**

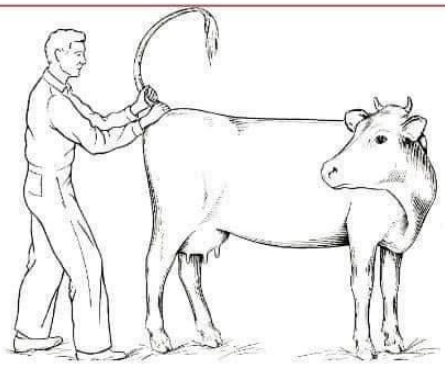
- This is done by grasping the nasal septum with thumb and forefinger of one hand and holding the horn or ear.



§ **Rope Squeeze:** This is a standard method of casting a cow. The rope for this restraint may be arranged on a cow while she is in the stanchion. She may then be led to the place where it is desired that she lie down and tension applied to the end of the rope.

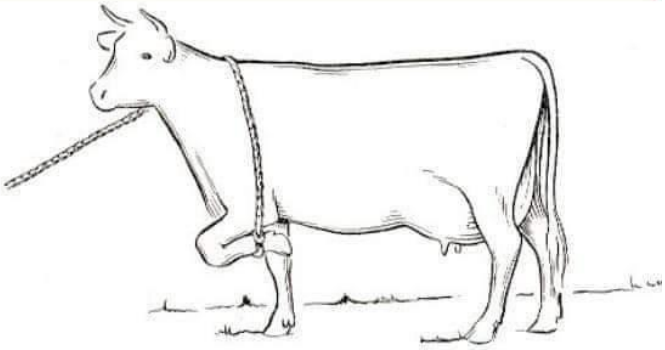
2. **Restraints that Divert Attention**

§ **Tail Restraint:** Standing on the side of the animal, the tail should be held close to the base of the tail.



3. Methods of Raising a Leg

§ **Front Leg Hopple:** In this way a front leg may be raised and held off the ground for examination or treatment. This restraint can also be used to make a cow stand still and to keep her from kicking with a hind leg.

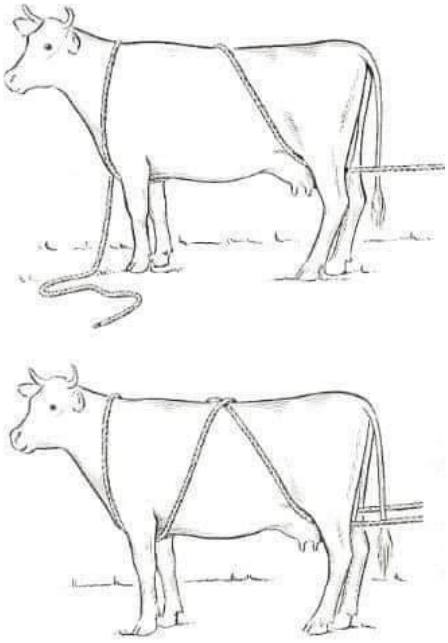


5. Casting Restraints

§ **Burley Method of Casting:**

While the cow is being held by a strong halter or by a nose lead a forty foot piece of rope is placed over her back with its center at the withers. The ends are carried between the forelegs and crossed at the sternum. One end is carried up each side of the animal's body and the two are crossed again over the back.

Each end passes downward between the rear legs going between the inner surface of the legs and under the udder or scrotum, as the case may be.

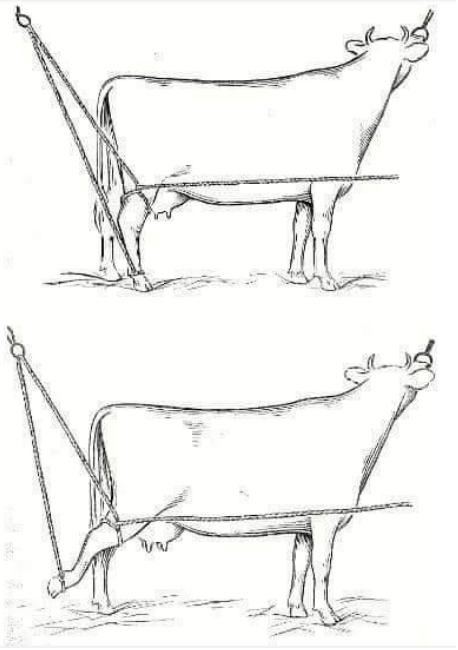


6. Cattle Chute: One of the most common methods of restraint for cattle is the squeeze chute.





§ **Raising Rear Leg Beam Hook Method**: By means of a nose lead the cow's head is pulled to the side opposite the leg which is to be lifted and made fast to a stanchion. A set of beam hooks is fastened to a beam above and somewhat behind the cow. A 30 foot piece of rope with an eye is used to make a loop around the



EPIDURAL ANAESTHESIA LOCATION IN RUMINANT AND EQUINE:

Ruminant	Equine
C1-C2	S5-C1 and C1-C2

SURGICAL INSTRUMENTS AND THEIR FUNCTIONS:

Instruments	Functions
Alligator forceps	Remove foreign body
Allis tissue forceps	To grab organ/slippery tissue
Babcock tissue forceps	Hold delicate tissue
Crile forceps	Stop bleeding before ligation
Kelly forceps	Stop bleeding before ligation
Halstead mosquito hemostatic forceps	Stop bleeding before smaller ligation
Brown adson thumb	Grasp & hold tissue. Muscle& skin.
Rat teeth thumb	Grasp & hold tissue. Muscle& skin.(teeth)
Suture wire cutting	Cut wire sutures
Bandage	Remove bandage
Littauer suture removal	Remove all kind of suture
Mayo dissecting	Used to cut body tissues & sutures
Met zenbaum dissecting	Cut delicate tissue
Scalpel blade	18-36 in number.
Needle point	It is triangular with cutting edges(close skin & sub cut tissue)
Taper needle	Close soft tissue/no cutting edge

Needle holder	1-myo Hagar(no cutting blade) 2-olsen-hegar(cutting blade)double function
Endotracheal tube	General anesthesia
Endoscope	Camera in front of endoscope
laryngoscope	View vocal cord intubation
Tonometer	Eye pressure
Balling gun	Administer capsule of bolus
Dental floats/ Rasper teeth	
Dental scalar	Remove plaque from teeth

PROCESS OF HEALING:

1. Regenerative (Complete retardation of original tissue)	2. Repair/ Substitution (Formation of scar)

TYPES OF CELLS:

Labile	Continuous dividing cell(epidermis epithelium)
Stable/ Quiescent cells	Under go division occasionally (liver+kidney+pancreas+fibroblast+endothelial)
Permanent	Non dividing cell,(neurons,muscle cell+cardial+skeletal cells)

TYPES OF WOUND HEALING:

1. 1st intention/ opposed edges (Surgery)	2. 2nd intention/ separated edges) (Burn/ Contamination)

STAGES OF WOUND HEALING:

Inflammatory phase	Reparative/ Proliferative phase	Maturation and remodeling (5-7 days)

IMPORTANT NERVE BLOCK AND THEIR INDICATION:

Nerve block	Indication
Lingual nerve block in cattle & buffalo	1-Removal of foreign body. 2-management of laceration,cut,renula etc.
Mandible alveolar N.B	1-management of molar incisor teeth &body of mandible in lower jaw. 2-suturing of wound of lower teeth& lips.
Mental N.B	1-suturing of wound of lower lip. 2-wiring around lower teeth & body of mandible.

Infra orbital N.B	1-removal of polyps. 2-extraction of teeth. 3-trephining of facial sinuses.
Supra orbital N.B	1-operation above upper eyelid. 2-forhead suturing of trephing of frontal sinuses.
Maxillary N.B	Management of upper lips, upper jaw & nose.
Corneal N.B	To desensitize the horn.
Auriculopalpabrel N.B	1-to relive spasm of eyelids. 2-for examination of eye.
Retro bulbar N.B	Management of eye ball & nictitating.
Petersons orbital N.B	Management of eye ball, eye lid & horn.

SURGICAL INSTRUMENTS:

Instruments are classified by their function:

- Cutting and Dissecting
- Grasping and holding
- Clamping and occulting
- Exposing and Retracting
- Suturing and stapling
- Viewing
- Sectioning
- Dilating and probing
- Measuring
- Microinstruments

1. CUTTING AND DISSECTING:

Cutting instruments have sharp edges .They are used to dissect, incise, seprate and or excise tissue.

- **Knife Handles:**
  It comes in various width and lengths, blades are attached with the slipping the slit in the blade into the groove on the handle.
- **Knife Blades:**
  Blades with numeric prefix of "1"(e.g. ,10, 11 ,12 ,15) fit #3 or #7 handles
  Blades with the numeric prefixs of "2"(eg 20, 21, 22, 23, 24) fit #4 handles.
  Blades wiith #10 used for large skin incision # 11 used for initial skin puncture of tiny deep incision #15 used for short shallow incision.
- **Other knife handles:**
  Long handles are used inside deep incision(e.g., open abdominal cases)
  Bevare knifes are used for small delicate cases.

➢ **SCISSORS:**

Blades of scissors may be straight, angle, or curved. Tips may be pointed or blunt, handles may be long or short , should be only used for their intended purpose.

- **Straight mayo(suture) scissors:**
  Used to cut tissue or for blunt dissection.
- **Bandage /Dressing scissors:**
  Cut bandage material.

➢ **Tissue/Dissecting Scissors**

- **Iris Scissors:**
  Used for delicate dissection, very small and sharp.
- **Metzenbaum Scissors:**
  Used to delicate tissue and also used for blunt dissection.
- **Tenotomy scissors:**
  Used for delicate dissection and cutting, commonly in ophthalmologic, neurological and plastic surgery procedures.

Other cutting instruments:

- **Ronguer:**
  Used as an essential surgical instrument while examination of challenging bone related problems and performing surgical procedures.
- **Osteotome:**
  An osteotome is an instrument used for cutting or preparing bone. Osteotomes are similar to a chisel but bevelled on both sides. They are used today in plastic surgery, orthopedic surgery and dental implantation.
- **Curette:**
  Used to clean out sockets after removing infected roots with its spoon shaped ends.
- **Freer Elevator:**
  Freer Elevator is designed for **surgical procedures** that need elevation and separation of the soft tissues, especially the periosteum. The Elevator is considered a multipurpose instrument used for height and used to scrape the bones and dissections.
- **Rasp:**
  Rasping nail clinches and hoof wall

2. Grasping and holding:

- These tissue are used to grasp tissues and hold it in place without injuring sorrounding tissues.Forcep can be ringed or the thumb variety.Most instruments set will include Kocher, allis, babcock, adson, debakey forceps, sponge sticksand towel clips.

➢ **Forceps:**

- **Kocher:**
  This instrument is used for holding fallopian tube in hysterectomy. The tips of the blades have teeth so that the tissue does not slip. The blades can either be straight or curved. This instrument is used in hysterectomy to clamp fallopian tube which are then transfixed. It is also used for salpingectomy in ectopic or oophorectomy in ovarian mass. This can also be used for clamping umbilical cord of new born at the time of delivery or for artificial low rupture of membranes (ARM).

Used to grasp heavy tissue or clamp large blood vessels to control bleeding.

- **Allis:**
  Finger ring, ratcheted forceps are used for firmly grasping organs and slippery or dense tissue during surgery.or grasp tissue that will be removed.
  Teeth curve to the inside and are designed to help decrease general pressure applied to the area.
- **Babcock:**
  Babcock Forceps are finger ring, ratcheted, non-perforating forceps used **to grasp delicate tissue in laser procedures**. They are frequently used with intestinal and laparotomy procedures.
- **Sponge stick:**
  Sponge holding forceps, available at Surgical Holdings, are designed **to grasp and hold sponges and/or swabs while conducting a medical procedure or surgery**. Sponge holding forceps are an essential tool in many different types of medical procedure and surgery.
- **Towel clip:**
  It is used in draping the operative area abdominal or vagina . The towels or sheets are fixed to the skin and each other with these clips.
- **Adsons:**
  Used to handle tissues and dressings during surgical procedures. This instrument comes with a toothed tip and serrated surface.
- **Tooth tissue :**
  Fine toothed forceps are usually used for **picking up skin edges during skin closure or manipulating the needle**. Grasped between thumb and forefinger in a pincer grip usually with the left hand, you will often use these to manipulate the needle and grasp tissue edges when suturing.
- **Ferris smith:**
  Ferris Smith Tissue Forceps are heavyweight forceps used with **fibrous tissue in colorectal procedures**. These thumb forceps feature a wide section of the handle for greater control handling firm tissue.
- **Russian forcep:**
  Used for **atraumatic tissue grasping during dissection**. Tissue Forceps:
  Non-toothed forceps used for fine handling of tissue and traction during dissection.
- **DeBakey Forcep:**
  Debakey forceps are very **useful to grasp soft tissues and organs during the surgical process**. The teeth on the jaws of the forceps allow for a stable grip on tissues. ... It allows large blood vessels to be separated from the underlying fascia and be better manipulated, according to the needs of the surgical procedure.

3. Clamping and occluding:

These instruments are used to apply pressure, some clamps are designed to crash the structure when apllied,others are noncrushing and are used to occlude or secure tissue.Most sets include mosquito, crile Kelly,tonsil, peon and right angle clamps.

- **Mosquito:**

A small, straight or curved hemostatic forceps used **to hold delicate tissue or compress a bleeding vessel**

- **Crile:**
  Crile Forceps are a surgery essential, commonly used to **clamp blood vessels, hold tissue for dissection, and clamp tissue prior to cauterization or ligation**.
  Surgeons rely upon this stainless-steel instrument to allow them to control blood loss and manipulate tissue during a variety of procedures.
- **Kelly:**
  Kelly hemostats can be **used to clamp larger vessels or grasp tissue**.
- **Tonsil:**
  Tonsil foceps were used **to grasp tonsils**, prior to their removal with a companion instrument known as a 'tonsillotome'.
- **Pean:**
  a hemostatic surgical forcep made of German stainless steel with fully serrated jaws. The peans are commonly used to occlude /clamp vessels to control bleeding. The serrated jaws prevent vessels from slipping through the hemostat.
- **Right angle forcep:**
  Right angle forceps are ratcheted, finger ring forceps commonly used for working in obscured surgical sites. They are most frequently used for **clamping, dissection, or grasping tissue**. Mixter right angle forceps are available in multiple lengths and have serrations the entire length of the jaw.

➢ **Non - Crushing clamps**

- **Bowel clamp:**
  Used for **atraumatic clamping of the bowels in abdominal surgery**
- **Vascular Clamp:**
  Used to clamp the blood vessels to block the flow of blood. This saves the patient from losing an unnecessary amount of blood during surgery.
- **Bulldogs:**
  A bulldog forceps, clamp or serrefine is a type of forceps which is used **in surgery**. It has serrated jaws and a spring action so that it will grip and hold sutures, tissues or vessels. The spring may be weak or the jaws sheathed in a soft material so that the item being gripped is not crushed too severely

4. Exposing and Retracting:

- Used to pull soft tissues and muscles aside to expose surgical site. They are of two types
- **Hand held**
- **Self retaining**.
  Most instrument set will include small hand held , army-navy, malleable, weitlaner, and rehardson retractors.

➢ **Hand held retractor:**

- **Skin hooks:**
  The skin hook is a simple and practical instrument for **use in dermatologic surgery**. This instrument has been utilized by plastic surgeons for a long time. By means of

the skin hook, undesirable trauma to the skin during surgical procedures may be reduced to a minimum.

- **Senn:**
  Senn retractors are double ended retractors used **in surgical procedures**. One end is typically L shaped, and the other has three bent prongs. The pronged ends come in sharp and dull tips."Often used in small bone and joint procedures, or throidectomy and dissection of neck tissue."
- **Cushing vein retractor:**
  A retractor is a surgical instrument used **to separate the edges of a surgical incision or wound**, or to hold back underlying organs and tissues so that body parts under the incision may be accessed.
- **Volkman:**
  These surgical instruments are **used to hold back, or retract organs and tissue** so the surgeon has access to the operative area. They spread open the skin, ribs and other tissue; and are also used separate the edges of a surgical incision.
- **Army-Navy:**
  Used to **retract shallow or superficial incisions**. From small wounds to abdominal operations.
- **Malleable (Ribbob):**
  Malleable Ribbon Retractors are used **to hold back tissues and organs during surgery**. They are suitable for abdominal, cardiovascular, ophthalmic, and neurosurgical procedures. Their malleable structure allows them to fit the form of the area. They can also be used to retract intestines.
- **Richardson:**
  Used for retraction of wound edges, and also used to retract deep abdominal or chest incision.
- **Kelly Retractor:**
  Used for exposing wound
- **Harrington:**
  To retract soft intra-abdominal visceral organs .The retractor has a heart-shaped blade, the tip of the blade is thicker to reduce trauma to organs.
- **Deaver:**
  The Deaver Retractor is a large, handheld retractor commonly used **to hold back the abdominal wall during abdominal or thoracic procedures**. It may also be used to move or hold organs away from the surgical site.

➢ **Self Retaining Retractors:**

- **Weitlaner:**
  It is used **to hold the incisions in the case of orthopaedic surgeries** as most of the incisions are shallow. Moreover, it is used in neurosurgical procedures to hold the skin of the scalp in procedures like craniotomy
- **Cerebellar:**
  Cerebellar Retractor is a specialized instrument that is used in neurosurgical procedures. This tool can be used for retraction of the cerebellum, it is protected

from damage by placement of cotton strips. There is a ratcheting mechanism that allows the ring handles to be locked at a certain width and held there.

- **Gelpi:**
  Gelpi retractors are used **in surgery** because they can reduce paravertebral muscle damage during retraction. No pleural injuries associated with their use in posterior spine surgery have been reported.
- **Balfour:**
  A Balfour retractor is a surgical instrument which is used by a surgeon to hold open abdominal incisions during a laparotomy. Balfour retractor, the device holds itself in place, leaving the hands of surgeon and assistants free to focus on surgical tasks.
- **Bookwalter:**
  The Bookwalter retractor is a high-quality table fixed retractor that allows for improved access to the operating field and stable retraction throughout long procedures. The Bookwalter retractor is a device used by surgeons to separate or hold back the ends of a surgical incision, mainly in the chest or the abdomen.

5. Suturing and stapling:

Needle holder size varies according to the type of needle used, most instruments sets will have Webster, crilewood and mayao hegar type needle holders.
Clip applier placed individual staples, available in reusable and disposable.

➢ **Needle Holders:**

- **Webster:**
  Webster needle holder is used used when significant force is needed to pass a needle or when long/deep passes are made through sub-cutaneous tissues.
- **Crilewood:**
  Crile-Wood Needle Holder is used to hold and guide small to medium size needles and suture materials.
- **Mayo Hegar:**
  Mayo-Hegar needle holder are used in **a multiple types of surgical procedures**. Due to its versatility, it is one of the most commonly used surgical tools for both human and animal procedures and is normally used with larger, heavier suture needles because of its wide jaws.
- **Castroviejo:**
  Castroviejo needle holders are mainly used in **microsurgery**. They are distinguished from other needle holders by the fineness of their tip and the mechanism of their handle which allows the jaws to be activated without effort.
- **Skin Stapler:**
  Skin staplers are used **externally to close skin under high tension**, such as on the skull or the trunk of the body. Surgical staples offer several advantages over sutures. They can be inserted quickly.
- **Weck Hemoclip Applier:**
  Used for applying clips to small vessels in all branches of surgery.
- **Liga clip Applier:**
  Used for ligating vessles in multiple specialties. Clip Appliers are designed with grooves on the inside jaw surfaces of **the applier to increase in-jaw clip security**.

LIGACLIP Multi-Patient Clip Appliers are reusable instruments for both open and endoscopic procedures that deliver and close metallic ligating clips.

- **Staplers:**
  Surgical staplers and staples are used both externally and internally. Removable skin staples are medical devices, used **externally to close wounds under high tension**, including wounds on the scalp or the trunk of the body.

6. Viewing:

Surgen can examine body cavities, hollow organs,or structures with viewing instruments. Procedures may be perform through them

- **Nasal speculum:**
  Nasal speculum is a medical tool used by the ENT physicians to examine the anterior part of the nose. Nasal speculum helps in enlarging the sinus area so that the exposed area can be clearly observed.
- **Vaginal Speculum:**
  Used for vaginal and cervical examination.
- **Rectal speculum:**
  Used for the examination of rectum and anus.
- **Rigid Endoscope:**
- Used in the majority of surgical endoscopic applications and **enable endoscopists to visualize the surface of organs**, their vessels, or pathological changes without large incisions of the body and delivering a view even more clear than with the naked eye.
- **Laproscopic Trocars:**
  Laproscopic trocars are sharp-pointed surgical instruments, used with a cannula **to puncture a body cavity and to provide intra-abdominal access**. In cases of ruminal tympany in cattle, a wide-bore trocar may be passed through the skin into the rumen to release trapped gas.

➢ **Laproscopic instruments:**

- **Thorascopic trocars:**
  Thorascopic Trocars are typically placed inside hollow cannulas and introduced inside body cavities to assist in draining fluids. Thorascopic trocars are now referred to as both the initial entry device as well as the hollow cannula used during the operation. These instruments play an important role in laparoscopic surgery.
- **Arthroscopic cannulas:**
  Arthroscopic cannulas are necessary to allow passage of suture and instruments through a single defect in the deltoid.
- **Cystoscopy sheath:**
  The cystoscope sheath is a sturdy outer covering, which protects the telescope, **allows water flow for irrigation and passage of various instruments**.
- **Camera and light cord:**
  Attaches to any standard laparoscopy or endoscopy system,for diagnostic and therapeutic procedures.
- **Flexible Bronchoscope:**
  A bronchoscope is used to view the air ways and check for any abnormalities.

- **Flexible uretroscope:**
  A flexible ureteroscope is **used to see beyond the bladder into the ureters**, the tubes that carry urine from the kidneys to the bladder.
- **Proctoscope:**
  A proctoscope is used **to look inside the anus and rectum**. A proctoscope is a thin, tube-like instrument with a light and a lens for viewing. It may also have a tool to remove tissue to be checked under a microscope for signs of disease.

7. **Suction:**

- **Yankauer:**
  This tool is used to **suction oropharyngeal secretions in order to prevent aspiration**. A Yankauer can also be used to clear operative sites during surgical procedures and its suctioned volume counted as blood loss during surgery.
- **Frazier:**
  The Frazier suction tubes are used **to remove fluid or debris from nasal cavities and in laryngeal suction**, where they irrigate and aspirate the throat.
- **Poole:**
  Poole Suction Tube is an instrument **used to remove large quantities of fluid from surgical sites during procedures**. The Poole suction tube has multiple holes so as to best prevent blockages caused by fat or tissue suctioned during a procedure.

8. **Probes and dilators**

Uterine dilators:
commonly used **to gently open the cervix before a gynecologic procedure** that requires the cervix to be open, allowing access to the uterus and fallopian tubes. Uterine dilation reduces the risk of injury to the cervix during such a procedure.

- **Uterine Sound:**
  To know the position of uterus and and the length of the uterine cavity prior to dilation of the cervix,to sound the uterine cavity to detect any foreign body.
- **Urethral Sound:**
  Used to dilate a strictured urethra or to locate stones in the bladder by “sounding” listening for the sound of the instrument hitting sound.
- **Lacrimal duct probe:**
  A lacrimal probe can be used **to identify the fistula and to dilate very small fistulas**. Stay sutures can be placed close to the margins of the fistula, and a pediatric-sized Foley catheter can be threaded through the tract and inflated to aid in retraction and dissection.
- **Vascular Tunnelers:**
  Vascular Tunneler is used to implant a vascular graft subcutaneously with an oversized tissue passageway.

9. **Measuring:**

- **Rulers:**
  Measuring Rulers are significant instruments widely used in orthopedic procedures. These measuring rulers are commonly used to get precise measurements.
- **Depth guage:**

That instrument is usually employed in situations where it is necessary to gauge the depth of holes drilled into bones for securing implants or other similar surgical uses**.**

10. Microinstruments:

Castroviejeo Needle holder:

Castroviejo needle holders are mainly used in **microsurgery**. They are distinguished from other needle holders by the fineness of their tip and the mechanism of their handle which allows the jaws to be activated without effort.

Micro scissors:

Micro scissors are used for the microdissection of soft and delicate tissues.

Bishop Herman Forcep:

Bishop herman forcep commonly used tool in variety of ophthalmologic procedures.The forcep are designed for grasping sterile surgical equipment or removing tissue and small foreign bodies in the periorbital area.

ANESTHETIC EQUIPMENTS

- **Endotracheal tubes:**
  Flexible tube placed in the trachea .Delivers anesthetic gases directly from the anesthetic machine to the lungs.
- **Laryngoscope:**
  Used to increase the visibility of the larynx while placing an ET tube.
- **Anesthetic Masks:**
  Cone-shaped devices used to administer oxygen and anesthetic gases to nonintubated patients Used for induction and maintenance of anesthesia in very small animals Plastic or of rubber type masks.
- **Anesthetic Chambers:**
  Clear, aquarium-like boxes used to induce general anesthesia Used in feral, vicious, or intractable animals to reduce stress.
- **Anesthetic Machine:**
  Used to deliver precise amounts of oxygen and volatile anesthetic under controlled conditions.

➢ **Principle operation of Anesthetic Machine:**

- **Carrier Gas:** Oxygen and nitrous oxide
- **Liquid inhalant anesthetic**: to be vaporized .Mixed gases delivered to patient and Exhaled gases removed from patient.

➢ **Components of Anesthetic Machine:**

- Compressed gas supply
- Anesthetic vaporizer .(Precision and non Presion)
- Breathing circuit (rebreathing or nonrebreathing)
- **Compressed Gas Supply:**

Oxygen :

Used to increase inspired air to at least 30% oxygen

Level necessary to maintain cellular metabolism under anesthesia

Used to carry vaporized anesthetic to patient

Cylinders (tanks) :

Contain large volume of gas under high pressure.
E tanks (small), attached directly to anesthetic machine.
H tanks (large), attached remotely to anesthetic machine.
Control valve (outlet port):
Located on top of the tank , Left loose (open), right tight (closed)
Pressure-reducing valve :
Reduces outgoing pressure to a usable level

- **Tank Pressure Guage:**
  Indicates the pressure of gas remaining in a compressed gas cylinder measured in pounds per square inch (psi) (United States) or kilopascals (kPa) . Determine the number of liters remaining in a tank .

➢ **Anesthetic Vaporizer** :
Converts liquid anesthetic agent to a gaseous state.

- **Types of Anesthetic Vaporizers** :
- **Nonprecision vaporizer**:
  Used to deliver low vapor pressure anesthetics and is rarely used □
- **Precision vaporizers**:
  Used to deliver a precise amount of anesthetic to the patient and is xpressed as a percent of total gases leaving the vaporizer.
- **Uses of vaporizer:**
  Specific-use vaporizers are color coded
  Isoflurane = purple
  Sevoflurane = yellow
  Halothane = red
  Desflurane = blue
  Induction and maintenance rates
  Isoflurane = 3-5% induction; 1.5-2.5% maintenance
  Sevoflurane = 4-6% induction; 2-4.5% maintenance
  Desflurane = 10-15% induction; 8-12% maintenance

➢ **Breathing Circuit:**
Carries anesthetic and oxygen from the fresh gas inlet to the patient and conveys expired gases away from the patient

- **Rebreathing or non-rebreathing:**
  Used on all but very small animals .Carbon dioxide removed from exhaled air and exhaled air is inhaled again with added oxygen and anesthetic.
  Parts of rebreathing system:
- Unidirectional valves
- Reservoir bag
- Pop-off (pressure relief) valve
- Carbon dioxide absorber canister
- Air intake valve
- Pressure manometer
- Corrugated breathing tubes

- Y-piece

➢ **Unidirectional Valves :**

Control the direction of gas flow
Inspiratory (inhalation)
Expiratory (exhalation)
Open and close as patient breathes
Monitor respiratory rate and depth

➢ **Pop-off Valve:**

Allows excess carrier and anesthetic gases to exit the breathing circuit and enter the scavenging system
Prevents excessive pressure or volume of gases in the circuit

- **Reservoir Bag (Rebreathing Bag)** :

Flexible air storage reservoir
Indicator of respiratory rate and depth
Confirms proper endotracheal tube placement
Allows delivery of anesthetic gases or pure oxygen to patient

➢ **Carbon Dioxide Absorber Canister** :

Contains absorbent granules
Primary absorbent ingredient: calcium hydroxide also water, sodium hydroxide, potassium hydroxide, calcium chloride, calcium sulfate
Granules react with carbon dioxide to form calcium carbonate
Heat and water produced
Becomes more acidic with more use
Granules must be replaced when depleted

➢ **Air Intake Valve:**

Negative pressure relief valve admits room air into the circuit if negative pressure is detected in the breathing circuit may be separate or incorporated into inspiratory unidirectional valve or pop-off valve. Negative pressure is indicated by a collapsed reservoir bag . Patient will develop hypoxemia

➢ **Pressure Manometer** :

Indicates the pressure of gases within the breathing circuit**.**

➢ **Breathing Tubes And Y-Piece:**

Corrugated breathing tubes or inspiratory and expiratory breathing tubes carry anesthetic gases to and from the patient ,connected to unidirectional valve and Y-piece three sizes: 50 mm, 22 mm, and 15 mm in diameter

Y-piece :

Connects breathing tubes and also Connects to mask or endotracheal tube.

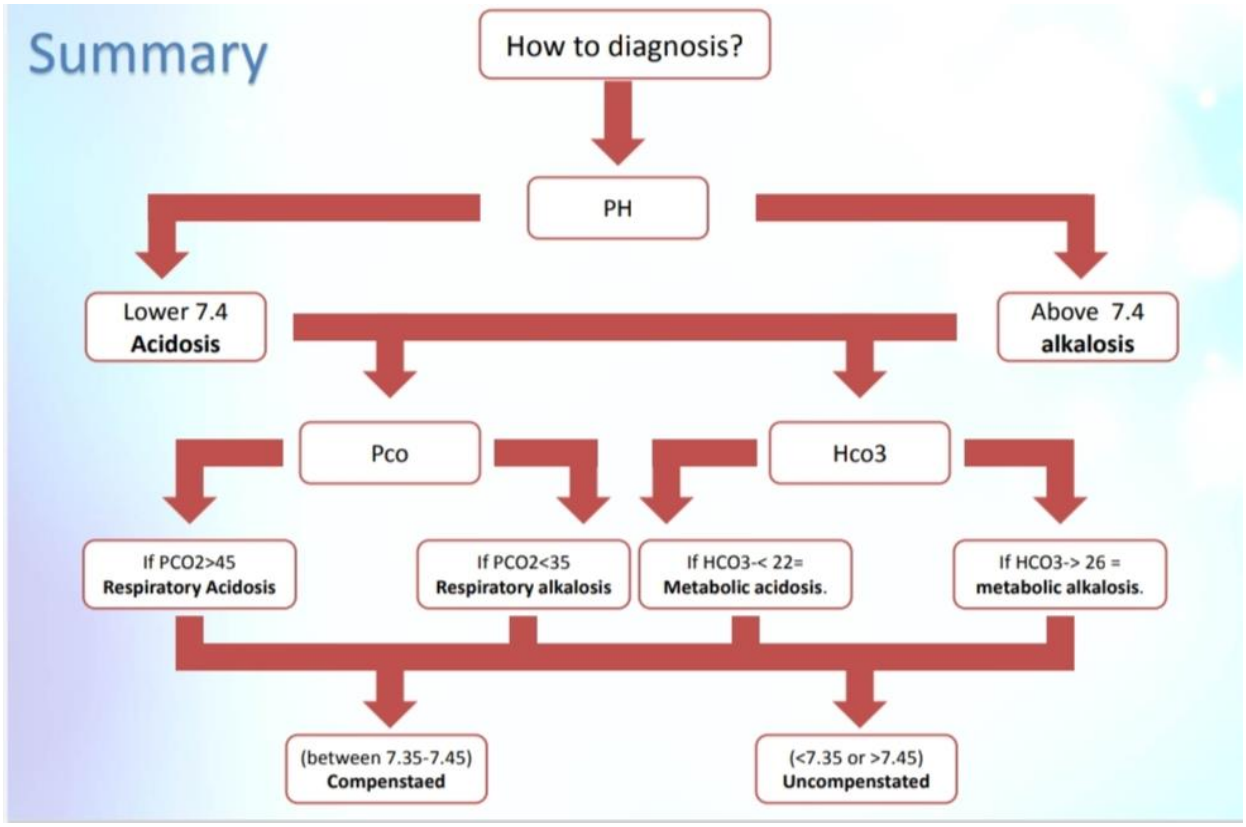
The difference between various head scans (X-ray, CT, MRI, MRA, PET scan) and what they show.

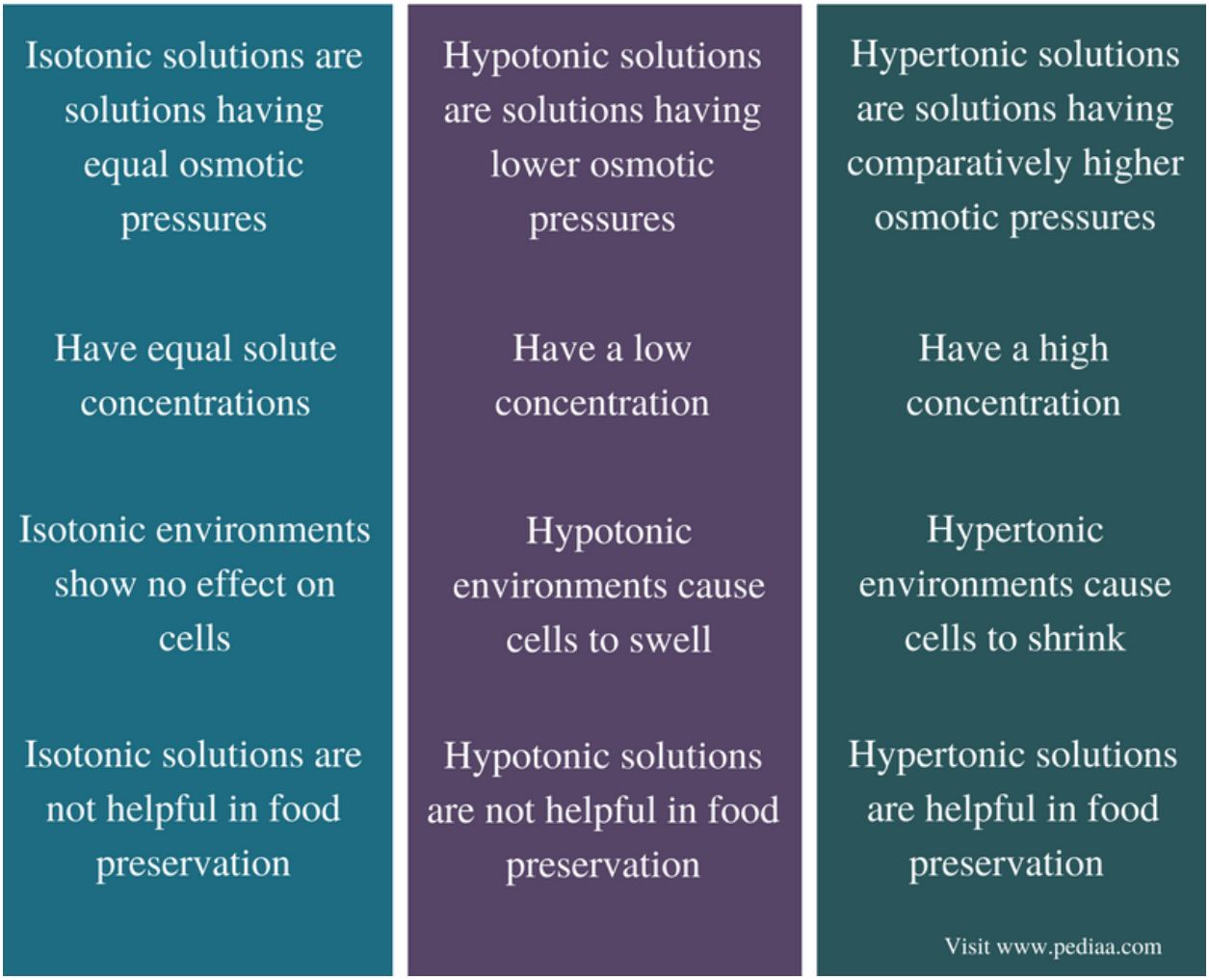
ACID BASE DISORDERS:

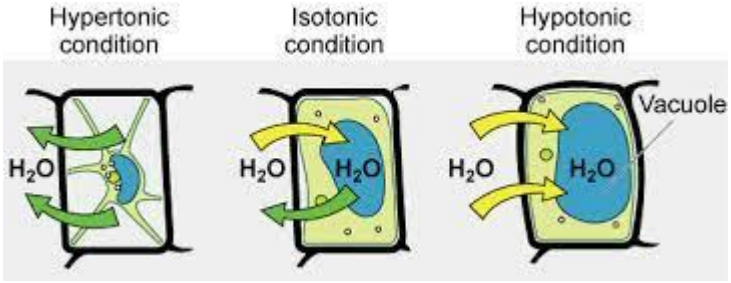
There are two types of Disorders:

1. Respiratory (Acidosis and Alkalosis)

2. Metabolic (Acidosis and Alkalosis)







FLUID THERAPY:

Indications of Fluid Therapy

1. Shock
2. Diarrhea-sever or prolonged
3. Anorexia as a result of pyrexia, digestive disorders and post operative condition
4. Rumen acidosis; Engorgement of grain leads to ketosis, dehydration
5. Vomition – severe or prolonged

6. Burns
7. Endocrine disorders
8. Alkalosis
9. Renal insufficiency
10. Abomasal displacement and torsion

The percentage of loss of fluid can be assessed by following rule of thumb:
- ✓ 4-6 % dehydration- No sign of dehydration
- ✓ 6-8 % dehydration- Moderate dehydration with sunken eyes, 2-4 second skin fold test
- ✓ 8-12 % dehydration- Sever dehydration skin fold test 20-45 seconds

Route of Administration:

The selection of the route of route for fluid therapy will depends on certain considerations:
a) Kind of disease and its severity
b) Extent of dehydration
c) Condition of the patient
d) Organic function of the patient
e) Type of electrolyte imbalance
f) Time and equipment available

Solutions Commonly Used:

1) Dextrose solution:

These are available in 2.5%, 5%, 10% and 25% concentration., but 5% solution is isotonic an used in routine practice.

Indications:
- ✓ Dehydration, depletion of liver glucose, ketosis, excess tissue catabolism
- ✓ Renal, hepatic, cardiac and gastrointestinal diseases
- ✓ To accelerate sodium excretion
- ✓ To increase intracranial pressure only hypertonic solution be used.

Contraindications:
- ✓ Never give more than 5% dextrose solution in ketosis.
- ✓ The higher percentage will increase ketosis rather than curing.
- ✓ Never give in case of milk fever
- ✓ Do not give following blood transfusion

2) Sodium Chloride (0.9% solution):

Indications:
- ✓ Excessive salivation called "Pantar" and digestive disturbances
- ✓ Vomition
- ✓ Pyloric obstruction
- ✓ Abomasal disorders
- ✓ Heat stroke
- ✓ Alkalosis due to fluid loss

Contraindications:
- ✓ Never give in ascitis and edema

3) Ringer's Solution
Indications:
- ✓ Dehydration
- ✓ Slight alkalosis or hypochloremia

Contraindications:
- ✓ Milk fever

4) Lactated Ringer's Solution
Indications:
- ✓ Slight acidosis
- ✓ Dehydration
- ✓ Burn, infection and peritoneal disorders
- ✓ To counteract the excess use of acidifying solution.

Contraindications:
- ✓ Liver dysfunction
- ✓ Congestive heart failure
- ✓ Anorexia due to shock
- ✓ Severe metabolic alkalosis
- ✓ Rumen acidosis

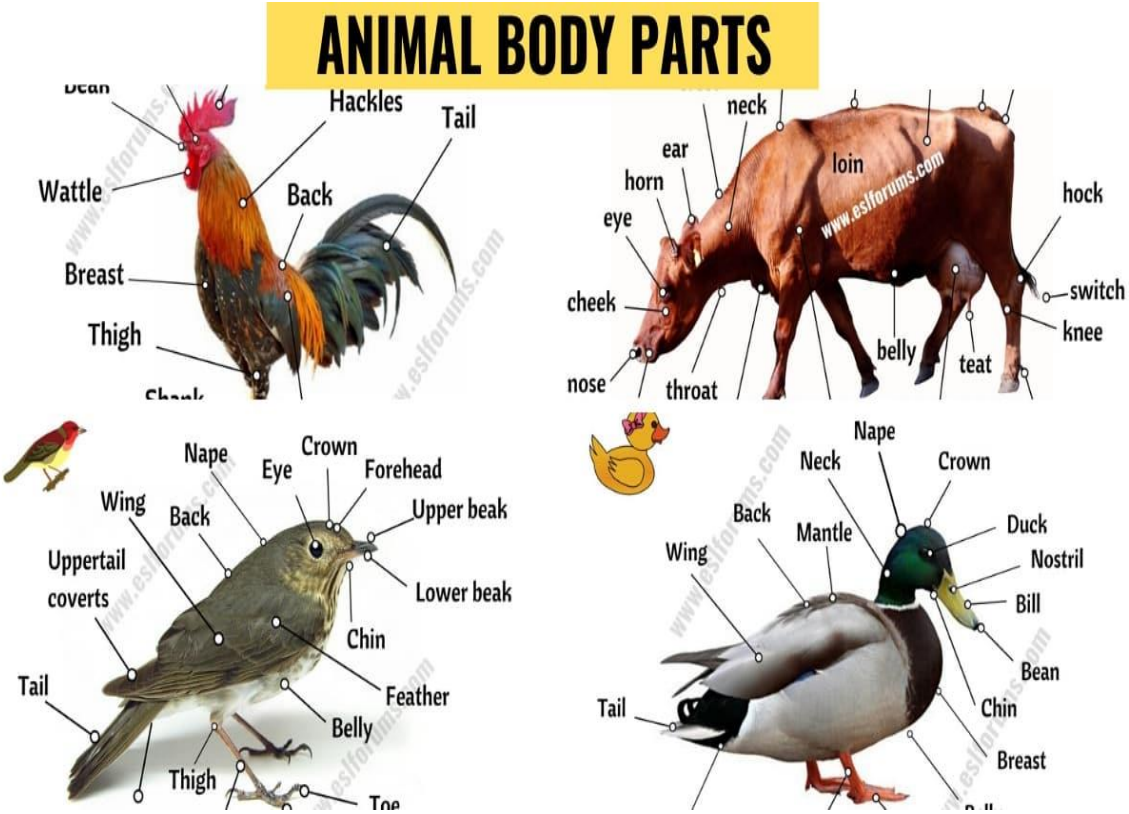
5) Sodium lactate Solution:
Indications:
- ✓ To induce quick alkalinity of urine, prior giving sulpha drugs
- ✓ To avoid renal damage and hemoglobinuria after blood transfusion
- ✓ In case of myohemoglobinuria in horses
- ✓ Metabolic acidosis not of anoxic origin
- ✓ At the dose rate of 25 ml/kg body weight Intravenously

Contraindications:
- ✓ Vomition
- ✓ Hepatic disorders
- ✓ Respiratory alkalosis
- ✓ Lactic acidosis

6) Acid Sodium Phosphate Solution
Indications:
Use in post parturient hemoglobinuria especially in buffaloes at dose rate of 60 gram acid sodium phosphate in 800 ml distilled water given intravenously very slowly. The same dose given SC 12 hours interval for 3 times.

BASICS OF ANATOMY:

- ✓ Ana means **up**,tome means **a cutting.**
- ✓ Anatomy refers to the science that deals with the form and structure of animals
- ✓ It has two types:
  A-Gross Anatomy
  B-Microscopic Anatomy.



- ✓ Smooth muscles are only non striated in nature.
- ✓ Skeletal muscles are only Voluntary in nature.
- ✓ There are 6 joins in forelimb and 7 joints in hind limb
- ✓ Cartilage has very limited repair capability and it is Avascular in nature.
- ✓ Splanchnology: It deals with the viscera of digestive, respiratory and urogenital system.
- ✓ Bile is produced and secreted by hepatocytes into a special duct called a **"bile canaliculus".**
- ✓ Bile is stored in Gall bladder and gall bladder send bile to duodenum in response to cholecystokinin.
- ✓ Eosinophil is pink staining.
- ✓ Bronchus: An airway whose walls are reinforced by the presence of cartilage plates.
- ✓ Hair has three layers I.e Cuticle, cortex and medulla.
- ✓ Hair follicle is composed of five layers.
- ✓ Heat is produced by muscular system.

BODY RECEPTORS:

1- Chemo receptors-stimulated by changes in the chemical concentration of substances.

2- Pain receptors-stimulated by tissue damage.

3- Thermo receptors.-stimulated by changes in temperature.

4- Mechano receptors-stimulated by changes in pressure or movement.

AXIAL SKELETON AND APPENDICULAR SKELETON:

Axial	Appendicular
Skull, Vertebral column, Thoracic Cage	All the bones that forms the limbs, shoulder and pelvic girdle

SPINAL NERVES:

Ox	**37 pairs**
Horse	**42 pairs**
Pig	**39 pairs**
Dog	**35 pairs**
Fowl	**36 pairs**

CRANIAL NERVES (MOTOR NERVES):

Oculomotor
Trochlear (Pathetic nerve, Origin behind Corpora Quadri gemina, Smallest nerve
Abducent
Accessory
Hypoglossal

SENSORY NERVES:

Olfactory (Origin from olfactory bulb)
Optic (Origin from optic commissure)
Cochlea vestibular

MIXED NERVES:

Trigeminal (Largest nerve, Origin from lateral aspects of pons)
Facial
Glosso pharyngeal
Vagus (Longest and Pneumo gastric nerve)

ORGAN ORIGIN:

Ecto-mesoderm	Teeth
Ectoderm	Nervous system, Epidermis, Neural tube of frog, Salivary Glands, Ear drum.
Mesoderm	Muscle cells, Connective tissue, Blood, Gonads, Excretory, Reproductive system
Endoderm	Gut, Internal organs, Yolk sac, Pharynx, Larynx, Bladder, Vagina, Urethra
Meso-endoderm	Allantois

BONES OF AXIAL AND APPENDICULAR SKELETON:

Animal	Skull	Vertebrae	Ribs	Clavicle	Forelimbs	Hind limbs	Visceral
Ox	34	51	26	0	48	48	2=210
Dog	34	53	26	2	80	76	1=271
Horse	34	57	36	0	40	40	0=208
Pig	30	52	28-30	0	80	80	1=270-72
Fowl	40	41	14	2	28	42	2=168
Camel							1

- ✓ Occipital+Ethmoid+Sphenoid+Vomar+Mandible+Hyoid are cranial bones, 1 and 1 in numbers.

VERTEBRAL FORMULAS IN DIFFERENT SPECIES:

Ox	$C_7, T_{13}, L_6, S_5, CY_{15-25}$
Horse	$C_7, T_{18}, L_6, S_5, CY_{16-22}$
Dog	$C_7, T_{13}, L_7, S_3, CY_{18-25}$
Pig	$C_7, T_{14-15}, L6_{6-7}, CY_{18-24}$
Fowl	$C_{14}, T_7, L\text{-}S_{14}, CY_4$

TYPES OF JOINTS IN BODY:

Ginglymus / Hinge joint	Knee joint,fetlock joint,elbow joint, femo-tibia articulation
Arthroidial/ Plane joint	Joint b/w adjacent carpals.
Trochoid/ Pivot joint	Atlanto-axial joint
Enarthroidal/ Ball & socket joint	Coxo-femoral joint
Suture/ Skull bone joint	Synarthrosis
Synsarcosis	B/w scapula & thorax
Syndesmosis	B/w split bone & cannon bone

Amphiarthrosis	B/w bodies of vertebrae
Gomphosis	Teeth in alveolar socket
Circumduction	B/w Shoulder & hip joint

POINTS OF BODY:

Tuber coxae	Point of hip/hook bone
Tuber ischia	Pin bone
Meta carpal	Cannon
First phalanx	Long pastern bone
2nd phalanx	Short pastern,os-corona
3rd phalanx	Coffin bone,os-ungulare
Olecranon process	Point of elbow

- ✓ Symphisis is slightly moveable joint.

BASICS OF HISTOLOGY:

- ✓ Greek **histos**, meaning **tissue** (web) and **logia**, **knowledge**
- ✓ Veterinary Histology is the science that focuses on the detailed morphology of tissues of domestic animals with the aid of microscope and correlates specific structures with function.

LEVELS OF ORGANIZATION:

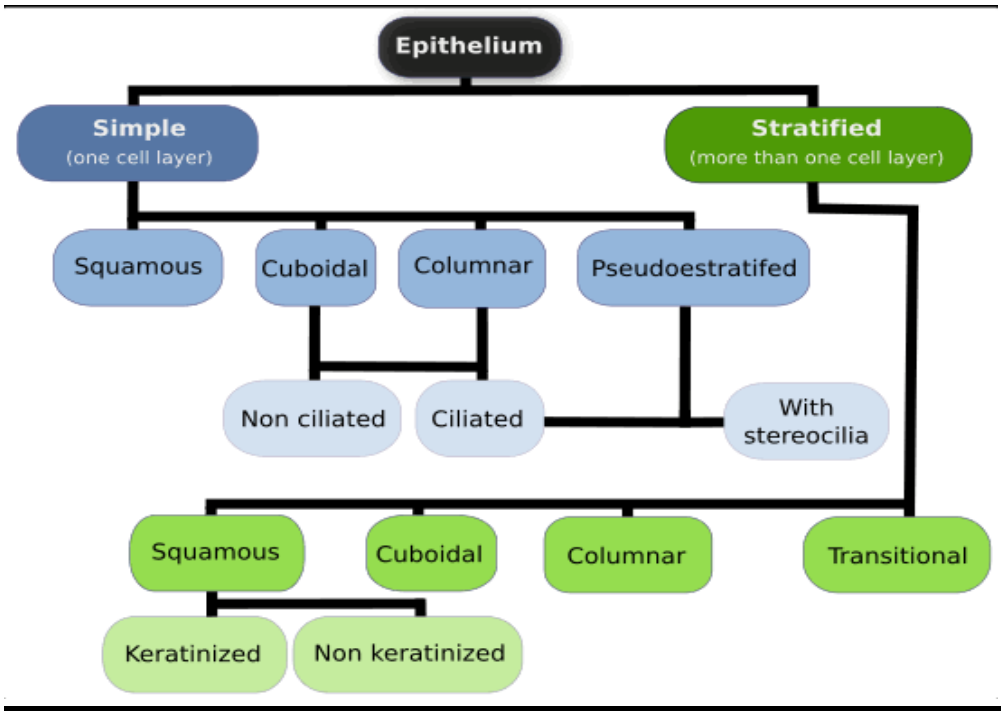
- ✓ Cells
- ✓ Tissues
- ✓ Organs
- ✓ Organ Systems
- ✓ Organism

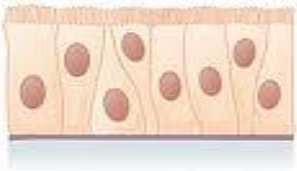
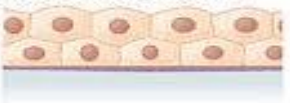
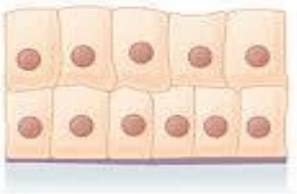
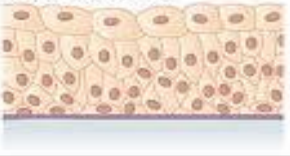
Methods used:

1. Making histological slide
2. Histochemistry
3. Immunohistochemistry
4. Microautoradiography

TYPES OF EPITHELIUM AND THEIR LOCATION:

Sr. No.	Simple Squamous Epithelium	Transitional Epithelium	Cuboidal Epithelium	Simple Columner Epithelium
01.	Peritoneum	Ureter and Renal Pelvis	Lens of Eye	Mucosa of Oviduct
02.	Parietal layer of Bowman's Capsule	Urinary Bladder	Ovary	Endometrium



Cells	Location	Function
Simple squamous epithelium 	Air sacs of lungs and the lining of the heart, blood vessels, and lymphatic vessels	Allows materials to pass through by diffusion and filtration, and secretes lubricating substance
Simple cuboidal epithelium 	In ducts and secretory portions of small glands and in kidney tubules	Secretes and absorbs
Simple columnar epithelium 	Ciliated tissues are in bronchi, uterine tubes, and uterus; smooth (nonciliated tissues) are in the digestive tract, bladder	Absorbs; it also secretes mucous and enzymes
Pseudostratified columnar epithelium 	Ciliated tissue lines the trachea and much of the upper respiratory tract	Secretes mucus; ciliated tissue moves mucus
Stratified squamous epithelium 	Lines the esophagus, mouth, and vagina	Protects against abrasion
Stratified cuboidal epithelium 	Sweat glands, salivary glands, and the mammary glands	Protective tissue
Stratified columnar epithelium 	The male urethra and the ducts of some glands	Secretes and protects
Transitional epithelium 	Lines the bladder, uretha, and the ureters	Allows the urinary organs to expand and stretch

BASICS OF VETERINARY MEDICINE:

FATHER OF ERA:

- ✓ Hippocrates (Father of Medicine) (بقراط)
- ✓ Socrates (سقراط)
- ✓ Plato (افلاطون)
- ✓ Aristotle (Father of Biology and Zoology) (ارسطو)
- ✓ Alexander (سکندراعظم)

PULSE RECORDING SITES:

Horse	Facial artery (Lower jaw)
Cattle	Coccygeal artery
Sheep, Goat, Dog, Cat	Femoral artery
fowl	Wing artery

Terminologies:

Sneezing	Rihinits,inflammation of nasal membrane.
Weezing	Constriction or edema of bronchial wall due to Ag+Ab reactions.
Snoring	Pharyngeal stenosis.
Snuffing	Stenosis of nasal passage
Roaring	Laryngeal stenosis
Grunting	A forced expiration against a closed glottis.
Purring	Tracheal abnormal sound.
Hiccups	Repeated short jerky inspiration due to irritation of phrenic nerve.

IMPORTANT POINTS:

- ✓ N.D cause hemorrhages in proventriculus+Greenish color of droppings
- ✓ Ranikhet in India.
- ✓ New castle is in USA.
- ✓ Off feed bird white droppings.
- ✓ 3 Systems involve in ND; a) Digestive. b) Respiratory.c) Nervous
- ✓ Bollinger bodies found in fowl pox.
- ✓ In BSE, disturbance of trigeminal nerve is present.
- ✓ In Black diseases, Clostridium type B, infectious necrotic hepatitis is present.
- ✓ Gas gangrene, Clostridium type A.
- ✓ Pulpy kidney(over eating disease),clostridium type D.
- ✓ In Black leg, myonecrosis, alpha beta toxins.
- ✓ Struck Clostridium type C.
- ✓ Lamb dysentery, beta toxins, Clostridium type B.
- ✓ Cold mastitis leptospirosis.
- ✓ California mastitis test, 3% surf solution used.
- ✓ Summer mastitis by Corynebacterium pyogens
- ✓ H.S is gram negative bacteria, type I or B, occur in monsoon season.

- ✓ Parenchyma tissue is functional tissue.
- ✓ Glander (present in world war 1 & 2 by Japanese)farcy pipes cord in equines,cutaneous vessels become firm and filled with exudate.
- ✓ Pyogenic bacteria are Corynebacterium, Pseudomonas, Streptoococcus, Staphlococcus.
- ✓ Parts of abscess are wall,membrane and pus.
- ✓ Pus is necrotized tissue+dead bacteria+leuckocytes+proteins.
- ✓ Malacia is the liquefaction necrosis of nervous tissue.
- ✓ Ulcer is the shallow area of necrosis.
- ✓ Degeneration is the accumulation of metabolites of cell.
- ✓ Infiltration is the normal prolificaton of cell.
- ✓ Pus is alkaline and yellow in color and donot clot.
- ✓ Acute abscess forms in 3-5 days after infection.
- ✓ Insipatted pus is solid art of pus.
- ✓ Local abscess donot cause fever and superficial pain.
- ✓ Deep abscess cause fever and pain.
- ✓ Chronic abscess (cold) are primary(repeated injury)& secondary(localized injury).
- ✓ Abscess treatment is Antibioma (sterile pus).
- ✓ Cellulitis/Philegmon is diffuse,supporative inflammation of loose connective tissue with necrotic tissue.
- ✓ Pustules are circumscribed cavity with us situated in epidermis.
- ✓ Furuncle/boil is supportive inflammation of hair follicle or sebaceous gland due to Staphylococcus aureus.
- ✓ A group of furuncles is known as furunculosis.
- ✓ Carbuncle is small boil drains outside with multiple opening.
- ✓ Acne is the abscess of sebaceous gland.
- ✓ Emphysema is collection of us in body cavity.
- ✓ Live fluke is fasciolosis and it is bottle jaw.
- ✓ Clostridial infection cause sudden death due to toxins,G+VE microbes and penicillin use for it.
- ✓ Cattle and sheep are resistant to CCPP.
- ✓ Tube precipitation test in CCPP.
- ✓ Phenol (carbolic acid) is cauterizing agent.
- ✓ Mycoplasmosis(CRD) causes by Mycoplasma gallisepticum becom CCRD with E.coli.
- ✓ Salmonella and mycoplasma both have vertical transmission.
- ✓ Miss shaped egg in IB.
- ✓ Tylosin is used against mycoplasma.
- ✓ Typhoid and paratyphoid caused by Sallmonela.
- ✓ Infectious coryza caused by Haemophilus paragallinarium.
- ✓ Salmonella pullorum cause bacillary white diarrhes.
- ✓ Bumble foot (pododermatitis) is caused by Staphlococcus bacteria.
- ✓ Esom salt MGSO4 is purgative.
- ✓ Oral cavity contain esophagus,stomach,S.I,L.I Rectum.
- ✓ Nasal cavity contains pharynx,layranxtracea,brochus,slungs.
- ✓ Lymphatic organs contains lymph nodes,spleen,tonsils.

- ✓ Heterakis egg transmit histomonas parasite (Black head disease).
- ✓ Argus percicus is a soft tick (Dermanysus gallinae+menopon gallinae).
- ✓ Round worm are ascarids.
- ✓ Hair worms are cailllaria.
- ✓ Caecal worms are heterakis.
- ✓ Tape worms are cestodes.
- ✓ IBD cause by Birna virus.
- ✓ N.D is caused by Abula + Rabula virus.
- ✓ Toxic granules in acute bacterial infection are Dohlis bodies.
- ✓ H2O2 is phagocytosis.
- ✓ Lamb dysentery in young sheep.
- ✓ Struck in adult sheep.
- ✓ Lamb hemorrhagic enteritis is CL.b.
- ✓ Necrotic hemorrhagic enteritis is CL.e.
- ✓ Braxy in sheep.
- ✓ Enterotoxemia in lamb.
- ✓ Causative agent of udder impetigo is Staphlococcus aureus.
- ✓ Collibacilosis/calf scour/white diarrhea is due to E.coli.
- ✓ Causative agent of Glassers disease is Haemophilus suis.
- ✓ Ist discovered bacterial disease is Anthrax.
- ✓ Pearls disease is T.B.
- ✓ Bacillus anthracis produce medusa head on blood agar.
- ✓ Froathy blood discharge from nostrils+anus is in B.Q.
- ✓ Botulism cause rapidly fatal motor nerve paralysis.
- ✓ Dogs are resistant to Botulism.
- ✓ Word veterinary used by THOMAS BROWNE in 1646.
- ✓ Veterinary in Latin word.
- ✓ Cow skull trepanation in NEOLITHIC.(3000-3400 BC)(STONE AGE)
- ✓ Veterinary practice as 2000 BC in BABYLONIA (IRAQ)AND EGYPT.
- ✓ 1st anti cancer vaccine in poultry is against Mareks disease.
- ✓ Veterinary public health by FAO/WHO in 1975.
- ✓ Feline distemper (panleukopenia).
- ✓ Diaphragmatic hernia common in buffaloes.
- ✓ Bloat induce fodders are berseem,lucerne,cloves.
- ✓ Enzooitc hematuria is due to prolonged ingestion of bracken fern.
- ✓ Pyelonephritis is due to Corynebacterium renale.
- ✓ Camel pox is orthopox.
- ✓ Pseudo FMD is blue tongue.
- ✓ Asphaxia is due to Strychnine poisoning.
- ✓ Warfarin is impair blood clotting.
- ✓ Feeding wheat bran to horses cause Osteodystrophic fibrosa.
- ✓ Osteomalacia occur in mature animal.
- ✓ Gangrene of extremities is due to Ergot.
- ✓ Vegetable oil reduces feedlot bloat.

- ✓ Ascariasis treatment is piperazine.
- ✓ Trifluropronazine is sedative.
- ✓ Diagnosis of lymphosarcoma is presence of immature lymphocytes.
- ✓ Lymptocytes in blood film occur in Aplastic anemia.
- ✓ Bone marrow smears in leukemic leukemia.
- ✓ Excessive administration oof chloramphenicol is aplastic anemia.
- ✓ Hemophilia type A cause haemorrhagic anemia.
- ✓ Incompatible blood transfusion is in hiccups and coughing.
- ✓ Leukopenia is viral infection, it is acute, endotoxins+anaphylaxis.
- ✓ Bracken fern poisoning cause reticulocytosis.
- ✓ Pharyngeal paralysis in Botulism Rabies.
- ✓ Pyelonephritis is ascending infection.
- ✓ Cystitis is frequent painful urination.
- ✓ Shock cause trauma and peripheral circulation.
- ✓ Abomasal displacement in ketosis.
- ✓ Chromatic force of monocyte in lipopolysaccharide of G-VE.
- ✓ Lymphocyte has IgA.
- ✓ Homograft rejection and immunoglobulins in monocytes.
- ✓ Pelvis flexure is predilection site for impactive colic.
- ✓ Plasma cells have immunoglobulins.
- ✓ Young immature neutrophil in Septicemia.
- ✓ White velvety deposition in mycotic stomatitis.
- ✓ Increase Histamine in ruminal atony.
- ✓ Common equine stomach neoplasm is squamous cell carcinoma.
- ✓ Snoring in pharyngeal obstruction.
- ✓ Ephemeral fever treatment is salicylates.
- ✓ Blood stain saliva in rinder pest(morbilivirus).
- ✓ Thymus atrophy in canine distemper.
- ✓ Blue tongue is infectious and non contagious in oral+intestine.
- ✓ Parvo virus isolation in feces.
- ✓ Edema of thymus is ICH.
- ✓ Moniliosis in yeast.
- ✓ Common stain of FMD is O.
- ✓ Malignant form of FMD is acute myocardial failure.
- ✓ Safest and best vaccine for rinder pest is cell culture vaccine.
- ✓ Abomasal lesion in Braxy.
- ✓ Idiopathic tetanus means no history of wound.
- ✓ Monday morning sickness in goat alpha toxin+lota+epilon+glycosuria+hyperglycemia in E.T.
- ✓ Glander is per acute involves lymph nodes+URT+Farcy.

BACTERIAL DISEASES:

Mastitis:

Causative agents:

Mastitis in cattle is an inflammation of the udder, normally caused by bacteria. Characterized by changes in udder and milk composition (physical, chemical, bacteriological), neutrophils in milk increase.

Staphylococcus aureus, Streptococcus aglactiae --- These are most prevalent bacteria, Others are: Corynebacterium bovis, Mycoplasma bovis

Major pathogens:

Staph aureus, Streptococcus aglactiae, E.coli (70% contribution by first two agents)

Minor pathogens:

Pseudomonas, fungal agents (aspergillus, yeast), corynebacterium

Less in buffalo but more severe (sphincter is tight)

48% decrease in milk yield during production, also medical changes.

Phases:

Invasion, Infectious, Inflammation

Tests:

California milk test (CMT), Surf field milk test (SFMT)

Abnormalities in udder:

Diffused --- > localized -- > fibrosis --- > gangrene

Abscess formation in calves, atrophy

Clinical Mastitis

Clinical mastitis are those infections we can see — the milk is abnormal, and we may see signs of inflammation in the udder (redness, heat, pain, swelling) or in the cow (off-feed, down in milk, fever).

Sub clinical Mastitis

Sub clinical mastitis is inflammation in the udder that DOES NOT cause any change in the milk, udder, or cow — the only way we know a cow has subclinical mastitis is if we run tests. The most common tests that we run are the California mastitis test (CMT) and somatic cell count (SCC) measurements.

Internal Teat Sealants

These products are non-antimicrobial, sterile intramammary pastes containing bismuth subnitrate in an oil vehicle. Syringes containing internal teat sealants are delivered to the quarter and they fill the teat canal, acting as a physical barrier to prevent bacteria from gaining deeper access to the mammary gland.

Need of Dry Cow Therapy:

For nearly 50 years, veterinarians and researchers have recommended that every quarter of every cow receives a licensed intramammary long-acting antimicrobial treatment at dry-off. This wasn’t a chance recommendation.

Study after study has found that using these medications at dry-off has 2 functions:

Curing existing mastitis infections that a cow may have picked up during her lactation Preventing new infections that may occur across the dry period, particularly the first few weeks. The early way of thinking was that it was difficult to tell whether a cow was infected at dry-off (historically, culturing every cow has been onerous and SCC testing

used to be less available). Also, aside from improving environmental hygiene and dry-off strategies, producers had little available to prevent new mastitis infections over the dry period.

Recently, new research has shed light on the necessity of dry cow antimicrobial therapy for certain farms. It may be that not every quarter of every cow needs a long-acting antimicrobial treatment at dry-off. With some new diagnostic techniques and the application of teat sealant products, many farms may be able to selectively treat only those cows that are most likely to have mastitis infections.

Diagnostic Tests:

Bulk tank somatic cell count (at herd level), California mastitis test (distinguish into peracute, acute, subacute and subclinical mastitis); it is negative when SCC is < 200,000 and positive when SCC > 200,000

Treatment:

Sulfonamide or ampicillin (parentral), Cefoperazone, Penicillin G and neomycin in combination (intra-mammary).

Strangle

Syn:

Equine distemper

It is highly contagious disease, mostly affect horses of young age, upper respiratory tract lymph nodes involve. 1-5 years of age foals affected. Emphysema in pleural space. Acute glutteral pouch involve+hypoglycemia+ glycosuria+ albuminuria. Bastard/metastatic strangles +puerperal hemorrhagica.

Causative agent:

Streptococus equi

Signs:

Acute onset of fever, anorexia, depression, submandibular and pharyngeal lymphadenopathy with abscessation and upon rupture, there is copious purulent discharge (it may be unilateral or bilateral)

Treatment:

Local treatment of abscess is penicillin @22000 IU/kg.

Anthrax

Syn:

Spleenic fever, Milzbrand, Charbon, Wool sorters disease and Malignant carbuncle.

Causative agent:

Bacillus anthracis

Anthrax is an infectious, septicemic, zoonotic disease which is caused by a Gram +ve rod shaped bacteria which can be killed by steam sterilization or burning.

Pathogenicity is due to four factors;

(i) Lethal factor

(ii) Edema factor

(iii) Protective antigen

(iv) poly-n-glutamate capsule.

Clinical Signs:

(a) Sudden death

(b) High fever, Ruminal stasis, Hemoglobinuria, Bloody diarrhea, Bloody milk, Dyspnoea, Collapse.

(c) Sever colic and bisket edema in horse

(d) Splenic fever (i.e. spleen is enlarged, dark black in color.

(e) Septicemia, edema at throat abdomen and flank region

(f) Exudation of tarry blood from the body orifices

(g) Failure of body to clot, absence of rigor mortis

Anthrax is a Greek word; mean "Coal" – dark color of lesions and blood.

Anthrax vaccine:

Stern strain (a non-capsulated toxigenic strain)

Source of anthrax: (i) alkaline soil (ii) Contaminated animal byproducts (skin hides, bones and wool)

Forms of Clinical Disease:

There are three forms of disease; (a) cutaneous (there is formation of papules, vesicles and pustules) (b) pulmonary form (c) intestinal form

Rx:

Penicillin @ 20,000 IU per kg bwt twice daily or Streptomycin @ 8-10 g/day or Oxytetracycline @ 5 mg/kg bwt per day IM

Tetanus

Causative agent:

Clostridium tetani; releases exotoxins which are: i) Tetanospasmin ii) Tetanolysin

Tetanus spasmin is neurotoxin and tetanolysin is haemolysin.

Signs:

- ✓ Condition is associated with the history of wound. Other signs are Nictating membrane third eyelid prolapse, body muscle stiffness, lock jaw, Saw horse. Birds are resistant to it.

Rx:

Penicillin (in larger doses), Chlorpromazine (muscle relaxant) @ 1 mg/kg IM, Hydration status maintained by passing stomach tube and pouring water directly into the stomach, to relieve the animal from urination, pass urine catheter. Other Treatment is chloral hydrate 30-45 gm, 200-300ml of 10% MgSO4.

Black Leg

Syn:

Black quarter, Myonecrosis

Causative agent:

Clostridium chauvoei, G +ve anaerobes, normal inhabitant of GIT and soil.

Cattle of 6 month to 2 years age and rapidly growing with high plan of nutrition are most susceptible.

It causes edematous swelling of muscles of neck, shoulder, back, gluteal muscles (myonecrosis or gangrenous necrosis) with crepitating sounds and foul odour from the

necrotic areas. There are two toxins; alpha toxin and beta toxin. Disease is characterized by myonecrosis, acute lameness, rising fever, muscle tremors, andultimately death of the animal. Autolysis proceeds rapidly.

Rx:

Procaine penicillin, IM @ 40000 IU repeat after every 6 hours. Benzyl penicillin IV @ 45000 IU repeat after every 4 hours.

Vaccination:

Clostridium chauvoie killed vaccine can be used.

Bacillary Haemoglobinuria

Causative agent:

Clostrodium hemolyticum; releases neurotoxins and hemolytic betatoxin

Signs:

Toxemia, Hemoglobinuria. Clinically high fever, hb-urea, dysentery, acute abdominal pain, dyspnoea. Cattle and sheep are mostly affected. Organism releases toxin "phospholipase C" a beta toxin. Abdominal pain is due to necrosis of intestinal epithelium. Severe anemia is de to loss of blood from urine and faeces.

Rx:

Penicillin, Tetracycline

Important Note:

Deworming with Triclabendzole to kill the liver flukes because liverflukes provides the anaerobic conditions.

Hemorrhagic Septicemia:

Syn:

Barbone disease,

Signs:

High temperature (106-107 F), Profused salivation, Warm painful swelling around throat and throat, Dyspnoea

Spp affected:

Cattle, buffalo, swine, sheep

It is an acute, subacute infectious disease of septicemic nature, which is characterized by

i) Acute gastroenteritis

ii) Subacute edema

iii) Serifibrinous pleuropneumonia

iv) Edema of intra-alveolar tissues

Causative agent:

Pasteurella spp., P.boviseptica (Cattle), P.bubalospetica (Buffalo). normal inhabitant of respiratory tract

Transmission:

Direct contact by droplets

PM lesions:

Subacute edema of head, neck thorax, and throat. Serofibrinous exudate in the subcutaneous tissues of the head, neck throat dewlap and thorax. Edema of buccal and pharyngeal mucosa. Swelling of tongue. Hemorrhagic gastroenteritis. Petechial hemorrhages on tissues and membranes. Enlargement of lymphatic glands. Hemorrhagic endocarditis and serofibrinous pericarditis.

Pectoral or thoracic form:

Thoracic cavity contains serous or sero-fibrinous exudate. Visceral pleura studded with hemorrhages. Lungs are edematous

Clinical sings:

Temp up to 106 F, constipation --> thin fluid containing flakes and blood mucus or fibrin. Head, neck, thorax, dewlap are swollen, hot, tense, and painful. Buccal membrane is edematous and tongue swollen, rendering deglutition difficult. Laboured respiration. Death from asphyxia or exhaustion due to the severe gastroenteritis.

Pneumonic form:

Prostration, Drooping ears, Congested hemorrhagic mucosa, Protruded tongue, Open mouth, Sever dyspnoeae.

Rx:

i) OTC, IV @ 1 mg/kg bwt for 3 days
ii) Chloramphenicol, IV @ 10 mg/kg bwt 3 times for 3 days
iii) Penicillin dihydrostreptomycin @ 30,000 IU, IM for 3 days
iv) Sulfamethazine @ 150 mg/kg bwt IV for 3 days
v) Ampicillin @ 10 mg/kg bwt, every 24 hours

Mass Medication:

a) Sulfamethazine @ 100 mg/kg bwt in drinking water daily for 7 days
b) OTC @ 3-5 mg/kg bwt in cattle feed for 7 days

H.S Vaccine produce immunity for 2 months. Bipolar organism and does not release exotoxins. Most susceptible age is 6 mo to 2 years

Brucellosis:

Syn:

Bang's disease (As causative agent was discovered by Bang (A scientist)

Zoonosis:

Man suffered from undulant fever (also called malta fever , severe headache and pain in back.

Causative agent:

Brucella abortus, B. mellitensis, B. suis

Intracellular bacteria which can survive and multiply within the cells of macrophage system.

Signs:

It causes abortion after 5th month of pregnancy

Rx:

No specific treatment

Vaccination:

Single 5 ml dose of B.abortus strain 19 is used for vaccination given SC at from 2 to 6 month of age.

Tuberculosis

Causative agent:

Mycobacterium tuberculosis (in human), Mycobacterium bovis (in bovine).m It is a zoonotic disease. It is also called chronic debilitating disease. Body weight loss, weakness, anorexia, generalized emaciation, low grade fever. In pulmonary form, chronic intermittent cough, dyspnoea, Tachypnoea. Lesions are called "tubercles". In advanced cases, all the lymph nodes may contain tubercles granulomatous lesions.

Signs:

Progressive emaciation, Fluctuation in temperature, Pharyngeal obstruction etc.

Test:

Tuberculin test (Single intradermal test)

Rx:

Isoniazid @ 20mg/kg bwt orally. Sreptomycin, Isoniazid, and Paraminosalvic may be used for treatment. Tuberculin Test: is used for diagnostic and screening purpose. Inject 0.05 ml tuberculin intradermally (I/D). +ve case is indicated by increased thickness of the skin.

Test and slaughter policy for eradication

Vaccination:

Bacillus Calmette Guerine (BCG)

Paratuberculosis

Syn:

Johne's disease

Causative agent:

It is caused by "Mycobacterium paratuberculosis". It is a chronic, progressive and diarrhoeal disease. It effects all the ruminants especially newborn calves, and young animals but the clinical disease occur over 2 years. Clinically watery diarrhea, emaciation and weight loss. The organism has long incubation period and disease remain subclinical and organisms shed through feces and milk.

Signs:

Progressive emaciation over several weeks, Chronic intractable diarrhea which does not respond to any treatment, Corrugation of intestinal wall. A distinguish characteristic of Johne's disease is that infection occurs in animal at an early age, usually under 30 days of age and clinical disease does not occur until 3-5 years of age.

Rx:

No effective treatment, Isoniazid @ 20mg/kg up to 100 days. Streptomycin is also effective. Vallee's vaccine is used for Paratuberclosis.

Tests:

(i) Johnin Intradermal Test

(ii) Johnin Intravenous Test

Johnin Test:

In this test, organism causes delayed type of hypersensitivity reaction. Inject Johnin extract I/D. If it will increase thickness, indicates positive case.

Vaccination:

Vallee's vaccine is used

Actinomycosis

Syn:

Lumpy Jaw

Causative agent:

Actinomyces bovis; normal inhabitant of the bovine mouth. It is caused by wound disease involving bones especially of maxilla and mandible. In maxilla lesions, there is suppurative ostitis.

Signs:

Initially pointless hard immovable swelling onmandible or maxilla usually at the level of central molar teeth. Eventually discharge small amounts of pus through one or more openings in skin.

Rx:

Surgical debridement, Iodides and/or Sulfonamide orally or parenterally

Actinobacillosis

Syn:

Wooden tongue

Causative agent:

Actinobicillus lignieresii; normal inhabitant of alimentary tract. A disease of zoonotic importance.

Etiology:

It is caused by; Actinobacillus ligneresii; Gram –ve rods. The causative agent penetrates through abrasions of the mouth. Disease is characterized by pyogranulomatous lesions on the tongue, gums, palate, and associated lymph glands. There is sulphur granules formation. Wooden tongue is the disease of cattle and sheep.

Signs:

Difficulty in swelling, nodules in cattle and of lips in sheep. Big head

Rx:
(i) Sodium iodide 1 g/12 kg can be given IV as 10% solution in one does both for cattle and sheep. At least one or two further treatments are at 10-14 days intervals are required for bony lesions.
(ii) Washing with tinc. Iodine and glycerin.
(iii) Penicillin can also be used.

Note:
Abortion occasionally occurs following the treatment of heavily pregnant cows with sodium iodide.

Infectious Foot Rot (in Sheep)
Causative agent:
Fusiformus necrophorum
Serious outbreaks in wet sand warm weather.
Signs:
Interdigital dermatitis, underrunning of horn, medial aspect of claw, strong smell of necrotic horn, very severe lameness, walk on knees.
Rx:
Topical treatment. Zinc sulfate solution 10% + sodium lauryl sulfate 2%. Sheep stood for one hour in a footbath. Thirsty sheep may drink the footbath solution and die due to zinc toxicity. Antibiotic treatment; Procaine penicillin single IM of 70,000 IU/kg or 70 mg/kg. Dihydrostreptomycin single IM or Long acting Oxytetracycline single IM dose @ 20mg/kg.

Postparturient Hemoglobinuria
It is due to dietary phosphorus deficiency.
Signs:
Hemoglobinuria, inappetance, reduced milk production, No fever.
Rx:
IV administration of 60 gm of Sodium acid phosphate in 300 ml distilled water and similar dose SC. Further SC injections at 12 hourly intervals on three occasions and similar daily doses by mouth.

Babesiosis
Syn:
Texas fever, Redwater fever, Cattle tick fever
Causative agent:
Babesia spp. [B.bigemina, B. equi]
Transmit by blood sucking ticks of Boophilus spp.
Signs:
Anemia, Hemoglobinuria, Jaundice, Fever, Heavy case fatality rate

Rx:

Imidocarb (Imizole) @ 2mg/kg bwt, Diminazine aceturate is affective in sheep @ 12 mg/kg bwt as a single dose.

Necrotic laryngitis:

Causative Agent:

Fusobacterium necrophorum is normal inhabitant of oral cavity and causes inflammation and
necrosis, injury of the mucosa of oral cavity, pharynx and larynx

Rx:

Debridement of ulcer, Application of solution of tincture of iodine, Oral administration of Sulfamethazine @ 150 mg/kg bwt for 3-5 days.

Glanders

Causative agent:

Pseudomonas mallei. It is Zoonotic in nature.

Signs:

Acute/ Chronic. Penumonia, Nodular ulcers in respiratory tract and on the skin. The disease is highly fatal, nasal discharge.

Rx:

Sulfadiazine and Sulfadimidine for 20 days. No vaccine is available.

Tests:

Mallein Test (Mallein reagent 0.1 ml is injected in lower aye lid and observed after 48 hrs. Swelling of eyelid confirms the +ve case).

Contagious Caprine Pleuropneumonia (CCPP)

A disease of Caprine (Goat), it is not transmissible to cattle and sheep.

Causative agent:

Mycoplasma capriculum

Signs:

Cough, Dyspnoea, Lagging, Lying down a lot, Fever 105-106 F and in terminated stages, there is mouth breathing tongue protrusion and frothy salivation.

Rx:

Tylosin tartrate @ 10mg/kg bwt or Oxatetracycline @ 15mg/kg bwt per day

Contagious bovine pleuro-pneumonia (CBPP)

Caused by Mycoplasma mycoids – small colony

Large colony caused CCPP (Contagious caprine pleuro-pneumonia)

Affect pleura + lungs

Peracute:

Sudden death is observed

Acute:

High temperature/fever (107 F), This infection may be sub acute

Rx:

Tylosine is more effective

Tick pyemia of lamb

Enzootic – Staphylococcus

Transmitted by ticks of Ixodes spp.

Affects lambs of 2 to 10 week age

It may cause sudden death

Contagious bovine pyelonephritis

Caused by Corynebacterium renale

Affects kidneys, and upper urinary tract

Clinical findings:

Hematuria, colic signs, forceful urination, pyouria, depression, decrease milk production

Rx:

Penicillin is the drug of choice

Caseous Lymphadenitis

Caused by Corynebacterium paratuberculosis (intracellular parasite of monocytes/ macrophages). External abscessation of lymph nodes (initially pus is greenish, but latter on calcification gives onion appearance to abscess and can distinguishable) Internal abscessation (liver, intestine, brain and spinal cord).

Listeriosis

Syn:

Circling disease, Silage sickness,

Caused by Gram +ve bacterial agent,; Listeria monocytogenes

There are four forms;

(a) Encephalitis (necrosis in brain)

(b) Septicemia (MOP in pregnant animals, placentitis, and ultimately death)

(c) Abortion

(d) Mastitis

Rx and Control:

Penicillin @ 44,000 IU / kg IM STD for 7 days

VIRAL DISEASES SUMMART CHART:

Disease	Synonyms	Causative Agents
Cow pox	Variola vaccine	Cow pox virus
Goat pox	Variola caprine	Capri pox virus
Sheep box	Variola ovine,ovine pox	Capri pox virus
Swine pox	Variola suilla/contagious impetigo	Sui pox virus
Horse pox	Variola equine/contagious pustular stomatitis	Equi pox virus
Pseudo pox	Milkers nodule/ring sore	Para pox virus
Contagious ecthyma	Contagious pustular dermatitis/ORF/scabby mouth	ORF virus of family pox viridae
Malignant catarrhal fever	MCF/catarrhal fever/gangrenous coryza	Gamma herpes virus
Blue tongue	Sour mouth/epizootic catarrh/pseudo FMD.	Orbivirus of family reoviridae
Pseudo rabies	Ajuskeys disease/mad itch/infectious bulbular paralysis	Herpes virus
Vesicular exanthema		Calci virus
African swine fever	African pig disease/wart hog disease	Irido virus
Transmissible gastroenteritis	T.G.E	Corona virus
Corona virus infection	CCV/viral gastroenteritis	Corona virus
Equine influenza	Equine distemper/typhoid fever/pink eye	Myxo virus
Pulmonary disease in horse	Chronic obstructive pulmonary disease(COPD)	Multifactorial
Respiratory disease complex	Kennel cough	Borderella bronchoseptica
Infectious canine hepatitis	Rubarths disease/contagious hepatitis	Adenovirus 1& 2family adenoviridae
Foot and Mouth Disease	FMD/apthous fever	Apthovirus/picornaviridae
Rinder pest	Cattle plague/	Rinder pest virus of family

	bovine typhus	paramyxoviridae
Peste des Petitis Ruminants	PPR/pseudo rinder pest/kata/goat plague	Morbilivirus of family paramyxoviridae
Canine Distemper	Hard pad disease/canine influenza	canine distemper virus family of paramyxoviridae
Maedi	Progressive interstitial pneumonia	Lento virus of family retroviridae
Bovine leucosis	Bovine lymphosarcoma	Bovine leukemia virus of family retroviridae
Feline Leukemia	Bovine Lymphosarcoma	Bovine leukemia virus of family retroviridae
Equine Infectious Anemia	EIA, Swamp fever	EIA Virus, Lentivirus of family Retroviridae
Vesicular Stomatitis	Mouth thrush/sporadic aptha/stomatitis	VS virus of family rhabdoviridae
Ephemeral fever	Three day sickness/stiff sickness	EF virus of family rhabdoviridae
Rabies	Mad dog/hydrophobia/jalatanka lyssa	Lyssa virus of family rhabdoviridae
Bovine viral diarrhea	BVD/mucosal disease	Pestivirus of family togaviridae
Swine fever	Hog cholera/pest porcine	Toga virus
Japanese encephalitis		Arbovirus of family togaviridae
Looping ill	Ovine encephalomyelitis	Flavivirus of family togaviridae
Infectious Bovine Rhino Tracheitis	IBR/red nose/necrotic rihinitis	Bovine herpes virus 1
Equine viral rhino pneumonia	Rhinopneumonitis/equine viral abortion	Equine herpes virus 1
Respiratory Disease Complex in cat	Cat flu fever/FVR	Feline herpes virus 1
Scrapie	Rida/trembling/du mouton	Prion
Bovine Spongiform Encephalopathy	BSE/ Mad cow disease	Prion
Feline Pan Leukopenia	Feline distemper/feline parvo	DNA virus

	virus/feline ataxia	

VIRAL DISEASES:

Foot and Mouth Disease (FMD)

Syn:

Apthous fever, Contagious aptha

FMD is an acute febrile highly contagious disease of cloven footed animals. It is characterized by vesicular eruption on the epithelium of buccal cavity, tongue, nares, muzzle, feet, teats and udder.

Etiology:

FMD virus (Apthovirus) --- > Family: Picornaviridae; strains found in Pakistan is Asia-1, O, A, C. Virus may remain viable the extent of one year in infected premises. Virus is resistant to common disinfectants. It is destroyed by sodium hydroxide (NaOH), formalin (1-2%) and sodium carbonate (4%). The virus can be preserved in glycerin phosphate buffer with a pH of 7.6. The virus is considered as smallest known virus of animal origin.

Transmission:

Through direct contact with infected animals from infective materials, food stuffs, feeding utensils, waterers etc. Spread through air ways like human influenza virus can also take place. The infection imposes a high spread during the cooler season when the air remains in damp condition. The virus initially settle in the cells of the dorsal surface of the soft palate or on the lateral wall of the pharynx following the entry.

Clinical Findings:

Morbidity:

100%, Incubation period : 2-8 days

Virus multiplies in blood stream -- > viremia --- > temp goes 104 to 106 F following peak temperature characteristic vesicles appear in the oral mucosa (vesicular stomatitis), interdigital space, udder etc. at this stage there is reduction in temperature, profuse salivation, and lameness. Drooling of slimy, ropy salivation and protrusion of the tongue are the futures in this stage; lesions appear in the junction of hoof. In severe case, there is shedding of the hooves. Suckling calf usually die as a result of myocarditis and myocardial degeneration.

Post-FMD complexes:

Penting --- > due to lesions in pituitary gland and thermoregulation (endocrine disturbance), anemia, slower growth of hairs, mastitis, diabetes mellitus.

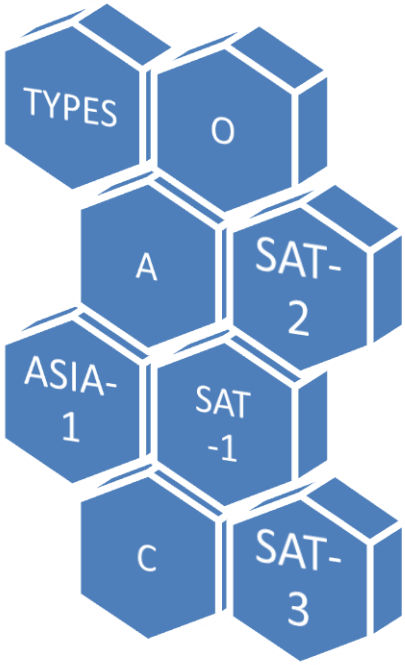
Treatment:

No specific treatment

Antiseptic mouth wash with potassium permagnate, sodium carbonate, boric acid and glycerin may be applied. Antiseptic may be given on foot lesions and mammary tissues.

Vaccination:

Commercially polyvalent vaccines is used which contain O,A, C, Asia-I strains.



Rinderpest

Syn:

Cattle plaque

It is an acute or sub-acute febrile, highly contagious disease of even toed ungulates. It is predominantly a fatal disease of cattle and buffalo characterized by necrotic stomatitis, gastroenteritis, dehydration and destruction of lymphocytes.

Etiology:

Rinderpest virus -- > Genus Morbillivirus -- > family Paramyxoviridae (Latine “morbi” denotes measles). Virus is spherical with diameter 100-300. virus can be destroyed by ultraviolet light and is heat sensitive.

Transmission:

Same like FMD. Feces is the main source of infection. Virus can remain alive upto 8 month in faces. Rinderpest virus got tremendous affinity to lymphoid tissue and epithelial tissues/cells of GIT and respiratory tract. Virus cause destruction of lymphocytes in tissues -- > leucopenia.

Clinical Findings:

Temperature 104 F, Photophobia, Muzzle is dry and there is profuse serious nasal and lacrimal discharge which later on muco-purulent. Necrotic lesion in mucous membrane of oral cavity. The chief sites of oral lesions comprise of inside of the lowerlip and adjacent gums, the cheeks near the commissures of lips; under the tip of the tongue and the back of the hard palate. There is diarrhea and abdominal pain.

Rx:

Rinderpest antiserum @ 1ml / kg IV. Antibiotic, symptomatic fluid and electrolyte therapy.

Peste Des Petitis Ruminants (PPR)

Syn:

Pseudorinderpest, Goat plaque, PPR (plaque as per French phrase in small ruminants); KATA

It is an acute to subacute highly contagious disease of small ruminants having resemblance to rinderpest characterized by fever (106-107 Degree Celsius), diarrhea, nasal discharge. The virus belong to Genus Morbillivirus and Family Paramyxoviridae same genus and family as that of Canine distemper, Rinderpest and Measle virus of human. The virus has tremendous affinity for epithelial cells of GIT and lymphoid tissue.

Transmission:

Some like rinderpest.

On entry, the virus invades the retropharyngeal lymph nodes and mucosa.

Bovine Viral Diarrhea (BVD)

Syn:

Mucosal disease complex

BVD is a subacute, acute, or inapparent contagious disease having the manifestation of high rise of temperature and diarrhea; Pathological features of which are comprised of erosion of the mouth, esophagus, rumen, abdomen and intestine. Diarrhea and erosion of GIT.

Etiology:

BVD virus --- > Genus: Pestivirus and Family: Togaviridae

Host susceptible:

Principally noted in Cattle, less in buffalo, deer and wild ruminants.

Transmission:

Occur during all seasons but more in rainy and winter season. Animal of all ages are affected but more in 6 to 24 months of age.

Blue Tongue

It is an infectious non-contagious (arthropod-borne) viral disease of domestic and wild animals. This is predominately a disease of sheep but occasionally cattle, and goat are affected. The disease is characterized by high fever (105-106 F), sneezing, coughing, salivation, catarrhal inflammation of buccal mucous membrane (stomatitis) and nasal mucous membrane (catarrhal rhinitis), cyanotic and bluish appearance of tongue; painful hoof; a pink line appear on coronet (coronitis + laminitis) which is absent in FMD. Also blood tained frothy saliva, edema of gums, ulcer formation at border of tongue, wry neck (head toward side lacrimation).

Etiology:

Blue tongue virus --- > Genus: Obrivirus --- > Family: Reoviridae

Transmission:

Spread by flies (culicoides) and mosquitoes (ades) and tick (ornithodorous)

Characteristic stomatitis, rhinitis, and lameness

Ephemeral Fever

Syn: Three days sickness

It is an arthropod transmitted disease of cattle characterized by high temperature, stiffness, lameness, muscular tremor with spontaneous recovery.

Cow Pox

It is a contagious eruptive skin disease of cattl . It is mild cutaneous disease where thelesions are mostly confined to udder and teat. The disease is transmissible to human beings.

Etiology:

Cowpox virus --- > Genus: Orthopoxvirus --- > Family Poxviridae

Transmission:

Direct contact, via hands of milkers and also by insects

Equine Influenza

It is an acute febrile highly infectious disease of horses characterized by general septicemia, respiratory problem accompanied by severe persistent + dry cough, nasal discharge.

Etiology:

Equine influenza virus --- > Genus Influenza virus --- > Family Orthomyxoviridae

Bovine Spongiform Encephalopathy

It is a progressive transmissible neurological disease of bovine characterized by sponge like destruction of brain.

Etiology:

BSE is caused by poorly understood type of infectious protein particle called “Prion”. It has also zoonotic importance.

Signs:

Initially abnormal gait particularly hind limb locomotion, behavioral disorders, ataxia, hyperesthesia, excessive salivation, pruritis, rubbing of head, incoordination, restricted stride.

Rx:

Not yet possible. Amphotericin B is somewhat helpful delay the disease but not cure.

Control:

Ban on feeding of ruminant derived protein feed.

Canine Distemper

It is an acute highly infectious viral disease of carnivores characterized by diphasic fever, ocular and nasal catarrh and frequent cutaneous eruptions. This infection is often manifested by bronchopneumonia, gastroenteritis and encephalitis.

Forms of CD

i) Pulmonary form: Oculo-nasal discharge, pharyngitis, bronchitis. Broncho pneumonia is the common feature. The pulmonary form is more prevalent than the digestive form.

ii) Digestive form: Loss of appetite, vomition, abdominal pain, semisolid or loose feces, hemorrhagic enteritis is common in young pups.
iii) Ocular form: Swollen eyelids, conjunctivitis, and purulent discharge from eyes.
iv) Nervous form: Restlessness, excitement, chewing movement, excessive salivation and convulsion. Lymphopenia is the distinct feature of canine distemper. The muscular spasm may be observed in the lips, alae nasii, cheeks, jaws, head, neck or limb muscles.
v) Cutaneous form: Appearance of rash, vesicles and pustules.

In some cases, skin of foot pads and nose become hard due to hyperkeratosis and the condition is described as "Hard Pad Disease". There may be vesiculo-pustular eruptions on the ventral aspect of the abdomen and on the inner parts of thighs.

Rx:

Symptomatic: Anti CD-serum to control the neurological damage. Sedative and anticonvulsants should be given

Rabies

It is an acute viral infection in man and other warm blooded animals. Cold blooded are carriers.

Etiology:

Rabies virus, Genus Lyssa virus and Family Rhabdoviridae. It is a bullet shaped virus; Alkali, sunlight and moderate heat destroy virus. It can be preserved in 50% glycerol.

Pathogenesis:

Following bite, virus is deposited in the depth of wound in the infected saliva. Then local replication of virus in the epithelial cells or myocytes; move centripetally to the central nervous system; nervous pathways **@ 3mm/hr** virus invade ganglion cells and then centrifugally to the peripheral nerves.

There are three forms;

Prodromal:

1-3 days; excited animal, stay in corners, dilated pupil, decrease corneal reflex

Furious:

7-10 days: Over-excited, hypersalivation, muscular in-coordination, tremors, convulsions.

Dumb/ Paralytic:

2-10 days: Ascending paralysis/centripetal paralysis, (H, FQ, Head and neck), No skin and GIT reflexes

Vaccination:

1st at 3 month, then after 1 month, then after 1 month, then after 1 year.

Reservoirs:

Bats and vampires

Stage of excitement:

In this stage, the excitability and irritability increases and dog become very much aggressive. This period may last for 1-7 days. At the onset the dog may hide in dark place due to photophobia. Change in bark due to paralysis of the vocal cord. There is drooling of saliva.

Dumb form:

This form is known as paralytic form. In this form, there is paralysis of lower jaw, tongue, larynx, and hind quarter. The dogs are not capable to bite man and other animals on their feces. Rabies in cats is usually more furious in form than dog. Cat used to bite man and other animals on their faces.

Rx:

No specific treatment for clinical rabies; we can use sedative some drugs like Vinkristin, Scopolamide hydrobromide give good response against rabies virus in experimental animals. The site of bite should be washed with running water and soap.

Equine Viral Arteritis

Caused by Togavirus

Characterized by edema, inflammation of vessels, fluid accumulation in body cavities, persistent high fever, lacrimation, spasm, abortion in pregnant animals,

Transmitted by Coitus

Postmortem findings

Multifocal myometritis

Parvovirus Infection in Dogs

It is characterized by enteritis and myocarditis with high morbidity rate (upto 100%) and frequent mortality (upto10%).

Etiology:

Parvovirus, Genus Parvovirus and Family Parvoviridae

Clinical Findings:

There are two forms of this disease:

a) Parvovirus enteritis:

Hemorrhagic gastroenteritis, serious in pups, vomition, and diarrhea. Polydipsia, Frothy yellow colored vomitus, retching and restlessness. The animal is then pass brownish semi-solid feces admixed with excess mucus followed by foeted haemorrhagic diarrhea.

b) Parvovirus myocarditis

Scrapie

Caused by Infectious proteinacious particles which are very strong stay against boiling

Characterized by pruritis, abnormal gait, mostly chronic. Causes vaculation of neurons in spinal cord, pons, mid brain etc. There are tremors, locomoter disorders, pruritis, change in behavior, hematoma on ear and swelling of face.

Scrapie scratch reaction:

Nibbling, licking, abnormality of gait

Canine Distemper

Syn:

Hard pad disease, Canine influenza

Characterized by diphasic fever, ocular and nasal catarrhal discharge, frequent cutaneous eruption. Due to secondary infection, there may be pneumonia, diarrhea.

Etiology:

RNA –virus, Canine distemper virus, family Paramyxoviridae; which is closely associated with Measles virus of human and rinderpest virus of cattle.

Clinical Findings:

Viremic diphasic fever, skin redness and pustules, gastroenteritis resulting to diarrhea, respiratory problem (pneumonia), reproductive form, nervous form

Vaccine:

1st : 6-8 wk 2nd: 12-14 wk

3rd: 16 wk

Dam vaccine:

1 month before whelping

Infectious Canine Hepatitis

Syn:

Canine adenovirus hepatitis, contagious hepatitis, Blue eye disease, Blue eyes subside in 7 to 10 days.

There are four forms of disease;

Mild: Mild increase in normal body temperature, mild photophobia, enlarged tonsils, sudden recovery is being observed.

Acute: Increased temperature, depression, sceleral edema, conjunctivitis, lacrimation

Per-acute: Sudden collapse, > 104 F temperature, vomiting, diarrhea, animal falls and rolls.

Chronic: Vomiting, purulent nasal discharge, recovered animal becomes carrier, nervous signs. Blueness eye due to cloudness of cornea.

Canine Parvovirus

Syn:

Panleukopenia of dog, Hemorrhagic enteritis

Caused by Parvo-virus, which causes depletion of B and T lymphocytes

Clinical findings:

Vomting, diarrhea, anorexia, dysentery, pale mucous membrane, gastroenteritis, myocarditis leading to heart arrest.

Forms:

1. Enteric
2. Congestive heart failure
3. Acute sudden heart failure

Vaccinations

2nd month

12-14 week

from vaccinated dam

1st at 15 day 2.5 month

from non-vaccinated dam

Vasicular Stomatitis

Caused by vasicular stomatitis virus of family Rhabdoviridae.

Attack on oral mucosa, and coronary band, vesicles develop, which upon rupture becomes ulcers. Nasopharynx ulcers, epistaxis, and hoof may shed.

Equine Herpes Virus Infection

EHV-1 (more pathogenic) than EHV-II

Infection may be associated with:

a) Respiratory system b) Reproductive system or c) Nervous system

Dermatophylosis

Syn:

Ringworm disease

Fungal infection Cause by Trichophytom equinum, T. versicosum and T. microsporum.

Clinical findings

Patchy alopecia, urticaria, pruritis, itching, pain

Use any skin ointments which may effective against fungal infection, Wash by 0.5% solution of Sodium hypochlorite.

Anhydrosis/ Puff/ DryCoat/ Non-Sweating Syndrome

Due to beta 2 adrenergic stimulation,

Caused by 1) hyperkeratinization 2) Hyperthyroidism 23) Heat exertion

Clinical Findings:

Decreased appetite, increased pulse rate, tachypnoea, increased temperature and alopecia.

RICKETTSIAL DISEASES:

Tick Borne Fever

C**ausative agen**t

Ehrlichia phagoctophilia

Signs:

Fever, Depression, Lethargy, Polypnoea, & Fall in milk production in cattle, Abortion

Rx:

Oxytetracycline @ 10mg/kg bwt IV

Q- Fever

Causative agent:

Coxiella burnetti

Signs:

Clinically inapparent and presents mainly as abortion in sheep

Rx:

Oxytetracycline is effective

Brucellosis

Predilaction sites are gravid uterus, mammary lymph node, mammary glands and testes.

Signs:

Swelling of mammary lymph nodes, Abortion, Mastitis, Orchitis

Tests:

Rose Bengal Test, Milk Ring Test

ENDOPARASITIC INFESTATIONS:

Schistosomiasis

Syn:

Nasa

It is a trematodal infection of cattle, sheep, goat, dog and man caused by schistosoma worms.

Causative Agent:

Schistosoma bovis

Life Cycle:

Cercariae penetrate the blood vessels and reach their site of predilection via general blood circulation.

Intestinal Schistosomiasis

Acute form of schistosomiasis, severe hemorrhagic lesions appear in the wall of posterior part of small intestine and caecum particularly. In chronic cases, there is scar formation and destruction of intestinal glands.

Rx:

Praziquantal @ 20 mg / kg orally for 3 days.

Clinical Findings:

In acute case, profuse diarrhea or dysentery. In chronic case, animal becomes emaciated,
anemic and eosinophilia.

Nasal Schistosomiasis

Adult parasites reach the veins of the nasal mucosa and cause rhinitis. Eggs cause small abscess on the mucous membrane of nasal cavity by sloughing out from the blood vessels. Fibrous tissue develops and leads to the large cauliflower like growth in the nasal passage. These are actually granulomatous growth. Common in cattle than buffaloes. The condition is known as “Nasal granuloma” in local language “Nasa”.

Clinical Findings:

Typical symptoms is rhinitis, muco-purulent discharge along with coryza, sneezing and dyspnoea.

Rx:

Tarter emetic @ 2 mg/kg SC or IM daily for 6 days. Copper sulphate is also effective to control the snails and is mixed in water.

Hepatic Fascioliasis

Syn:

Liver fluke disease

Etiology:

Fasciola hepatica – in hilly areas (sheep liver fluke)

Fasciola gigantica – in plane areas (deer liver fluke)

Life Cycle:

Female lays egg 20,000 per day. Life cycle completes in 16 to 17 weeks.

Pathogenesis:

The metacercariae penetrate through intestinal wall and reach peritoneal cavity. They migrate towards liver through peritoneum and then migrate through the liver parenchyma.

Acute fascioliasis

It is found in sheep, not in cattle. Acute form is rare and chronic is common. In acute form, there is traumatic hepatitis due to migration of large no. of metacercariae.

Chronic fascioliasis

In cattle, depletion of calcium in bile duct is common pathogenesis. In cattle bile duct protrude out of the surface of the liver giving it a clay pipe appearance which is popularly known as "Pipe stem".

Clinical Findings:

Acute fascioliasis in sheep most common occur as sudden death with discharge of frothy blood, through nostrils and anus.

In chronic form, animal becomes lazy, mucous membranes become pale and skin becomes dry, subsequently edema starts developing particularly below the mandibles and the condition is known as bottle jaw condition. In cattle, most common symptoms are digestive disturbance, emaciation, constipation with animal feeding very difficulty in defecation and followed by diarrhea in extreme cases.

Diagnosis:

Symptoms, if parasite in bile duct, findings of eggs in feces.

Rx:

Triclabendazole in specific compound for use in sheep and cattle @ 10 mg/kg in sheep and 12 mg/k g in cattle. It is highly effective against all stage of fluke.

Rafaxonide @ 7.5 mg/kg (80-90% effective).

Albendazole @ 7.5 mg/kg in sheep and 10 mg/kg in cattle. It is effective against adult fluke.

Dictylocaulus viviparous is the only lungworm of cattle which is most of the time, treated with albendazole successfully. Trichinella spiralis is found in muscles.

TAPEWORM INFESTATIONS

Cestodes (tapeworms) belonging to Moniezia spp. in ruminants and Anocephala spp in horses. Transmission by ingestion of infected free living pasture mites.

Signs:

Large no. of tapeworms form closely packed ball in the intestine and may cause obstruction, digestive disturbances including constipation, mild diarrhea and dysentery.

Animal occasionally move in circle and show masticatory movement of jaw. In horses, enlargement of belly and symptoms of colic may appear.

Rx:

Albendazole @ 10 mg/kg for horses; Praziquantal @ 1 mg/kg

Coenurosis

Syn:

Gid, Sturdy

This condition is caused by invasion of the brain and spinal cord by the intermediate stage (i.e. coenurus cerebralis) of cestode "Taenia multiceps" which inhabits the intestine of dogs and wild canidae (intermediate hosts). Disease can occur in sheep, goat, cattle, horses and wild ruminants. Clinically it is primarily a disease of sheep and occasionally cattle.

Clinical Findings:

Ataxia, Muscle tremor, Excitability and Collapse. The most obvious sign in slowly developing partial or complete blindness in one eye. Dullness, Head pressing, incomplete mastication, and periodic epileptiform convulsion are the usual signs. If the cyst is present on the spinal cord, there is gradual development of paresis and eventually inability to rise. If cyst is in brain then there is deviation of head and circling. The neurological signs are referred to as "Gid or Staggers".

Diagnosis:

On the basis of signs and particularly characteristic gid movement

Rx:

Surgical drainage of cyst

Haemonchosis in Ruminants

Syn:

Barber pole disease

It is one of the most pathogenic blood sucking nematodal infestation of sheep gat and cattle.

It occurs in the forestomach (abomasum) of the above animals.

Etiology:

Haemonchus placei; in cattle (10,000 eggs per day)

Haemonchus contortus; in sheep and goat

Life Cycle:

No intermediate host. It is more common in hot and humid season.

Clinical Findings:

Acute form of disease is found in lambs and young sheep. There is loose bloody diarrhea; mucus membrane and conjunctiva are extremely pale due to severe anemia.

VETERINARY DISEASES:

Botulism

Syn:

Forage poisoning, Shaker foal syndrome

It is caused by Clostridium botulinum type D. Forage spoiled, spoiled food, hay spoiled in humid environment. Organism produce toxins that paralysis "motor paralysis". Paralysis of masticatory, perineal muscles, tone of GIT also decreased. Animal becomes recumbent.

DDx:

Tetanus

Rx:

Botulinum toxoids are available

Enterotoxemia

Syn:

Pulpy kidney disease

Etiology:

It is caused by Clostridium perfrigens type D, G +ve organism

Signs:

Clinically, high fever, diarrhea (pasting of hind quarter), Mostly affects lambs, kids and calves. Lush green fodders, grains, highly milk feed are pre-disposing to this disease. The toxins are produced in the intestine and absorbed in blood. The kidney becomes pulp like so called "pulpy kidney" because organism affects the kidney and causes necrosis of glomeruli. Glucosurea, high fever, diarrhea, convulsion and death within 2 to 4 hours.
The organism causes sloughing of the intestine and diarrhea.

Rx:

Penicillin, OTC (orally @ 10 mg)

Vaccination:

1st at 3 months, 2nd at 4 months, 3rd repeat twice a year.

Dam vaccination:

2 month before parturition.

Malignant Edema

It is caused by "Clostridium septicum".

Wounds are source of infection. Malignant edema in sheep is called as "Braxy or big head" because of edema on head. Malignant edema should be differentiated from black quarter on the basis of gas gangrene in BQ.

Rx:

Penicillin, OTC, and washing of wounds with hydrogen peroxide (H2O2).

Listeriosis

Syn:

Circling disease

Causative agent:

It is caused by "Listeria monocytogenes" which is acquired through contaminated silage. Clinically it is characterized by encephalitis, meningitis, enteritis, abortion in last trimester and uveitis/opthalmitis. Microabscesses in encephalitis is characteristic feature of this disease. It has zoonotic impact and effect mainly sheep and then other ruminants.

Rx:

Cholortetracycline or penicillin, IV is very effective.

DDx:

(i) Scrapie (ii) Gid (iii) Polioencephalomalacia

EQUINE DISEASES:

(a) Viral Diseases

(i) Equine influenza
(ii) Equine rhinovirus --- > upper respiratory tract
(iii) African horse sickness (Genus: orbivirus) --- > respiratory/circulatory
(iv) Vesicular stomatitis --- > vesicles on foot and mouth
(v) Equine infectious anemia (Family: Retroviridae, Genus: Lentiviru) HIV like

(b) Bacterial Diseases

(i) Tetanus --- > Clostridium tetani
(ii) Anthrax ---- > Bacillus anthracis (iii) Strangles ---- > Streptococcus equi
(iv) Glanders ---- > Pseudomonas mallei
(v) Contagious equine metritis --- > Taylorella equigenitalis
(vi) Mastitis --- > Streptococcus zooepidermicus

Colic

Abdominal pain; visceral pain

Types of Colic

i) **Spasmodic coli:**

Increase in peristaltic movement and spasm occur. Intermittent bouts of pain with period of relaxation.

ii) **Impactive colic:**

Impaction of bowl with dry and partial digested feed contents, bowl lumen distended and its motility reduced. There is continuous pain.

iii) **Obstructive colic:**

Passage of ingesta is obstructed either due to torsion or intussusceptions. There is continuous pain.

iv) **Flatulent colic:**

It is also known as intestinal tympany or tympanitis. There is distension of bowl lumen due to gas accumulation

v) Idiopathic colic;

No obvious cause or lesion is found.

PROTOZOAL DISEASES

Trypanosomiasis

Nagana

Caused by Trypanosoma congolense, It is transmitted by tsetse flies.

Signs:

Intermittent fever, anemia, weight loss, edema of limbs and genitalia.

Surra

Caused by Trypanosoma evansi. It is transmitted by biting flies.

Sings:

Same as above + progressive paralysis of hindquarter in chronic cases.

Dourine:

Caused by Trypanosoma equiperdum. It is transmitted by coitus (mating)

Signs:

Genital and ventral edema, progressive emaciation, ascending motor paralysis.

Babesiosis

Caused by B. equi and B. caballi

DDx:

Equine infectious anemia, Equine viral arteritis, Purpoura hemorrhagica.

Immunological Disorders

i) Severe Combined Immune Deficiency (SCID) is a hereditary disease. It is caused by EHV-I in which immuno-suppression can occur.

Vaccines for Horses

i) Influenza vaccines
ii) EHV-I
iii) Tetanus [Clostridium tetani]
iv) Strangles

DDx of Cattle Diseases in which Redwater Comes:

(i) Enzootic hematuria:

Grazing "Broken" There is hematuria with no pus.

(ii) Enzootic bovine pyelonephritis: Intermittent hematuria, + pyuria caused by "Corynebacterium renale" in cattle.

(iii) Babesiosis: Fever 41 C

(iv) Postparturient hemoglobinurea:

Low phosphate or low copper diet [no temperature increase from normal]

(v) Bacillary hemoglobinurea; Diarrhea + Abdominal pain + Fever 41 C. Mortality is 100 %.

(vi) Leptospirosis: Caused by L. pomona. Mostly calves are affected. There is mucosal petechiation.

(vii) Chronic copper poisoning: Hemoglobinurea, acute death

Infectious stages of diseases:
STAGE 1: INCUBATION PERIOD.
STAGE 2: PRODROMAL PERIOD.
STAGE 3: ACUTE PERIOD.
STAGE 4: CONVALESCENCE PERIOD.

METABOLIC DISORDERS:

Postparturient Hemoglobinurea

Due to low phosphate and copper dietary intake

Rx:
(i) 5 liter blood
(ii) Na-acid phosphate, IV 60gm+ 300 ml water for 5 days
(ii) Dicalcium phosphate orally

Milk Fever

Normal serum calcium level: 4-5 mg/dL

Abnormal level: 1.5-3.0 mg/dL

Condition is characterized by hypocalcemia, weakness, depression of consciousness.

Low calcium level around parturition due to (i) Ca released in clostrum (ii) Intestine can not absorb sufficient calcium (iii) Skeleton can not fulfill the requirement of Ca+. Along with Hypocalcemia, hypophosphatemia and hypomagnecimia may be accompanied.

Clinical Stages:

There are three clinical stages.

Stage I:

Excitement with hypersensitivity

Muscular tremors of hind and forelimbs

Stage II:

Sternal recumbancy

Rectal temperature is subnormal (97-101 F)

Cow unable to rise

Stage III:

Stage of lateral recumbancy

Heart sounds inaudible

Treatment:

Ca-borogluconate, IV 25% solution @ 500-1000 ml.

Characteristic position in milk fever



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A- Prodromal stage: (short excitatory phase)

Characterized by:

- 1- Excitement and restlessness (vigrous-1icking of skin).
- 2- Hypersensitivity and fine muscle tremors.
- 3- Disinclination to eat or move.

(B) Stage of sternal recumbency (Semicomatused stage):

Characterized by:

- 1-Depression of consciousness.
- 2- Animal in sternal recumbency and the head turned laterally toward flank region resting on the shoulder region.
- 3- Dry muzzle dilated pupi1e cold extremities and loss of anal reflex, as well as, atonic digest-ive tract manifested by constipation.
- 4- Temp. normal or even subnormal.

(C) Stage of lateral recumbency (comatosed stage):

Characterized by:

- 1- Animal almost comatozed.
- 2- Animal in lateral recumbency,
- 3- Complete flaccidity of limbs and loss of nervous reflexes.

Clinical signs:

- 3 stages of the disease are known including:
- A) prodromal stage (excitative short phase)
- B) Stage of sternal recumbency (semicomatosed stage).
- C) Stage of lateral recumbency (comatosed stage).

Prevention:

- 1- Feeding high phosphorus low calcium ration in two last months of pregnancy (dry period).
- 2- Single 1/V or S/C-10 million I.U crystalline vitamin-D 8-10 digs before calvinng repeated if the cow not calving.
- 3- Diet rich calcium after parturition only.

Treatment:

- Most untreated cases die within 1-2 days.

(A) Nursing care:

- I- Stomach tube feeding to avoid aspiration pneumonia.
- 2- Turning of animal from side to side to avoid tympany.
- 3- Massage of limbs and bony prominants to avoid limb dysfunction.

(B) Chemotherapy:

- Calcium borogluconate is the drug of choice or the classical **treatment.**
- 400-800 ml C-B-.G 25% slow I/V or S/C repeated after 12 hrs.if no r response, then repeated after 24 hrs. And if no further response use alternative therapy containing phosphorus and magnesium e.g calphomag (Virbac).

Paralytic Myoglobinurea

Called "Azoturia"

A Metabolic Disorder of horses occurring during exercise after a period of inactivity on full ration. It is characterized by myoglobinurea and muscular degeneration.

Tympany / Bloat

Tympany is an accumulation of gas in the rumen while bloat is the accumulation of gas along with frothy foams.

Caused by leguminous plants + high concentrate ration + ruminal stasis

Treatment:

(i) T.T. oil
(ii) Linseed oil (iii) Carbacol Injection 1-3 ml S/C (iv) Nux vomica powder
(v) As last resort, use Trocar and Canula to save the life of the animal

POULTRY DISEASES:

DISEASES OF CHICKEN CORRELATED WITH AGE
BROILERS, PULLETS, LAYERS

Typical losses to 7 weeks of age are 4-5%. Losses in the first two weeks account for 30-50% of total mortality.

BROODING PERIOD (0-2 Weeks)

1. Mortality

a) Management
b) Starvation/ dehydration – floor temperature, water management
c) Navel and yolk sac infection: *Salmonellosis*, *E.coli*, *Staphylococcus*, *Proteus* etc.
d) Vaccine contamination
e) Improper incubation conditions: small, weak hatchling or increased susceptibility to infections.

2. Respiratory disease

a) Aspergillosis (Brooder pneumonia)
b) Vaccine problems (Respiratory reaction)

3. CNS Diseases

a) Avian Encephalomyelitis
b) Encephalomalacia
c) Poor vaccine placement (Pox, MDV)
d) Spiking Mortality

4. Nutritional Deficiencies

a) Rickets
b) Others

Rickets	Osteomalacia	Osteoporosis
Serum calcium or phospate is low	Serum calcium is low	No biochemical changes in blood
Males& females equally affected unless it is X linked	Females more often	Females more often=postmenopause & senility
↓ed mineralizat. of Bone & cartilage at growth plate	Decreased mineralization of bone	No defect in mineralization

5. Eye diseases
a) Ammonia burns
b) Mycotic Keratoconjunctivitis

GROWING PERIOD (2-8)
1. Mortality
a) Coccidiosis
b) Aspergillosis
c) Ascites
d) Marek's disease
e) Clinical infectious bursal disease (IBD)
f) Inclusion body hepatitis / Aplastic anemia
g) Ulcerative enteritis
h) Necrotic enteritis
i) Chicken infectious anemia
j) Gangrenous dermatitis
k) Blackhead

2. Respiratory disease
a) Mycoplasmosis
b) Newcastle disease
c) Infectious bronchitis

d) Infectious laryngotracheitis
e) Colisepticemia
f) Avian influenza

3. Lameness
a) Tibial dyschondroplasia
b) Long bone distortion (Valgus-varus deformaties)
c) Infectious synovitis
d) Bumblefoot
e) Viral arthritis
f) Osteomyelitis
g) Staphylococcosis/ Other septic arthrides
h) Spondylolisthesis
i) Rickets
j) Ionophore/ 3-Nitro-toxicity

4. Skin disease
a) Gangrenous dermatitis
b) Fowl pox
c) Exudative diathesis
d) Skin leukosis

5. CNS disease
a) Avian encephalomyelitis
b) Nutritional encephalomalacia
c) Newcastle disease
d) Marek's disease

6. Others
a) Roundworms
b) Toxicities - Mycotoxin, Botulism, Ionophore, 3- Nitro, etc.
c) Crop Mycosis
d) Cellulitis
e) Swollen head syndrome
f) Inflammatory process
g) Immunosuppression – IBD, CIA

PULLET PERIOD (8-20 weeks)
1. Neoplastic diseases
a) Marek's disease
b) Avian leukosis

2. Respiratory diseases
a) Infectious coryza

b) Infectious laryngotracheitis
c) Mycoplasmosis
d) Infectious bronchitis
e) Newcastle disease
f) Avian influenza

3. Systemic diseases
a) Fowl cholera

LAYERS (>20 weeks)
1. Neoplasia
a) Lymphoid luekosis
b) Carcinoma
c) Sarcoma
d) Marek's disease

2. Respiratory diseases
a) Newcastle disease
b) Avian influenza
c) Infectious bronchitis
d) Mycoplasmosis
e) Infectious coryza
f) Laryngotracheitis

3. Egg production drops
a) Newcastle disease
b) Avian influenza
c) Avian encephalomyelitis
d) Infectious bronchitis
e) Mycoplasmosa gallisepticum
f) Infectious coryza

4. Nutrition / Management
a) Salpingitis / Peritonitis
b) Cage layer fatigue
c) Fowl mites
d) Fatty liver hemorrhagic syndrome
e) Parasitism: Capillariasis, Heterakis, Roundworm etc.
f) Uterovaginal prolapsed
g) Fowl cholera

SPORADIC DISEASES
1. Tuberculosis
2. Botulism

3. Streptococcosis
4. Arbovirus infection
5. Pullorum/Typhoid
6. Other parasitic diseases

Infectious Bursal Disease (IBD)

Syn:

Gumboro

Etiology:

Birnaviridae

Mostly young birds of 6 weeks age are affected. Broiler less affected than layers. At early age, bursa of fibricious does not produce B-cells – immunosuppression

Disease span is:

6-7 days. Morbidity may reaches up to 100%

Clinical Signs:

Birds sit with closed eyes and are unable to stand, drink and eat.

Whitish or yellowish diarrhea on vent -- > bird pick their vent -- > blood mixed diarrhea -- > shivering

Lesions:

Dehydration is noted by shrunken lesions on legs, claws and joints are visible.

Hemorrhages on thigh and pectoral muscles.

Bursa of fibricious swollen -- > yellow exudate

Opening bursa, hemorrhages and exudate

Kidney suffer from nephrosis -- > swollen and large

Spleen becomes enlarged

Treatment:

No specific treatment

Diuretics, Sugar in water to provide energy

Newcastle Disease (ND)

Etiology:

Avulavirus (Family Paramyxoviridae). ND virus is of three types depending upon pathogenecity.

Velogenic:

Highly pathogenic – 100% mortality

Mesogenic:

Intermediate pathogenic – mostly affect young birds

Lentogenic

Produce subclinical signs only in young birds

No vertical transmission because infected eggs do not hatch.

Clinical Signs:

Increased respiration – fluid from nostrils

Mouth paralysis -- > increased temperature

Greenish diarrhea --- > Opisthotoc torticollis

Wings drop --- > lame bird

Lesions:

Hemorrhages on ventricular surfaces, caecal surface, serosa of the intestine and gizzard.

Petechial hemorrhages ulcers on intestine

In caecum, caecal tonsils become congested.

Congestion in trachea

Vaccine:

Valogenic --- > Rockin strains

Mesogenic --- > Mukteswar strains

Lentogenic --- > Lasota strains

FORMS OF NEW CASTLE DISEASE:

1.	Velogenic (high 100% Pathogenic.)
2.	Mesogenic (Moderate)
3.	Lentogenic (Sub clinical)

Avian Influenza

Syn:

Fowl plaque

Etiology:

Orthomyxovirus; three types of virus A, B, C

Two antigens;

HA (Hemagglutinin) NA (Neuraminidase)

No vertical transmission

Clinical Signs:

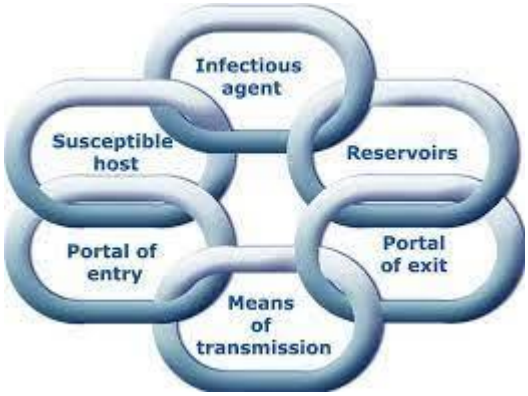
Mild respiratory signs, sneezing. Swelling of head and face. Diarrhea.

Lesions:

Congestive, hemorrhagic necrotic changes in skin. Hemorrhages on digestive tract.

Vaccination:

No vaccination because there is frequent mutations.



ZOONOTIC DISEASES:

- ✓ Type of disease passes from an animal or insect to a human. Some don't make the animal sick but will sicken a human.
- ✓ Zoonotic diseases range from minor short-term illness to a major life-changing illness. Certain ones can even cause death.

Types:

The types of zoonosis include those caused by:

- ✓ Virus
- ✓ Bacteria
- ✓ Fungus
- ✓ Parasites

Examples of Zoonotic diseases

- ✓ Animal flu
- ✓ Anthrax
- ✓ Bird flu
- ✓ Bovine tuberculosis
- ✓ Brucellosis
- ✓ *Campylobacter* infection
- ✓ Cat scratch fever
- ✓ Cryptosporidiosis
- ✓ Ysticercosis
- ✓ Dengue fever
- ✓ Ebola
- ✓ Encephalitis from ticks
- ✓ Enzootic abortion
- ✓ Erysipeloid
- ✓ Fish tank granuloma
- ✓ Giardiasis
- ✓ Glanders
- ✓ Hemorrhagic colitis
- ✓ Hepatitis E
- ✓ Hydatid disease

- ✓ Leptospirosis
- ✓ Listeria infection
- ✓ Louping ill
- ✓ Lyme disease
- ✓ Lymphocytic choriomeningitis
- ✓ Malaria
- ✓ ORF infection
- ✓ Parrot fever
- ✓ Pasteurellosis
- ✓ Plague
- ✓ Q fever
- ✓ Rabies
- ✓ Rat-bite fever
- ✓ Ringworm
- ✓ Rocky Mountain spotted fever
- ✓ *Salmonella* and *E. coli* infections
- ✓ Streptococcal sepsis
- ✓ Swine flu
- ✓ Toxocariasis
- ✓ Toxoplasmosis
- ✓ Trichinellosis
- ✓ Tularemia
- ✓ West Nile virus
- ✓ Zoonotic diphtheria

VENEREAL DISEASES:

Campylobacter fetus (Vibrio),	Bovine Herpes Virus (a form of IBR)
Trichomonas fetus	Ureaplasma,
Mycoplasma	Bovine Viral Diarrhea (BVD)
Hemophilus somnus	Trypanosomiasis

H
O

RIZONTALLY TRANSMITTED DISEASES:

Rickettsiae driven diseases (like typhus)	Rabies

Strongyloides	Toxoplasma gondii
Histoplasmosis	Coccidiomycosis
FMD	

VERTICALLY TRANSMITTED DISEASES:

Bovine Viral Diarrhea Virus (BVDV)	Bovine Leukosis Virus (BLV)
Blue Tongue Virus (BTV)	Mycobacterium paratuberculosis
Bovine spongiform	Neospora caninum
Foot-and-Mouth Disease Virus	Salmonella

DISEASES TRANSMITTED BY TICKS:

Lyme disease	Amblyomma
Theileriosis	Rhipicephalus appendiculatus
Babesiosis	Rhipicephalus
Anaplasmosis	Dermacenter
Ehrlichiosis	Rihipicehalous
Mycoplasma hemocanis	Rihipicehalous
Fransicella tullaransis	Various ticks
Hepatozoan canis	Rihipicehalous
Cytauxzoon felis	Dermacenter

DISEASES TRANSMITTED BY FLIES AND MOSQUITO:

Three day sickness (Ephemral fever)	Flies and mosquito
Blue tongue	Culicuides
Rift Valley Fever	Mosquitoes
Trypanosomiasis	Tse-tse fly, Tabanid flies
Leshmaniasis	Phlebotomus Sand fly
Tularemia	Deer fly
African horse sickness	Culicode midges

WHY DO FLIES SPREAD DISEASES?

Because flies eat garbage, manure, and other bacteria-laden materials, and because **they spit up their stomach contents on surfaces** for which humans and animals may make frequent contact, these winged devils are capable of transmitting different diseases.

VECTORS:

A vector is a living organism that transmits an infectious agent from an infected animal to a human or another animal. Vectors are frequently arthropods, such as mosquitoes, ticks, flies, fleas and lice.

Vectors can transmit infectious disease either actively or passively:

- **Biological Vectors:**
  A vector that transmits the parasite only after it complete its body of host, such as mosquitoes and ticks that carry its pathogens that can multiply within their bodies and be delivered to new host, usually by biting.
- **Mechanical Vectors:**
  A vector that just transport the parasitic stages of life for a parasitic moist fecal matter outside the body, such as flies can pick up infectious agents on the outside of their bodies and transmit them through physical contact.

Disease transmitted by vectors are called *vector borne diseases*. Many vector borne diseases are **zoonotic diseases**, i.e Diseases that can be transmitted directly or indirectly between animals and humans. These include for example Lyme disease, tick-borne encephalitis, west Nile virus, Leishmaniosis and Crimean-congo haemorrhagic fever.

VECTOR BORNE DISEASES:

Disease	Causative agent	Host	Vector
Babesiosis	Babesia microti, B. *bovis* B. *divergens*	Wild rodents. Cattle Cattle and other mammals	Bites of infected *ixodes* ticks
Chagas' disease(American trypanosomiasis	Trypanosomiasis *cruzi*	Dogs, cats, bats, rodents, armadillos, wild and domestic animals	Fecal material of tratoma bug, including Reduviidae(also calledcone-nosed, kissing or assassian bug); contaminates bite wounds abrasions, or mucous membrane
Leishmaniasis Visceral (kala-azar)	Leishmania *donovani* and other species	Wild canids and dogs	Bite of infected phlebotomine sand flies
Leishmaniasis Cutaneous and mucosal	L. *trpoica*, L. *braziliensis* complex	Canids, marsupials, sloths, wild mammals, rodents	

Malaria of non human primates	Atleast 20 species of *plasmodium*	Monkeys, chimpanzees	Anopheline mosquitoes
Trypanosomiasis (African sleeping sickness)	Trypanosoma *brucei*, Trypanosome *brucei Rhodesiense*, T. *brucei gambiense*	Wild and domestic dogs, ruminants, hyenas, carnivores	Bite of infected *tsetse fly(*glossina spp)
Gastrodiscoidiasis	Gastrodiscoides *hominis*	Swine, rat	Snails(metacercariae encyst on plants)
Schistosomiasis (Bilharziasis)	Schistosoma *japonicum* S. *hematobium* S.*mansoni* S.*mattbeei* S.mekongi Schistomes cercariae	Cattle, buffalo, swine, dogs, cats rodents. People are the only reservoir Baboons, rodents, cattle, dogs Cattle. Dogs, monkeys Birds, mammals	Penetration of unbroken skin by cercariae larva from infected snails in water.
Yellow fever	Yellow fever virus (flavivirus)	Monkeys, baboons	Bites of mosquitoes (*Aedes aegypti* in urban cycles, *Haemagogus* spp in jungle cycles in south America, *Aedes* spp in jungle cycles in Aferica)
Western Equine encephalomyelitis	WEE Virus(alpha virus)	Wild birds, domestic fowl, horses, mules, Donkeys, bats, reptiles, amphibians	Mosquitoes(*culex tarsalis i*n USA, other *culex* and *Aedes* spp outside USA)
Western Nile Fever	Western Nile Virus (flavivirus)	Wild birds horses	Bites of mosquitoes (culex univittatus, C.pipiens, C modestus)
Wesselsbron fever	Wesselsbron virus (flavivirus)	Sheep	Bites of mosquetos (*Aedes, Mansonia, culex* spp)
Vescular stomatitis	Vascular stomatitis virus(Indiana and newly Jersey strains)	Swine, cattle, horse, bats, rodents, other wild mammals	Insect bites including bites mosquitoes and biting flies(*Phlebotomous* spp)

Venezuelan equine encephalomyelitis	VEE virus(alpha virus)	Rodents. equids	Bites of mosquitoes(*Monsonia Ades, Culex* spp)
Sindbis virus disease	Sindbis virus(alpha virus)	Birds	Bites of mosquitos(*culex* spp)
St.Louis encephalitis	St.Louis encephalitis virus (flavivirus)	Wild birds, domestic fowl	Bites of mosquitoes(*Culex tarsalis, C.pipiens-quiquefasciatus complex, C nigripalpus*)
Rift Valley fever	Rift Valley fever virus(phlebovirus)	Sheep, goats, cattle, camels	Bites of mosquitoes(*Ades* spp)
Omsk hemorrhagic fever	Omsk hemorrhagic fever virus(flavivirus)	Rodents, muskrats	Bits of ticks (*Dermacentor* spp)
Murray valley encephalitis	Murray Valley encephalitis virus (flavivirus)	Wild birds	Bites of mosquitoes(*Culex annulirostris*)
Louping ill	Louping ill virus(flavivirus)	Sheep, goats, grouse, small rodents)	Bites of ticks(ixodes ricinus)
Kyasanur forest disease	Kyasanur forest virus (flavivirus)	Rodents, monkey	Bites of ticks(*Haemaphysalis spinigera*)
Japanese B encephalitis	Japanese encephalitis virus(flavivirus)	Swine. Wild birds, horses	Bites of mosquitoes *culex tritaeniorhynchus,* other *Culex* spp)
Far eastern tick-borne encephalitis(Russian spring-summer encephalitis)	Far eastern (Russian spring- summer encephalitis)virus (flavivirus)	Birds,small mammals, sheep)	Bites of ticks(i*xodes persulcatus* and *ixodes ricinus*)
Eastern equine encephalomyelitis	EEE virus (alpha virus)	Wild birds, domestic fowl, horses, mules donkeys)	Mosquitoes(*Culiseta melanura and ades spp)*
Crimean-Congo hemorraghic fever	Crimean-Congo hemorraghic fever virus(bunyavirus)	Cattle, rodents, sheep, goats birds)	Bites of (*Hyalomma and Boophilus* spp)
Colorado tick fever	Colorado tick fever virus	Ground squirrels, chipmunks Porcupines and	Bites of ticks(Dermacentor andersoni)

		sma54ll rodents	
Central European tick borne encephalitis	Central European tick borne encephalitis virus(flavivirus)	Rodents hedgehogs, birds, goats, sheep	Bites of Ixodes ticks;may be milk borne)
California group infections LaCross encephalitis	California group of bunyaviruses	Groud squirrels, other rodents	Bites of mosquitoes (*Ades* ssp)
Myiasis	Cordylobia anthrophage (Tumbu fly), *Dermatobia hominis* (human botfly) *Gastterophilus* spp (equine botfly)	Mammals	Tambufly. Human botfly, equine botfly
Ascariasis(Mange)	Mites of *sarcopetes, Cheyletiella, Dermanyssus,* and *ornothonyssus* spp	Domestic animals	Mites
Theleriosis	Theleria spp	Domestics and wild ruminants	The tick vector is *Rhipicephalus appendiculatus*
Bartonelliosis	B.henselae	Feline	Cat flea (*Ctenocephalides)*
Lyme disease	Borrelia bacterium	Dog , horses and sometime cattle	By the bite of tick The blacklegged tick (or *deer tick, Ixodes scapularis*)
Murine Typus	*Rickettsia typhi*	Rats cats andopossum	By bite of fleas *Xenopsylla cheopis*
Feline hemoplasmosis	Mycoplasma *haemofelis*	Cats	Ticks, mites , fleas and mosquitoes

NOTIFIABLE, SELF LIMITING AND OPHISTHOTONUS POSTURE DISEASES:

Notifiable Diseases	Self Limiting Diseases	Ophisthotonus posture
Anthrax and Glander	Coccidiosis and Ketosis	Milk fever and Botulism

RX OF MASTITIS:

Rx of acute mastitis:			Modern Rx of mastitis
For lactating:	For dry animals:	For G-ve bacteria	TMP + Sulphadiazine (Penicillin G intramammary for 5 days at evening)
Ampicillin + Na-cloxacillin	Benzathin + Cloxacillin	Furazolidone in procaine penicillin	

LIST OF DRUGS USED IN VETERINARY MEDICINE:

Sr. No.	Category/ Disease	Drugs
1.	Babesiosis	Imidocarb @ 3mg / kg bwt Diminazine @ 3-5 mg/kg bwt S/C Quinuronium sulphate [Acaprine, Babesan] @ 0.4 mg/kg bwt Tryan Blue [Fresh] @ 1-4 gm / kg bwt
2.	**Trypanocidals**	Suramine [Antrypol] @ 10 mg/kg bwt Diminazine @ 8 mg / kg bwt S/C
3.	**Anti- Giardia, Amoeba, Blantidtium, Trichomonas**:	Metronidazole; in canine dysentery @ 50 mg/kg wt orally for 5 days In bovine trichomoniasis: @ 75 mg/kg bwt IV 3 times a day
4.	**Anitcoccidals**:	Suplhaquinoxalone Amprolium
5.	**Ascaricidals:**	Piperazine
6.	**Roundworms**	Thiabendazole
7.	**Broadspactrum Antiparasitic**	Albendazole, Fenbendazole, Mebendazole, [Benzimedazole] Tetramizole Levamisole Pyrental pamoate [Combantrin] Against Nematodes, Ectoparasites and Heartworms: Ivermectin Taenicidal: Niclosamides Flukicidals (Against Liver Fluke]: Oxyclosanide @ 15 mg /kg bwt orally Broodicidal: Preziquantal @ 5 mg/kg bwt for cat/dog mostly
8.	**ANTIFUNGALS**	Nystatin [Against Streptomyces Neursei] Griseofulvin [Against Penicillium spp] Synthetic: Coltrimazole Amphotericin B Ketokenazole
9.	**ANTIVIRALS**	Cyclovir

		Acyclovir Amantadine Ribavirale (Against Hepatitis B & C viruses)
10.	**RESPIRATORY STIMULANTS**	Nikthamide Amphetamine Sulphate
11.	**ANEMETICS**	Dimenhydramine Diphenhydramine Cyclizine Meclizine
12.	**EMETICS**	Apomorphine Conc. HCl solution
13.	**ECBOLICS (**Uterine Contractors)	Oxytocin PGF2 alpha (Dinoprost)
14.	**DIURETICS**	Furosemide (Lasix) Mannitole

PARASITIC DRUG OF CHOICE:

Parasite	Hydatid cyst	Amoeba	Trematode	Cestode	Nematode	Schistosoma.	Triclabendazole
DOC	Albendazole	Metronidazole	Oxyclozanide	Piperazine Sulphate	Oxfendazole	Praziquantel and tartar emetic	Liver fluke

DRUG AGAINST TICKS / INTERMITTENT ECTOPARASITES:

DOC in round worms	Piprazine
DOC in caecal worms	Phenothiazine
DOC of hair worm	Phenothiazine,hygromycine,methridin
DOC of tracheal worms	Tincture iodine + Lugols iodine, Barium tartarate

DRUG OF CHOICES:

Disease/ Organism	**DOC**
Fasciolosis	Trichlabendazole
Cestodes + Trematodes	Praziquantal
Psittacosis	Chlortetracycline
Babesiosis	Berenil/diaminoaceturate
L.tropica	Batberium sulphate
Theleriosis	Butalex/buparvaquinone
L.donovani	Antimonyl compound
Surra	Quinapyramine sulphatequinapyramine chloride

Coccidiosis	Amprolium
Ring worm	Grisoflavin
Liver fluke	Oxyclozanide
Round worm	Piperazine
Hook worm	Parental palmoate
Pullurum	Sulpha
Meningitis	Penicillin/chloramphenicol
Salmonellosis	Sulpha
Listeriosis	Ampicillin
Clostridia	Penicillin
Blue breast in rabbits	Penicillin
Respiratory diseases	Sulpha
Pyelonephritis	Procaine penicillin
Endocarditis	Penicillin

DIFFERENCE B/W DIARRHEA AND VOMITING:

Diarrhea	Vomiting
Metabolic acidosis due to loss of Bicarbonate from Intestine	Metabolic Alkalosis due to loss of HCl
pH will decrease	pH will increase

VACCINATION SCHEDULE:

Disease	Age	Interval	Month
FMD	3rd month	Every six month	Jan-Feb, June-July
BQ	6th Month	Every year	Aug-Sep
HS	6th Month	Every Year	Sep-Oct
Anthrax	6th Month	Every Year (Affected area only)	April - May
Brucellosis	4-8th month of Heifer	--	Mar - April

IMPORTANT DISEASES WITH PECULIAR FEATURES:

1-Azooturia	a-Sarcastic b-Horse attempting to lift the hind quarter c-Glutted muscle d-Increase serum creatinine phosphokinase enzyme e-Monday morning disease

	f-Increase nitrogen & myoglobin in urea
2-Fatty Cow Syndrome	Pregnancy toxemia
3-Downer Cow Syndrome	a-Post parturient recombency b-Frog leg like leg c-Unknown etiology d-Paresis
4-Ventricular fibrillation	a-Sudden calcium infusion b-In-coordinated twitching of ventricle
5-Pimply gut	O radiatum
6-Casonis intracutaneous test	Echinococcus
7-Babesiosis	Horse bacillary fever
8-Percarditis	a-Grape like masses b-Pain on pressure of cardiac area + cardiac dullness c-Edema of brisket
9-Myocarditis	a-Cardiac murmur b-Tigor heart
10-Endocarditis	a-Cardiac murmur b-Cauliflower like growth

NAMES OF VITAMINS:

Vit A	Retinol
Vit C	Ascorbic acid
Vit E	Tocopherol (Apha, Beta, Gamma and Delta)
Vit D	Ergocalciferol/ Cholecalciferol
Vit K	Phylloquinone
Vit K_2	Menaquinone
Vit K_3	Menadiones

VITAMINS & MINERALS RELATED ISSUES:

Molybdenum Increase	Nutritional anemia
Folic acid Deficiency	Glossitis,macrocytic anemia
Mg decrease/ Potassium Increase	Green grass/ Grass tetany
Pantothenic acid Deficiency	Goose steeping gait in pig , ulcerative colitis,dermatitis,scabs and crust around eyes,beaks,vent
Sulphur Increase	Red water
Thiamine (B_1) Deficiency	Polioencephalomalacia, polyneuritis
Copper Deficiency	peat score, molybdenum, Enzootic ataxia
B_2 (Riboflavin) Deficiency	Clubbed down feather, curled toe paralysis
Choline Deficiency	Slipped tendon

Vit B_{12} Deficiency	Ketosis
Copper Deficiency	Demyelination & cerebral ataxia+keratinization+pigmentation of hair
Mn, B_{12} and B_3 Deficiency	Perosis
Vit E Deficiency	Nutritional encephalomalacia+infertility+muscular dystrophy + anemia
Zn Deficiency	Scaliness in feet, Enlargement of hock joint, bone shortening poor feather
Vit E, Deficiency	Exudative diathesis
Thiamine in N. D Deficiency	Star grazing posture
Riboflavin/ Niacin Deficiency	Black tongue
Niacin Deficiency	Bowed leg
Biotin Deficiency	Pellagra/ 3 D **(Dementia, Diarrhea and Dermatitis)**

Important points:

- ✓ Threonine is tryptophan.
- ✓ Methionine is glycine.
- ✓ Isoleucine is valine.
- ✓ Neurobian is B1+B6+B12.
- ✓ Methycobal is B12.

VITAMIN B-COMPLEX RELATED ISSUES:

Vitamin B	Function	Deficiency Diseases
Thiamine-B_1	Carbohydrates into Energy	Beri beri
Riboflavin-B_2	Immune booster	Ariboflavinosis
Niacin/ Nicotinic Avid-B_3	Decrease cholesterol	Pellagra
Pantothenic acid-B_5	Anti stress Hormone and Metabolism	Paresthesia
Pyridoxine-B_6	Vision and brain	Anemia
Biotin-B_7 (Vitamin H)	(Skin, Hair, Nail and Sugar)	Dermatitis,enteritis
Folate/ Folic Acid-B_9	Pregnancy and baby	Megablastic anemia
Cobalmin-B_{12}	RBCs, DNA and nerve	Megablastic anemia

VITAMIN DEFICIENCY DISEASES:

Vitamin	Disease
Vit A	Night blindness
Vit C	Beri-beri, Scurvy and Gum swelling
Vit D	Rickets and osteomalacia
Vit B1	Beri beri

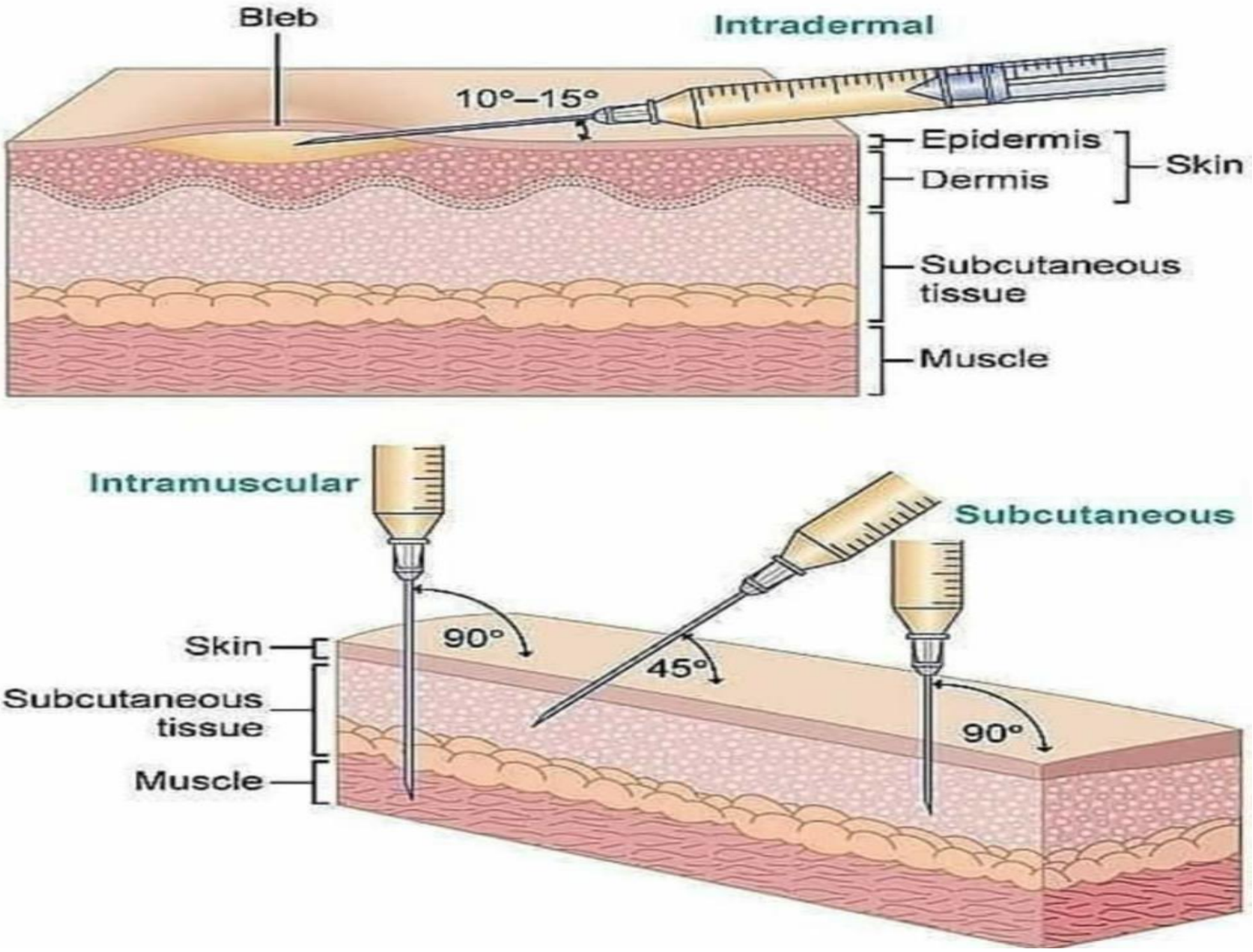
Vit B17 (Amegdalin)	Cancer
Vit E	Less fertility
Vit K	Non-clotting of Blood

HYPERSENSITIVITY REACTION AND ITS TYPES:

Immediate Type	Delayed Type
Anaphylaxis	Arthus reaction
Graft Rejection	Tuberculin reaction
Acute Allergic reaction	----------
Exp: Type I, Type II, Type III, Type V	**Exp: Type IV**

Sr. No.	Type	Synonym
1.	**Type-I**	Anaphylaxis or anaphylactic hypersensitivity
2.	**Type-II**	Cytotoxic H.S
3.	**Type-III**	Immunocomplex Mediated H.S
4.	**Type-IV**	Cell Mediated H.S
5.	**Type-V**	Stimulating H.S

Injection Sites





Bulla
Circumscribed collection of free fluid, >1 cm



Macule
Circular flat discoloration, <1 cm brown, blue, red or hypopigmented



Nodule
Circular, elevated, solid lesion, >1cm



Patch
Circumscribed flat discoloration, >1cm



Papule
Superficial solid elevated, ≤0.5 cm, color varies



Plaque
Superficial elevated solid flat topped lesion, >1 cm



Pustule
Vesicle containing pus (inflammatory cells)



Vesicle
Circular collection of free fluid, ≤1 cm



Wheal
Edematous, transitory plaque, may last few hours



Scale
Epidermal thickening: consists of flakes or plates of compacted desquamated layers of stratum corneum



Crust
Dried serum or exudate on skin



Fissure
Crack or split



Excoriation
Linear erosion



Erosion
Loss of epidermis (superficial); Part or all of the epidermis has



Lichenification
Thickening of the epidermis seen with



Scar
Thickening; permanent fibrotic changes that occu

BASICS OF MICROBIOLOGY:

Endospore Forming Bacteria:

1. Bacillus anthrasis
2. Bacillus subtilis
3. Sporosarcina
4. Clostridium

- ✓ Louis Pasture: Father of Microbiology and gave **"Germ Theory of Disease".**
- ✓ Robert Koch: Father of Bacteriology and gave **"Koch's postulates".**
- ✓ Optimum temperature requirement for bacterial growth is 37 Degree Celsius.
- ✓ **Psychrophile (15 Degree C), Mesophile (25-40 Normally 37 Degree C) and Thermophiles (45 Degree C)**

Immunology:

- ✓ Immunology is study of defense mechanism of body against infection and diseases.
- ✓ Phagocytosis (cellular immunity) is term given by E.Metenchikoff.

IMMUNE RESPONSE:

Humoral (B cells or B-lymphocytes)	Cell mediated (T cells or T lymphocytes)

LYMPHOID ORGAN:

1-Primary.	2-Secondary
Thyums, Bursa of fabricious, Bone marrow, Skin, Payers patches.	Lymph nodes, Spleen, Milk spots, Tonsils, Bone marrow

DIFFERENCE B/W GRAM +VE AND GRAM -VE ORGANIMS:

Gram +ve	Gram -ve
Thick Walled	Thin Walled
Absence of Outer Membrane	Presence of Outer Membrane
Violet Color on Staining	Red Color on Staining

DIFFERNCE B/W ENDOTOXIN AND EXOTOXIN:

Endotoxin	Exotoxin
Heat Stable	Heat Labile
Made up of Lipopolysaccharide	Made up of Protein
Cannot convert into harmless	Convert into harmless by Formaldehyde
Present in Gram -ve	Mostly present in G +ve
E.g: H.S	E.g: BQ

DIFFERENCE B/W ACID FAST STAINING AND GRAM STAINING:

Acid Fast Staining	Gram Staining
Carbol Fuschin	Crystal Violet
Methylene Blue	Saffranine

INCLUSION BODIES:

Sr. No.	Inclusion Bodies	Disease
1.	Itra-Nuclear	ICH, Herpes Virus and Adeno Virus
2.	Intra-Cytoplasmic	Rabies, Goat Pox, Canine Distemper, Pox Virus (Acidophilic) and Cow Pox (Eosinophilic)
3.	Both	Rinderpest and Canine Distemper Virus

DIFFERENCE B/W ACIDIC STAIN AND BASIC STAIN:

Sr. No.	Stain Type	Examples
01.	Acidic Stain	Picric Acid, Eosin, Rose Bangal and Acid Fuschin
02.	Basic Stain	Methylene Blue, Basic Fuschin, Crystal Violet and Malachite Green

STAINING:

Sr. No.	Items	Color
1.	Mucin	**Blue**
2.	Mucin with H & E	**Blue**
3.	Mucin with Alcian Blue	**Blue**
4.	Collagenous fibers with Malori Trichromic Stain	**Blue**
5.	Pseudomucin	**Pink**
6.	Hydropic Degenerated Vesicles	**Pink**
7.	Glycogen color with Best's Caramine Stain	**Light Pink**
8.	Elastic Fibers with Resorein or Orcein	**Light Pink**
9.	Iodine to Amyloid	**Mahogany Brown**
10.	Fat Color by Osmic Acid	**Black**
11.	Iodine by weak 1% H_2SO_4 color to Amyloid	**Black** or **Blue**
12.	Methyl Violet to Amyloid	**Rose Red** and surrounding **Blue**
13.	Congo red to Amyloid	**Red**
14.	Dye to assess extent of Amyloidosis IV Injection	**Congo Red**
15.	Mucin with PAS	**Purplish red**
16.	Fat Color by Sudan III	**Yellow**/ **Orange**

17.	Fat Color by Sudan IV	**Red Color**
18.	Passive Hyperaemia Organs Color	**Bluish Red**
19.	Muscle fibers with H & E	**Reddish Violet**.

STAINS:

- ✓ Diene's stain: Mycoplasma
- ✓ Fontana stain: Spirochaetes
- ✓ Machiavelli stain: Chlamydia

GOLD STANDARD TESTS:

Sr. No.	Disease	Test
1.	Glander	CFT
2.	Leptospira	Microscopic Agglutination Test
3.	Rinderpest	Virus Neutralization Test and CFT
4.	PPR	Virus Neutralization test
5.	Rabies	FAVN and FAT
6.	Swine Fever	FAVN
7.	Influenza	HI
8.	FMD	ELISA
9.	Blue Tongue	ELISA and PCR
10.	African Horse Sickness	ELISA and CFT
11.	IBD	AGID
12.	Equine Infectious Anemia	Coggin's Test

AEROBIC, ANAEROBIC AND SPORE FORMING BACTERIA:

Aerobic	Anaerobic
E. Coli	Actinomyces
Citrobacter	Bifidobacterium
Klebsiella	Fusobacterium
Proteus	Propionibacterium
Salmonella	Clostridium
Achromobacter	Bacteroides
	Prevotella

SPORE FORMING BACTERIA:

Clostridium spp
Bacillus spp
Actinomyces
Sporobacterium

MICROORGANISM SHAPE:

Sr. No.	Organism	Shape
01.	**Rabies**	Bullet Shape
02.	**Tetanus**	Drum Stick
03.	**Cl. Chovei**	Pear shape/Tannis Racket
04.	**Rhabdo Virus**	Bullet Shape
05.	**Bacillus**	Rod shape.
06.	**Cestodes**	Ribbon Shape
07.	**Flate worm**	Leaf shape.
08.	**IB Virus**	Crown Like Projection
09.	**Stilesia hepatica**	Dumble shape
10.	**Seteria Digitata**	Lemon shape egg
11.	**Trichuris spp.**	Barrel shape egg

ORGANISM AND TOXINS:

Staphylococcus aureus	α-toxin,
Escherichia coli	Heat-stable toxin
Botulinum	Neurotoxins
Tetanus	Neurotoxins
S. aureus	Toxic-shock syndrome toxin
Corynebacterium diphtheria	Diphtheria Toxin (Dtx)
Pseudomonas aeruginosa	Exotoxin A
Shiga toxin	Cytotoxic necrotizing factor type 1

EPIDEMIOLOGICAL TERMS:

Terminology	Definition
Zoonoses	Diseases naturally transmitted between vertebrate animals and humans
Anthropozoonoses	Diseases in animals that can be transmitted to man
Zooanthroponoses	Diseases affecting humans that can be transmitted to animals
Amphixenoses	Diseases that are exchanged between animals and human occasionally
Euzoonoses	Diseases in which humans are an obligatory host of the pathogen
Cyclozoonoses	Diseases that require two different vertebrate hosts but no invertebrate vector
Pherozoonoses	Pherozoonoses isosymptomatic, similar symptoms are observed in animals and humans
	Pherozoonoses anisosymptomatic, symptoms are different in animals and humans
Cryptozoonoses	Zoonotic diseases in which symptoms are only evident in humans
Saprozoonoses	Diseases that depend upon inanimate reservoirs and vertebrate hosts
Emerging parasitic zoonoses	Zoonotic diseases caused either by new parasites or by old known species in an area where the disease was previously unknown

STAPHYLOCOCCI	STREPTOCOCCI
1. They are arranged in grape like clusters.	1. They are arranged in chain.
2. They may be coagulase positive or negtive.	2.They are always coagulase negative.
3. Infection cause by them is associated with purulent discharge.	3. Infection caused by them is associated with serous discharge. (because of streptodornase)
4. They produces golden yellow coloured pigment in culture media.	4. They don't produce any pigment in the culture media.
5. No diseases are associated with them related to molecular mimicry.	5. Infection by Lancefield Group A streptococcus is associated with molecular mimicry,causing Rheumatic fever and Glomerulonephritis.

BASICS OF PATHOLOGY:

Inflammation of area	Terminology
Arteries	Arteritis
Bile Duct	Cholangitis
Blood Vessel	Vaculitis
Bone	Ostetitis
Bone Marrow	Osteomyelitis
Brain	Encephalitis
Bursa	Bursitis
Caecum	Typhlitis
Colon	Colitis
Cornea	Keratitis
Dura Matter Of Brain	Pachymenengitis
Ear	Otitis
Endocardium	Endocarditis
Eustachian Tube	Eustachitis
Eye Lid	Blepharitis
Eye	Ophthalamitis
Gall Bladder	Cholecystitis
Glans Penis	Balanitis
Heart	Carditis
Intestine	Enteritis
Iris	Iritis
Joints	Artritis
Kidney	Nephritis
Lacrimal Gland	Dacryoadenitis
Ligaments	Desmitis
Lip	Cheilitis
Liver	Hepatitis
Lung	Pneumonitis
Lymph Nodes	Lymphadenitis
Renal Pelvis	Pyelits
Salivary Glands	Sialadenitis
Sinus	Sinusitis
Skin	Dermatitis
Spermatic Cord	Funiculitis
Stomach	Gastritis
Testicles	Orchitis
Vagina	Vaginitis
Vertebrae	Spondylitis
Lymph Vessels	Lymphadenitis
Meninges	Meningitis

Mouth	Stomatitis
Muscle	Myositis
Myocardium	Myocarditis
Nasal Cavity	Rhinitis
Nerve	Neuritis
Ovary	Oophoritis
Oviduct	Salpingitis
Pancreas	Pancreatitis
Pericardium	Pericarditis
Perostium	Periostits
Pleura	Pleuritis
Prepuce	Posthitis
Renal Glomeruli	Glomerulits
Spinal Nerve Root	Radiculitis
Spleen	Splenitis
Tongue	Glossitis
Trachea	Tracheitis
Tympanum	Tympanitis
Uterus	Metritis
Vas Deferens	Vasitis
Vein	Phlebitis
Vessels	Vasculits
Fats	Steatitis
Fascia of a Muscle	Fasciitis

PATHOGNOMIC LESIONS OF DISEASES:

1-VIRAL DISEASES:

Diseases	Pathognomic Signs	Gross Lesions
1-F.M.D	Lesions on coronary band and teats	Tiger heart in Calf
2-Rinder pest	Grains like bumps in nostrils and inside cheeks and lips	Zrebra strippings in intestine.
3-Peste des petitis ruminants	pneumonia	Zebra strippings in mucosa of colon and rectum
4-Malignant catarrhal fever	Nervous signs(head and eye)	swollen,friable, and hemorrhagic lymph nodes and other lymphatic tissues,nonsuppurative vasculitis and encephalitis
5-Blue tongue	Bluish appearance of mucous membrane of	

	tongue,coronitis	
6-Ephemeral fever	Polyphasic fever(102-107.6) degree forenhite	Edema of lymph nodes of lungs,polyserositis
7- Infectious bovine rhinotracheitis	Rhinitis(red nose)	
8- Pseudo rabies	Intense local pruritis	Pulmonary edema,increase cerebriospinal fluid
9- Pseudo cow pox	Horse shoe shapped ring	
10-swine fever	Nervous signs	Turkey egg appearance of kidney Oval button ulcers in caecum and colon
11-African swine fever	Hyperemia of distal limb	Paint brush hemorrhage on serosa of stomach
12-contagious ecthyma	Scaby mout	Rete ridges and hyper keratosis of skin

BACTERIAL DISEASES:

Diseases	Pathognomic signs	
1-Anthrax	Absence of clotting of blood and rigor mortis,blood oozing from natural orifices	
2-Hemorrhagic septicemia	Edematous swelling of throat area	
3-Brucellosis	Abortion,orchitis,mastitis	
4-Tetanus	Temp 108-110f,lock jaw	
5-Paratuberculosis	diarrhea	chronic, contagious granulomatous enteritis
6-Leptospirosis	Spontaneous abortion	nonsuppurative hepatitis
7-Listeriosis	Fascial paralysis,circling	Meningoencephalitis,csf is doudy
8-Colibacilosis	gingivitis and petechial haemorrhages on the tongue	Abomasitis
9-Salmonelosis	diarrhea that may contain blood or mucus.	
10-Actinomycosis	Lumpy jaw	Suppurative necrosis with sulfur granules
11-Actinobacilosis	Wooden tongue	Club like rosette
12-Dermatophylosis	Lumpy wool	

13-Swine erysipelas	diamond-skin,purple skin lesions	Cauliflower looking lesions on the heart valves
14-Compylobacteriosis	Watery to mucoid diarrhea	Darting motility in fresh stool
15-Glanders	crater-shaped ulcers around the muzzle and limbs	
16-Strangles	Swollen lymph nodes around the jaw	Retropharyngeal and brain abscess
17-Exudative epidermitis	Greasy lesions	Distention of the kidneys and ureters with mucus
18- Infectious kerato conjuctivitis	marked pain when the eye is exposed to direct sunlight corneal ulceration, pink eye	
19-Black quarter	lameness	Black colour of leg muscle,crepitus sound
20-Enterotoxemia	Some times bloody diarrhea	pulpy kidney
21-Bacillary hemoglobinuria	Port wine colour urine	Quick rigor mortis
22-Botulism	Limber neck.excess urates in loose dropping	
23-Contagious Bovine Pleuropneumonia	Poly arthritis	thoracic cavity may contain up to 10 L of clear yellow or turbid fluid mixed with fibrin flakes
24-Contagious Caprine Pleuropneumonia		excess of straw-colored pleural exudate and acute fibrinous pneumonia

POULTRY DISEASES:

Diseases	Pathogonomic signs	
1-New Castle (ND)	Greenish Diarrhea, Torticollis	Haemorrhages on tip of Proventriculus and Cecal tonsils
2-Infectious Bronchitis (I.B)	Urate Plug, Tracheal Rales Misshapen egg	Exudate in trachea and nasal cavity Caseous plug in bronchi Kidney pale swollen Salpingitis
3-Avian Influenza (A.I)	Cyanosis of wattle and comb Ruffled feather Edema of head	Pin point haemorrhages on base of Shank and on All viscera
4-Infectious	Blood tinge nasal	Tracheitis Blood in trachea

Laryngotracheitis (ILT)	discharge Conjunctivitis	Conjunctivitis
5-CRD CCRD	Slow FCR Lameness	Yellow colour cheesy flakes around lungs, Synovitis Pericarditis Peri-hepatitis
6-Infectious bursal disease 7-7-Gumboro	Young birds Sudden mortality at 3-7 days Swollen/ atrophy of bursa	Haemorrhages on thigh & musculature Atrophy of bursa
7-Marek's Disease	Tumor on feather Blindness Grey iris Paralysis	Grey white neoplastic lesions on liver spleen
8-Infectious Coryza	Facial swelling Upper respiratory signs Swollen wattle Purulent ocular/nasal discharge	Catarrhal inflammaton of nasal sinus ,Conjunctivitis
9-Fowl Cholera	Swollen joints	Focal hepatitis Star shape white foci on live
10-Typhoid	Thirst Yellow diarrhea	Engorgement of kidney &spleen Enteritis of anterior small intestine
11-Pullorum Disease	Whitish diarrhea White fecal pasting on vent	Grey nodules in the liver, spleen, lungs, heart, gizzard, and intestine. Firm, cheesy material in the ceca.
12-Necrotic Enteritis	Closed eyes Immobility Golden yellowish diarrhoea	Golden yellow intestine
13-Aspergillosis	Ocular discharge	plaques in lungs, air sacs, trachea, peritoneal cavity, may have greenish surface. Conjunctivitis/keratitis
14-HPS (Angara)	Yellow mucoid droppings	Excessive strawcoloured fluid in pericardium (up to 10 ml) Enlarged, pale friable liver and kidney Lungs edematous
15-Avian Nephriti		Visceral Gout
16-Egg Drop Syndrome	Loss Of Color In Pigmented Eggs .Production Of Thin Shelled Or Shell Less Eggs.	Inactive Ovaries And A Decrease In The Size Of Oviduct
17- Salmonellosis (Para Typhoid)	White Areas Of Dead Tissue In The Lungs , Liver And Heart .	Unabsorbed Yolk
18-Candidiasis		Layer Of White Cheesy Material In The Crop.
19-Tape Worm Infection	anemic	Large Tapeworm May Block The

		Whole Intestine .
20-Coccidiosis	Diarrhea Which May Mucoid Or Bloody	Caeca Greatly Enlarge And Distended With Clotted Blood, e Lumen Is Filled With Blood
21- Ascites		Presence Of Large Amount Of Fluid In Abdominal Cavity ,Under Then Covering of the heart
22-Cage Layer Fatigue	Posterior Paralysis,and Death	Fracture In Leg And Wing Bones , And Thoracic Spine ,Sternum Is Often Deformed.
23-Cannibalism	Feather Pulling, And Vent, Head, And Toe Picking	
24-Oophoritis: (Follicular Regression)		Turgid Yellow Ovules Become Wrinkled
25-Bumble Feet	Abscess In The Foot Pad	Tendonitis , Septic Arthritis

PARASITIC DISEASES:

Diseases	Pathogonomic signs	
1-Schistosomiasis a-(nasal) b-(intestinal)	Cauli flower like growth in nose, rhinitis,	Nasal granuloma. In the intestinal and hepatic forms, adult flukes are found in the portal, mesenteric, and intestinal submucosal and subserosal veins
2-Fasciolosis (hepatica)	Jaundice,froty blood through nostrils and anus,bottle jaw	Pipe stem liver,biliary obstruction
3-Coenurosis	unilateral blindness,convulsions	Water filled Cyst in brain and spinal cord
4-Echinococcosis		Hydrated cysts in liver and lung

PROTOZOAN AND RICKETTSIAL DISEASES:

Diseases	Pathognomonic Signs	
1-Anaplasmosis		Petechial hemorrhages in pericardium and epicardium
2-theleriosis	Corneal opacity	
3-Babesiosis	haemoglobinuria	CNS involvement due to adhesion of parasitized erythrocytes in brain capillaries,reddish brown urine in gall bladder
4-Trypanosomiasis	Testicular degeneration	
5-leshminiasis	epistaxis	Glomerulonephritis
6-Toxoplasmosis	icterus, seizures	chorioretinitis

FUNGAL DISEASES:

Diseases	Pathogonomic Signs
1-Dermatohylosis	Mycotic dermatitis,hair matted together
2-Rhinosporidiosis	vascular friable polyps in nose and external eye

MOST IMPORTANT PATHOGNOMONIC LESIONS:

1. Ground Glass appearance of cytoplasm and parboiled swelling of organ- Cloudy Swelling
2. The short estrous cycle in large domestic animals is a pathognomonic sign for uterine infection.
3. Pathognomic sign of fracture is Crepitation.
4. Pathognomic signs of sharp molar is Quidding, Halitosis, and Salivation.
5. Haemorrhages at the tip of the proventricular gland is the pathognomic lesions seen in Ranikhet disease.
6. Bronze discoloration of the liver in poultry is the pathognomonic feature of Salmonellosis/ Fowl typhoid.
7.The pathognomonic finding in ICH is
8. Pinpoint hemorrhages on the tip of proventriculus glands is a pathognomonic feature of Newcastle disease.
9. Hemorrhages at the junction of glandular stomach & mechanical stomach is a characteristic feature of Infectious Bursal Disease.
10. Cornmeal liver is present in Fowl Cholera.
11. Splenomegaly is a pathognomonic feature of Spirochetosis.
12. Continuous lacrimation in canine is a characteristic sign of Obstruction of the lacrimal duct.
13. ‘Butchers jelly' by Hypoderma lineatum.

14. ‘Sweet itch’ by Culicoides robertsi.
15. Big Liver Disease- ALC.
16. Angara Disease- Hydropericardium.
17. Milk spots by Ascaris suum
18. Swimmer’s itch by Schistosoma spp.
19. Dollar spots by T. equiperdum

CLASSIFICATION ACCORDING TO EXUDATE:

Serous
Hemorrhagic
Catarrhal
Purulent
Fibrinous

CLASSIFICATION ACCORDING TO SEVERITY:

Acute (exudative)	Chronic (proliferative)

COMMON TERMINOLOGIES:

Granuloma	Small nodular lesions of chronic inflammation with some special feature.
Hemorrhage	Escape of blood from an artery,capillary,vein to the outside(external)or inside (internal)body cavity /tissue.
Epistaxis	Bleeding from nose
Hematemesis	Blood in vomit
Hemoptysis	Blood in sputum
Enterorhagia	Bleeding from intestine
Hematuria	Blood in urine
Hemothorax	Blood in thoracix cavity
Hemopericardium	Blood in pericardium
Hemosalpinx	Blood in oviduct
Anemia	Reduction in quantity and number of Rbcs
Oligocythemia	Increase in total blood volume
Thrombosis	Intravascular clotting of blood
Necrobiosis	Programmed cell death
Apoptosis	Physiological cell death
Necrosis	Cell death due to an injury
Pustules	Visible collection of pus,can be below the epidermis
Abscess	Localized collection of pus caused by pyogenic bacteria

- ✓ Coccidiosis is transmittion via infected droppings with Emeria tenalla, cause caecal coccidiosis (blood in droppings).

GRANULAR AND A-GRANULAR CELLS OF BLOOD:

Granular cells	Agranular cells
Neutrophils	Lymphocytes
Eosinophils	Monocytes
Basophils	

INFECTION AND BODY CELL RELEASE:

Sr. No.	Infection	Cell Release
01.	Viral Infection	Lymphopenia
02.	Chronic Infection	Monocytosis
03.	Acute Local Peritonitis	Leucopenia
04.	Acute Diffuse Peritonitis	Leucocytosis and Neutrophilia

DIFFERENCE B/W ACUTE AND CHRONIC INFLAMMATION:

Acute inflammation	Chronic inflammation
Removal of cause	Removal of inflammatory part
Ice pack application/cold water	Warm application
Application of astringent lotion	Antiseptic application for septic wound
Injection of anti-inflammatory	Manage
Injection of antibiotics	Apply counter irritants
Complete rest	Try Biers hyperemia
	Anodyne preparation

DIFFERENCE B/W EXUDATE AND TRANSUDATE:

Sr. No.	Parameters	Exudate	Transudate
1.	Origin	Acute Inflammation	Cardinal/ Renal failure
2.	Mechanism	Increase membrane permeability	Hydrostatic pressure increase and Osmotic pressure decrease
3.	Gross Changes	Turbid (WBCs), Colored (RBCs),Flaex (Fibrin)	Clear
4.	Specific Gravity	1.018	1.015
5.	Coagulability	Fibrin present	Fibrin absent
6.	Chemical Changes	3-4 % protein	<1 % protein
7.	Cytology	WBCs and RBCs present	Cells absent

8.	Organism	Present	Absent

DROPPED JAW, BOTTLE JAW AND LUMPY JAW:

Sr. No.	Type of Jaw	Disease
1.	Dropped Jaw	Tetanus
2.	Bottle Jaw	Fascioliosis
3.	Lumpy Jaw	Actinomycosis

GANGRENE AND ITS TYPES:

Invasion and putrification of necrotic tissue by saprophytic bacteria.

Sr. No.	Type of Gangrene	Condition
01.	Dry Gangrene	Extremities I.e Ear, tail and nail etc
02.	Moist Gangrene	Internal Organs
03.	Gas Gangrene	Spore forming Bacteria

CARDINAL SIGNS OF INFLAMMATION:

Sr. No.	Sign	Synonym
01.	Redness	Rubort
02.	Heat	Calor
03.	Swelling	Tumour
04.	Pain	Dolar
05.	Loss of Function	Laesa function

TYPES OF ANEMIA:

Macrocytic normochromic	Microcytic Hypochromic	Macrocytic hypochromic	Microcytic normochromic	Normocytic normochromic	Macrocyti hypochromic
Cobalt	Cu , Me poisoning	Babesia.Anaplasmosis, Bacillary hemoglobin-urea	Stomach worm & nephritis	Bracken fern poisioning & radiation injury	Sweet clover poisoning

Morphology Classification

- Microcytic (MCV <80 μ^3):
  - Iron deficiency, thalassemia, chronic disease/inflammation, sideroblastic anemia, lead poisoning.
- Normocytic (MCV 80-100 μ^3):
  - Acute blood loss, chronic disease, hypersplenism, bone marrow failure, hemolysis.
- Macrocytic (MCV >100 μ^3):
  - B12 or folate deficiency, hemolysis with reticulocytosis, chemotherapy, hypothyroidism, MDS.

DIFFERENCE AMONG PERACUTE, ACUTE, SUBACUTE AND CHRONIC:



Main Types of Inflammation

- Peracute: Very rapid onset and fulminating
- Acute: Rapid onset and of shorter duration
- Subacute: Slower onset with longer duration than acute
- Chronic: Slowest onset with protracted duration

Acute disease	Chronic disease
(i) These disease lasts for short period of time	(i) Lasts for long period of time
(ii) Can cause major effects on general health in a very short time	(ii) It takes a long time to major effects on general health
(iii) Feeling of tiredness	(iii) One feels tired all the time
(iv) Weight loss does not occur	(iv) Weight loss does occur
(v) Examples : Common cold, flu and asthma	(v) Examples : Diabetes, cancer and arthritis

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Acute diseases	**Chronic diseases**
Equine influenza	Surra
Anthrax	Johnes disease
SARS	Hyperthyroidism(cats)
Rinder pest	Hypothyroidism(dogs)
Acute hepatopancreatic necrosis	Inflammatory bowl disease
Avian influenza	Diabetes melitus
Acute septicemia	Skin allergy
Mastitis	Hepatitis
Black quater	Liver diseases
Fowl cholera	Chronic kidney diseases
Bovine respiratory diseases complex	Arthritis
Ruminal acidosis	Cancer

Acute pink eye	Chronic respiratory diseases
Infectious bovine rhinotracheitis	Mastitis

Peracute Diseases	Sub acute Diseases
H.S	Rheumatoid arthristis
Anthrax	T.B
Ammonia toxicity	Subacute endocarditis
Polioencephalomalacia	Glanders
CBPP	CBPP

Classification of Jaundice

III. **According to site of disease:**

Class of Jaundice	Type of Bilirubin raised	Causes
Pre-hepatic or Hemolytic	Unconjugated	✓ Hemolysis/Abnormal RBCs ✓ Antibodies ✓ drugs and toxins Thalessemia ✓ Hemoglobinopathies
Hepatic or Hepatocellular	Unconjugated and conjugated	✓Viral hepatitis ✓Toxic hepatitis ✓ Intrahepatic cholestasis ✓Gilbert's Syndrome ✓Crigler-Najjar Syndrome
Post-hepatic or Obstructive	Conjugated	✓Extrahepatic cholestasis ✓Gallstones ✓Tumors of the bile duct ✓Carcinoma of pancreas ✓Carcinoma of ampulla of Vater

TYPES OF JAUNDICE

PRE HEPATIC	HEPATIC	POST HEPATIC
Excessive amount of bilirubin is presented to the liver due to excessive hemolysis	Impaired cellular uptake, defective conjugation or abnormal secretion of bilirubin by the liver cell	Impaired excretion due to mechanical obstruction to bile flow
Elevated unconjugated bilirubin in serum	Both conjugated and unconjugated bilirubin may be elevated in serum	Elevated conjugated bilirubin in serum

BASICS OF BIOCHEMISTRY:

CARBOHYDRATES:

Monosacchrides	Mannose, Glucose, Fructose and Galactose
Disaccharides	Cellobiose and Maltose
Trisaccharides	Raffinose and Sugar beets
Pentosans	Xylan

METABOLIC WATER:

Substrate	Metabolic Water Produced
1 gram glucose or sugar	**0.60 grams**
1 gram starch	**0.56 grams**
1 gram protein	**0.40 grams**
1 gram fat	**1.07 grams**

OXIDATION REDUCTION REACTION:

1. **Oxidation**: Addition of Oxygen+Removal of Hydrogen+Removal of Electrons
2. **Reduction**: Removal of Oxygen+Addition of Hydrogen+Addition of Electrons

PAIRED TERMINOLOGIES:

- ✓ Species is a group of living organisms consisting of similar individuals capable of exchanging genes or interbreeding.
- ✓ A breed is a specific group of domestic animals having homogeneous appearance (phenotype), homogeneous behavior, and/or other characteristics that distinguish it from other organisms of the same species.

- ✓ Bottle Jaw: Fascioliosis
- ✓ Swollen Neck: Haemonchosis

- ✓ Angle of Lourch: Tuber Coxas
- ✓ Angle of Croup: Tuber Socrale

- ✓ Wolfian Duct: Males (Mesonephric ducts)
- ✓ Mullerian Duct: Females (Paramesonephric ducts)

- ✓ Age of Puberty in Riverine Buffalo (50 Chromosomes): 15-18 months
- ✓ Age of Puberty in Swamp Buffalo (48 Chromosomes): 20-24 months

- ✓ Cow Cervix Length: 5-10 cm
- ✓ Cow Vagina Length: 25-30 cm

- ✓ Gizzard (Muscular): Non-glandular stomach
- ✓ Proventriculus: Glandular stomach

- ✓ Acetic acid and Propionic Acid: Structural Carbohydrates (Milk Fat and Milk Lactose synthesis, respectively)
- ✓ Butyric Acid: Non-structural carbohydrates-Cell content (Ketone Bodies and Fat Synthesis).

- ✓ Emboli: Artery
- ✓ Thrombi: Vein

- ✓ Femur: Thigh bone
- ✓ Tibia: Shin bone

- ✓ Prostate Secretions: Acidic
- ✓ Fructose/ Seminal Vesicle: Alkaline

- ✓ Exudate: Acidic
- ✓ Pus: Alkaline

- ✓ Persistent froth: Reticulo pericarditis
- ✓ Recurrent froth: Diaphragmatic hernia

- ✓ DS RNA Virus: Reo Virus
- ✓ SS DNA Virus: Parvo Virus

- ✓ Largest Exocrine gland: Liver
- ✓ Smallest Exocrine gland: Goblet (Unicellular gland)

- ✓ Largest Endocrine gland: Pancreas/ Thyroid
- ✓ Smallest Endocrine gland: Oinea

- ✓ Bolinger Bodies/ Borrel Bodies: Fowl Pox
- ✓ Guarneri Bodies: Small pox
- ✓ Negri Bodies: Rabies
- ✓ Dohle's bodies: Neutrophil

- ✓ Vagus nerve: Longest
- ✓ Sciatic nerve: Largest

- ✓ Southern Bloat Technique: DNA
- ✓ Northern Bloat Technique: RNA
- ✓ Western Bloat Technique: Protein

- ✓ Merocrine Secretions: No Change to cell E.g, Sweat gland and Salivary gland.
- ✓ Apocrine Secretions: Part of cell is released E.g, Skin, Breast, Eyelid, Ear, Mammary Gland.
- ✓ Holocrine Secretions: Destroys the whole cell E.g, Sebaceous gland and Meibomian gland.

- ✓ Epimysium: CT that surrounds the entire muscle tissue
- ✓ Perimysium: CT that surrounds the each bundle of muscle fiber
- ✓ Endomesium: CT that surrounds each single muscle fiber/myofiber/muscle cell.

- ✓ Incubating Temperature: 99.5 Degree F or 37.5 Degree Celsius
- ✓ Setter temperature: 99.5-100 Degree F/80-85 Degree F or 27-30 Degree Celsius
- ✓ Hatcher Temperature: 98-99 Degree F/ 70-75 Degree F or 21-24 Degree Celsius.

- ✓ In-season flock: Pullets grown during the period when most of the days have decreasing light.
- ✓ Out-season flock: Pullets grown during the period when most of the days have increasing light.

- ✓ Isoschizomers: Recognize and cut the same sequence.
- ✓ Neoschizomers: Recognize the same sequence but cut sites vary.

- ✓ Relaxed: The plasmid is maintained in the host cell in multiple copies.
- ✓ Stringent: The plasmid is maintained in the host cell in a limited number of copies.

- ✓ Nose ring: Two years age
- ✓ Horn ring: Three years age

- ✓ In a sex linked cross involving silver and gold, the silver gene carrying Female parent is used.
- ✓ In a sex linked cross involving silver and gold, the gold gene carrying Male parent is used.
- ✓ In a sex linked cross involving feathering gene, a late feathering Female parent is used.

- ✓ Blue eye: Paramyxoviridae in pigs/ Infectious Canine Hepatitis.
- ✓ Pink eye: New Forest Disease/ Infectious Keratocunjunctivitis/ Equine Influenza.

- ✓ Pharmacokinetics: Effect of body on the drug.
- ✓ Pharmacodynamics: Effect of drug on the body

- ✓ Alkaline Phosphatase: The enzyme which is used to remove the phosphate group at the 5' end of the DNA molecule.
- ✓ Polynucleotide kinase: The enzyme which is used to add the phosphate group at the free 5' end of the DNA molecule.

- ✓ Intron: The non coding sequence within gene that do not translate into protein.
- ✓ Exon: The coding sequence within gene that translate into protein.

- ✓ Telocentric (I-shape): Terminal centromere
- ✓ Acrocentric (J-shape): Sub-terminal centromere or centromere is located close to the end giving a very short arm and exceptionally long arm.

- ✓ Sub-Metacentric (L-shape): Centromere is located near to the center
- ✓ Metacentric (V-shape): Centromere in the center & chromosome with equal arms.

- ✓ Crossing over occurs between non sister chromatids of homologous pair of chromosomes.
- ✓ The interchange of chromosome segments in none homologous chromosomes is called translocation.

- ✓ LD bodies: Leishmaniosis
- ✓ KB bodies: Theileriosis

- ✓ Cattle Buffalo placentome: 70-90
- ✓ Sheep Goat placentome: 80-120
- ✓ Deer placentome: 5-6

- ✓ Median plane: Sagital plane
- ✓ Transverse plane: Frontal plane

- ✓ Dorsal spine is largest: Thoracic vertebra

- ✓ Transverse spine is largest: Lumber vertebra

- ✓ Largest Vertebra: Lumber
- ✓ Strongest Vertebra: Sacral

- ✓ Tendon: Muscle to bone
- ✓ Ligament: Bone to Bone

- ✓ Adduction: Towards centre.
- ✓ Abduction: Away from centre.

- ✓ Flexion: Brachialis muscle
- ✓ Extension: Trcep brachi

- ✓ Muscle insertion in the form of thin sheet: Aponeurosis
- ✓ Muscle insertion in the form of fibrous cord: Tendon

- ✓ Eustachian duct: Naso-pharynx
- ✓ Stenson's duct: Parotid Salivary duct (Largest salivary gland)
- ✓ Wharton's duct: Sub-mandibular salivary duct

- ✓ No. Of Tracheal cartilages in animals: 50-60
- ✓ No. Of tracheal cartilages in fowl: 115

- ✓ 'Yes' joint: Occipito-atlantal articulation
- ✓ 'No' joint: Atlanto-axial joint

- ✓ Retractor occuli muscle: Optic nerve
- ✓ Superficial obliqus muscle: Trochlear muscle

- ✓ Individual entire development: Ontogeny
- ✓ Ancestral history of species: Phylogeny

- ✓ Papillae with taste buds: Fungiform and Circum Vallate
- ✓ Papillae large in size: Circum vallate.

- ✓ Head of femur: Acetabular cavity
- ✓ Head of Humerus: Glenoid Cavity

- ✓ Falciform ligament: Liver to diaphragm
- ✓ Caudate ligament: Liver to kidney

- ✓ Reticular Groove: Cardia
- ✓ Oesophagus to stomach: Cardia

- ✓ Os-cardis: Bone of Heart

- ✓ Os-penis: Bone of penis
- ✓ Os-phrenic: Bone of Diaphragm in camel
- ✓ Os-rostrale: Bone of snout in pig
- ✓ Os-opticus: Bone of eye in Birds

- ✓ Round Ligament of Uterus
- ✓ Proper Ligament of Ovary

- ✓ Round Ligament of Uterus
- ✓ Proper Ligament of Ovary

- ✓ CIDR= 1.95 gm P4
- ✓ PRID= 1.55 gm P4 + Estradiol benzoate (10 mg)

- ✓ Rabies forms: 2
- ✓ Strangle complications: 2
- ✓ Glander forms: 3 (In human, 4 forms)
- ✓ Enterotoxemia forms: 5

- ✓ Listeriosis= Silage Disease
- ✓ Leptospirosis= Rice Workers Disease.

- ✓ Babes Nodules= Rabies
- ✓ Milker Nodules= Pseudo Cowpox/ Parapox virus.

- ✓ Pearl diseases= T.B
- ✓ White Diamond Disease: Swine erisepales

- ✓ Horse pox= Variola Equine/ Contagious pustular stomatitis
- ✓ Goat pox= Variola Caprina
- ✓ Cow Pox= Variola Vaccina

- ✓ Rinder Pest= Cattle Plague
- ✓ PPR= Goat Plague.
- ✓ CBPP= Lung Plague.

- ✓ Rabies Virus speed= 3-4 mm/hr
- ✓ Sperm Speed= 3 mm/ Sec

- ✓ Temporary Teat Engorgmnet= Buff (Doki)
- ✓ Waxing of Teat= Mare

- ✓ Embryo Holding medium- 0.4 % BSA
- ✓ Embryo Flushing medium- 0.1 % BSA

- ✓ Avg. Time for MBR test by superior quality semen: 3-6 minutes.

- ✓ Time for MBR test by Poor quality semen: 30 minutes.

- ✓ Broiler ration should contains 22-24% crude protein
- ✓ Layer ration should contains 16-18% crude protein.

- ✓ In Bovines, the Heart is formed & starts beating on the 22nd day of pregnancy.
- ✓ In Chicken, the beginning of the heart is started and beats on the 2nd day of Embryo development.

- ✓ Comfortable temp for optimal growth of chicken? 65-85 degrees F
- ✓ Eggs for hatching should be stored from 15-17 Degree Celsius

- ✓ Ovulation fossa: Mare
- ✓ Ovarian Bursa: Bitch

- ✓ Bulbus glandis: Dog
- ✓ Corona glandis: Stallion.

- ✓ Horse Urine pH: Basic
- ✓ Ruminants urine pH: Basic
- ✓ Dog Urine pH: Acidic

- ✓ PABA: Para Amino Benzoic Acid
- ✓ GABA: Gamma Amino Butyric Acid

- ✓ Giblet: Gizzard, Heart and Liver
- ✓ Pluck: Lungs, Heart and Liver

- ✓ Phagocytosis: Ingestion of Solid material
- ✓ Pinocytosis: Ingestion of surrounding Fluid

- ✓ Sign (Objective Evidence): Manifestation of disease that the physician perceives.
- ✓ Symptom (Subjective Evidence): Manifestation of disease apparent to the patient himself/ Owner of patient

- ✓ Steroid: Steroids inhibit phospholipase A2 and block inflammatory cascade.
- ✓ NSAID: NSAIDs stop the action of Cyclo oxygenase, thus preventing PG synthesis.

- ✓ Auscultation: Action of listening to sounds with a stethoscope as a part of diagnosis
- ✓ Percussion:It is a method of tapping on a surface to determine the underlying structures.

- ✓ Tympany: Accumulation of free gases, knows as tympanites, cause outside the rumen.
- ✓ Bloat: Over distension of rumenoreticulum wirh froathy gases of fermentation that makes bubble, known as bloat, cause within rumen.

- ✓ Pseudorabies: Aujeszky’s disease

- ✓ Pseudo FMD: Blue Tongue

- ✓ Burn: Injury of skin and subcutaneous tissue by dry heat
- ✓ Scald: Injury of skin and subcutaneous tissue by moist heat.

- ✓ Petechial Haemorrhage: Pinpoint on Skin and Subcut
- ✓ Echymosis Haemorrhage: Spots on Skin and Subcut
- ✓ Perpura Haemorrhage: Haemorrhage all over the body

- ✓ Kv= Quality of beam
- ✓ mA= Quantity of beam

- ✓ Azoospermia: Devoid of sperms
- ✓ Necrospermia: Immotile Sperms.

- ✓ Growth of Cilia: Metestrous
- ✓ Uterine Glands Atrophy: Diestrous

- ✓ Bleeding from ear and nose: Brain
- ✓ Bleeding from ear and eye: Head

- ✓ Leptospirosis= G -ve
- ✓ Listeria= G +ve

- ✓ Listeria= Brain
- ✓ Leptospirosis= Urine

- ✓ Rani Khet- India Area
- ✓ New Castle- USA State Area

- ✓ Eye worm of Ruminants: Thelazia spp.
- ✓ Eye worm of Poultry: Oxyspirura mansoni

IMMUNOGLOBULINS SUMMARY:

1. Ig with maximum synthesis per day.....IgA
2. Immunoglobulin that is heat-labile.....IgE.
3. Immunoglobulin mediating the prausnitz Kastner reaction .....IgE
4. Homocytotropism is seen in which Ig...IgE.
5. Ig with minimum serum concentration..IgE
6. Ig with the shortest half-life .....IgE
7. Ig with minimum synthesis per day.....Ig E
8. Allergic Reaction.. IgE
9. Mast cells adsobs- IgE
10. Ig that crosses the placenta ..... IgG.
11. Ig in secondary immune response....IgG
12. Ig with minimum sedimentation coefficient ....IgG
13. Ig with maximum serum concentration..IgG
14. Ig with the longest half-life ..... IgG
15. Ig responsible for hypersensitivity pneumonitis .....IgG
16. Binary exposure to antigen results in a sudden increase in.....IgG
17. Major Share....IgG
18. Ig present in milk ..... Ig G and Ig A
19. Ig in sero-mucinous glands..IgG and IgA
20. Ig in primary immune response.....IgM
21. Ig with maximum molecular weight ....IgM
22. Ig with max. sedimentation rate ..IgM
23. Primary Response.... IgM
24. Blood group antibodies belong to which type of immunoglobulins ....Ig M
25. Rheumatoid factor belongs to which type of immunoglobulins ..... Ig M (antibody against Fc fragment of IgG)
26. Ig appearing first in life.....Ig M
27. Ig to fix complements via classical pathway..... Ig G1 and Ig M
28. Ig to fix complements via alternate pathway .....Ig A, Ig D, Ig G4
29. Arthus Reaction IgG+IgM
30. Colostrum IgG+IgM+IgA

VETERINARY TESTS:

1. Liptek test- Abomasal displacement.
2. Dick test- Scarlet fever (Strept. Pyogens)
3. FAT- Reference test for rabies.
4. Cuboni test- Pregnancy diagnosis in mare.
5. Vaginal Mucos Agglutination test- Campylobacter diagnosis.
6. Poll test- TRP.
7. Rossett Inhibition Test- Early Pregnancy Diagnosis.
8. Galactose tolerance test and BSP Test- Detect Liver Damage.
9. Hotis test- Strep. Aglactiae Mastitis.
10. Test for residual chlorine- Orthitolidine.
11. Hobel Test (Mouse Test)- Rabies
12. Knott Test- Mirofilariae in dog
13. Rose Waaler Test- Rheumatoid factor test.
14. Van den Bergh test- Jaundice.
15. Stormont test- TB.
16. ABR Test- How the inner ear, called the cochlea, and the brain pathways for hearing are working.
17. CAMP Test- Identify Streptococcus agalactiae.
18. Coggins test- EIA
19. Milk Ring Test- Brucellosis
20. Rose Bengal Test- Brucellosis
21. Strauss Test- Glander+ Brucella+ Blue Tongue.
22. Merchuric Chloride test- Mercuric salt is tested for its solubility in water.
23. Formol Gel Test- Detect the greatly increased serum proteins in visceral leishmaniasis.
24. Ascoli Precipitation Test/ Mac Fadyean's test- Anthrax
25. Mallein Test- Glander
26. Agglutination test- Vibriosis
27. Oxidase test- Identify bacteria that produce cytochrome c oxidase, an enzyme of the bacterial electron transport chain.
28. Catalase test- Identify organisms that produce the enzyme, catalase which detoxifies hydrogen peroxide.
29. Phosphatase test- Milk Pasteurisation.
30. Ames test- Mutagen Carcinogenicity
31. Old Hen Test- Organo Phosphorus Induced Delayed neurotoxicty.
32. White Side Test- Somatic cell counting and nonspecific bacterial genital infections of repeat breeding cattle.
33. Surf Field Test- Mastitis
34. Test Cross- Dominant character due to homo or heterozygosity.
35. Organoleptic test- Immediate milk quality test
36. Alcohol Test- Ethanol in blood.
37. Sediment test- Body information.
38. Booling test- Meat Color/ Odor.
39. Tuberculin test-TB
40. Hamster egg penetration test- Functional integrity of sperm membrane

41. Hyposmotic swelling test- Evaluate the functional integrity of the sperm's plasma membrane.
42. Coagulation test- Blood ability to clot.
43. Hensa test- Buffalo and Cow Milk adulteration.
44. Weil Felix test- Rickettsial infections.
45. Rectal pinch test- John's disease
46. Skinfold test- Dehydration
47. Bulk tank test- Mastitis
48. California test- Mastitis
49. Johnin Intradermal and Intravenous test- Paratuberculosis
50. Fetal slip membrane test- Pregnancy Dx.
51. Rig test- Cryptorchidism.
52. Aschein Zondek test and Friedmoat test- Pregnancy Diagnosis.
53. Riding test- Heat Dx.
54. Delvo test- Presence of Antibiotics and Sulpha residues in milk
55. Sand Ball and Tail to hair test- Presence of Trypanosomiasis.
56. Provocative Scrapie test-
57. Charm test- Detect penicillin residues in Milk.
58. Coomb's test- Detect the presence of Antibody on RBCs or in serum.

GLOSSARY:

Nutrition:
It is study of quality and quantity of nutrients for particular stage of animal for particular goal. Goal is to optimize productivity.

Nutrients:
The chemical substance found in feed that can be used and are necessary for maintenance, production and health of the animal. Main classes of nutrients are: Water, Carbohydrates, Protein, Fat, Minerals and Vitamins.

Forages:
Vegetative portion of plant fed to the animal in fresh, dried or ensiled state. Although most forages are roughages. Conversely many forages such as corncobs, straws are not forages.

Roughages:
These are feeding stuffs which are bulky and contain more than 18% crude fiber and less conc. of energy yielding nutrients. These as high in fiber content and low in energy (1-1.4 mcal/kg of DM). Example: Straws, green fodder, hay, silage

Concentrate:
These are feed stuffs which contain less than 18% crude fiber and less bulky and rich in energy yielding nutrients. These are low in fiber contents and high in energy (1.5-2.2 Mcal/kg of DM). Example: Cereal grains, oil seeds.

Health:
Health is a state of an individual living in complete harmony with his environment/ surroundings.

Disease:
It is a condition in which an individual shows an anatomical, chemical or physiological deviation from the normal.

Illness:
Illness is the reaction of an individual to disease in the form of illness.

Hemostasis:
It is the mechanism by which body keeps equilibrium between health and disease.

Pathology:
It is the study of the anatomical, chemical and physiological alteration from the normal as a result of disease in animals.

Oncology:
It is the study of cancer/tumor/neoplasms.

Etiology:
It is the study of causation of disease.

Diagnosis:
It is an art of precisely knowing the cause of a particular disease

Symptoms:
Any subjective evidence of disease of animal characterized by an indication of altered bodily or mental state as told by owner (complaints of the patients).

Signs:
These are indication of the existence of something, any objective evidence of disease, perceptible to veterinarian.

Syndrome:
A combination of symptoms caused by altered physiological process involve a no. of causative agents.

Lesion:
It is a pathological alteration in structure/ function that can be detectable.

Pathogenesis:
It is the progressive development of a disease process. It starts with the entry of causal agent in body and ends either with the recovery or death.

Incubation period:
It is the time that elapses between the action of a cause and manifestation of disease.

Course of disease:
It is the duration for which the disease process remains till fate either in the form of recovery or death of animal.

Prognosis:
It is an estimate by a clinician of probable severity or outcome of disease.

Morbidity rate:
It is the percentage/proportion of affected animals out of the total population in a particular disease outbreak.

Mortality rate:
It is the percentage/proportion of animals out of total population, died due to disease in a particular disease outbreak.

Case fatality rate:
It is the percentage/proportion of animals died among the affected animals.

Biospsy:
It is the examination of tissue received from the living animals.

Infection:
It is the invasion of the tissues of the body by pathogenic organisms resulting in the development of a disease.

Infestation:
It is the superficial attack of any parasite/organism on the surface of the body.

Pathogenicity:
It is the capability of an organism for producing a disease.

Virulence:
Virulence is the degree of invasiveness of pathogenic organism.

USE OF CHEMICALS IN VETERINARY PRACTICE:

1. Antibiotics
2. Anthelmentics
3. Disinfectant (Formaldehyde)
4. Sterilization
5. Antiseptic (Chlorine, Alcohol and Iodine)

MOST IMPORTANT PARAMETERS:

- ✓ 1cm = 10mm
- ✓ 1meter = 100cm
- ✓ 1Km = 1000meter
- ✓ 1kg = 1000grams
- ✓ 1gram= 1000milligram(mg)
- ✓ 1Quintal = 100Kg
- ✓ 1Metric ton = 1000Kg
- ✓ 1Pound = 454gm
- ✓ 1litre = 1000ml
- ✓ I00 ml= 1 IU
- ✓ 1kilo litre=1000litres
- ✓ 1Gallon = 3.79litres
- ✓ 1Barrel oil= 163.65litres
- ✓ 1cusec = 1 cubic feet of water flows through a point in one second(this measure used only for flowing water)
- ✓ 1TMC = 100 crore cubic feet water(this measure used only for reserved water)
- ✓ 1 inch = 2.54cm
- ✓ 1 feet =12 inch= 30.48cm
- ✓ 1Yard(????)= 3feet
- ✓ 1mile = 1.609 km
- ✓ 1Natical mile=1.852km
- ✓ 1 cent= 435.6 sq feet
- ✓ 1acre = 100 cents
- ✓ 1Hectare = 2.471 acres
- ✓ 1kilo byte(KB)=1024bytes
- ✓ 1Megabyte(MB)=1024KB
- ✓ 1Gigabyte(GB)=1024MB
- ✓ 1Tera Byte (TB)=1024GB
- ✓ 1Million=10 Lakhs
- ✓ 1Billion=1000 Million= 100 crore
- ✓ 1 Trillion= 1000 Billion=1Lakh crore
- ✓ 1 Karat = 4.16 '/. gold
- ✓ 24 karat gold=99.5 '/. gold(pure gold)
- ✓ 22 karat gold= 91.6 '/. gold + 8.4 '/. other metal Normally this 22 Karat gold is called as 916, KDM, HALL MARK
- ✓ 18 Karat gold=75 '/. gold+25 '/. other metal
- ✓ 12 karat gold = 50 '/. gold+ 50 '/. other metal
- ✓ 1 Ream = 500 papers
- ✓ 1gross = 12 dozens = 144 articles.
- ✓ 1 Gunta = 121 Sq yards.
- ✓ 1 Gunta = 101.171 Sq Meter.
- ✓ 1 Gaj = 1 Yard
- ✓ 1 Yard = 36 inch
- ✓ 1 Yard = 3 feet

- ✓ 1 Yard = 0.9144 meter =1 mtr.
- ✓ 1 sq Yard = 0.83612 Sq meter.
- ✓ 1 sq Yard = 9 sq feet.
- ✓ 1 Sq yard = 1296 Sq inch.
- ✓ 1 Meter = 1.0936 Yards.
- ✓ 1 Meter = 39.370 inch.
- ✓ 1 Meter = 3.280 feet.
- ✓ 1 Sq meter = 1.1959 Sq yard.
- ✓ 1 Sq meter = 1550 Sq inch.
- ✓ 1 Sq Meter = 10.763 Sq feet.
- ✓ 1 feet = 0.304 meter.
- ✓ 1 feet = 0.333 yards.
- ✓ 1 feet =12 inch
- ✓ 1 Sq feet = 0.111 Sq Yard.
- ✓ 1 Sq feet = 0.09290 Sq Meter.
- ✓ 1 Sq feet = 144 Sq inch.
- ✓ 1 inch = 2.54 vv
- ✓ 1 Inch = 0.0254 meter.
- ✓ 1 Inch = 0.0277 yards.
- ✓ 1 Inch = 0.0833 feet.
- ✓ 1 Sq Inch = 0.00064516 Sq Meter.
- ✓ 1 Sq Inch = 0.00077160 Sq Yards.
- ✓ 1 Sq Inch = 0.00694444 Sq feet.
- ✓ 1 Acre = 4046.86 Sq Meter.
- ✓ 1 Acre = 4840 Sq yards.
- ✓ 1 Acre = 43560 Sq feet

COMMON FALSE MANAGEMENT ASSOCIATED DISORDERS

1. Anaemia - Deficiency of haemoglobin in the blood
2. Analgesic - Medicine which alleviates pain
3. Arthralgia - Pain in a joint
4. Cephalalgia - Headache
5. Nephralgia - Pain in the kidney
6. Neuralgia - Nerve pain
7. Myalgia - Muscle pain
8. Otalgia - Ear ache
9. Gastralgia - Pain in the stomach
10. Pyoderma - Skin infection with pus formation
11. Leucoderma - Defective skin pigmentaion
12. Hysterodynia - Pain in the uterus
13. Hysterectomy - Excision of the uterus
14. Nephrectomy - Excision of a kidney
15. Adenectomy - Excision of a gland
16. Cholecystectomy - Excision of gall bladder
17. Thyroidectomy - Excision of thyroid gland
18. Arthritis - Inflammation of a joint
19. Bronchitis - Inflammation of the bronchi
20. Carditis - Inflammation of the heart
21. Cervicitis - Inflammation of the cervix
22. Colitis - Inflammation of the colon
23. Colpitis - Inflammation of the vagina
24. Cystitis - Inflammation of the urinary bladder
25. Enteritis - Inflammation of the intestines
26. Gastritis - Inflammation of the stomach
27. Glossitis - Inflammation of the tongue
28. Hepatitis - Inflammation of the liver
29. Laryngitis - Inflammation of the larynx
30. Metritis - Inflammation of the uterus
31. Myelitis - Inflammation of the spinal cord
32. Nephritis - Inflammation of the kidney
33. Pharyngitis - Inflammation of the pharynx
34. Blepharitis - Inflammation of the eyelids
35. Cholelithiasis - Stone in the gall bladder
36. Nephrolithiasis - Stone in the kidney
37. Osteomalacia - Softening of bones through deficiency of calcium or D vitamin
38. Adenoma -Benign tumour of glandular tissue
39. Myoma - Tumour of muscle
40. Diplopia - Double vision
41. Thrombosis - Formation of a blood clot
42. Pyloromyotomy - Incision of pyloric sphincter muscle
43. Hedrophobia - Fear of water(Rabies in humans)
44. Neuroplasty - Surgical repair of nerves

45. Pyloraplasty - Incision of plastic pylorus to widen passage
46. Hemiplegia - Paralysis of one side of the body
47. Nephroptosis - Downward displacement of the kidney
48. Amenorrhoea - Absence of menstrual discharge
49. Dysmenorrhoea - Painful menstruation
50. Leucorrhoea - Whitish vaginal discharge
51. Menorrhoea - Menstrual bleeding
52. Haemorrhage - Escape of blood from a vessel
53. Arrhythmia - Any deviation of normal rhythm of heart
54. Cholestasis - Diminution in the flow of bile
55. Haemostatis - Arrest of bleeding
56. Neurasthenia - Nervous debility
57. Cystostomy - Surgical opening made into the bladder
58. Cystotomy - Incision into the urinary bladder
59. Hypertrophy - Increase in the size of tissues
60. Haematuria - Blood in the urine
61. Glycosuria - Presence of sugar in the urine
62. Albuminuria - Albumin in Urine

SOME MEDICAL FACTS:

1. Largest artery — Aorta
2. Ph of Blood — 7.4
3. Number of Pumps in Heart — 2
4. Largest Organ — Skin
5. Largest gland — Liver
6. Biggest cell — female Ovum
7. Smallest cell — male Sperm
8. Smallest Bone — Stapes
9. First transplanted Organ —kidney
10. Life Span of RBC — 120 days
11. Life Span of WBC — 10 to 15 days
12. Largest Endocrine gland — Thyroid
13. Largest Lymphatic Organ — Spleen
14. Largest part of Brain — Cerebrum
15. Largest & Strongest Bone — Femur
16. Smallest Muscle — Stapedius (Middle Ear)
17. Viscosity of Blood — 4.5 to 5.5
18. Universal Donor Blood Group — O
19. Universal Recipient Blood Group — AB
20. Largest WBC — Monocyte
21. Smallest WBC — Lymphocyte
22. Increase RBC count called — Polycethemia
23. Blood Bank in the Body is — Spleen
24. Non Nucleated Blood cell is — RBC
25. RBC produced in the — Bone Marrow
26. River of Life is Called — Blood
27. Fluid part of Blood is — Plasma

PRESCRPTION WRITING SHORTS:

- ✓ Rx = Treatment
- ✓ Hx = History
- ✓ Dx = Diagnosis
- ✓ q = Every
- ✓ qd = Every day
- ✓ qod = Every other day
- ✓ qh = Every Hour
- ✓ S = without
- ✓ SS = One & half
- ✓ C = With
- ✓ SOS = If needed
- ✓ AC = Before Meals
- ✓ PC = After meals
- ✓ BID = Twice a Day
- ✓ TID = Thrice a Day
- ✓ QID = Four times a day
- ✓ OD = Once a Day
- ✓ BT = Bed Time
- ✓ hs = Bed Time
- ✓ BBF = Before Breakfast
- ✓ BD = Before Dinner
- ✓ Tw = Twice a week
- ✓ SQ = sub cutaneous
- ✓ IM = Intramuscular
- ✓ ID = Intradermal
- ✓ IV = Intravenous

Important medical abbreviations

Abbreviations	Definition	Example
SID	Once a day	Give medication SID
BID	Twice a day	Apply to wound BID
TID	Three times a day	Feed 1/2 cup TID
QID	Four times a day	Apply one drop to right eye QID
PRN	As needed	Apply topical medication to affected area PRN
Ad lib	As much as desired	Provide access to water ad lib
NPO	Nothing by mouth	Patient must be NPO 12 hours prior to surgery
ID	Intradermal	Administer injection ID
IM	Intramuscular	Administer injection IM
IO	Intraosseous	Administer fluids IO
IP	Intraperitoneal	Administer medication IP
SC or SQ	Subcutaneous	Administer medication SQ

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vettechprobs
hx – history
tx – treatment
sx – surgery
ROBO – rolled over by owner
BAR – bright, alert, responsive
WNL – within normal limits
NSF – no significant findings
ADR - Ain't doin' right
HBC – hit by car
NPO – nothing per os
ART – achieving room temp
HWT – heartworm test
HWP – heartworm prevention
N/C – new client or no charge
TGH – to go home
RTG – ready to go
DMITO - dog more intelligent than owner
OWB - (Written on patient's records) Owner will bite
WPWW - Will Poop Without Warning
TTJ- transfer to jesus
Princess or precious = sos or spawn of Satan
SBI- something bad inside
Sunshine club - difficult clients
M.A.I - money an issue
AG - anal gland expression
TNT - nail trim
Snow White - a female client with LOTS of small children with her

KNOWLEDGE ABOUT BLOOD:

1. Which is known as ‘River of Life’?
Answer: Blood
2. Blood circulation was discovered by?
Answer: William Harvey
3. The total blood volume in an adult?
Answer: 5-6 Litres
4. The pH value of Human blood?
Answer: 7.35-7.45
5. The normal blood cholesterol level?
Answer: 150-250 mg/100 ml
6. The fluid part of blood?
Answer: Plasma
7. Plasma protein fibrinogen has an active role in?
Answer: Clotting of blood
8. Plasma protein globulins functions as?
Answer: Antibodies
9. Plasma proteins maintain the blood pH?
Answer: Albumins
10. Biconcave discs shaped blood cell?
Answer: RBC (Erythrocytes)
11. Non nucleated blood cell?
Answer: RBC (Erythrocytes)
12. Respiratory pigments present in RBC?
Answer: Haemoglobin
13. Red pigment present in RBC?
Answer: Haemoglobin
14. RBC produced in the?
Answer: Bone marrow
15. Iron containing pigment of Haemoglobin?
Answer: Haem
16. Protein containing pigment of Haemoglobin?
Answer: Globin
17. Graveyard of RBC?
Answer: Spleen
18. Blood bank in the body?
Answer: Spleen
19. Life span of RBC?
Answer: 120 Days
20. Total count is measured by an instrument known as?
Answer: Haemocytometer
21. A decrease in RBC count is known as?
Answer: Anemia
22. An increase in RBC count is known as?
Answer: Polycythemia

23. A high concentration of bilirubin in the blood causes?
Answer: Jaundice
24. The disease resistant blood cell?
Answer: WBC (leucocytes)
25. Which WBC is known as soldiers of the body?
Answer: Neutrophils
26. Largest WBC?
Answer: Monocyes
27. Smallest WBC?
Answer: Lymphocytes
28. Antibodies producing WBC?
Answer: Lymphocytes
29. Life span of WBC?
Answer: 10-15 days
30. Blood cell performs an important role in blood clotting?
Answer: Thrombocytes (Platelets)
31. Vessels is called?
Answer: Thrombus
32. Anticoagulant present in Blood?
Answer: Heparin
33. A hereditary bleeding disease?
Answer: Haemophilia
34. Bleeder's disease?
Answer: Haemophilia
35. Christmas disease?
Answer: Haemophilia
36. A type of Anemia with sickle shaped RBC?
Answer: Sickle cell anemia
37. Viscosity of Blood?
Answer: 4.5 to 5.5
38. Instrument used to measure haemoglobin?
Answer: Haemoglobinometer
39. Who demonstrated blood groups?
Answer: Karl Landsteiner
40. Who demonstrated Rh factor?
Answer: Karl Landsteiner
41. Blood group which is called Universal donor?
Answer: O
42. Blood group which is called Universal recipient?
Answer: AB

PREFIX AND SUFFIX AND COMPOUND WORDS:

Prefix - Meaning:

1. Adeno - Glandular
2. An - Not
3. Anti - Against
4. Aorto - Aorta
5. Artho - joint
6. Bleph - Eyelid
7. Broncho - Bronchi
8. Cardio - Heart
9. Cephal - Head
10. Cerebro - Brain
11. Cervico - Cervix
12. Cholecysto - Gall Bladder
13. Coli - Bowel
14. Colpo - Vagina
15. Entero - Intestine
16. Gastro - Stomach
17. Glosso - Tongue
18. Haema - Blood
19. Hepa - Liver
20. Hystero - Uterus
21. Laryngo - Larynx
22. Leuco - White
23. Metro - Uterus
24. Myelo - Spinal cord
25. Myo - Muscle
26. Nephro - Kidney
27. Neuro - Nerve
28. Odonto - Tooth
29. Orchido - Testis
30. Osteo - Bone
31. Oto - Ear
32. Pharyngo - Pharynx
33. Pio - Pus
34. Pneumo - Lung
35. Ren - Kidney
36. Rhin - Nose
37. Spleno - Spleen
38. Thyro - Thyroid Gland
39. Urethro - Urethra
40. Vesico – Bladder

Suffix - Meaning:

1. -aemia : Blood
2. -algia : Pain
3. -derm : skin
4. -dynia : pain
5. -ectomy : removal
6. -Itis : inflammation
7. -lithiasis : Presence of Stone
8. -malacia : softening
9. -oma : tumour
10. -opia : eye
11. -osis : Condition,excess
12. -otomy : incision of
13. -phobia : fear
14. -plasty : surgery
15. -plegia : peralysis
16. -ptosis : falling
17. -rhoea : excessive discharge
18. -rhage : to burst forth
19. -rhythmia : rhythm.
20. -stasis : stoppage of movement
21. -sthenia : weakness
22. -stomy : outlet
23. -tomy : removal
24. -trophy : nourishment
25. -uria : urine

COMPOUNDED WORDS - MEANING:

1. Anaemia - Deficiency of haemoglobin in the blood
2. Analgesic - Medicine which alleviates pain
3. Arthralgia - Pain in a joint
4. Cephalalgia - Headache
5. Nephralgia - Pain in the kidney
6. Neuralgia - Nerve pain
7. Myalgia - Muscle pain
8. Otalgia - Ear ache
9. Gastralgia - Pain in the stomach
10. Pyoderma - Skin infection with pus formation
11. Leucoderma - Defective skin pigmentaion
12. Hysterodynia - Pain in the uterus
13. Hysterectomy - Excision of the uterus
14. Nephrectomy - Excision of a kidney
15. Adenectomy - Excision of a gland
16. Cholecystectomy - Excision of gall bladder
17. Thyroidectomy - Excision of thyroid gland
18. Arthritis - Inflammation of a joint
19. Bronchitis - Inflammation of the bronchi
20. Carditis - Inflammation of the heart
21. Cervicitis - Inflammation of the cervix
22. Colitis - Inflammation of the colon
23. Colpitis - Inflammation of the vagina
24. Cystitis - Inflammation of the urinary bladder
25. Enteritis - Inflammation of the intestines
26. Gastritis - Inflammation of the stomach
27. Glossitis - Inflammation of the tongue
28. Hepatitis - Inflammation of the liver
29. Laryngitis - Inflammation of the larynx
30. Metritis - Inflammation of the uterus
31. Myelitis - Inflammation of the spinal cord
32. Nephritis - Inflammation of the kidney
33. Pharyngitis - Inflammation of the pharynx
34. Blepharitis - Inflammation of the eyelids
35. Cholelithiasis - Stone in the gall bladder
36. Nephrolithiasis - Stone in the kidney
37. Osteomalacia - Softening of bones through deficiency of calcium or D vitamin
38. Adenoma -Benign tumour of glandular tissue
39. Myoma - Tumour of muscle
40. Diplopia - Double vision
41. Thrombosis - Formation of a blood clot
42. Pyloromyotomy - Incision of pyloric sphincter muscle
43. Hedrophobia - Fear of water(Rabies in humans)
44. Neuroplasty - Surgical repair of nerves

45. Pyloraplasty - Incision of plastic pylorus to widen passage
46. Hemiplegia - Paralysis of one side of the body
47. Nephroptosis - Downward displacement of the kidney
48. Amenorrhoea - Absence of menstrual discharge
49. Dysmenorrhoea - Painful menstruation
50. Leucorrhoea - Whitish vaginal discharge
51. Menorrhoea - Menstrual bleeding
52. Haemorrhage - Escape of blood from a vessel
53. Arrhythmia - Any deviation of normal rhythm of heart
54. Cholestasis - Diminution in the flow of bile
55. Haemostatis - Arrest of bleeding
56. Neurasthenia - Nervous debility
57. Cystostomy - Surgical opening made into the bladder
58. Cystotomy - Incision into the urinary bladder
59. Hypertrophy - Increase in the size of tissues
60. Haematuria - Blood in the urine
61. Glycosuria - Presence of sugar in the urine
62. Albuminuria - Presence of albumin in the urine

BACTERICIDAL AND BACTERIOSTATIC:

Bactericidal	Bacteriostatic
Penicillin	Tetracycline
Cephalosporin	Chloramphenicol
Aminoglycosides	Macrolides
Trimethoprim/ Sulphonamide	Lincosamides
Nitro furans	Spectinomyce
Metronidazole	Sulfonamides
Quinolones	

ANTIBIOTICS DRUGS:

1- **Aminoglycosides**
1 - Amikacin
2 - Gentamiin
3 - Neomycin
4 - Strepyomycin
5 - Tobramycin

2 - **Cephalosporins**
1 - Cefaclor
2 - Cefadroxil
3 - Cefazolin
4 - Cefdinir
5 - Cefepime
6 - Cefotaxime
7 - Cefotetan
8 - Cefoxitin
9 - Cefpodoxime
10 - Ceftazidime
11 - Ceftibuten
12 - Ceftriaone
13 - Cefuroxime
14 - Cephalexin

3 - **Fluoroquinolones.**
1 - Ciprofloxacin
2 - Gemifloxacin
3 - Levofloxacin
4 - Moxifloxacin
5 - Norfloxacin
6 - Ofloxacin.

4 - **Macrolides**

1 - Aziothromycin
2 - Clarithromycin
3 - Erythromycin

5 - **Lincosamides**
1 - Clindamycin
2 - Lincomycin
3 - Vencomycin

6 - **Monobactam**
1- Aztreonam

7 - **Penicllins**
1 - Amoxicillin
2 - Ampicillin
3 - Penicillin G
4 - Penicillin V
5 - Piparcillin

8 - **Penicillinase -Resistant Penicillin**
1 - Dicloxacillin
2 - Nafecillin
3 - Oxacillin

9 - **Sulfonsmides**
1 - Sulfamethoxazole
2 - Sulfadiazine
3 - Sulfasalazine
4 - Sulfisoxazole
5 - Trimethprim/ Sulfamethoaxzole

10 - **Tetracyclines**
1 - Demeclocycline
2 - Doxycycline
3 - Minocycline
4 - Tetracycline

11- **Antimycobacterials.**
1 - Anti tuberculosis Agents
2 - Leprostatics : Clofzimine And Thalidomide.

12 - **Antifungal Medication**
1 - Amphoteracin B
2 - Fluconazole
3 - Itraconazole
4 - Ketoconazole

5 - Voriconaxole

13 - **Antiviral Medication**
1 - Acyclovir
2 - Foscarnet
3 - Valacyclovir

STUDY OF SUBJECT:

- ✓ Study of blood = Haematology
- ✓ Study of liver= Hepatology
- ✓ Study of fungi = Mycology
- ✓ Study of Algae= Phycology
- ✓ Study of virus= Virology
- ✓ Study of Kidney= Nephrology
- ✓ Study of Cancers = Oncology
- ✓ Study of Universe= Cosmology
- ✓ Study of Fruits= Pomology
- ✓ Study of birds= Ornithology
- ✓ Study of bones= Osteology
- ✓ Study of Egg= Oology
- ✓ Study of Dream= Oneirology
- ✓ Study of Hair= Trichology
- ✓ Study of Eyes= Opthalmology
- ✓ Study of Soil= Pedology
- ✓ Study of Languages= Philology
- ✓ Study of brain= Encephlology
- ✓ Study of Nails= Cosmetology
- ✓ Study of Air= Aerology
- ✓ Study of Earth= Geology
- ✓ Study of fish = Ichthyology
- ✓ Study of insects = Entomology
- ✓ Study of Heart= Cardiology
- ✓ Study of Skin= Dermatologist

MEDICAL TERMINOLOGY:

1. Hyperorexia- Increased appetite
2. Polyorexia- Increased food intake
3. Inappetance- Partial absence of appetite
4. Anorexia- Complete absence of appetite
5. Anophagia- Decreased food intake
6. Allotriophagia- Abnormal appetite
7. Osteophagia- Chewing of bone
8. Infantophagia- Eating of young
9. Coprophagia- Eating of feces

LIVESTOCK LAWS:

- ✓ Punjab Animals Slaughter Control Act 2016
- ✓ The Punjab Animals Feed Stuff and Compound Feed Act 2016
- ✓ Punjab Animals Slaughter Control Acts 1963
- ✓ The Cattle Trespass Act, 1871
- ✓ The Dourine Act, 1910
- ✓ The Glanders and Farcy Act 1899
- ✓ The Prevention of Cruelty of Animals Act 1890
- ✓ The Punjab Livestock Breeding Act 2014
- ✓ The Punjab Milk Boards Ordinance 1963
- ✓ Animal Quarantine Ordinance 1979
- ✓ Punjab Halal Development Agency Act 2016
- ✓ Punjab Animals Feed Stuff and Compound Feed Rules 2017
- ✓ Punjab Animal Health Act 2019

START AND STOP CODON:

1. Start Codon:

First codon of mRNA transcript translated by a ribosome. Start codon makes the site at which translation into protein sequence begins.

Examples:

AUG is the codon for Methionine.

2. Stop Codon:

Stop codon signals the termination of translation process of current protein.

Examples:

UAA, UAG, UGA

COMMON PHYSIOLOGICAL VALUES OF VARIOUS DOMESTICATED LIVESTOCK

Particulars	Cattle	Buffalo	Sheep	Goat	Horse
Rectal Temperature (F^o)	101.5	101.0	103.0	103.0	100.5
Normal Pulse Rate/min	50-70	48-65	70-80	70-80	30-40
Normal Resp. Rate/min	20-30	20-28	20-30	20-30	10-16
Age at First Service (mo)	15-18	24-27	8-12	8-12	16-18
Gestation Period (days)	283	304	150	150	340

Particulars	Dog	Cat	Camel	Fowl
Rectal Temperature (F^o)	101-102	101.0	99.5	107.0
Normal Pulse Rate/min	70-130	110-130	30-50	120-160
Normal Resp. Rate/min	15-30	20-30	10-12	15-50
Age at First Service (mo)	6-12	6-15	--	--
Gestation Period (days)	63	56	370	21 (hatch)

Fetlock joint	Metacarpus and proximal phalanx
Pastern joint	Proximal phalanx and middle phalanx
Coffin joint	Middle phalanx and distal phalanx

LIVER BIOPSY

Animal	Site
Horse	Right dorsal half, 11th – 14th ICS
Cattle & Sheep	Upper third right side, 11th ICS
Dog, Cat, Pig	Last 2 ICS close to the costal arch

SITE FOR PULSE RECORDING

Horse	Lower jaw, where the external maxillary artery is continued across the face as facial artery.
Cattle	Coccygeal artery on either side of the under aspect of tail.
Sheep, Goat, Dog, Cat	Femoral artery inside the hind limb, more than the half way between stifle and hip joint
Fowl	Wing artery on the inner aspect of the wing

SUTURE MATERIALS

Category	Example
Absorbable (organic)	Catgut, Fascia lata, Kangaroo tendon, cargile membrane, Amniotic membrane, Polyglycolic acid (Dexon)
Non-absorbable a) Organic b) Inorganic c) Synthetic	 Cotton, Silk, Silk warm gut, Horse hair, Linen, Umbilical tape Metallic, suture wire, steel wire, Wound clips, Pin suture. Nylon, Terelene, Vetafil, Polyester, Surgilene

ABSORBABLE SUTURE MATERIAL

Name	**Obtained from**
Cat gut	Submucosa of small intestine of sheep
Kangaroo tendon	Tendon of tail of Kangaroo
Cargile membrane	Bovine caecum
Collagen	Bovine flexor tendon filament
Amniotic membrane	Foetal amnion

VETERBRAL COLUMN IN DIFFERENT ANIMALS

Species	**Cervical**	**Thoracic**	**Lumber**	**Sacral**	**Coccygeal**
Horse	7	18	6	5	15-20
Cattle	7	13	6	5	18-20
Sheep	7	13	6-7	4	16-18
Hog	7	14-15	6-7	4	20-23
Chicken	14	7	14 (fused)	6	--
Human	7	12	5	5	4

TYPE OF MUSCLES

Sr.	Type of Muscles	Example
1.	Flexor	Biceps brachii for elbow
2.	Extensor	Triceps brachii for elbow
3.	Sphincter striated	Obricularis oculi
4.	Cutaneous muscle	Cutaneous trunchi muscle
5.	Antagonistic	Biceps brachii and brachialis
6.	Synergists	Triceps brachii and Anconeus

SENSATION AND NERVE ENDINGS

Sensation	Receiving nerve endings
Touch	Meissner's corpuscles, Morke's discs
Deep pressure	Vater Pacinian corpuscles
Heat	Corpuscles of Ruffini
Cold	Krause end bulbs
Muscle sense	Neurotendinous organ

TASTE PERCEPTION IN TONGUE

Area of tongue	Type of taste
Base	Bitter
Lateral side	Sour, salt
Tip	Sweet or salt

CARCASS CHARACTERISTICS

Parameters	Horse	Ox
Side length	Unusual	Lesser
Muscular development of hind quarter	Great	Less
Thoracic cavity	Longer	Shorter
Ribs	18 pairs	13 pairs
Superior spinous process of first six vertebrae	Markedly developed	Less developed
Extension of ulna	Up to half the length of radius	Articulates with carpus
Articulation among last three lumber vertebral	Articulates	Do not articulate
Flesh	Dark bluish red, sweet taste, fibrous	Lacks blue tinge

CARCASS CHARACTERISTICS

Features	Sheep	Goat
Back and withers	Round and well fleshed	Sharp, little flesh
Thorax	Barrel shaped	Flattened laterally
Tail	Fairly broad	Thin
Radius	1 1/4 times length of metacarpus	Twice as long as metacarpus
Scapula	Short and broad	Long
Flesh	Pale red and fine	Dark red of coarse with goaty odour

IODINE VALUE OF MEAT

Animal species	Iodine value of meat
Horse	71-86%
Ox	38-46%
Sheep	35-46%
Pig	50-70%
Good Lard	66%

REFRACTIVE INDEX OF MEAT

Animal species	R.I. of meat fat
Horse	53.5
Ox	< 40
Pig	< 51.9

BLOOD AS % OF LIVE BODY WEIGHT

Species	Cattle	Calves	Pigs	Sheep	Lamb
% of blood	3-4	5-6	3-4	4-4.5	3.5-4

TIME OF EGG FORMATION IN THE CHICKEN

Part of egg formed	Site of formation	Time of formation
Yolk	Ovary	7-9 hours
Thick mucin	Infundibulum	15-30 minutes
Albumin	Magnum	2-3 hours
Shell membrane	Isthmus	1.5 hours
Watery solution	Shell gland	3-5 hours
Shell	Shell gland	19-20 hours
Bloom (mucus)	Vagina	1-10 minutes

COMPOSITION OF MILK AND EGG

Nutrients	Milk	Egg
Water (gm)	87	60
Protein (gm)	4	12
Fat (gm)	3.5	4
Calcium (mg)	118	48
Phosphorus (mg)	93	180
Iron (mg)	Trace	2.1
Vit. A (IU)	140	1080
Riboflavin (mg)	0.17	0.27
Niacin (mg)	0.11	0.07

DUNG AND URINE EXCRETION IN DIFFERENT SPECIES

Type of Animals	Dung (kg)	Urine (kg)
Horse	16.10	3.6
Cattle	23.50	9.0
Sheep	1.13	0.60
Pig	2.70	1.5
Poultry	0.04	--

APPEARANCE OF CONJUNCTIVA IN DIFFERENT SPECIES

Animals	Conjunctival colour
Cattle, buffalo, and sheep	Light pink
Horse	Pale pink/ roseate
Pigs	Reddish
Dogs	Pale Pink
Cats	Pale

PERCENT CAPACITY OF RUMINANT STOMACH

Animals	Rumen	Reticulum	Omasum	Abomasum
Cattle and buffalo	55	5-7	26-30	13-14

Sheep and goat	62	11	5	22-23

RUMINANT STOMACH

Part of Stomach	Remarks
Rumen	Largest compartment of stomach, on left side. Dorsal sac is the largest sac of rumen. Lined by glandless stratified squamous epithelium (Turkish towel appearance). Rumen liquor has pH 5.8 to 6.8, also known as Paunch. Temperature inside rumen is 39 C.
Reticulum	Also called Honey comb. Made up of stratified squamous epithelium located immediately behind diaphragm in opposition to heart. Inside temperature is 39 C.
Omasum	Primary function is to remove water by about 50% and to absorb VFAs. (Absent in Camel)
Abomasum	True stomach. Glandular. pH = 2.0. Enzymatic digestive site. It is similar to fundic region of non-ruminant stomach.

PANCREATIC JUICE

Sr.	Parameters	Components
1	Enzyme precursors	Trypsinogen, Chymotrypsinogen
2	Active enzyme	Elastase, Amylase, Lipase
3	Cations	Sodium, Potassium, Calcium
4	pH	7.5-8.0

DIGESTIVE JUICE

Secretion	Source	Target	Contents	Remarks
Saliva	Salivary glands	Mouth	Amylase, salt, mucus water	Break starch into dextrin and maltose
Gastric juice	Gastric glands	Stomach	HCl, Pepsinogen, Renin, Mucus	Milk curdling, Protein protease and peptones
Bile	Liver	Small intestine	Bile pigments, bile salts	Fat emulsification, Neutralize chyme
Pancreatic juice	Pancreas	Small intestine	Lipase, Amylase, Trypsinogen and Chymotrypsin	Fat degradation, starch and dextrin break into maltose, amino acids liberation
Intestinal juice	Duodenal glands and goblet cells	Small intestine	Enterokinase, Peptidase, Maltase, Sucrose, and Lactase	Splits amino acid, maltose, sucrose and lactose

VARIATION IN THE COLOUR OF FECES

Animals	Conditions	Colour
Calves	Unweaned	Yellowish-brown or grey
	Diarrhea	Whitish or yellowish
Adult	Grazing on pasture	Dark green (loose)
Cattle/Buffalo	Stall feeding	Brownish
	Constriction of bile duct	Pale to grey
	Acute acidosis (Grains overload)	Yellowish-brown
	Excess bile	Yellowish
	Poor ration, Ketosis	Brown-black surface
	Occult blood	Dark brown or tarry
	Blood from caudal intestine	Dark red

CLASSES OF CARBOHYDRATES

Class	Examples
1. Monosaccharides	
a) Pentoses ($C_5H_{10}O_5$)	Arabinose, Xylose, Ribose
b) Hexoses ($C_5H_{12}O_6$)	Glucose, Fructose, Galactose, Mannose.
2. Disaccharides ($C_{12}H_{22}O_{11}$)	Sucrose, Maltose, Lactose, Cellobiose
3. Trisaccharides ($C_{18}H_{32}O_{16}$)	Raffinose
4. Polysaccharides	
a) Pentosans ($C_5H_{10}O_{16}$)x	Araban, Xylan
b) Hexosans ($C_6H_{10}O_5$)x	Dextrin, Starch, Cellulose, Insulin, Glycogen
c) Mixed type	Mimicellulose, Pectingums, Mucilages

MINERALS

Macro-Minerals	Micro-Minerals
Calcium	Iron
Phosphorus	Copper
Sodium	Zinc
Potassium	Cobalt
Magnesium	Iodine
Sulphur	Manganese
Chlorine	Selenium Molybdenum Fluorine, Vanadium, Chromium, Silicone

QUARANTINE PERIOD

Disease	Species	Period (OIE guidelines)
Infectious bronchitis	Fowl	28 days
Rabies	Dog, Cat	4 months
Bacterial infection	Zebra	2 weeks
Fowl cholera	Poultry	14 days
Fowl typhoid	Poultry	28 days
African swine fever	Swine	40 days
Newcastle Disease	Birds	21 days
Fowl Plaque	Wild birds	21 days
Chlamydiosis	Pigeon	45 days

DENTAL FORMULA

Species	Deciduous or Temporary 2 x [ICPM/ICPM]	Permanent 2 x [ICPM/ICPM]
Horse	3030 = 24 3030	3133 = 40 (Male) 3133 3033 = 36 (Female) 3033
Cattle, Buffalo, Sheep, Goat	0030 = 20 4030	0033 = 32 4033
Cat	3130 = 26 3120	3131 = 30 3121
Dog	3130 = 28 3130	3142 = 42 3143
Pig	3130 = 28 3130	3143 = 44 3143
Camel	1130 = 26 3120	1133 = 34 3123

MOUTH GAGS

Name	Animals
Varnell's gag	Horse
Butler's gag	Horse
Haussman's gag	Horse
Probang (Wooden) gag	Cattle
Spring gag	Dog

DIFFERENCE BETWEEN SMALL INTESTINE DIARRHEA AND LARGE INTESTINE DIARRHEA

Characteristics	Small Intestine	Large Intestine
Frequency of defecation	No change	Increased
Fecal volume	Increased	Decreased
Urgency	Absent	Present
Tenesmus	Absent	Present
Mucus in feces	Absent	Present
Blood in feces	Dark black (Malena)	Red (Fresh)
Weight loss	May be	Rare

TYPES OF PLACENTA

Sr.	Type of Placenta	Gross Shape	Example
1	Epitheliochorial	Diffuse	Horse, Donkey, Pig
2	Syndesmochorial	Cotyledonary	Cattle, Sheep, Goat
3	Endotheliochorial	Zonary or Discoid	Dog, Cat, Ferret
4	Hemochorial	Zonary or Discoid	Primates
5	Hemoendothelial	Shperoidal or Discoid	Rat, Rabbit

VACCINATION SCHEDULE FOR CATTLE/BUFFALO

Disease	Vaccine	Time for vaccination	Dose rate	Route
Haemorrhagic Septicemia (HS)	HS oil based vaccine (NIAB)	Preferably in May/June	3 ml (L.A) 2 ml (S.A)	SC
Black Quarter	Polyvalent BQ vaccine	March/April	5 ml	SC
Anthrax	Anthrax spore vaccine	August	1 ml SC	SC
Rinderpest	Rinderpest vaccine	1st injection at 6 mo & 2nd at 2 yrs of age	1 ml	SC
FMD	FMD (VRI)	February/March and September/October	1 ml / 100 kg	SC
	FMD (Marial)	At start of winter season	3 ml (L.A) 2 ml (S.A)	SC
Rabies	Rabies	Post exposure	32 ml daily for 14 days	SC or IM

VACCINATION SCHEDULE FOR SHEEP/GOAT

Disease	Vaccine	Time for vaccination	Dose rate
Enterotoxemia	Enterotoxemia	January and July	2-3 ml
Anthrax	Anthrax	February or Rainy season	0.5 ml
Sheep Pox / Goat Pox	Sheep Pox / Goat Pox	March and September	1 ml S/C or 0.5 ml IM
FMD	FMD	February and August	1-3 ml S/C
Pleuro-pneumonia	Pleuro-pneumonia	October/November	1 ml S/C

VACCINATION SCHEDULE FOR POULTRY

A) LAYERS

Vaccine	Age	Route of Administration
Marek's disease	1 day	Intransal/ Intra ocular
Ranikhet disease (F-strain)	4-10 days	Intra ocular / Intra nasal
Gumboro disease	18-21 days	Intra ocular / Drinking water
Infectious bronchitis	24-48 days	Intra ocular / Drinking water
Ranikhet disease (Booster)	5 – 6 weeks	Intra ocular/ Drinking water
Ranikhet (R_2B strain)	8-9 weeks	Subcutaneous
Fowl pox	10-11 weeks	Scratching
Infectious bronchitis (booster)	14-16 weeks	Intra ocular / Drinking water
Fowl pox (Booster)	16-17 weeks	Scratching / Intramuscular
Ranikhet disease (Killed)	18-19 weeks	Intramuscular / Subcutaneous
Ranikhet and Bronchitis (Combined vaccine)	40 wks & above	Drinking water
Gumboro disease	45-50 weeks	Intramuscular / Subcutaneous

B) BROILER

Vaccine	Age	Route of administration
Marek's disease	1 day	Subcutaneous
Ranikhet disease (F-strain)	4-10 days	Intra ocular / Intra nasal
Gumboro disease	18-21 days	Intra ocular / Drinking water
Ranikhet disease (Booster)	30-35 days	Intra ocular / Intra nasal / Drinking water

VACCINATION SCHEDULE FOR DOG

Age	Vaccination
6-8 week	Hexa dog (CD, canine hepatitis, parvovirus, leptospirosis, parainfluenza)
9-12 week	Booster dose of hexa dog
12 + week	Rabies vaccination
13-16 week	Repeat hexa dog + rabies vaccination
Repeat it annually	

VACCINATION SCHEDULE FOR CAT

Age	Vaccination
8 week	Falovac® (Feline rhinotrachitis, feline panleukopenia, feline calcivirus)
12 week	Repeat Falovac®
16 week	Rabies + booster of Falovac®

POULTRY DISEASES:

Disease	Etiology	Signs	Postmortem	Diagnosis	Treatment	Vaccine/Control
New Castle (ND)	Paramyxo Type-I	Greenish Diarrhea Gasping Torticollis	Haemorrhages on tip of Proventriculus and Cecal tonsils	Signs Postmortem ELISA H.I	**ND colon** 1cc/2-6 L H_2O **Aspirin** 1cc/1L H_2O **Novacoc** 1cc/2-6L H_2O	NDV 100 ml **1st week** eye drop **3rd week** 0.5cc inj. **10th week** 1cc inj. **Every 2 month** 1cc inj.
Infectious Bronchitis (I.B)	Corona Virus	Urate Plug Gasping Tracheal Rales Misshapen egg	Exudate in trachea and nasal cavity Caseous plug in bronchi Kidney pale swollen Salpingitis	Postmortem ELISA H.I AGPT **DDx** ND ILT A.I	**E-sel** 1cc/6L H_2O **Aspirin** 1cc/1L H_2O **Novacoc** 1cc/2-6L H_2O **Flushing** to remove pug	Avoid Overcrowding Biosecurity Vaccine
Avian Influenza (A.I)	Orthomyxo virus type A	Cyanosis of wattle and comb Ruffled feather Edema of head Diarrhea	Pin point haemorrhages on base of proventriculus Shank and on All viscera	Signs Postmortem **DDx** ND ILT I.B Mycoplasma Coryza	No treatment Supportive	Biosecurity Strict Quarantine All in and all out Vaccine
Infectious Laryngo-tracheitis (ILT)	Herpes virus	Sneezing Dyspnea Blood tinge nasal discharge Conjunctivitis	Tracheitis Blood in trachea Conjunctivitis	No response to treatment Blood tinge nasal discharge	No treatment Supportive	Biosecurity Strict Quarantine All in and all out Vaccine
CRD CCRD	Mycoplasma Myco + E.coli	Tracheal rale Drooling Sneezing Emaciation Slow FCR Lameness	Tracheitis Exudate in trachea Yellow colour cheesy flakes around lungs Salpingitis Synovitis Pericarditis Peri-hepatitis	Signs Postmortem	Tilmycosin 260mg/ml 1 ml / 12.2 kg	Purchase of uninfected chicks All in and all out Biosecurity Preventative medication
Infectious bursal disease Gumboro	Birna virus	Young birds Sudden mortality at 3-7 days Swollen/ atrophy of bursa	Haemorrhages on thigh & musculature Atrophy of bursa Swollen, pale kidney	Signs Postmortem	No treatment Supportive	Control Stress Control dehydration Feed dilution Multi vitamins Flushing
Marek's Disease	Herpes virus	Tumor on feather Blindness Grey iris Paralysis	Grey white neoplastic lesions on liver spleen	Signs Postmortem	No treatment Supportive	Vaccine At 0 day in hatchery
Infectious Chicken Anemia	Gyro virus Circoviridae	Poor growth Pale birds Sudden mortality at 13-16 days	Pale bone marrow Atrophy of thymus & bursa Discoloured liver & kidney	Signs Postmortem	No treatment Supportive	Vaccine Live in breeder flock 6 weeks before incubation of eggs

Infectious Coryza	Haemophilus paragallinarum	Facial swelling Upper respiratory signs Swollen wattle Purulent ocular/nasal discharge	Catarrhal inflame. Of nasal sinus Conjunctivitis Haemorrhagic trachea	Signs Postmortem	Tylosin 10-20mg/kg Tylo + Doxy 10 : 20 1 g/2L H_2O	Avoid stress Prophylactically Tylo + Doxy 10 : 20 1 g/2L H_2O
Fowl Cholera	Pasteurella multocida	Swelling of neck Diarrhea Nasal discharge Coughing Swollen joints	Focal hepatitis Star shape white foci on liver Enteritis	Signs Postmortem **DDx** Strept aurius Coryza Mycoplasma	Sulpha drugs Penicillin Norfloxacin	Biosecurity Strict Quarantine All in and all out Avoid Rodents
Typhoid	Salmonella gallinarum Affect any age	Ruffled feathers Thirst Yellow diarrhea Reluctance to move.	Pale green enlarged liver with small necrotic foci& congestion Engorgement of kidney &spleen Enteritis of anterior small intestine	Signs Postmortem	Florfenicol 1cc/2 L H2O	Biosecurity Strict Quarantine All in and all out
Pullorum Disease	Salmonella pullorum	Whitish diarrhea White fecal pasting on vent	Grey nodules in the liver, spleen, lungs, heart, gizzard, and intestine. Firm, cheesy material in the ceca.	Signs Postmortem	Florfenicol 1cc/2 L H_2O	Biosecurity Strict Quarantine All in and all out
Necrotic Enteritis	Clostridium perfringens	Closed eyes Immobility Golden yellowish diarrhoea	Small intestine (usually middle to distal) thickened and distended Golden yellow intestine	Signs Postmortem	**Co-Amoxy clave** Clavulanic acid + Amoxicillin	Penicillin in feed Probiotics Biosecurity Strict Quarantine All in and all out
Aspergillosis	Aspergillus fumigatus	Silent gasping Nervous signs (rare) Ocular discharge	Yellow to grey nodules or plaques in lungs, air sacs, trachea, peritoneal cavity, may have greenish surface. Conjunctivitis/keratitis.	Signs Postmortem	Amphotericin B Nystatin 1cc/1 L H_2O	Nystatin in feed Dry Litter Hygiene Good quality feed
HPS (Angara)	Adeno virus	Sudden increase in mortality Yellow mucoid droppings	Excessive straw-coloured fluid in pericardium (up to 10 ml) Enlarged, pale friable liver and kidney Lungs edematous	Signs Postmortem	Flushing Stralka 1cc/1 L 4h x 3 day	Biosecurity Strict Quarantine All in and all out

CAMEL BREEDS AND CHICKEN BREEDS:

E) Camel

Breed Name	Synonym	Utility	Geographic Distribution
Bagri	Booja	Meat, Milk, Draught	Punjab
Bikarai	Mehra, Mereeha	Meat, Milk, Draught	Punjab
Brahvi		Meat, Milk, Draught	Balochistan
Brela	Thalocha	Meat, Milk, Draught	Punjab
Campbelpuri		Meat, Milk, Draught	Punjab
Dhatti	Thari	Meat, Milk, Draught	Sindh
Gaddi		Meat, Milk, Draught	KPK
Gulmani		Meat, Milk, Draught	KPK
Kacchi		Meat, Milk, Draught	Balochistan
Kala-chitta		Meat, Milk, Draught	Punjab
Khader		Meat, Milk, Draught	KPK
Kharai		Meat, Milk, Draught	Sindh
Kharani		Meat, Milk, Draught	Balochistan
Larri	Sindhi	Meat, Milk, Draught	Sindh
Lassi		Meat, Milk, Draught	Balochistan
Makrani		Meat, Milk, Draught	Balochistan
Maya		Meat, Milk, Draught	KPK
Pishin		Meat, Milk, Draught	Balochistan
Rodbari		Meat, Milk, Draught	Balochistan
Sakrai		Meat, Milk, Draught	Sindh
Pak Bactrian		Meat, Milk, Draught	Northern Area

G) Chicken

Breed Name	Synonym	Utility	Geographic Distribution
Aseel		Sport	All over Pakistan
Desi		Egg, Meat	All over Pakistan
Lyallpur Silver Black		Egg, Meat	UAF
Naked Neck	Ring Neck, Ghoni	Egg, Meat	All over Pakistan